# Studies in Algebra and Group Theory
# Harvard University

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# Contents

Lecture 1.

# Set Theory

## 1.1 Ordered Pairs

We begin with a discussion of ordered pairs. Consider a set of numbers $\{1,2,3,4\}$. For any set there is no particular order that triumphs over another, i.e., $\{1,2,3,4\} = \{1,3,2,4\} = \{3,4,2,1\} = \ldots$, an obvious truth. Say we want $\{1,2,3,4\}$ to be in the order of $\{3,2,1,4\}$ for some arbitrary reason; we may then consider the collection,

$$\mathfrak{C} = \{\{3\},\{3,2\},\{3,2,1\},\{3,2,1,4\}\}.$$

It becomes obvious what is happening here: the method by which we get order is dependent on the number of iterations in which a particular element appears in any number of sets in a given collection (in comparison to the other elements in the other sets of the respective collection).

Then consider this process but for a nested collection (a collection within a collection) which contains two sets, namely the arbitrary elements $a, b$ in opposing orders, and notice that it gives us the definition of a 2-tuple (ordered pair),

$$\mathfrak{C}_1 = \{\{a\},\{a,b\}\} \coloneqq (a,b) \text{ and } \mathfrak{C}_2 = \{\{b\},\{b,a\}\} \coloneqq (b,a).$$

**Note:-**

This form of generating or representing ordered pairs is called the Kuratowski of ordered pairs.

This, of course, naturally extends to greater $n$-tuples. There are some obvious consequences of this that will not be gone over here.

The **Cartesian product** is a set of ordered pairs which can be given by taking power sets.

**Definition 1.1.1: Cartesian Product**

The Cartesian product is the cross product of two sets given by

$$A \times B = \{(a,b) \mid a \in A, b \in B\},$$

where, by the Kuratowski encoding,

$$(a,b) \coloneqq \{\{a\},\{a,b\}\} \in \mathcal{P}(\mathcal{P}(A \cup B)).$$

## 1.2 Relations

The review or ordered pairs leads us to our next discussion on relations. A relation in more colloquial terms is a dynamic or characteristic of a given element shared with another given element such that they are arranged in the form of ordered pairs. Think of equality, then think of equality as equating $(2+3, 1+4)$, or perhaps something more interesting, the set of all $(x,y)$ such that $x$ is a man, $y$ is an enby (slay), and $x$ is married to $y$ (many such cases). We can then think of relations as a set of ordered pairs $R \subset A \times B$. Consider the following examples:

1. $R \coloneqq \{(x,y) \in \mathbb{N}^2 \mid y - x > 0\} \iff xRy \equiv y > x$, the relation of greater than;
2. $R \coloneqq \{(a,B) \in A \times \mathcal{P}(A) \mid a \in B\} \iff aRB \equiv a \in B$, the relation of belonging (being an element of);
3. $R \coloneqq \{(c,d) \in \{\text{all cats}\} \times \{\text{all dogs}\} \mid c, d \text{ live in the same house}\}$.

Naturally, there are an infinite number of possible relations, however there is a particular class of these which we are interested in for the time being—that which relates two equivalent elements (the definition of what equivalence means is, for now, notably ambiguous).

If a relation $R$ exists s.t. $xRx$ for any $x \in X$, then we say $R$ is reflexive. If $xRy \iff yRx$, it is symmetric, and if $xRy, yRz \implies xRz$, it is transitive. If a given relation $R$ satisfies all three of these properties (reflexivity, symmetry, transitivity), then we say it is an **equivalence relation**.

**Definition 1.2.1: Equivalence relations**

An equivalence relation, usually denoted $\sim$, is a relation that is reflexive, symmetric, and transitive $\forall(a,b) \in A \times B$.

**Question 1**

Consider each of the three qualifiers (axioms) for the existence of an equivalence relation. Find an explicit relation that has one property but not the other for each of the three axioms.

**Solution**

I apologize for the terse abstraction involved in this first example, but let

$$R \coloneqq \{(1,1),(2,2),(3,3)(1,2),(2,3) \in \{1,2,3\}^2\},$$

notice that $(1,1),(2,2),(3,3)$ is true for $R$, thus reflexivity holds; however, it is quick to see that $(1,2)$ exists in $R$ but $(2,1) \notin R$, thus symmetry cannot hold. For transitivity, notice that we have $(1,2)$ and $(2,3)$, so we can set up the first step, but $(1,3) \notin R$! Thus transitivity fails. This example is a reminder that we need not think of concrete properties that can be applied in front of our eyes (shapes, dogs, even equality!) in order to have or manipulate some sort of mathematical relation. Next, consider a the relation $R \coloneqq \{(a,b) \mid a \neq b\}$. First, $aRa$ is categorically untrue thus reflexivity means nothing to us. Symmetry, however, tells us that if $aRb$ then $bRa$, which is true! Transitivity may seems sneakily true at first, but notice that $aRb, bRa \implies aRa$ is not true (we simply replaced $c$ with another instance of $a$). Finally, transitivity as a singular property is pretty straightforward. Consider $x > y$, notice that $x > x \implies x < x$ is categorically untrue since $x$ can never be greater or less than itself, additionally, $x > y$ never implies $y > x$. For transitivity however, notice that $x > y, y > z \implies x > z$ is true!

Returning to our discussion of relations and partitions we can have this thing called an induced relation. Let $R \coloneqq X/\mathfrak{C}$, we say that if $x, y \in \mathfrak{C}$ and $xX/\mathfrak{C}y$, then $X/\mathfrak{C}$ is the relation induced by the partition $\mathfrak{C}$.

**Proposition 1.2.1** Relationship between equivalence relations, classes, and set partitions

Explicitly the relationship between the two is given as follows: if $R$ is an equivalence relation on $X$, then the set of equivalence classes form a partition of $X$ that induces the relation $R$, and if $\mathfrak{C}$ is a partition of $X$, then the induced relation is an equivalence relation whose set of equivalence classes is exactly $\mathfrak{C}$.

## 1.3 Functions

First I want to remind us of some notation and terminology things (mathematicians are **very** specific):

**Note:-**

To say a function is on $X$ into $Y$, means that the function is from $X$ to $Y$

**Definition 1.3.1: Functions**

{We all know what functions are at some level, but here were are going to be extraordinarily explicit (though you likely have already heard it in this form, regardless). A function on $X$ into $y$ is a relation $(R)$ given by a letter like $f, g, h, \varphi, \ldots$, such that:

1. $\operatorname{dom} f = X$;
2. if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$ (uniqueness of $y$ for each $x$); additionally, $f(x) = y$ is conventional notation (instead of $(x, y) \in f$ or $xfy$);
3. $y$ is called the **value** that $f$ **assumes** at the **argument** $x$; i.e., $f$ **maps** **sends**/ **transforms**/ $x$ to/ into $y$. We often use the terms map, mapping, transformation, correspondence, operator, and function interchangeably.

We denote this correspondence via

$$f : X \to Y.$$

Finally, we note that the set of all functions from $X$ to $Y$ is a subset of the power set $\mathcal{P}(X \times Y)$ and is denoted $Y^X$.

Something to note is that a function is not an active entity despite action-invoking words such as transforms; it merely is. The use of the word function to describe the undefined object that is somehow active in relation to a graph is nevertheless a static set, similar to that of a directory of names that correspond to addresses. Reminder that while the $\operatorname{dom} f = X$, it need not be the case that $\operatorname{ran} f = Y$, instead the range is reserved for the subset of all $y \in Y$ that have a corresponding $x$ such that $f(x) = y$, i.e., is mapped to by some $x$. We call this property being **onto** and we say $f$ maps $X$ **onto** $Y$; more commonly however we say that $f$ is **surjective**—we explicitly write this as $\forall y \in Y, \exists x \in X : f(x) = y$—and we denote this surjectivity via

$$f : X \twoheadrightarrow Y,$$

and if $A \subset X$, we may consider all $y \in Y$ such that there is an $x \in A$ such that $f(x_A) = y$ (here I use $x_A$ as crude shorthand for $f(x)$ s.t. $x \in A$, where Halmos uses the more primitive $f(A)$), the set of those $y \in Y$ is known as the **image** of $f$ under $A$. If it is the case that $X \subset Y$, the function $f$ defined by $f(x) = x$, $\forall x \in X$, we call this the **inclusion map** **embedding**/ **injection**/ or we say $f$ is **one-to-one**; we explicitly write this as $\forall a \neq b$, $f(a) \neq f(b)$. We denote injectivity via

$$f : A \hookrightarrow Y.$$

A special case of injectivity comes when we inject $X \hookrightarrow X$, relation-wise this is equality in $X$, language-wise, we call this the **identity map** on $X$.

**Proposition 1.3.1** Restriction and extensions

Consider a function $f : Y \to Z$ and a set $X \subset Y$. We say that there is a **natural way** of constructing a new function $g : X \to Z$; let $g(x) = f(x), \forall x \in X$, we then say $g$ is a **restriction** of $f$ to $X$ and $f$ is the **extension** of $g$ to $Y$. We denote this as $g = f|X$ where we can express the definition as $(f|X)(x) = f(x) \forall x \in X$ and $\operatorname{ran}(f|X) = f(x_X)$.

There are a few other terminologies that follow from all that has been mentioned so far, and I will go through them rather quickly:

**Definition 1.3.2: Projection**

$f$ (also pr is a **projection** from $X \times Y$ onto $X$ if if $f(x, y) = x$; similarly, if $R = X \times Y$, then instead of being a projection of $R$ onto $X$, it is now the range of the projection $f$ (ran(pr)).

**Definition 1.3.3: Bijection**

Expounding on our notion of inclusion mappings and surjections from earlier, **bijectivity**—one-to-one correspondence-ness—is a function that is both injective and surjective.

Bijections are derived from an equivalence relation $R$ in $X$ with $aRb \iff f(a) = f(b)$ for $a, b \in X$; so given a function/ set $g(y) = \{x \in X \mid f(x) = y, \forall y \in Y\}$, the equivalence relation $R$ tells us that $g(y)$ is an equivalence class of the relation $R$; that is to say, we define the function as $g : Y \twoheadrightarrow X/R$, notice that for $g$, if $c \neq d$, $c, d \in Y$, due to the nature of equivalence classes (remember, all the classes are disjoint), $g(c) \neq g(d)$; notice this is the exact explicit definition of injectivity, thus this is a surjective-injective, i.e., bijective mapping. Although there is no universially agreed on notation for bijection, we shall denote this mapping as $g : Y \leftrightarrow X/R$.

**Definition 1.3.4: Characteristic functions**

If $A \subset X$ the characteristic function of $A$ is the function$\mathcal{X} : X \to 2$ given by

$$\mathcal{X}(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in X - A \end{cases}$$

, however, we may write $\mathcal{X}_A$ to indicate that it depends on $A$. The function that assigns to each $A \subset X$, i.e., $\mathcal{P}(X)$, the characteristic function of $A \in 2^X$ is a bijection, i.e., $f : \mathcal{P}(X) \leftrightarrow 2^X$.

**Question 2**

Concerning bijectivity (point iii.), the examples have all the inclusion maps as one-to-one, however, except in a few trivial special cases, the projections are not. What are these special cases?

Solution

Consider a function's projection as the cannonical map $\gamma : X \to X/R$, the following cases are trivial examples of non-injective projection maps:

1. $X = \emptyset$, vacuously fulfilled (it matters not what $R$ is);
2. $R \coloneqq \{(x, x) \mid x \in X\}$, so every equivalence class is a singleton.

**Question 3**

Prove:

1. $Y^{\emptyset}$ has exactly one element, namely $\emptyset$, regardless of if $Y$ is empty or not.
2. If $X$ is non-empty, then $\emptyset^X$ is empty.

Solution

1. Given that $Y^{\emptyset} \subset \mathcal{P}(\emptyset \times Y)$, and the definition of Cartesian sets has **and** as a qualifier, i.e., $A \times B = \{(a, b) \mid a \in A \textbf{ AND } b \in B\}$, we can see that $\emptyset \times Y$ contains in fact no elements which satisfy this, and thus, it is the empty set. So $Y^{\emptyset} \subset \mathcal{P}(\emptyset) = \{\emptyset\}$, thus the only element that can be in $Y^{\emptyset}$ is the singleton $\{\emptyset\}$. This is a vacuous satisfaction and is non-intuitive; some intuition, perhaps, may be gained but thinking "<u>the only thing nothing can map to is nothing</u>," or by taking the cardinality of these two sets with each other given by $|\emptyset| \cdot |Y| = 0|Y| = 0$ (though this is mainly for intuition).
2. $\emptyset^X$ means that something must map to the emptyset, meaning there must be an element within the emptyset, but this cannot be true! Thus, by definition, the set must be empty.}

## 1.4 Families

Forget literally everything you know about notational convention! Why? Idk, that's just how mathematicians decided to approach this topic—the family (<u>awwwwww</u>). Consider a function $x$ (yes, this is strange) s.t. $x : I \to X$. We call $I$ the <u>**index set**</u>, and thereby each $i \in I$ is an <u>**index**</u>; the range of the function is then called the <u>**indexed set**</u> (why, this is so stupid?!); the function has a special name, being called a <u>**family**</u>, and the value of $x$ at a given $i$ is called a <u>**term**</u> of the family (denoted $x_i$). Other even more bewildering notation may be used such as $\{x_i\}$ being a family in $X$ or, more horrifically, a family $\{A_i\}$ of subsets of $X$, being a function $A : I \to \mathcal{P}(X)$ (now that I think of it, this notation, sickeningly, is becoming somewhat preferable).

Given that $\operatorname{ran}(\{A_i\} = \bigcup_{i \in I} A_i)$, we notice that every arbitrary collection of sets is in fact the range of some arbitrary family: consider $I$ and $X$ equal to $\mathfrak{C}$ s.t. $\{A_i\} : \mathfrak{C} \to \mathfrak{C}$. I am realizing it is the terminology which is so obtuse, that what I do not like.

We may think of $\bigcup_{i \in I} A_i$ as $\bigcup_{i=1}^{n} A_i$ (similar to what we have seen for summations and products), where $n$ is the order of $i$ and we assume 1 is the first element in $i$.

**Example 1.4.1**

For some index set $i = \{1, 2, 3\}$, and an indexed set $A_i$ we have $\bigcup_{i=1}^{3} = A_1 \cup A_2 \cup A_3$. Another example may be given as follows: let $I = \mathbb{N}$, and $A_i = \{-i, 0, i\}$, that is to say for every $i$ we have

$A_1 = \{-1,0,1\}, A_2 = \{-2,0,2\}, \ldots$, and so on. Notice that the infinite union of $A_i$ produces $\mathbb{Z}$ and the infinite intersection produces $\{0\}$.

**Example 1.4.2**

Let us now consider an uncountably infinite indexing set it a plane $I = [-1,1], A_i = \{i\} \times [0,1] \subset \mathbb{R}^2$. Notice this is going to be a subset of ordered pairs, i.e., $\{(i,y) \mid 0 \leqslant y \leqslant 1\}$. Geometrically, the union over all $i \in I$ of $A_i$ gives us a rectangle in $\mathbb{R}^2$ defined by $[-1,1] \times [0,1]$; similarly the intersection of these sets, notice, are intersections of finite parallel lines, thus there exists no element in common with these sets, thus it is $\emptyset$.

**Theorem 1.4.1** Associative law of unions

Let $\{I_j\} : J \to K \mid K = \bigcup_{j \in I}$ and $\{A_k\} : K \to X$, then

$$\bigcup_{k \in K} A_k = \bigcup_{j \in J} \left( \bigcup_{i \in I_j} A_i \right).$$

**Question 4**

Prove the theorem above.

Sorry! I am leaving this one as an exercise for the reader (sorrryyyyyy:) )

De Morgan's laws apply as standard for index/ indexed sets and their unions. From such, it is easy to see that if $\{A_i\}, \{B_i\}$ are families of sets, then $\bigcup_i A_i \cap \bigcup_j B_j = \bigcup_{i,j}(A_i \cap B_j)$, a similarly for $\bigcap_{i,j}(A_i \cup B_j)$.

Let us consider a special type of family: let $I = \{a, b\}$, $a \neq b$ and let $Z$ be the set of all families $z$ indexed by $I$ such that $z_a \in X, z_b \in Y$. Now consider $f : Z \leftrightarrow X \times Y$ such that $f(z) = (z_a, z_b)$. This is the **generalized Cartesian product**! There is further detail one can go into about families of sets and sets of all given families of sets, and even sets of those sets (including their Cartesian products), but these are so disgusting as to induce spontaneous retching in even the most tough-stomached mathematicians. So we will avoid it, if possible.

## 1.5 Inverses and Composites

Consider a function $f : X \to Y$, a natural mapping that might arise (when thinking on power sets) is some function $F : \mathcal{P}(X) \to \mathcal{P}(Y)$; just as each element of $X$ is mapped to an element $Y$ under $f$, each subset of $X$ maps to some subset of $Y$ under $F$, so if $A \subset X$ then $\exists(A \mapsto \mathrm{Im}(f_A))\forall A \in \mathcal{P}(X)$. Cool? I guess? What is to be done about this? In all honesty, I am not sure this factoid has much use other than being a pedagogical tool (though I am almost certainly wrong and I am sure someone will rush to correct me, wherein I might toil and languish over my incorrectness and stupidity, suffering immensely—but probably not, because I simply do not care), but for our cases we use it to introduce its behavior in the inverse.

**Note:-**

Regarding notation—as we all too often are—$\mathrm{Im}(f_A)$ is the same thing as $f(A)$ when regarding images of subsets. I prefer the former notation as it feels a bit more natural to me.

**Definition 1.5.1: Inverses and inverse images**

Given a function $f : X \to Y$ we say $f^{-1}$ is the **inverse** of $f$ given by $f^{-1} : \mathcal{P}(Y) \to \mathcal{P}(X)$ such that if $B \subset Y$, then

$$\mathrm{Im}(f_B^{-1}) = \{x \in X \mid f(x) \in B\},$$

is the **inverse image** of $B$ under $f$

This definition results in some immediate necessary and sufficient conditions: first, for $f : X \to Y$ to exist, the inverse image under $f$ of each non-empty subset of $Y$ must be a non-empty subset of $X$; second, for $f$ to be injective is that the inverse image under $f$ of each singleton in the range of $f$ be a singleton in $X$.

**Note:-**

We often use $f^{-1}$ for another purpose: for some $f : X \to Y$, $f^{-1} : \mathrm{ran}(f) \to X \implies f^{-1}(y) = x \iff f(x) = y$. This will be the most common use case for $f^{-1}$ as the inverse function $g : Y \to X$

There are some connections between images and inverse images which I will quickly enumerate, but I will not provide proof for (for the proofs see Halmos,~pp. 59): Let $f$ be a function such that $f : X \to A$ for each of the following,

1. $B \subset Y \implies f(f^{-1}(B)) \subset B$;
2. $f : X \twoheadrightarrow Y,\ B \subset Y \implies f(f^{-1}(B)) = B$;
3. $A \subset X \implies A \subset f^{-1}(f(A))$;
4. $f : X \hookrightarrow Y,\ A \subset X \implies f^{-1}(f(A)) = A$;
5. If $\{B_i\}$ is a family of subsets of $Y$, then $f^{-1}(\bigcup_{i\in I} B_i) = \bigcup_{i \in I} f^{-1}(B_i)$ and $f^{-1}(\bigcap_{i\in I} B_i) = \bigcap_{i\in I} f^{-1}(B_i)$;
6. $f^{-1}(Y - B) = X - f^{-1}(B)$ for each $B \subset Y$.

We now move onto **composites**.

**Definition 1.5.2: Composition**

Let $f : X \to Y$ and $g : Y \to Z$ it is clear that we can make a map from $X \to Z$ since the range of $f$ can be the domain of $g$. We can represent this new function—i.e., the **composite function**—as either $h : X \to Z$ or $g \circ f$, and often simply just $gf$. We read this right to left.

Notice that function composition is not always commutative, nor need it be. Also notice that this remains true under the special case that $f : X \to Y$ and $g : Y \to X$; functional composition, however, is always associative. There will be a strong proof for this later on (in Groups). Notice that if $g : X \to Y$ for standard $f$, if it is bijective we have an inverse function and identity map given by $h$. Functional composition under inverses appears something like this: if $f : X \to Y$ and $g : Y \to Z$, and $f^{-1} : \mathcal{P}(Y) \to X$ and $g^{-1} : \mathcal{P}(Z) \to \mathcal{P}(Y)$. Then $g \circ f$ has an inverse given by $f^{-1} \circ g^{-1}$. Notice this is not true for $g^{-1} \circ f^{-1}$.

### 1.5.1 Inverse/ converse and composite relations

The main idea of this comes out of our exploration of inverse functions. Functions are relations after all. Consider the **converse relation** of $R$ on $X \times Y$ s.t. $xRy$ and $yR^{-1}x$. Similar things can be done for composites by generalizing composition to relations, observe: let $S$ be a relation on $Y \times Z$ and $R$ be a relation on $X \times Y$ such that $xRy$ and $ySz$. Now let $T$ over $X \times Z$ be given by $S \circ R$ such that $xTz$ exists $\iff \exists y \in Y$ such that $xRy$ and $ySz$. Congrats! You made a $\{$*composite relation*$\}$. If it is the case that $R$ and $S$ are functions s.t. $R(x) = y$, $S(y) = z$ and $S(R(x)) = z \iff xTz$ which implies that functional composition is a special case of a greater abstraction called the $\{$*relative product*$\}$(??).

Lecture 2.

# Group Theory Basics

Groups are the central object of abstract algebra, capturing the essence of symmetry and invertible operations. The definition is deceptively simple—three axioms—yet groups appear everywhere: in geometry (symmetries), number theory (modular arithmetic), topology (fundamental groups), physics (gauge theories), and even chemistry (molecular symmetries). This lecture develops the foundations rigorously.

## 2.1 The group axioms

**Definition 2.1.1: Groups**

A **group** is a set $G$ together with a map

$$m : G \times G \to G$$

called a **law of composition** (also **binary operation**). Writing $ab$ for $m(a, b)$, the law of composition must satisfy three axioms:

1. **Associativity:** $\forall a, b, c \in G$, $(ab)c = a(bc)$.
2. **Identity:** $\exists e \in G$ such that $ea = ae = a$ for all $a \in G$.
3. **Inverses:** $\forall a \in G$, $\exists b \in G$ such that $ab = ba = e$.

We write $(G, m)$ or simply $G$ for the group, and call $|G|$ the **order** of $G$.

The axioms have immediate but important consequences that we now prove carefully.

**Proposition 2.1.1** Uniqueness of identity

The identity element $e$ is unique.

***Proof:*** Uniqueness of identity

Suppose $e$ and $e'$ are both identities. Then $e = ee'$ (since $e'$ is an identity) and $ee' = e'$ (since $e$ is an identity). Thus $e = e'$. □

**Proposition 2.1.2** Uniqueness of inverses

For each $a \in G$, the inverse is unique. We denote it $a^{-1}$ (or $-a$ in additive notation).

***Proof:*** Uniqueness of inverses

Suppose $b$ and $c$ are both inverses of $a$, so $ab = ba = e$ and $ac = ca = e$. Then:

$$b = be = b(ac) = (ba)c = ec = c.$$

□

**Proposition 2.1.3** Cancellation laws

In a group $G$:

1. **Left cancellation:** $ab = ac \implies b = c$.
2. **Right cancellation:** $ba = ca \implies b = c$.

***Proof:*** Proof of cancellation

For left cancellation: if $ab = ac$, multiply both sides on the left by $a^{-1}$: $a^{-1}(ab) = a^{-1}(ac)$, so $(a^{-1}a)b = (a^{-1}a)c$, giving $eb = ec$, i.e., $b = c$. Right cancellation is similar. □

**Proposition 2.1.4** Inverse of products

For all $a, b \in G$: $(ab)^{-1} = b^{-1}a^{-1}$. Note the reversal of order!

***Proof:*** Proof of inverse formula

We verify $(ab)(b^{-1}a^{-1}) = e$: $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e$. Similarly for the other side. □

**Proposition 2.1.5** Inverse of inverse

For all $a \in G$: $(a^{-1})^{-1} = a$.

***Proof:*** Proof of double inverse

By definition, $a^{-1}$ satisfies $a \cdot a^{-1} = e$. This says $a$ is an inverse of $a^{-1}$, so by uniqueness, $(a^{-1})^{-1} = a$. □

**Note:-**

It is customary to use additive notation ($+$, $0$, $-a$) for abelian groups and multiplicative notation ($\cdot$, $1$ or $e$, $a^{-1}$) in general. This does not mean that multiplication implies non-commutativity!

### 2.1.1 Group variants and related structures

Groups sit in a hierarchy of algebraic structures with fewer or more axioms.

**Definition 2.1.2: Abelian groups**

A group $G$ is **abelian** (or **commutative**) if $ab = ba$ for all $a, b \in G$.

**Definition 2.1.3: Monoid**

A **monoid** is a set with an associative binary operation and an identity element, but inverses are not required. Example: $(\mathbb{N}, +)$ with identity $0$, or $(\mathbb{Z}, \times)$ with identity $1$.

**Definition 2.1.4: Semigroup**

A **semigroup** is a set with an associative binary operation—no identity or inverses required. Example: $(\mathbb{Z}_{>0}, +)$.

**Note:-**

There's a pattern here: Group = Monoid + Inverses = Semigroup + Identity + Inverses. Category theory provides a unifying framework: a group is a category with one object where every morphism is invertible; a monoid is a category with one object (morphisms need not be invertible).

## 2.2 Fundamental examples

### 2.2.1 Integers and modular arithmetic

**Example 2.2.1** (The integers)
The set $(\mathbb{Z}, +)$ is an abelian group with identity 0 and inverse $-n$ for each $n$. This is the prototypical infinite cyclic group.

**Example 2.2.2** (Integers mod $n$)
$\mathbb{Z}/n\mathbb{Z} = \{0, 1, 2, \dots, n-1\}$ with addition mod $n$:

$$a + b := \begin{cases} a+b & \text{if } a+b < n \\ a+b-n & \text{if } a+b \geqslant n. \end{cases}$$

Identity is 0, inverse of $a$ is $n - a \pmod{n}$. This is an abelian group of order $n$.

**Proposition 2.2.1** Units mod $n$
The set $(\mathbb{Z}/n\mathbb{Z})^\times = \{a \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}$ forms a group under multiplication mod $n$. Its order is $\varphi(n)$, Euler's totient function.

***Proof:*** Proof that units form a group

**Closure:** If $\gcd(a, n) = \gcd(b, n) = 1$, then $\gcd(ab, n) = 1$ (if a prime $p$ divides both $ab$ and $n$, it divides $a$ or $b$, contradicting coprimality).

**Identity:** 1 is coprime to $n$.

**Inverses:** If $\gcd(a, n) = 1$, then by Bézout's identity, $\exists r, s$ with $ra + sn = 1$. Thus $ra \equiv 1 \pmod{n}$, so $r$ is a multiplicative inverse. □

### 2.2.2 Multiplicative groups of fields

**Example 2.2.3** (Multiplicative groups)
$\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, and $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ are abelian groups under multiplication, with identity 1 and inverse $1/x$.

**Example 2.2.4** (The circle group)
The unit circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is a group under multiplication. It is isomorphic to $\mathbb{R}/\mathbb{Z}$ via $t \mapsto e^{2\pi i t}$. This is a compact abelian group and plays a role in Fourier analysis and representation theory.

**Example 2.2.5** (Roots of unity)
For each $n \geqslant 1$, the $n$th roots of unity $\mu_n = \{z \in \mathbb{C} : z^n = 1\} = \{e^{2\pi i k/n} : k = 0, 1, \dots, n-1\}$ form a cyclic group of order $n$ under multiplication. We have $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$.

### 2.2.3 Symmetric and permutation groups

**Definition 2.2.1: Permutation group**

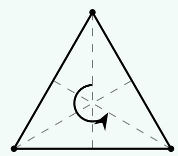
A **permutation** of a set $X$ is a bijection $\sigma : X \to X$. The set of all permutations of $X$ forms a group $\mathrm{Sym}(X)$ under composition, with identity $\mathrm{id}_X$ and inverse $\sigma^{-1}$.

**Definition 2.2.2: Symmetric group**

The **symmetric group** on $n$ letters is $S_n = \mathrm{Sym}(\{1, 2, \ldots, n\})$. It has order $n!$ and is non-abelian for $n \geqslant 3$.

**Example 2.2.6** (Symmetries of a triangle)

$S_3$ can be visualized as symmetries of an equilateral triangle: 3 rotations (including identity) and 3 reflections.



Labeling vertices $1, 2, 3$, rotations give permutations $e, (123), (132)$ and reflections give $(12), (13), (23)$. Thus $|S_3| = 6$.

### 2.2.4 Matrix groups

**Definition 2.2.3: General linear group**

The **general linear group** $\mathrm{GL}_n(\mathbb{F})$ is the group of invertible $n \times n$ matrices over a field $\mathbb{F}$, under matrix multiplication. A matrix is invertible iff its determinant is nonzero.

**Definition 2.2.4: Special linear group**

The **special linear group** $\mathrm{SL}_n(\mathbb{F}) = \{A \in \mathrm{GL}_n(\mathbb{F}) : \det(A) = 1\}$ is a subgroup of $\mathrm{GL}_n(\mathbb{F})$.

**Note:-**

Matrix groups are our first examples of Lie groups—groups that are also smooth manifolds. They are, in my opinion, the most exciting part of representation theory: a representation of $G$ is a homomorphism $\rho : G \to \mathrm{GL}_n(\mathbb{F})$, making $G$ act on $\mathbb{F}^n$ by linear transformations.

**Example 2.2.7** (Non-abelian example)

In $\mathrm{GL}_2(\mathbb{R})$, let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Then:

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = BA.$$

So $\mathrm{GL}_2(\mathbb{R})$ is non-abelian.

## 2.3 Constructions: Products and sums

How do we build new groups from existing ones?

### 2.3.1 Direct products

**Definition 2.3.1: Direct product**

Given groups $G$ and $H$, the **direct product** $G \times H$ is the Cartesian product with componentwise operation:

$$(a, b) \cdot (a', b') = (aa', bb').$$

Identity is $(e_G, e_H)$, inverse of $(a, b)$ is $(a^{-1}, b^{-1})$.

**Proposition 2.3.1** Direct product is a group

The direct product $G \times H$ satisfies all group axioms. Moreover, $|G \times H| = |G| \cdot |H|$ (for finite groups).

***Proof:*** Verification of group axioms for $G \times H$

**Associativity:** $((a, b)(a', b'))(a'', b'') = (aa', bb')(a'', b'') = (aa'a'', bb'b'')$. Similarly, $(a, b)((a', b')(a'', b'')) = (a, b)(a'a'', b'b'') = (aa'a'', bb'b'')$. Equal by associativity in $G$ and $H$.

**Identity:** $(e_G, e_H)(a, b) = (e_G a, e_H b) = (a, b)$, and similarly on the right.

**Inverses:** $(a, b)(a^{-1}, b^{-1}) = (aa^{-1}, bb^{-1}) = (e_G, e_H)$. □

**Example 2.3.1**

We have $\mathbb{Z}/2 \times \mathbb{Z}/3 \cong \mathbb{Z}/6$ (since $\gcd(2, 3) = 1$), but $\mathbb{Z}/2 \times \mathbb{Z}/2 \ncong \mathbb{Z}/4$ (the former has no element of order 4).

This generalizes to arbitrary (even infinite) collections of groups.

**Definition 2.3.2: Arbitrary direct products**

For a family of groups $\{G_i\}_{i \in I}$, the **direct product** is:

$$\prod_{i \in I} G_i = \{(a_i)_{i \in I} : a_i \in G_i\}$$

with componentwise operations.

**Definition 2.3.3: Direct sum**

The **direct sum** (for abelian groups) is the subgroup:

$$\bigoplus_{i \in I} G_i = \{(a_i)_{i \in I} \in \prod_{i \in I} G_i : a_i = e_{G_i} \text{ for all but finitely many } i\}.$$

**Note:-**

For finite index sets, $\prod = \bigoplus$. For infinite sets, they differ crucially. "O-plus" notation will become important when we move onto linear algebra.

**Example 2.3.2** (Power series vs polynomials)

Taking $G_i = (\mathbb{R}, +)$ for all $i \in \mathbb{N}$:

$$\prod_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[[x]] \quad \text{(formal power series under addition)}$$

$$\bigoplus_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[x] \quad \text{(polynomials under addition)}$$

A power series $\sum_{i=0}^{\infty} a_i x^i$ can have infinitely many nonzero terms; a polynomial cannot.

**Question 1**

1. Prove that $\mathbb{Z}/m \times \mathbb{Z}/n \cong \mathbb{Z}/mn$ if and only if $\gcd(m, n) = 1$.
2. Find all groups of order 4 up to isomorphism.
3. Prove that $G \times H \cong H \times G$ for any groups $G, H$.

**Solution**

**(1)** ($\Leftarrow$) Assume $\gcd(m, n) = 1$. The element $(1, 1) \in \mathbb{Z}/m \times \mathbb{Z}/n$ has order $\operatorname{lcm}(m, n) = mn$ (since the order of $(a, b)$ is $\operatorname{lcm}(\operatorname{ord}(a), \operatorname{ord}(b))$). Thus $\mathbb{Z}/m \times \mathbb{Z}/n$ contains an element of order $mn = |G|$, so it's cyclic, hence $\cong \mathbb{Z}/mn$.
($\Rightarrow$) If $d = \gcd(m, n) > 1$, then for any $(a, b) \in \mathbb{Z}/m \times \mathbb{Z}/n$, we have $\frac{mn}{d} \cdot (a, b) = (\frac{mn}{d}a, \frac{mn}{d}b) = (0, 0)$ since $m \mid \frac{mn}{d}$ and $n \mid \frac{mn}{d}$. So every element has order dividing $\frac{mn}{d} < mn$, meaning no element has order $mn$. But $\mathbb{Z}/mn$ has an element of order $mn$, so they're not isomorphic.
**(2)** Let $G$ have order 4. Every element has order dividing 4, so orders are 1, 2, or 4.
Case 1: $G$ has an element $a$ of order 4. Then $G = \{e, a, a^2, a^3\} = \langle a \rangle \cong \mathbb{Z}/4$.
Case 2: Every non-identity element has order 2. Then $G = \{e, a, b, c\}$ with $a^2 = b^2 = c^2 = e$. What is $ab$? It's not $e$ (else $a = b^{-1} = b$), not $a$ (else $b = e$), not $b$ (else $a = e$). So $ab = c$. Similarly $ba = c$ (check: $(ab)(ba) = a(bb)a = a^2 = e$, so $ba = (ab)^{-1} = ab$ since $ab$ has order 2). So $G$ is abelian, and $G \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four-group).
**(3)** Define $\varphi : G \times H \to H \times G$ by $\varphi(g, h) = (h, g)$. This is clearly a bijection. It's a homomorphism: $\varphi((g_1, h_1)(g_2, h_2)) = \varphi(g_1g_2, h_1h_2) = (h_1h_2, g_1g_2) = (h_1, g_1)(h_2, g_2) = \varphi(g_1, h_1)\varphi(g_2, h_2)$.

## 2.4 Subgroups

**Definition 2.4.1: Subgroup**

A **subgroup** of a group $G$ is a nonempty subset $H \subseteq G$ such that:

1. **Closure:** $a, b \in H \implies ab \in H$;
2. **Inverses:** $a \in H \implies a^{-1} \in H$.

We write $H \leqslant G$ (or $H < G$ if $H \neq G$, a **proper subgroup**).

**Proposition 2.4.1** Subgroup criterion
A nonempty subset $H \subseteq G$ is a subgroup iff $a, b \in H \implies ab^{-1} \in H$.

***Proof:*** One-step subgroup test
($\Rightarrow$) If $H \leqslant G$ and $a, b \in H$, then $b^{-1} \in H$ (inverses) and $ab^{-1} \in H$ (closure).

($\Leftarrow$) Assume the condition. Since $H \neq \emptyset$, pick $a \in H$. Then $e = aa^{-1} \in H$. For any $b \in H$, $b^{-1} = eb^{-1} \in H$. For $a, b \in H$, we have $b^{-1} \in H$, so $ab = a(b^{-1})^{-1} \in H$. $\square$

**Example 2.4.1** (Subgroups of $\mathbb{Z}$)

Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z} = \{\dots, -2n, -n, 0, n, 2n, \dots\}$ for some $n \geqslant 0$. These form a chain: $\mathbb{Z} = 1\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z} \supset 8\mathbb{Z} \supset \cdots$ (not all subgroups are comparable, though: $2\mathbb{Z}$ and $3\mathbb{Z}$ are incomparable).

**Theorem 2.4.1** Subgroups of $\mathbb{Z}$

Let $H$ be a subgroup of $(\mathbb{Z}, +)$. Then either $H = \{0\}$ or $H = n\mathbb{Z}$ where $n$ is the smallest positive element of $H$.

***Proof:*** Classification of subgroups of $\mathbb{Z}$

If $H = \{0\}$, we're done. Otherwise, $H$ contains some nonzero element $a$; then $-a \in H$ too, so $H$ contains a positive element. Let $n$ be the smallest positive element of $H$.

We show $H = n\mathbb{Z}$. Clearly $n\mathbb{Z} \subseteq H$ since $H$ is closed under addition: $n \in H \implies 2n = n + n \in H \implies 3n \in H \implies \dots$, and $-n \in H \implies -2n \in H \implies \dots$

For $H \subseteq n\mathbb{Z}$: take any $m \in H$. By division algorithm, $m = qn + r$ with $0 \leqslant r < n$. Then $r = m - qn \in H$ (since $m, qn \in H$). By minimality of $n$, we must have $r = 0$, so $m = qn \in n\mathbb{Z}$. $\square$

**Proposition 2.4.2** Greatest common divisor

Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then:

1. $\mathbb{Z}a + \mathbb{Z}b = \{ra + sb : r, s \in \mathbb{Z}\} = d\mathbb{Z}$;
2. $d$ divides both $a$ and $b$;
3. If $n$ divides both $a$ and $b$, then $n$ divides $d$;
4. $\exists r, s \in \mathbb{Z}$ such that $d = ra + sb$ (Bézout's identity).

***Proof:*** Proof of GCD properties

The set $S = \mathbb{Z}a + \mathbb{Z}b$ is a subgroup of $\mathbb{Z}$ (verify: closed under addition and negation). By the theorem, $S = d\mathbb{Z}$ for some $d \geqslant 0$. Since $a = 1 \cdot a + 0 \cdot b \in S$ and $b \in S$, and $S = d\mathbb{Z}$, we have $d \mid a$ and $d \mid b$. Since $d \in S$, we have $d = ra + sb$ for some $r, s$. If $n \mid a$ and $n \mid b$, then $n \mid (ra + sb) = d$. These properties characterize $d = \gcd(a, b)$. $\square$

**Corollary 2.4.1** Coprimality criterion

$a$ and $b$ are coprime (i.e., $\gcd(a, b) = 1$) iff $\exists r, s \in \mathbb{Z}$ with $ra + sb = 1$.

**Corollary 2.4.2** Euclid's lemma

If $p$ is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

***Proof:*** Proof of Euclid's lemma

Suppose $p \nmid a$. Since $p$ is prime, $\gcd(p, a) = 1$. By Bézout, $\exists r, s$ with $rp + sa = 1$. Multiply by $b$: $rpb + sab = b$. Since $p \mid rpb$ and $p \mid ab$, we have $p \mid b$. $\square$

### 2.4.1 Generated subgroups and cyclic groups

**Definition 2.4.2: Subgroup generated by a set**

For $S \subseteq G$, the **subgroup generated by** $S$, denoted $\langle S \rangle$, is the smallest subgroup of $G$ containing $S$. Explicitly:

$$\langle S \rangle = \{s_1^{\epsilon_1} s_2^{\epsilon_2} \cdots s_k^{\epsilon_k} : k \geqslant 0,\ s_i \in S,\ \epsilon_i \in \{1, -1\}\}.$$

**Definition 2.4.3: Cyclic groups**

A group $G$ is **cyclic** if $G = \langle a \rangle$ for some $a \in G$. The element $a$ is called a **generator**.

**Definition 2.4.4: Order of an element**

The **order** of $a \in G$, denoted $\mathrm{ord}(a)$ or $|a|$, is the smallest positive integer $n$ such that $a^n = e$, or $\infty$ if no such $n$ exists.

**Proposition 2.4.3** Structure of cyclic groups

Let $G = \langle a \rangle$.

1. If $\mathrm{ord}(a) = \infty$, then $G \cong \mathbb{Z}$ via $n \mapsto a^n$.
2. If $\mathrm{ord}(a) = n < \infty$, then $G = \{e, a, a^2, \ldots, a^{n-1}\}$ and $G \cong \mathbb{Z}/n\mathbb{Z}$.

***Proof:*** Proof of cyclic group structure

Define $\varphi : \mathbb{Z} \to G$ by $\varphi(k) = a^k$. This is a homomorphism: $\varphi(k + l) = a^{k+l} = a^k a^l = \varphi(k)\varphi(l)$. It's surjective since $G = \langle a \rangle$.

**Case 1:** $\mathrm{ord}(a) = \infty$. If $\varphi(k) = \varphi(l)$ with $k > l$, then $a^{k-l} = e$, contradicting infinite order. So $\varphi$ is injective, hence an isomorphism $\mathbb{Z} \xrightarrow{\sim} G$.

**Case 2:** $\mathrm{ord}(a) = n$. Then $\ker(\varphi) = \{k \in \mathbb{Z} : a^k = e\}$. Since $a^n = e$, we have $n\mathbb{Z} \subseteq \ker(\varphi)$. Conversely, if $a^k = e$, write $k = qn + r$ with $0 \leqslant r < n$. Then $a^r = a^{k-qn} = a^k(a^n)^{-q} = e$. By minimality of $n$, $r = 0$, so $k \in n\mathbb{Z}$. Thus $\ker(\varphi) = n\mathbb{Z}$, and $G \cong \mathbb{Z}/n\mathbb{Z}$ by the First Isomorphism Theorem (this will be introduced later but to oversimplify: surjectivity is guaranteed iff the kernal only contains the identity). $\square$

**Proposition 2.4.4** Subgroups of cyclic groups

Every subgroup of a cyclic group is cyclic. If $G = \langle a \rangle$ has order $n$, the subgroups of $G$ are exactly $\langle a^d \rangle$ for divisors $d \mid n$, and $|\langle a^d \rangle| = n/d$.

***Proof:*** Subgroups of cyclic groups

Let $H \leqslant \langle a \rangle$. If $H = \{e\}$, it's cyclic. Otherwise, let $d$ be the smallest positive integer with $a^d \in H$. We claim $H = \langle a^d \rangle$.

Clearly $\langle a^d \rangle \subseteq H$. For the reverse: take $a^k \in H$. Write $k = qd + r$ with $0 \leqslant r < d$. Then $a^r = a^k(a^d)^{-q} \in H$. By minimality of $d$, $r = 0$, so $a^k = (a^d)^q \in \langle a^d \rangle$.

If $|G| = n$, then $a^d$ has order $n/\gcd(d, n)$. The subgroups correspond to divisors via $d \leftrightarrow \langle a^{n/d} \rangle$ of order $d$. $\square$

**Question 2**

1. Find all subgroups of $\mathbb{Z}/12\mathbb{Z}$ and draw the subgroup lattice.
2. Prove that $\mathbb{Z}/p\mathbb{Z}$ has no proper nontrivial subgroups for $p$ prime.

3. How many generators does $\mathbb{Z}/n\mathbb{Z}$ have?

4. Prove: if $G$ is a group where every element satisfies $x^2 = e$, then $G$ is abelian.

Solution

**(1)** Divisors of 12: 1, 2, 3, 4, 6, 12. Subgroups: $\langle 0\rangle = \{0\}$, $\langle 6\rangle = \{0, 6\}$, $\langle 4\rangle = \{0, 4, 8\}$, $\langle 3\rangle = \{0, 3, 6, 9\}$, $\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$, $\langle 1\rangle = \mathbb{Z}/12$.
Lattice (larger subgroups higher):

$$\begin{tikzcd}
& \mathbb{Z}/12 \arrow[dl, dash] \arrow[dr, dash] & \\
\langle 2\rangle \arrow[d, dash] \arrow[drr, dash] & & \langle 3\rangle \arrow[d, dash] \arrow[dll, dash] \\
\langle 4\rangle \arrow[dr, dash] & & \langle 6\rangle \arrow[dl, dash] \\
& \{0\} &
\end{tikzcd}$$

**(2)** By Lagrange's theorem, the order of any subgroup divides $p$. So subgroups have order 1 or $p$, meaning $\{0\}$ or $\mathbb{Z}/p$.
**(3)** $a$ generates $\mathbb{Z}/n$ iff $\mathrm{ord}(a) = n$ iff $\gcd(a, n) = 1$. The count is $\varphi(n)$, Euler's totient.
**(4)** For any $a, b \in G$: $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$ (using $x^{-1} = x$ since $x^2 = e$). So $G$ is abelian.

## 2.5 Homomorphisms

**Definition 2.5.1: Homomorphism**

A **homomorphism** $\varphi : G \to H$ between groups is a map satisfying

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

**Proposition 2.5.1** Basic properties of homomorphisms
Let $\varphi : G \to H$ be a homomorphism.

1. $\varphi(e_G) = e_H$.
2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for all $a \in G$.
3. $\varphi(a^n) = \varphi(a)^n$ for all $n \in \mathbb{Z}$.
4. If $K \leqslant G$, then $\varphi(K) \leqslant H$.
5. If $L \leqslant H$, then $\varphi^{-1}(L) \leqslant G$.

***Proof:*** Proofs of homomorphism properties
**(1)** $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)\varphi(e_G)$. Multiply both sides by $\varphi(e_G)^{-1}$: $e_H = \varphi(e_G)$.
**(2)** $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_H$. So $\varphi(a^{-1})$ is the inverse of $\varphi(a)$.
**(3)** For $n > 0$: induction. For $n = 0$: part (1). For $n < 0$: $\varphi(a^n) = \varphi((a^{-1})^{-n}) = \varphi(a^{-1})^{-n} = (\varphi(a)^{-1})^{-n} = \varphi(a)^n$.
**(4)** For $\varphi(a), \varphi(b) \in \varphi(K)$ (with $a, b \in K$): $\varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \in \varphi(K)$ since $ab^{-1} \in K$.
**(5)** For $a, b \in \varphi^{-1}(L)$: $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in L$ (since $L \leqslant H$), so $ab^{-1} \in \varphi^{-1}(L)$. $\square$

**Note:-**

The homomorphism condition can be expressed as a commutative diagram: the two paths $G \times G \xrightarrow{m_G} G \xrightarrow{\varphi} H$ and $G \times G \xrightarrow{\varphi \times \varphi} H \times H \xrightarrow{m_H} H$ give the same result.

$$\begin{tikzcd} G \times G \arrow[r, "\varphi \times \varphi"] \arrow[d, "m_G"'] & H \times H \arrow[d, "m_H"] \\ G \arrow[r, "\varphi"'] & H \end{tikzcd}$$

**Definition 2.5.2: Types of homomorphisms**

Let $\varphi : G \to H$ be a homomorphism.

1. $\varphi$ is a **monomorphism** (or **embedding**) if it's injective.
2. $\varphi$ is an **epimorphism** if it's surjective.
3. $\varphi$ is an **isomorphism** if it's bijective. We write $G \cong H$.
4. An isomorphism $G \to G$ is called an **automorphism**. The set of all automorphisms of $G$, denoted $\mathrm{Aut}(G)$, forms a group under composition.

**Example 2.5.1** (Important(ish) homomorphisms)

1. **Reduction mod $n$:** $\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $\pi(a) = a \mod n$. Surjective, kernel $n\mathbb{Z}$.
2. **Quotient mod $n$:** If $n \mid m$, then $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $a \mapsto a \mod n$. Example: last-two-digits to last-digit, $\mathbb{Z}/100 \to \mathbb{Z}/10$.
3. **Determinant:** $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$. Homomorphism since $\det(AB) = \det(A)\det(B)$. Kernel is $SL_n(\mathbb{R})$.
4. **Exponential:** $\exp : (\mathbb{R}, +) \to (\mathbb{R}_{>0}, \times)$, $x \mapsto e^x$. Isomorphism with inverse ln.
5. **Complex exponential:** $\exp : (\mathbb{R}, +) \to (S^1, \times)$, $t \mapsto e^{2\pi i t}$. Surjective, kernel $\mathbb{Z}$. Induces $\mathbb{R}/\mathbb{Z} \cong S^1$.

### 2.5.1 Isomorphisms and invariants

**Definition 2.5.3: Isomorphic groups**

Groups $G$ and $H$ are **isomorphic**, written $G \cong H$, if there exists an isomorphism $\varphi : G \to H$. Isomorphic groups are "the same" for all group-theoretic purposes.

What properties are preserved by isomorphisms? These are called invariants.

**Proposition 2.5.2** Isomorphism invariants

If $G \cong H$, then:

1. $|G| = |H|$ (same order);
2. $G$ is abelian iff $H$ is abelian;
3. $G$ is cyclic iff $H$ is cyclic;
4. For each $n$, $G$ and $H$ have the same number of elements of order $n$;
5. For each $n$, $G$ and $H$ have the same number of subgroups of order $n$.

***Proof:*** Why these are invariants

Let $\varphi : G \xrightarrow{\sim} H$ be an isomorphism.

**(1)** Bijections preserve cardinality.

**(2)** If $G$ is abelian and $h_1, h_2 \in H$, write $h_i = \varphi(g_i)$. Then $h_1 h_2 = \varphi(g_1)\varphi(g_2) = \varphi(g_1 g_2) = \varphi(g_2 g_1) = \varphi(g_2)\varphi(g_1) = h_2 h_1$.

**(3)** If $G = \langle a \rangle$, then $H = \varphi(G) = \varphi(\langle a \rangle) = \langle \varphi(a) \rangle$.

**(4)** $\operatorname{ord}(\varphi(a)) = \operatorname{ord}(a)$ since $\varphi(a)^n = \varphi(a^n) = e_H \iff a^n = e_G$.

**(5)** $\varphi$ induces a bijection between subgroups of $G$ and subgroups of $H$. □

**Example 2.5.2** (Non-isomorphic groups of same order)

$\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ both have order 4, but:

- $\mathbb{Z}/4$ has an element of order 4 (namely, 1);
- $\mathbb{Z}/2 \times \mathbb{Z}/2$ has no element of order 4 (all non-identity elements have order 2).

So they're not isomorphic. These are the only groups of order 4 up to isomorphism.

**Example 2.5.3** (All groups of order 2 are isomorphic)

If $|G| = 2$, write $G = \{e, x\}$. Then $x \neq e$, so $x^2 \neq x$ (else $x = e$). Thus $x^2 = e$, and $G = \langle x \rangle \cong \mathbb{Z}/2$.

**Question 3**

1. Prove that $\operatorname{Aut}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.
2. Find $\operatorname{Aut}(\mathbb{Z})$.
3. Prove that if $\varphi : G \to H$ is an isomorphism, then $\varphi^{-1} : H \to G$ is also an isomorphism.
4. Prove that "is isomorphic to" is an equivalence relation on groups.

**Solution**

**(1)** An automorphism $\varphi$ of $\mathbb{Z}/n$ is determined by $\varphi(1)$, since $\varphi(k) = k\varphi(1)$. We need $\varphi(1)$ to generate $\mathbb{Z}/n$, i.e., $\gcd(\varphi(1), n) = 1$, i.e., $\varphi(1) \in (\mathbb{Z}/n)^\times$. Conversely, any $a \in (\mathbb{Z}/n)^\times$ defines an automorphism $\varphi_a : k \mapsto ka$. The map $a \mapsto \varphi_a$ is an isomorphism $(\mathbb{Z}/n)^\times \to \operatorname{Aut}(\mathbb{Z}/n)$ (check it's a homomorphism and bijection).
**(2)** An automorphism of $\mathbb{Z}$ is determined by $\varphi(1)$. We need $\langle \varphi(1) \rangle = \mathbb{Z}$, so $\varphi(1) = \pm 1$. Thus $\operatorname{Aut}(\mathbb{Z}) = \{\mathrm{id}, -\mathrm{id}\} \cong \mathbb{Z}/2$.
**(3)** $\varphi^{-1}$ is a bijection. For $h_1, h_2 \in H$, let $g_i = \varphi^{-1}(h_i)$. Then $\varphi^{-1}(h_1 h_2) = \varphi^{-1}(\varphi(g_1)\varphi(g_2)) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$.
**(4)** Reflexive: $\mathrm{id}_G : G \to G$ is an isomorphism. Symmetric: if $\varphi : G \to H$ is an isomorphism, so is $\varphi^{-1} : H \to G$ by part (3). Transitive: if $\varphi : G \to H$ and $\psi : H \to K$ are isomorphisms, so is $\psi \circ \varphi : G \to K$.

## 2.6 The order of an element and its properties

We've introduced order; now we develop its theory more fully.

**Proposition 2.6.1** Powers and order

Let $a \in G$ have finite order $n$. Then:

1. $a^k = e \iff n \mid k$.
2. $a^i = a^j \iff n \mid (i-j)$.
3. The elements $e, a, a^2, \ldots, a^{n-1}$ are all distinct.

***Proof:*** Properties of order

**(1)** ($\Leftarrow$) If $k = mn$, then $a^k = (a^n)^m = e^m = e$. ($\Rightarrow$) Write $k = qn + r$ with $0 \leqslant r < n$. Then $e = a^k = a^{qn+r} = (a^n)^q a^r = a^r$. By minimality of $n$, $r = 0$, so $n \mid k$.

**(2)** $a^i = a^j \iff a^{i-j} = e \iff n \mid (i-j)$ by (1).

**(3)** If $a^i = a^j$ for $0 \leqslant i < j < n$, then $a^{j-i} = e$ with $0 < j - i < n$, contradicting minimality of $n$. ☐

**Proposition 2.6.2** Order of powers

Let $\operatorname{ord}(a) = n$. Then $\operatorname{ord}(a^k) = \frac{n}{\gcd(k,n)}$.

***Proof:*** Order of powers

Let $d = \gcd(k, n)$. We show $\operatorname{ord}(a^k) = n/d$.

First, $(a^k)^{n/d} = a^{kn/d} = (a^n)^{k/d} = e$, so $\operatorname{ord}(a^k) \mid n/d$.

Conversely, let $m = \operatorname{ord}(a^k)$. Then $a^{km} = e$, so $n \mid km$. Write $k = d \cdot k'$ and $n = d \cdot n'$ where $\gcd(k', n') = 1$. Then $dn' \mid dk'm$, so $n' \mid k'm$. Since $\gcd(k', n') = 1$, we have $n' \mid m$, i.e., $n/d \mid m$. Thus $m = n/d$. ☐

**Corollary 2.6.1** Generators of cyclic groups

In $\langle a \rangle$ of order $n$, the element $a^k$ is a generator iff $\gcd(k, n) = 1$.

**Example 2.6.1**

In $\mathbb{Z}/12$, the generators are $1, 5, 7, 11$ (the four elements coprime to 12). The element 4 has order $12/\gcd(4, 12) = 12/4 = 3$.

**Question 4**

1. In a group $G$, if $a^{12} = e$ and $a^8 = e$, what are the possible orders of $a$?
2. Prove: if $\operatorname{ord}(a) = n$ and $\operatorname{ord}(b) = m$ with $\gcd(n, m) = 1$ and $ab = ba$, then $\operatorname{ord}(ab) = nm$.
3. Find the order of $(2, 3)$ in $\mathbb{Z}/6 \times \mathbb{Z}/8$.

**Solution**

**(1)** $\operatorname{ord}(a)$ divides both 12 and 8, so $\operatorname{ord}(a) \mid \gcd(12, 8) = 4$. Possible orders: 1, 2, 4.
**(2)** Let $k = \operatorname{ord}(ab)$. Then $e = (ab)^{nm} = a^{nm} b^{nm} = (a^n)^m (b^m)^n = e$ (using commutativity), so $k \mid nm$.
Now $e = (ab)^{kn} = a^{kn} b^{kn} = b^{kn}$ (since $a^n = e$), so $m \mid kn$. Since $\gcd(m, n) = 1$, $m \mid k$. Similarly, $n \mid k$. Since $\gcd(m, n) = 1$, $mn \mid k$. Thus $k = mn$.
**(3)** $\operatorname{ord}(2, 3) = \operatorname{lcm}(\operatorname{ord}(2), \operatorname{ord}(3)) = \operatorname{lcm}(6/\gcd(2, 6), 8/\gcd(3, 8)) = \operatorname{lcm}(3, 8) = 24$.

Lecture 3.

# Group Relations

We say that two sets have the same **cardinality** if $\exists f : X \hookrightarrow Y$, and we write $|X| \leqslant |Y|$. This allows us to inter-relate cardinality and function relations via a special theorem.

**Theorem 3.0.1** Schröder-Berstein Theorem

If $\exists f : X \hookrightarrow Y$ and $g : Y \hookrightarrow X$ then $|X| = |Y|$.

The Schröder-Berstein theorem provides some immediate results for allowing explicit bijections.

**Proposition 3.0.1** Cardinality of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ to any power

Using Schröder-Berstein we can create explicit bijections $f : \mathbb{Z} \leftrightarrow \mathbb{N}$ given by $n \mapsto \begin{cases} 2n & \text{if } n \geqslant 0; \\ -(2n+1) & \text{if } n < 0. \end{cases}$

**Proposition 3.0.2** $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}| = \mathbb{N}^n$

We create the explicit bijection via $f : \mathbb{N} \times \mathbb{N} \leftrightarrow \mathbb{N}$, $(k, 1) \mapsto 2^k(2l + 1) - 1$. It follows then that $\mathbb{N}^a, \mathbb{Z}^b, \mathbb{Q}^c, \forall a, b, c \in \mathbb{N}$ permits a similar bijection.

***Proof:*** Proof of Prop. 3.1.2

Notice that any natural number can be written as $2^k(m)$ where $m$ is odd, i.e., $m = 2n + 1$. A bijection is automatically created here with $2^k(2l+1)$, and the additional $-1$ at the end of the statement is there to shift the entire bijection down one. So instead of $(0, 0) \mapsto 2^0(2(0)+1) = 1$ we get $(0, 0) \mapsto 2^0(2(0)+1)-1 = 0$. $\square$

**Proposition 3.0.3** Countable vs uncountable infinites

$|\mathbb{R}| > |\mathbb{N}|$.

***Proof:*** Proof by Cantor

Given by Cantor's diagonal argument, no map $f : \mathbb{N} \to \mathbb{R}$ be surjective, because:

$$\begin{cases} f(0) & = a_{00}a_{01}a_{02}a_{03}\dots \\ f(1) & = a_{10}a_{11}a_{12}a_{13}\dots \\ f(2) & = a_{20}a_{21}a_{22}a_{23}\dots \end{cases}$$

then let $y = b_0b_1b_2b_3$ where $b_j \neq a_{jj}$ for each $j$. Now look at the $j^{\text{th}}$ digit, $y \neq f(j), \forall j \in \mathbb{N}$, so $f$ can't be surjective. $\square$

**Proposition 3.0.4** Cardinality of mappings

$|\mathbb{R}^{\mathbb{N}}| = |\mathbb{R}|$

***Proof:*** Proof by Cantor's diagonal used for mappings

We can use the fact that $|(0,1)| = |\mathbb{R}|$ in addition to Cantor's diagonal argument as a decimal expansion—$t = 0.t_1t_2t_3\ldots$—with $t_i$ as an integer between 0 and 9. So now to show $\exists f \mid f : (0,1) \leftrightarrow (0,1)^{\mathbb{N}}$, we create some $s_j \in (0,1)^{\mathbb{N}}$ and construct the bijection via:

$$\begin{cases} s_1 &= 0.t_1t_3t_5\ldots \\ s_2 &= 0.t_2t_6t_{10}\ldots \\ s_3 &= 0.t_4t_{12}t_{20}\ldots \end{cases}$$

We say that $s_1$ is made up of all $t_i$ that are odd; $s_2$ is made up of all $i$ which are odd such that $i \cdot 2 \forall i \in s_2$; $s_3$ is made up of all $i$ which are odd such that $i \cdot 2^2, \forall i \in s_3$, $i \cdot 2^3 \in s_3$ and so on (the general form being $i \cdot 2^n \mid n = j - 1$). Therefore there exists a bijection between $(0,1) \leftrightarrow (0,1)^{\mathbb{N}}$. □

**Proposition 3.0.5** There does not exist a largest cardinality

Let $\mathcal{P}(S)$ be the standard powerset. It follows then that $\mathcal{P}(S) = \{0,1\}^S$ and $|\mathcal{P}(S)| > |S|$ for any $S$.

***Proof:*** Proof of non-existence of largest cardinality

Recall that $\{0,1\}^S = \{\text{all } f \text{ such that } f : S \to \{0,1\}\}$. There must exist a bijection since $|\mathcal{P}(S)| = 2^{|S|}$ and $|\{0,1\}^S| = 2^{|S|}$. Now to show $|\mathcal{P}(S)| > |S|$ we use proof by contradiction: suppose $\exists \alpha : S \leftrightarrow \mathcal{P}(S)$ and let $A \subset S$ defined by $A \coloneqq \{s \in S \mid s \neq \alpha(s)\}$, now for $A = \alpha(t)$ for some $t \in A$ we notice that $t \notin \alpha(t) = A$ means that $t \in A$, but to say $t \in \alpha(t) \implies t \notin A$, thus we have a contradiction. □

Another option for this proof, though less formal, can be found by showing that $\mathcal{P}(S) = \{0,1\}^S$ and that subsets of $S$ have $f$ such that $f \mapsto f^{-1}$; let $A \subset S$ then

$$A \mapsto \left( \not\Vdash_A : x \mapsto \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases} \right)$$

Note that if $S$ is finite then $|S| = n$ and $\mathcal{P}(S) = 2^n$. This naturally leads us into our next theorem regarding the infinitude of the cardinality of $S$ in relation to $\mathcal{P}(S)$.

## 3.1 Quotient groups

Consider $\mathbb{Z}/n$. Under the equivalence relation congruence mod $n$ we can take the sum of the various subsets that mod $n$ partitions $\mathbb{Z}$ into as the law of composition given by $(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = a + b + n\mathbb{Z}$. Notice that we can let $T$ be the set of equivalence classes under mod $n$, and thus we have a homomorphic map from $\mathbb{Z} \to T$. Finally, $T = \mathbb{Z}/n$.

Why stop at integers? What if we want to make congruence mod $H$ where $H$ is a subgroup $H \subset G$? Under what cases, given that we can make mod $H$ equivalence classes (we will define this in a bit) is the quotient map $G \to T$ a homomorphism? This question is the question of quotient groups, and its answer—normality—is of note and importance to much of group theory (i.e. it is foundational material).

**Example 3.1.1** (Quotient subgroups of $S_3$)

Note that $S_3$ is isomorphic to $\mathbb{Z}/6$, thus we may think of of the subgroups as such:

$$S_3 \cong \mathbb{Z}/6 \cong \mathbb{Z}/3 \times \mathbb{Z}/2$$

and since $\mathbb{Z}/6$'s elements (03) form $\mathbb{Z}/2$ with $\{e, \mu\} \mapsto (03) \mapsto \{\pm 1\}$, we have similar happening for (024).

$$\begin{tikzcd}
& & S_3 \arrow[dll, dash] \arrow[dl, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\langle \rho \rangle \arrow[drr, dash] & \langle \mu \rangle \arrow[dr, dash] & & \langle \rho\mu\beta \rangle \arrow[dl, dash] & \langle \rho^2\mu \rangle \arrow[dll, dash] \\
& & \{e\} & &
\end{tikzcd}$$

### 3.1.1 Congruence mod $H$ and cosets

For any two elements $a, b \in G$ we define $a \sim b \iff ab^{-1} \in H$, this is to say that $a \in bH$. We thus have $a \sim a$ since $e \in H$, $a \sim b \implies ab^{-1} \in H$, and $ba^{-1} = (ab^{-1})^{-1} \in H$, thus $b \sim a$. Finally, $a \sim b, b \sim c$ meaning that $ab^{-1}$ and $bc^{-1}$ are both in $H$, so $ac^{-1} = (ab^{-1})(bc^{-1}) \in H$ since $H$ is closed under the group operation, thus $a \sim c$ and transitivity holds. We have shown congruence mod $H$ is indeed an equivalence relation.

**Definition 3.1.1: Cosets**

Given a subgroup $H \subset G$ and an element $a \in G$, the **left coset** of $H$ containing $a$ is the set $aH = \{ah \mid h \in H\}$. Similarly, the **right coset** is $Ha = \{ha \mid h \in H\}$. The equivalence classes under congruence mod $H$ are precisely the left cosets of $H$.

**Proposition 3.1.1** Cosets partition $G$

The left cosets of $H$ form a partition of $G$. Every element $g \in G$ belongs to exactly one coset, namely $gH$, and two cosets $aH, bH$ are either identical or disjoint.

***Proof:*** Proof of coset partition

This follows immediately from the fact that cosets are equivalence classes: $g \in gH$ since $g = ge$ and $e \in H$. For disjointness, suppose $aH \cap bH \neq \emptyset$, so there exists $c \in aH \cap bH$. Then $c = ah_1 = bh_2$ for some $h_1, h_2 \in H$, which gives $a = bh_2h_1^{-1} \in bH$. For any $ah \in aH$, we have $ah = bh_2h_1^{-1}h \in bH$, so $aH \subseteq bH$. By symmetry $bH \subseteq aH$, thus $aH = bH$. $\square$

Here is the key observation that makes quotient theory work: all cosets have the same size!

**Proposition 3.1.2** Cosets are equinumerous

For any $a \in G$, the map $\lambda_a : H \to aH$ given by $h \mapsto ah$ is a bijection. In particular, $|aH| = |H|$ for all $a \in G$.

***Proof:*** Proof of coset bijection

Surjectivity is immediate from the definition of $aH$. For injectivity, suppose $ah_1 = ah_2$. Left-multiplying by $a^{-1}$ gives $h_1 = h_2$. (This is just left-translation, which is always a bijection in a group—a fact we will revisit when discussing group actions.) $\square$

**Theorem 3.1.1** Lagrange's Theorem

Let $G$ be a finite group and $H \subset G$ a subgroup. Then $|H|$ divides $|G|$, and the quotient $|G|/|H| = [G : H]$ equals the number of distinct left cosets of $H$ in $G$.

***Proof:*** Proof of Lagrange's Theorem

The left cosets of $H$ partition $G$ into disjoint subsets, each of cardinality $|H|$. If there are $k$ distinct cosets, then $|G| = k \cdot |H|$, so $|H| \mid |G|$ and $k = [G : H]$. □

**Definition 3.1.2: Index**

The **index** of a subgroup $H$ in $G$, denoted $[G : H]$, is the number of distinct left cosets of $H$ in $G$. For finite groups, $[G : H] = |G|/|H|$.

**Note:-**

Lagrange's theorem has immediate corollaries: the order of any element divides $|G|$ (since $\langle g \rangle$ is a subgroup), and groups of prime order are cyclic (the only subgroups have order 1 or $p$). The converse of Lagrange is false in general: $A_4$ has order 12 but no subgroup of order 6. However, for abelian groups and more generally for solvable groups, partial converses exist (feel free to use the ol' google machine for more info).

A question naturally arises then: when is $G/H$ a group? If we try to define $(aH)(bH) = abH$, we need this to be well-defined, meaning if $aH = a'H$ and $bH = b'H$, then $abH = a'b'H$. This does not always work!

**Example 3.1.2**

Consider $G = S_3$ and $H = \{e, (12)\}$. We have three left cosets: $H = \{e, (12)\}$, $(13)H = \{(13), (132)\}$, and $(23)H = \{(23), (123)\}$. Take representatives $a = (13)$ and $a' = (132)$ from the same coset; both satisfy $aH = a'H = (13)H$. Now take $b = (23)$. We compute:

$$\begin{aligned} ab &= (13)(23) = (132), \quad \text{so } abH = (132)H = (13)H \\ a'b &= (132)(23) = (13), \quad \text{so } a'bH = (13)H \end{aligned}$$

Okay, this worked (yay)! But now try $b = (13)$ and $b' = (132)$ (same coset), with $a = (23)$:

$$\begin{aligned} ab &= (23)(13) = (123), \quad \text{so } abH = (123)H = (23)H \\ ab' &= (23)(132) = (12), \quad \text{so } ab'H = (12)H = H \end{aligned}$$

Disaster! We got $(23)H \neq H$, so coset multiplication is not well-defined for this $H$.

What went wrong? The issue is that $H = \{e, (12)\}$ is not "symmetric" enough—left cosets and right cosets differ. For instance, $(13)H = \{(13), (132)\}$ but $H(13) = \{(13), (123)\}$.

### 3.1.2 Normal subgroups and quotient groups

**Definition 3.1.3: Normal subgroup**

A subgroup $N \subset G$ is **normal** (written $N \trianglelefteq G$) if any of the following equivalent conditions hold:

1. $gNg^{-1} = N$ for all $g \in G$ (conjugation preserves $N$);
2. $gN = Ng$ for all $g \in G$ (left cosets equal right cosets);
3. $gNg^{-1} \subseteq N$ for all $g \in G$ (one inclusion suffices);
4. $N$ is the kernel of some homomorphism $\varphi : G \to H$.

***Proof:*** Equivalence of normality conditions

(1) $\Rightarrow$ (2): If $gNg^{-1} = N$, then $gN = Ng$ by right-multiplying by $g$.

(2) $\Rightarrow$ (3): If $gN = Ng$, then for $n \in N$, we have $gn \in gN = Ng$, so $gn = n'g$ for some $n' \in N$, giving $gng^{-1} = n' \in N$.

(3) $\Rightarrow$ (1): We have $gNg^{-1} \subseteq N$ by assumption. For the reverse, take $n \in N$. We want $n \in gNg^{-1}$, i.e., $g^{-1}ng \in N$. But applying (3) with $g^{-1}$: $g^{-1}N(g^{-1})^{-1} = g^{-1}Ng \subseteq N$, so $g^{-1}ng \in N$.

(4) $\Rightarrow$ (1): If $N = \ker(\varphi)$, then for $n \in N$ and $g \in G$: $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g)e_H\varphi(g)^{-1} = e_H$, so $gng^{-1} \in \ker(\varphi) = N$.

(1) $\Rightarrow$ (4): This is the content of the next theorem—every normal subgroup is a kernel! $\square$

**Theorem 3.1.2** Quotient groups exist for normal subgroups

If $N \trianglelefteq G$, then $G/N$ is a group under the operation $(aN)(bN) = abN$, called the **quotient group**. The map $\pi : G \to G/N$ given by $\pi(g) = gN$ is a surjective homomorphism (the **canonical projection**) with $\ker(\pi) = N$.

***Proof:*** Proof of quotient group structure

**Well-definedness:** Suppose $aN = a'N$ and $bN = b'N$, so $a' = an_1$ and $b' = bn_2$ for some $n_1, n_2 \in N$. Then

$$a'b' = an_1bn_2 = a(n_1b)n_2.$$

Since $N$ is normal, $n_1b = bn_3$ for some $n_3 \in N$ (because $bN = Nb$ implies $n_1 \in Nb$, so $n_1 = bn_3$... wait, let me be more careful). We have $n_1b \in Nb = bN$, so $n_1b = bn_3$ for some $n_3 \in N$. Thus $a'b' = abn_3n_2 \in abN$, so $a'b'N = abN$.

**Group axioms:** Associativity: $(aN\cdot bN)\cdot cN = abN\cdot cN = (ab)cN = a(bc)N = aN\cdot bcN = aN\cdot(bN\cdot cN)$. Identity: $eN = N$ satisfies $N \cdot aN = eaN = aN$. Inverses: $(aN)^{-1} = a^{-1}N$ since $aN \cdot a^{-1}N = aa^{-1}N = N$.

**The projection:** $\pi(ab) = abN = aN \cdot bN = \pi(a)\pi(b)$, so $\pi$ is a homomorphism. Surjectivity is clear. Finally, $\ker(\pi) = \{g \in G : gN = N\} = \{g \in G : g \in N\} = N$. $\square$

**Question 1**

Let $G$ be a group and $Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ the center of $G$.

1. Prove that $Z(G) \trianglelefteq G$.
2. Prove that if $G/Z(G)$ is cyclic, then $G$ is abelian.
3. Find $Z(D_4)$ and describe $D_4/Z(D_4)$.

**Solution**

**(1)** We verify normality via condition (1). For any $z \in Z(G)$ and $g \in G$, we have $gzg^{-1} = zgg^{-1} = z \in Z(G)$ (using that $z$ commutes with everything). Thus $gZ(G)g^{-1} \subseteq Z(G)$ for all $g$, so $Z(G) \trianglelefteq G$.
**(2)** Suppose $G/Z(G) = \langle aZ(G)\rangle$ is cyclic. Then every element of $G$ has the form $a^kz$ for some $k \in \mathbb{Z}$ and $z \in Z(G)$. Take any two elements $g = a^iz_1$ and $h = a^jz_2$. Then:

$$gh = a^iz_1a^jz_2 = a^ia^jz_1z_2 = a^{i+j}z_1z_2$$

(since $z_1$ commutes with $a^j$ and $z_2$), and similarly $hg = a^{j+i}z_2z_1 = a^{i+j}z_1z_2 = gh$ (since $z_1, z_2$ commute with everything and with each other). Thus $G$ is abelian.
**(3)** Recall $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$ with $r^4 = s^2 = e$ and $srs = r^{-1}$. For $z \in Z(D_4)$, we need $zr = rz$ and $zs = sz$.
For rotations: $r^k \in Z(D_4)$ requires $sr^k = r^ks$. We have $sr^k = r^{-k}s$ (by induction using $srs = r^{-1}$), so $r^{-k}s = r^ks$, giving $r^{2k} = e$, so $k \in \{0, 2\}$. Thus $e, r^2$ are central among rotations.
For reflections: $sr^j \in Z(D_4)$ requires $r(sr^j) = (sr^j)r$, i.e., $rsr^j = sr^{j+1}$. But $rsr^j = r \cdot r^{-j}s = r^{1-j}s$,

so we need $r^{1-j}s = sr^{j+1}$, giving $r^{1-j} = r^{-(j+1)} = r^{-j-1}$, so $r^2 = e$, contradiction since $r$ has order 4. No reflections are central.
Therefore $Z(D_4) = \{e, r^2\} \cong \mathbb{Z}/2$. The quotient $D_4/Z(D_4)$ has order $8/2 = 4$. The cosets are $\{e, r^2\}, \{r, r^3\}, \{s, sr^2\}, \{sr, sr^3\}$. One can verify this is $\cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (the Klein four-group), not $\mathbb{Z}/4$, since every non-identity element has order 2 in the quotient.

As a final remark before moving onto our next topic, I wish to introduce the concept of simple groups. Simply put, no pun intended, a simple group is a group that cannot be decomposed into further normal subgroups other than the trivial subgroup and itself. Formally:

**Definition 3.1.4: Simple groups**

We say a group is simple if it has no normal subgroups other than itself and the trivial subgroup.

## 3.2 Group actions and symmetry

The notion of a group "acting" on a set is one of the main "substances" which underlies much of our study of mathematics. When we study symmetries of geometric objects, solutions to polynomial equations, or even the structure of groups themselves, we are studying group actions. This section develops the theory systematically—pay close attention, as these ideas will reappear throughout when we discuss representation theory (where groups act on vector spaces via linear maps).

### 3.2.1 Group actions

**Definition 3.2.1: Group action**

A **(left) group action** of a group $G$ on a set $X$ is a map $G \times X \to X$, written $(g, x) \mapsto g \cdot x$, satisfying:

1. $e \cdot x = x$ for all $x \in X$ (identity acts trivially);
2. $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$ (compatibility).

We say $G$ **acts on** $X$, or that $X$ is a **G-set**.

**Note:-**
There is an equivalent, perhaps more fun, formulation: a group action is a homomorphism $\rho : G \to \mathrm{Sym}(X)$, where $\mathrm{Sym}(X)$ is the group of all bijections $X \to X$. Given an action, define $\rho(g)(x) = g \cdot x$; the axioms ensure each $\rho(g)$ is a bijection (with inverse $\rho(g^{-1})$) and that $\rho$ is a homomorphism. Conversely, any such homomorphism defines an action. This perspective is crucial: group actions are the same as representations of G into symmetric groups.

**Example 3.2.1**
Some fundamental examples of group actions:

1. **Left multiplication:** $G$ acts on itself by $g \cdot h = gh$. This action is always faithful (only $e$ acts trivially) and transitive (any element can be sent to any other).
2. **Conjugation:** $G$ acts on itself by $g \cdot h = ghg^{-1}$. The kernel is $Z(G)$, the center.
3. **Conjugation on subgroups:** $G$ acts on the set of its subgroups by $g \cdot H = gHg^{-1}$. Normal subgroups are precisely the fixed points!
4. **Coset action:** If $H \subset G$, then $G$ acts on $G/H$ (left cosets) by $g \cdot (aH) = (ga)H$.

5. **Linear actions:** $GL_n(\mathbb{R})$ acts on $\mathbb{R}^n$ by matrix multiplication. This is the prototype for representation theory.

**Definition 3.2.2: Orbits and stabilizers**

Let $G$ act on $X$. For $x \in X$:

1. The **orbit** of $x$ is $G \cdot x = \{g \cdot x : g \in G\} \subseteq X$.
2. The **stabilizer** (or **isotropy group**) of $x$ is $G_x = \mathrm{Stab}_G(x) = \{g \in G : g \cdot x = x\} \subseteq G$.

**Proposition 3.2.1** Stabilizers are subgroups

For any $x \in X$, the stabilizer $G_x$ is a subgroup of $G$.

***Proof:*** Proof that stabilizers are subgroups

Identity: $e \cdot x = x$, so $e \in G_x$. Closure: if $g, h \in G_x$, then $(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x$, so $gh \in G_x$. Inverses: if $g \in G_x$, then $x = e \cdot x = (g^{-1}g) \cdot x = g^{-1} \cdot (g \cdot x) = g^{-1} \cdot x$, so $g^{-1} \in G_x$. □

**Proposition 3.2.2** Orbits partition $X$

The orbits of a group action form a partition of $X$. Define $x \sim y$ iff $y = g \cdot x$ for some $g \in G$; this is an equivalence relation whose equivalence classes are exactly the orbits.

***Proof:*** Proof that orbits partition

Reflexivity: $x = e \cdot x$, so $x \sim x$. Symmetry: if $y = g \cdot x$, then $x = g^{-1} \cdot y$, so $y \sim x$ implies $x \sim y$. Transitivity: if $y = g \cdot x$ and $z = h \cdot y$, then $z = h \cdot (g \cdot x) = (hg) \cdot x$, so $x \sim z$. □

The following theorem is one of the most useful results in all of group theory. It connects the size of an orbit to the index of a stabilizer.

**Theorem 3.2.1** Orbit-Stabilizer Theorem

Let $G$ act on $X$ and let $x \in X$. There is a bijection between the orbit $G \cdot x$ and the set of left cosets $G/G_x$, given by $g \cdot x \mapsto gG_x$. In particular, if $G$ is finite,

$$|G \cdot x| = [G : G_x] = \frac{|G|}{|G_x|}.$$

***Proof:*** Proof of Orbit-Stabilizer

Define $\varphi : G/G_x \to G \cdot x$ by $\varphi(gG_x) = g \cdot x$.

**Well-defined:** If $gG_x = hG_x$, then $h = gk$ for some $k \in G_x$, so $h \cdot x = (gk) \cdot x = g \cdot (k \cdot x) = g \cdot x$.

**Injective:** If $g \cdot x = h \cdot x$, then $h^{-1}g \cdot x = x$, so $h^{-1}g \in G_x$, giving $gG_x = hG_x$.

**Surjective:** Every element of $G \cdot x$ has the form $g \cdot x = \varphi(gG_x)$. □

**Example 3.2.2**

Consider $S_4$ acting on $\{1, 2, 3, 4\}$ in the natural way. Take $x = 1$. The orbit is $S_4 \cdot 1 = \{1, 2, 3, 4\}$ (any element can be sent to any other). The stabilizer is $(S_4)_1 = \{\sigma \in S_4 : \sigma(1) = 1\}$, which consists of all permutations of $\{2, 3, 4\}$, so $(S_4)_1 \cong S_3$ and $|(S_4)_1| = 6$. Check: $|S_4 \cdot 1| = 4 = 24/6 = |S_4|/|(S_4)_1|$. ✓

### 3.2.2 Cayley's theorem and applications

**Theorem 3.2.2** Cayley's Theorem

Every group $G$ is isomorphic to a subgroup of $\mathrm{Sym}(G)$. If $|G| = n$, then $G$ embeds into $S_n$.

***Proof:*** Proof of Cayley's Theorem

Consider the left multiplication action of $G$ on itself: $g \cdot h = gh$. This gives a homomorphism $\rho : G \to \mathrm{Sym}(G)$ where $\rho(g)$ is the bijection $h \mapsto gh$.

**Injectivity:** If $\rho(g) = \rho(g')$, then for all $h \in G$, $gh = g'h$. Taking $h = e$: $g = g'$. Thus $\ker(\rho) = \{e\}$, so $\rho$ is injective.

By the First Isomorphism Theorem, $G \cong \mathrm{Im}(\rho) \subset \mathrm{Sym}(G)$. If $|G| = n$, then $\mathrm{Sym}(G) \cong S_n$. $\square$

**Note:-**

Cayley's theorem is simultaneously profound and useless. Profound: every abstract group is "really" a permutation group; the group axioms capture exactly the structure of bijection composition. Useless: the embedding is almost always far larger than necessary. $S_n$ has order $n!$, so embedding a group of order $n$ into $S_n$ wastes enormous space. Better embeddings come from finding smaller faithful actions. 'Tis fun nonetheless.

**Proposition 3.2.3** Improved Cayley embedding

Let $H \subset G$ with $[G : H] = n$. The action of $G$ on $G/H$ gives a homomorphism $\rho : G \to S_n$ with $\ker(\rho) = \bigcap_{g \in G} gHg^{-1}$, the largest normal subgroup of $G$ contained in $H$.

***Proof:*** Proof of improved embedding

The action $g \cdot (aH) = (ga)H$ is well-defined (check!). The kernel consists of all $g$ fixing every coset: $gaH = aH$ for all $a$, i.e., $a^{-1}ga \in H$ for all $a$, i.e., $g \in aHa^{-1}$ for all $a$. Thus $\ker(\rho) = \bigcap_{a \in G} aHa^{-1}$. $\square$

**Example 3.2.3**

Let $G = S_4$ and $H = S_3$ (permutations fixing 4). Then $[G : H] = 4$, and the action on $G/H$ identifies with the natural action on $\{1, 2, 3, 4\}$. The kernel is trivial (since the natural action is faithful), recovering the identity $S_4 \hookrightarrow S_4$. But: if we take $G = A_5$ and $H = A_4$ (stabilizer of a point), then $[G : H] = 5$ and we get $A_5 \hookrightarrow S_5$. Since $|A_5| = 60 < 120 = |S_5|$, this is a non-surjective embedding, showing $A_5$ as a proper subgroup of $S_5$.

**Question 2**

Let $G$ be a group of order $2p$ where $p$ is an odd prime.

1. Prove that $G$ has a unique subgroup of order $p$, which is therefore normal.
2. Conclude that $G \cong \mathbb{Z}/2p$ or $G \cong D_p$ (the dihedral group of order $2p$).

**Solution**

**(1)** By Cauchy's theorem (or direct counting), $G$ has an element $a$ of order $p$, so $H = \langle a \rangle$ is a subgroup of order $p$. Since $[G : H] = 2$, we have $H \trianglelefteq G$ (index 2 subgroups are always normal: $G = H \sqcup gH = H \sqcup Hg$ for any $g \notin H$, so $gH = Hg$).
For uniqueness: suppose $K$ is another subgroup of order $p$. Since $|H| = |K| = p$ is prime, $H \cap K$ is a subgroup of both with order dividing $p$, so $|H \cap K| \in \{1, p\}$. If $|H \cap K| = p$, then $H = K$. Otherwise, $H \cap K = \{e\}$, and since elements of $H$ and $K$ (excluding $e$) have order $p$, we have $|H \cup K| = 1 + (p-1) + (p-1) = 2p-1$ elements of orders 1 or $p$. But then $G$ has only one element not in $H \cup K$, call

it $b$. We must have $b^2 \in H \cup K$ (since $b \notin H \cup K$), but $b$ has order dividing $2p$... this gets complicated. Easier: if $H \neq K$, then $HK$ is a subgroup (since $H \trianglelefteq G$) of order $|H||K|/|H \cap K| = p^2 > 2p = |G|$, contradiction.
**(2)** Let $H = \langle a \rangle \cong \mathbb{Z}/p$ be the unique subgroup of order $p$. Pick $b \in G \setminus H$; then $b$ has order 2 (if $|b| = p$, then $b \in H$ by uniqueness; if $|b| = 2p$, then $G = \langle b \rangle$ is cyclic and $G \cong \mathbb{Z}/2p$).
Assume $|b| = 2$. Since $H \trianglelefteq G$, conjugation by $b$ gives an automorphism of $H \cong \mathbb{Z}/p$. We have $bab^{-1} = a^k$ for some $k$. Since $b^2 = e$: $a = b^2ab^{-2} = b(bab^{-1})b^{-1} = ba^kb^{-1} = (bab^{-1})^k = a^{k^2}$, so $k^2 \equiv 1 \pmod p$, giving $k \equiv \pm 1 \pmod p$.
If $k \equiv 1$: then $bab^{-1} = a$, so $ab = ba$, and $G = \langle a, b \rangle$ is abelian of order $2p$ with $|a| = p, |b| = 2$, so $G \cong \mathbb{Z}/p \times \mathbb{Z}/2 \cong \mathbb{Z}/2p$.
If $k \equiv -1$: then $bab^{-1} = a^{-1}$, i.e., $ba = a^{-1}b$. This is exactly the dihedral relation! We have $G = \langle a, b : a^p = b^2 = e,\ bab^{-1} = a^{-1} \rangle \cong D_p$.

## 3.3 The isomorphism theorems

The isomorphism theorems we have already introduced somewhat. They describe how quotients, subgroups, and homomorphisms interact. These results will reappear essentially unchanged when we study rings, modules, and vector spaces—they are instances of a more general phenomenon that category theory makes precise.

### 3.3.1 Kernels, images, and the First Isomorphism Theorem

**Definition 3.3.1: Kernel**

The **kernel** of a group homomorphism $\varphi : G \to H$ is

$$\ker(\varphi) = \{g \in G : \varphi(g) = e_H\} = \varphi^{-1}(\{e_H\}).$$

**Definition 3.3.2: Image**

The **image** of a group homomorphism $\varphi : G \to H$ is

$$\mathrm{Im}(\varphi) = \varphi(G) = \{h \in H : \exists g \in G,\ \varphi(g) = h\}.$$

**Proposition 3.3.1** Basic properties of kernels and images

Let $\varphi : G \to H$ be a homomorphism.

1. $\ker(\varphi)$ is a normal subgroup of $G$.
2. $\mathrm{Im}(\varphi)$ is a subgroup of $H$ (not necessarily normal).
3. $\varphi$ is injective $\iff \ker(\varphi) = \{e_G\}$.
4. $\varphi(g_1) = \varphi(g_2) \iff g_1^{-1}g_2 \in \ker(\varphi) \iff g_1 \ker(\varphi) = g_2 \ker(\varphi)$.

***Proof:*** Proofs of kernel/image properties

**(1)** First, $\ker(\varphi)$ is a subgroup: $\varphi(e) = e$, so $e \in \ker(\varphi)$; if $\varphi(a) = \varphi(b) = e$, then $\varphi(ab) = ee = e$; if $\varphi(a) = e$, then $\varphi(a^{-1}) = \varphi(a)^{-1} = e^{-1} = e$. For normality: if $k \in \ker(\varphi)$, then $\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = \varphi(g) \cdot e \cdot \varphi(g)^{-1} = e$.

**(2)** Subgroup: $e = \varphi(e) \in \mathrm{Im}(\varphi)$; if $\varphi(a), \varphi(b) \in \mathrm{Im}(\varphi)$, then $\varphi(a)\varphi(b) = \varphi(ab) \in \mathrm{Im}(\varphi)$; $\varphi(a)^{-1} = \varphi(a^{-1}) \in \mathrm{Im}(\varphi)$.

**(3)** If $\ker(\varphi) = \{e\}$ and $\varphi(g_1) = \varphi(g_2)$, then $\varphi(g_1^{-1}g_2) = e$, so $g_1^{-1}g_2 = e$, giving $g_1 = g_2$. Conversely, if $\varphi$ is injective and $k \in \ker(\varphi)$, then $\varphi(k) = e = \varphi(e)$, so $k = e$.

**(4)** $\varphi(g_1) = \varphi(g_2) \iff \varphi(g_1)^{-1}\varphi(g_2) = e \iff \varphi(g_1^{-1}g_2) = e \iff g_1^{-1}g_2 \in \ker(\varphi)$. The last equivalence with cosets is the definition of coset equality. □

**Theorem 3.3.1** First Isomorphism Theorem

Let $\varphi : G \to H$ be a homomorphism. Then:

1. $\ker(\varphi) \trianglelefteq G$;
2. The map $\bar{\varphi} : G/\ker(\varphi) \to H$ defined by $\bar{\varphi}(g \cdot \ker(\varphi)) = \varphi(g)$ is a well-defined injective homomorphism;
3. $\mathrm{Im}(\bar{\varphi}) = \mathrm{Im}(\varphi)$, so $G/\ker(\varphi) \cong \mathrm{Im}(\varphi)$.

In slogan form: the image of a homomorphism is isomorphic to the domain modulo the kernel.

***Proof:*** Proof of First Isomorphism Theorem

Part (1) was proved above. For (2): $\bar{\varphi}$ is well-defined because $g_1 \ker(\varphi) = g_2 \ker(\varphi)$ implies $\varphi(g_1) = \varphi(g_2)$ by property (4). It's a homomorphism: $\bar{\varphi}(g_1K \cdot g_2K) = \bar{\varphi}(g_1g_2K) = \varphi(g_1g_2) = \varphi(g_1)\varphi(g_2) = \bar{\varphi}(g_1K)\bar{\varphi}(g_2K)$ (writing $K = \ker(\varphi)$). It's injective: $\bar{\varphi}(gK) = e \implies \varphi(g) = e \implies g \in K \implies gK = K$.

For (3): $\mathrm{Im}(\bar{\varphi}) = \{\bar{\varphi}(gK) : g \in G\} = \{\varphi(g) : g \in G\} = \mathrm{Im}(\varphi)$. Since $\bar{\varphi}$ is an injective homomorphism onto $\mathrm{Im}(\varphi)$, it's an isomorphism $G/\ker(\varphi) \xrightarrow{\sim} \mathrm{Im}(\varphi)$. □

**Note:-**

The First Isomorphism Theorem is sometimes called the "fundamental theorem of homomorphisms." It says that every homomorphism $\varphi : G \to H$ factors as $G \twoheadrightarrow G/\ker(\varphi) \xrightarrow{\sim} \mathrm{Im}(\varphi) \hookrightarrow H$: a surjection, followed by an isomorphism, followed by an inclusion. This factorization is unique and natural—in the categorical sense, which you'll see later.

### 3.3.2 The Second and Third Isomorphism Theorems

**Theorem 3.3.2** Second Isomorphism Theorem (Diamond Theorem)

Let $G$ be a group, $H \subset G$ a subgroup, and $N \trianglelefteq G$ a normal subgroup. Then:

1. $HN = \{hn : h \in H, n \in N\}$ is a subgroup of $G$;
2. $H \cap N \trianglelefteq H$;
3. $H/(H \cap N) \cong HN/N$.

***Proof:*** Proof of Second Isomorphism Theorem

**(1)** $HN$ is a subgroup: $e = e \cdot e \in HN$. For closure: $(h_1n_1)(h_2n_2) = h_1(n_1h_2)n_2$. Since $N$ is normal, $n_1h_2 = h_2n_3$ for some $n_3 \in N$, so $(h_1n_1)(h_2n_2) = h_1h_2n_3n_2 \in HN$. For inverses: $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$ since $hn^{-1}h^{-1} \in N$ by normality.

**(2)** $H \cap N$ is a subgroup of $H$ (intersection of subgroups). For normality in $H$: if $x \in H \cap N$ and $h \in H$, then $hxh^{-1} \in H$ (since $H$ is a subgroup) and $hxh^{-1} \in N$ (since $N \trianglelefteq G$), so $hxh^{-1} \in H \cap N$.

**(3)** Define $\varphi : H \to HN/N$ by $\varphi(h) = hN$. This is a homomorphism: $\varphi(h_1h_2) = h_1h_2N = (h_1N)(h_2N) = \varphi(h_1)\varphi(h_2)$.

Surjectivity: Any element of $HN/N$ has the form $hnN = hN$ (since $n \in N$), which equals $\varphi(h)$.

Kernel: $\ker(\varphi) = \{h \in H : hN = N\} = \{h \in H : h \in N\} = H \cap N$.

By the First Isomorphism Theorem, $H/(H \cap N) = H/\ker(\varphi) \cong \mathrm{Im}(\varphi) = HN/N$. □

**Note:-**

The name "Diamond Theorem" comes from the lattice diagram: draw $G$ at top, $HN$ below it, $H$ and $N$ below $HN$ on left and right, and $H \cap N$ at the bottom. The isomorphism $H/(H \cap N) \cong HN/N$ says the "left edge" quotient equals the "right edge" quotient. This is a manifestation of a general principle called the modular law in lattice theory.

**Theorem 3.3.3** Third Isomorphism Theorem

Let $G$ be a group and $N \subset M$ be normal subgroups of $G$ (so $N \trianglelefteq G$ and $M \trianglelefteq G$ with $N \subset M$). Then:

1. $M/N \trianglelefteq G/N$;
2. $(G/N)/(M/N) \cong G/M$.

***Proof:*** Proof of Third Isomorphism Theorem

**(1)** Elements of $G/N$ are cosets $gN$; elements of $M/N$ are cosets $mN$ for $m \in M$. We verify $M/N \trianglelefteq G/N$: for $gN \in G/N$ and $mN \in M/N$, we have $(gN)(mN)(gN)^{-1} = gmg^{-1}N$. Since $M \trianglelefteq G$, $gmg^{-1} \in M$, so $gmg^{-1}N \in M/N$.

**(2)** Define $\pi : G/N \to G/M$ by $\pi(gN) = gM$.

Well-defined: If $g_1N = g_2N$, then $g_1^{-1}g_2 \in N \subset M$, so $g_1M = g_2M$.

Homomorphism: $\pi(g_1N \cdot g_2N) = \pi(g_1g_2N) = g_1g_2M = (g_1M)(g_2M) = \pi(g_1N)\pi(g_2N)$.

Surjective: For any $gM \in G/M$, we have $\pi(gN) = gM$.

Kernel: $\ker(\pi) = \{gN : gM = M\} = \{gN : g \in M\} = M/N$.

By the First Isomorphism Theorem, $(G/N)/\ker(\pi) = (G/N)/(M/N) \cong \mathrm{Im}(\pi) = G/M$. ☐

**Note:-**

The Third Isomorphism Theorem says "quotients of quotients are quotients": $(G/N)/(M/N) \cong G/M$. The $N$'s "cancel." This is exactly what you'd hope for, and it's a sign that quotient groups are a natural and well-behaved construction.

### 3.3.3 The Correspondence Theorem

The final structural theorem describes the relationship between subgroups of a quotient and subgroups of the original group.

**Theorem 3.3.4** Correspondence Theorem (Fourth Isomorphism Theorem)

Let $N \trianglelefteq G$ and $\pi : G \to G/N$ be the canonical projection. There is a bijection

$$\{\text{subgroups of } G \text{ containing } N\} \longleftrightarrow \{\text{subgroups of } G/N\}$$

given by $H \mapsto H/N = \pi(H)$ with inverse $\bar{H} \mapsto \pi^{-1}(\bar{H})$. Moreover:

1. $H_1 \subset H_2 \iff H_1/N \subset H_2/N$, and $[H_2 : H_1] = [H_2/N : H_1/N]$;
2. $H \trianglelefteq G \iff H/N \trianglelefteq G/N$.

***Proof:*** Proof of Correspondence Theorem

Given $H \supset N$, we have $H/N = \{hN : h \in H\}$, which is a subgroup of $G/N$. Given a subgroup $\bar{H} \subset G/N$, define $H = \pi^{-1}(\bar{H}) = \{g \in G : gN \in \bar{H}\}$. This contains $N$ (since $eN = N \in \bar{H}$) and is a subgroup.

These operations are inverses: starting with $H \supset N$, we get $\pi^{-1}(H/N) = \{g : gN \in H/N\} = \{g : gN = hN \text{ for some } h \in H\} = \{g : g \in hN \text{ for some } h \in H\} = HN = H$ (since $N \subset H$).

Starting with $\bar{H} \subset G/N$: $\pi(\pi^{-1}(\bar{H})) = \{gN : gN \in \bar{H}\} = \bar{H}$.

**(1)** $H_1 \subset H_2 \implies H_1/N \subset H_2/N$ is clear. Conversely, if $H_1/N \subset H_2/N$ and $h \in H_1$, then $hN \in H_1/N \subset H_2/N$, so $hN = h'N$ for some $h' \in H_2$, giving $h \in h'N \subset H_2$. The index statement follows since the correspondence preserves cosets.

**(2)** If $H \trianglelefteq G$, then for $gN \in G/N$ and $hN \in H/N$: $(gN)(hN)(gN)^{-1} = ghg^{-1}N \in H/N$ since $ghg^{-1} \in H$. Conversely, if $H/N \trianglelefteq G/N$, then for $g \in G, h \in H$: $ghg^{-1}N = (gN)(hN)(gN)^{-1} \in H/N$, so $ghg^{-1} \in H$. ☐

**Question 3**

Let $G = \mathbb{Z}$ and $N = 12\mathbb{Z}$.

1. List all subgroups of $G/N \cong \mathbb{Z}/12$.
2. For each, find the corresponding subgroup of $\mathbb{Z}$ containing $12\mathbb{Z}$.
3. Which of these are normal in $\mathbb{Z}$? (Trick question!)
4. Verify the Third Isomorphism Theorem: check that $(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$.

**Solution**

**(1)** $\mathbb{Z}/12$ is cyclic of order 12, so its subgroups correspond to divisors of 12: they are $\langle\bar{0}\rangle, \langle\bar{6}\rangle, \langle\bar{4}\rangle, \langle\bar{3}\rangle, \langle\bar{2}\rangle, \langle\bar{1}\rangle$ of orders 1, 2, 3, 4, 6, 12 respectively.
**(2)** The corresponding subgroups of $\mathbb{Z}$ are those containing $12\mathbb{Z}$, i.e., of the form $d\mathbb{Z}$ where $d \mid 12$:

$$\langle\bar{0}\rangle \leftrightarrow 12\mathbb{Z}, \quad \langle\bar{6}\rangle \leftrightarrow 6\mathbb{Z}, \quad \langle\bar{4}\rangle \leftrightarrow 4\mathbb{Z},$$
$$\langle\bar{3}\rangle \leftrightarrow 3\mathbb{Z}, \quad \langle\bar{2}\rangle \leftrightarrow 2\mathbb{Z}, \quad \langle\bar{1}\rangle \leftrightarrow \mathbb{Z}.$$

**(3)** All subgroups of $\mathbb{Z}$ are normal because $\mathbb{Z}$ is abelian!
**(4)** We have $4\mathbb{Z}/12\mathbb{Z} = \{\bar{0}, \bar{4}, \bar{8}\} \cong \mathbb{Z}/3$. The quotient $(\mathbb{Z}/12)/(4\mathbb{Z}/12\mathbb{Z})$ has order $12/3 = 4$. Explicitly, the cosets are $\bar{0} + 4\mathbb{Z}/12\mathbb{Z} = \{\bar{0}, \bar{4}, \bar{8}\}$, $\bar{1} + \cdots = \{\bar{1}, \bar{5}, \bar{9}\}$, $\bar{2} + \cdots = \{\bar{2}, \bar{6}, \overline{10}\}$, $\bar{3} + \cdots = \{\bar{3}, \bar{7}, \overline{11}\}$. This is cyclic of order 4, generated by $\bar{1} + 4\mathbb{Z}/12\mathbb{Z}$, hence $\cong \mathbb{Z}/4$.
The Third Isomorphism Theorem says $(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$, which checks out!

## 3.4 The symmetric group

The symmetric group $S_n$ is arguably the most important finite group. By Cayley's theorem, every finite group embeds in some $S_n$. But $S_n$ is also interesting in its own right: its structure illuminates concepts like conjugacy classes, normal subgroups, and simplicity. Moreover, symmetric groups connect to linear algebra through the sign homomorphism and determinants, and to Galois theory through their role in permuting roots of polynomials.

### 3.4.1 Cycle structure and decomposition

**Definition 3.4.1: Symmetric group**

The **symmetric group** $S_n$ is the group of all bijections $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$ under composition. It has order $n!$.

**Definition 3.4.2: Cycle notation**

A **k-cycle** (or cycle of length $k$) is a permutation $(a_1\, a_2\, \cdots\, a_k)$ that acts by $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_k \mapsto a_1$ and fixes all other elements. A 2-cycle is called a **transposition**. Two cycles are **disjoint** if they have no elements in common; disjoint cycles commute.

**Theorem 3.4.1** Cycle Decomposition Theorem

Every permutation $\sigma \in S_n$ can be written as a product of disjoint cycles, uniquely up to the order of the factors. The decomposition is obtained by writing out the orbits of $\sigma$ acting on $\{1, \dots, n\}$.

***Proof:*** Existence and uniqueness of cycle decomposition

**Existence:** Consider the action of $\langle \sigma \rangle$ on $\{1, \dots, n\}$. The orbits partition $\{1, \dots, n\}$. For each orbit $\{a_1, a_2, \dots, a_k\}$ where $\sigma(a_i) = a_{i+1}$ (indices mod $k$), define the cycle $c = (a_1\, a_2 \cdots a_k)$. Fixed points give 1-cycles (usually omitted). Since orbits are disjoint, these cycles are disjoint.

To verify $\sigma$ equals the product of these cycles: for any $i$, if $i$ is in the orbit giving cycle $c$, then $c(i) = \sigma(i)$ and all other cycles fix $i$, so the product sends $i \mapsto \sigma(i)$.

**Uniqueness:** Suppose $\sigma = c_1 c_2 \cdots c_r = d_1 d_2 \cdots d_s$ are two decompositions into disjoint cycles. For any $i$, exactly one $c_j$ moves $i$ (the others fix it), and that $c_j$ must generate the orbit of $i$ under $\sigma$. Similarly for the $d$'s. Thus $c_j$ and the $d_k$ containing $i$ describe the same orbit, hence are the same cycle (up to cyclic rotation of the notation). □

**Definition 3.4.3: Cycle type**

The **cycle type** of $\sigma \in S_n$ is the partition of $n$ given by the lengths of the cycles in its disjoint cycle decomposition (including 1-cycles for fixed points). We write it as $(1^{m_1} 2^{m_2} \cdots n^{m_n})$ where $m_k$ is the number of $k$-cycles.

**Example 3.4.1**

In $S_6$, the permutation $\sigma = (1\,3\,5)(2\,4)$ has cycle type $(1^1 2^1 3^1)$ since there's one fixed point (6), one 2-cycle, and one 3-cycle. The identity has cycle type $(1^6)$. A single 6-cycle has type $(6^1)$.

### 3.4.2 Conjugacy in $S_n$

Conjugacy classes are fundamental to representation theory. In $S_n$, they have an elegant characterization.

**Theorem 3.4.2** Conjugacy in $S_n$

Two permutations in $S_n$ are conjugate if and only if they have the same cycle type.

***Proof:*** Proof of conjugacy classification

**Key computation:** Let $\tau \in S_n$ and $(a_1\, a_2 \cdots a_k)$ be a cycle. Then

$$\tau(a_1\, a_2 \cdots a_k)\tau^{-1} = (\tau(a_1)\, \tau(a_2) \cdots \tau(a_k)).$$

Proof: For any $i$, if $i = \tau(a_j)$, then $\tau(a_1 \cdots a_k)\tau^{-1}(i) = \tau(a_{j+1}) = \tau(a_{j+1})$. If $i \notin \{\tau(a_1), \dots, \tau(a_k)\}$, then $\tau^{-1}(i) \notin \{a_1, \dots, a_k\}$, so the cycle fixes $\tau^{-1}(i)$, and conjugation gives $\tau(\tau^{-1}(i)) = i$.

**Same type ⇒ conjugate:** Let $\sigma = (a_1 \cdots a_{k_1})(b_1 \cdots b_{k_2}) \cdots$ and $\rho = (c_1 \cdots c_{k_1})(d_1 \cdots d_{k_2}) \cdots$ have the same cycle type. Define $\tau$ by $\tau(a_i) = c_i, \tau(b_j) = d_j$, etc. Then by the key computation, $\tau\sigma\tau^{-1} = \rho$.

**Conjugate ⇒ same type:** If $\rho = \tau\sigma\tau^{-1}$, write $\sigma$ as a product of disjoint cycles. By the key computation, $\rho$ is the product of the conjugated cycles, which have the same lengths. Disjointness is preserved since $\tau$ is a bijection. □

**Proposition 3.4.1** Number of conjugacy classes in $S_n$

The number of conjugacy classes in $S_n$ equals $p(n)$, the number of partitions of $n$. For example, $p(1) = 1, p(2) = 2, p(3) = 3, p(4) = 5, p(5) = 7, p(6) = 11$.

**Note:-**

This fact that conjugacy classes correspond to partitions is one reason why the representation theory of $S_n$ is so whimsical. The irreducible representations of $S_n$ are also indexed by partitions of $n$, and the character table can be computed combinatorially. This is a whole subject in itself.

### 3.4.3 The sign homomorphism and alternating group

Every permutation can be written as a product of transpositions. The number of transpositions is not unique, but its parity is!

**Proposition 3.4.2** Transposition generation

$S_n$ is generated by transpositions. More precisely, every $k$-cycle can be written as a product of $k-1$ transpositions:

$$(a_1\, a_2\, \cdots\, a_k) = (a_1\, a_k)(a_1\, a_{k-1})\cdots(a_1\, a_2).$$

***Proof:*** Verification of transposition decomposition

Apply the right side to $a_1$: $(a_1\, a_2)$ sends $a_1 \mapsto a_2$, then subsequent transpositions fix $a_2$. So $a_1 \mapsto a_2$. ✓

Apply to $a_j$ for $2 \leqslant j \leqslant k-1$: transpositions $(a_1\, a_2), \ldots, (a_1\, a_{j-1})$ fix $a_j$. Then $(a_1\, a_j)$ sends $a_j \mapsto a_1$. Then $(a_1\, a_{j+1})$ sends $a_1 \mapsto a_{j+1}$. Subsequent transpositions fix $a_{j+1}$. So $a_j \mapsto a_{j+1}$. ✓

Apply to $a_k$: all transpositions except $(a_1\, a_k)$ fix $a_k$, which sends $a_k \mapsto a_1$. ✓ □

**Theorem 3.4.3** Well-definedness of parity

If $\sigma = \tau_1 \cdots \tau_r = \tau_1' \cdots \tau_s'$ are two expressions of $\sigma$ as products of transpositions, then $r \equiv s \pmod 2$.

***Proof:*** Proof of parity well-definedness

Consider the polynomial $P(x_1, \ldots, x_n) = \prod_{1\leqslant i<j\leqslant n}(x_i - x_j)$. For $\sigma \in S_n$, define $\sigma \cdot P = \prod_{i<j}(x_{\sigma(i)} - x_{\sigma(j)})$. Each factor $(x_{\sigma(i)} - x_{\sigma(j)})$ equals $\pm(x_k - x_l)$ for some $k < l$, so $\sigma \cdot P = \pm P$.

For a transposition $\tau = (a\, b)$ with $a < b$: the factor $(x_a - x_b)$ becomes $(x_b - x_a) = -(x_a - x_b)$. For $i < a < b < j$, the factor $(x_i - x_j)$ is unchanged. For $i < a$, the factors $(x_i - x_a)$ and $(x_i - x_b)$ are swapped, contributing no sign change. Similarly for other cases. Careful counting shows $\tau \cdot P = -P$ (the key is that one factor picks up a sign, and swaps come in pairs).

Thus if $\sigma = \tau_1 \cdots \tau_r$, then $\sigma \cdot P = (-1)^r P$. If also $\sigma = \tau_1' \cdots \tau_s'$, then $(-1)^r = (-1)^s$, so $r \equiv s \pmod 2$. □

**Definition 3.4.4: Sign of a permutation**

The **sign** (or **signature**) of $\sigma \in S_n$ is $\operatorname{sgn}(\sigma) = (-1)^r$ where $\sigma = \tau_1 \cdots \tau_r$ is any expression as a product of transpositions. Equivalently, $\operatorname{sgn}(\sigma) = (-1)^{n-c(\sigma)}$ where $c(\sigma)$ is the number of cycles (including 1-cycles). A permutation is **even** if $\operatorname{sgn}(\sigma) = 1$ and **odd** if $\operatorname{sgn}(\sigma) = -1$.

**Theorem 3.4.4** Sign is a homomorphism

The map $\operatorname{sgn} : S_n \to \{1, -1\}$ is a surjective group homomorphism.

***Proof:*** Proof that sign is a homomorphism

If $\sigma = \tau_1 \cdots \tau_r$ and $\rho = \tau_1' \cdots \tau_s'$, then $\sigma\rho = \tau_1 \cdots \tau_r \tau_1' \cdots \tau_s'$, a product of $r + s$ transpositions. Thus $\operatorname{sgn}(\sigma\rho) = (-1)^{r+s} = (-1)^r(-1)^s = \operatorname{sgn}(\sigma)\operatorname{sgn}(\rho)$. Surjectivity: any transposition has sign $-1$. □

**Definition 3.4.5: Alternating group**

The **alternating group** $A_n = \ker(\mathrm{sgn}) = \{\sigma \in S_n : \mathrm{sgn}(\sigma) = 1\}$ is the group of even permutations.

**Proposition 3.4.3** Properties of $A_n$

We have:

1. $A_n \trianglelefteq S_n$ with index $[S_n : A_n] = 2$, so $|A_n| = n!/2$.
2. $S_n/A_n \cong \mathbb{Z}/2$.
3. $A_n$ is generated by 3-cycles. In fact, $A_n = \langle (1\,2\,3), (1\,2\,4), \ldots, (1\,2\,n) \rangle$.
4. For $n \geqslant 5$, $A_n$ is **simple** (has no proper nontrivial normal subgroups).

***Proof:*** Proof of $A_n$ properties (1)-(3)

**(1)-(2)** follow from the First Isomorphism Theorem: $S_n/\ker(\mathrm{sgn}) \cong \mathrm{Im}(\mathrm{sgn}) = \{1, -1\} \cong \mathbb{Z}/2$.

**(3)** Every even permutation is a product of an even number of transpositions. We can pair them: $(a\,b)(c\,d)$. If $\{a, b\} \cap \{c, d\} = \emptyset$, then $(a\,b)(c\,d) = (a\,c\,b)(a\,c\,d)$ (verify!). If $\{a, b\} \cap \{c, d\} = \{b\} = \{c\}$, then $(a\,b)(b\,d) = (a\,b\,d)$. Thus $A_n$ is generated by 3-cycles. For the specific generating set:

$$(a\,b\,c) = (1\,2\,a)(1\,2\,b)(1\,2\,c)(1\,2\,a)^{-1}(1\,2\,b)^{-1}$$

(if $a, b, c$ all differ from 1,2; similar manipulations handle other cases). □

**Note:-**

The simplicity of $A_n$ for $n \geqslant 5$ is a historically revolutionary result (it regards the non-existence of the quintic formula). It implies that $A_n$ has no quotients except $\{e\}$ and itself, which means $A_n$ is "irreducible" in a sense. Via Galois theory, this simplicity is precisely why the general quintic polynomial has no solution in radicals: the Galois group of the "generic" degree-$n$ polynomial is $S_n$, and solvability by radicals requires a composition series with abelian quotients. For $n \geqslant 5$, $A_n$ blocks this because it's simple and non-abelian.

**Note:-**

The sign homomorphism is intimately related to the determinant: for a matrix $A \in GL_n(\mathbb{R})$, we have $\det(A) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}$. The sign ensures the determinant is alternating in the rows (swapping rows negates the determinant). This is no coincidence: the determinant is the unique alternating multilinear function on $n$ vectors with $\det(I) = 1$, and "alternating" is encoded by the sign.

**Question 4**

1. Compute the cycle type and sign of $\sigma = (1\,2\,3)(4\,5)(1\,4)(2\,5\,3)$ in $S_5$. (First simplify!)
2. How many elements of $S_7$ have cycle type $(1^1 2^1 4^1)$?
3. Prove that the only normal subgroups of $S_3$ are $\{e\}, A_3, S_3$.
4. Let $H = \{e, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\} \subset S_4$. Prove $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ and $H \trianglelefteq S_4$. What is $S_4/H$?

**Solution**

**(1)** First, multiply the cycles right-to-left. Trace each element:

$$1 \mapsto 4 \mapsto 5 \mapsto 5 \mapsto 4 \mapsto 1 : \text{wait, let's be systematic.}$$

Write $\sigma = (1\,2\,3)(4\,5)(1\,4)(2\,5\,3)$. Reading right-to-left: start with $(2\,5\,3)$, then $(1\,4)$, then $(4\,5)$, then $(1\,2\,3)$.

Trace 1: $1 \xrightarrow{(2\,5\,3)} 1 \xrightarrow{(1\,4)} 4 \xrightarrow{(4\,5)} 5 \xrightarrow{(1\,2\,3)} 5$. So $1 \mapsto 5$.
Trace 2: $2 \xrightarrow{(2\,5\,3)} 5 \xrightarrow{(1\,4)} 5 \xrightarrow{(4\,5)} 4 \xrightarrow{(1\,2\,3)} 4$. So $2 \mapsto 4$.
Trace 3: $3 \xrightarrow{(2\,5\,3)} 2 \xrightarrow{(1\,4)} 2 \xrightarrow{(4\,5)} 2 \xrightarrow{(1\,2\,3)} 3$. So $3 \mapsto 3$ (fixed).
Trace 4: $4 \xrightarrow{(2\,5\,3)} 4 \xrightarrow{(1\,4)} 1 \xrightarrow{(4\,5)} 1 \xrightarrow{(1\,2\,3)} 2$. So $4 \mapsto 2$.
Trace 5: $5 \xrightarrow{(2\,5\,3)} 3 \xrightarrow{(1\,4)} 3 \xrightarrow{(4\,5)} 3 \xrightarrow{(1\,2\,3)} 1$. So $5 \mapsto 1$.
Thus $\sigma = (1\,5)(2\,4)$, cycle type $(1^1 2^2)$. Sign: two 2-cycles = two transpositions, so $\operatorname{sgn}(\sigma) = (-1)^2 = 1$ (even).
**(2)** Cycle type $(1^1 2^1 4^1)$ means one fixed point, one 2-cycle, one 4-cycle. Choose the fixed point: 7 ways. Choose 4 elements for the 4-cycle from the remaining 6: $\binom{6}{4} = 15$ ways. Arrange them in a 4-cycle: $(4-1)! = 6$ ways. The remaining 2 elements form the 2-cycle: 1 way. Total: $7 \cdot 15 \cdot 6 \cdot 1 = 630$.
**(3)** $S_3$ has order 6. By Lagrange, subgroups have order 1, 2, 3, or 6. Normal subgroups must be unions of conjugacy classes. Conjugacy classes in $S_3$: $\{e\}$ (size 1), $\{(12), (13), (23)\}$ (size 3), $\{(123), (132)\}$ (size 2).
A normal subgroup of order 2 would need size-1 and size-1 classes, but $\{e\}$ has size 1 and we'd need another element with a size-1 class—none exist. A normal subgroup of order 3 must be $\{e\} \cup \{(123), (132)\} = A_3$. ✓
**(4)** Check $H$ is a subgroup: $(1\,2)(3\,4) \cdot (1\,3)(2\,4) = (1\,4)(2\,3)$, etc.—closure holds (verify all products give elements of $H$). Each non-identity element has order 2. So $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four-group).
Normality: $H$ consists of $e$ plus all products of two disjoint transpositions. These form a conjugacy class in $S_4$ (cycle type $(2^2)$), so $H$ is a union of conjugacy classes, hence normal.
$S_4/H$ has order $24/4 = 6$. The quotient is non-abelian (since $S_4$ is non-abelian and $H$ is not all of $S_4$), so $S_4/H \cong S_3$. Explicitly: $S_4$ acts on the three partitions $\{\{1,2\},\{3,4\}\}, \{\{1,3\},\{2,4\}\}, \{\{1,4\},\{2,3\}\}$, and $H$ is the kernel.

## 3.5 Exact sequences

The isomorphism theorems reveal a pattern: kernels and images of homomorphisms are intimately related. Exact sequences provide elegant jargon to express these relationships, and form the foundation of homological algebra—a subject that permeates modern mathematics from algebraic topology to representation theory to algebraic geometry. We introduce the basic concepts here.

### 3.5.1 Exact sequences and their meaning

**Definition 3.5.1: Exact sequence**

A sequence of groups and homomorphisms

$$\cdots \to G_{i-1} \xrightarrow{\varphi_{i-1}} G_i \xrightarrow{\varphi_i} G_{i+1} \to \cdots$$

is **exact at $G_i$** if $\operatorname{Im}(\varphi_{i-1}) = \ker(\varphi_i)$. The sequence is **exact** if it is exact at every group (except possibly the endpoints).

The condition $\operatorname{Im}(\varphi_{i-1}) = \ker(\varphi_i)$ says: "everything that comes from the left dies when we go right." This is stronger than $\operatorname{Im}(\varphi_{i-1}) \subseteq \ker(\varphi_i)$ (which just says $\varphi_i \circ \varphi_{i-1} = 0$, i.e., the composition is the zero map).

**Example 3.5.1** (Exactness encodes injectivity and surjectivity)
Consider short sequences with trivial groups $\{e\}$:

1. $\{e\} \to A \xrightarrow{\varphi} B$ is exact at $A$ iff $\ker(\varphi) = \operatorname{Im}(\{e\} \to A) = \{e_A\}$, i.e., $\varphi$ is **injective**.
2. $A \xrightarrow{\varphi} B \to \{e\}$ is exact at $B$ iff $\operatorname{Im}(\varphi) = \ker(B \to \{e\}) = B$, i.e., $\varphi$ is **surjective**.

3. $\{e\} \to A \xrightarrow{\varphi} B \to \{e\}$ is exact iff $\varphi$ is an **isomorphism**.

### 3.5.2 Short exact sequences

The most important type of exact sequence is the short exact sequence, which captures the structure of quotient groups.

**Definition 3.5.2: Short exact sequence**

A **short exact sequence** (SES) is an exact sequence of the form

$$\{e\} \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to \{e\}$$

where exactness means: $\iota$ is injective, $\pi$ is surjective, and $\mathrm{Im}(\iota) = \ker(\pi)$.

**Note:-**

We often write a short exact sequence as $1 \to N \to G \to Q \to 1$ (using 1 for the trivial group in multiplicative notation) or $0 \to N \to G \to Q \to 0$ (in additive notation for abelian groups).

**Proposition 3.5.1** Short exact sequences and quotients

A sequence $1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$ is exact if and only if:

1. $\iota$ identifies $N$ with a normal subgroup of $G$ (namely $\iota(N) = \ker(\pi) \trianglelefteq G$);
2. $\pi$ induces an isomorphism $G/\iota(N) \cong Q$.

In other words, short exact sequences encode the statement "$G/N \cong Q$" or "$G$ is an extension of $Q$ by $N$."

***Proof:*** Proof of SES characterization

($\Rightarrow$) Suppose the sequence is exact. Then $\iota$ is injective, so $N \cong \iota(N) \subseteq G$. Exactness at $G$ gives $\iota(N) = \ker(\pi)$, which is normal in $G$. By the First Isomorphism Theorem, $G/\ker(\pi) \cong \mathrm{Im}(\pi) = Q$ (using surjectivity of $\pi$).

($\Leftarrow$) If $\iota(N) \trianglelefteq G$ and $G/\iota(N) \cong Q$, take $\pi : G \to G/\iota(N) \cong Q$ to be the quotient map (composed with the isomorphism). Then $\ker(\pi) = \iota(N) = \mathrm{Im}(\iota)$, giving exactness. $\square$

**Example 3.5.2** (Fundamental examples of short exact sequences)

1. **Integers mod $n$:** $0 \to n\mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$, where $\iota$ is inclusion and $\pi$ is reduction mod $n$.
2. **Determinant:** $1 \to SL_n(\mathbb{R}) \xrightarrow{\iota} GL_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R}^* \to 1$. The kernel of det is exactly $SL_n$.
3. **Sign:** $1 \to A_n \xrightarrow{\iota} S_n \xrightarrow{\mathrm{sgn}} \{\pm 1\} \to 1$. The alternating group is the kernel of the sign homomorphism.
4. **Center:** For any group $G$, we have $1 \to Z(G) \xrightarrow{\iota} G \xrightarrow{\pi} G/Z(G) \to 1$. The quotient $G/Z(G)$ is called the inner automorphism group (it acts on $G$ by conjugation).
5. **Product decomposition:** $1 \to N \xrightarrow{\iota} N \times Q \xrightarrow{\pi} Q \to 1$, where $\iota(n) = (n, e)$ and $\pi(n, q) = q$. This is a "trivial" extension.

### 3.5.3 Splitting and semidirect products

When does a short exact sequence "come from" a direct product? This leads to the important notion of splitting.

**Definition 3.5.3: Split exact sequence**

A short exact sequence $1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$ is **split** (or **splits**) if any of the following equivalent conditions holds:

1. There exists a homomorphism $s : Q \to G$ with $\pi \circ s = \mathrm{id}_Q$ (a **section** or **right inverse**);
2. There exists a homomorphism $r : G \to N$ with $r \circ \iota = \mathrm{id}_N$ (a **retraction** or **left inverse**);
3. $G \cong N \rtimes Q$ for some semidirect product structure.

**Note:-**

For abelian groups, split exact sequences correspond exactly to direct products: if $1 \to A \to B \to C \to 1$ splits with $A, B, C$ abelian, then $B \cong A \times C$. For non-abelian groups, the situation is more subtle—we get semidirect products, which we'll study in detail later.

**Example 3.5.3** (Split vs non-split)

1. **Splits:** $1 \to A_3 \to S_3 \to \mathbb{Z}/2 \to 1$. A section $s : \mathbb{Z}/2 \to S_3$ sends the generator to any transposition, e.g., $s(1) = (12)$. Thus $S_3 \cong A_3 \rtimes \mathbb{Z}/2$.
2. **Does not split:** $0 \to \mathbb{Z}/2 \xrightarrow{\times 2} \mathbb{Z}/4 \xrightarrow{\text{mod } 2} \mathbb{Z}/2 \to 0$. If it split, we'd have $\mathbb{Z}/4 \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, but $\mathbb{Z}/4$ has an element of order 4 while $\mathbb{Z}/2 \times \mathbb{Z}/2$ doesn't. The sequence is exact but not split.

**Proposition 3.5.2** Splitting lemma (partial)

If $1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$ has a section $s : Q \to G$, then every element $g \in G$ can be written uniquely as $g = \iota(n) \cdot s(q)$ for some $n \in N$ and $q \in Q$.

***Proof:*** Proof of splitting decomposition

**Existence:** Given $g \in G$, let $q = \pi(g) \in Q$. Then $\pi(g \cdot s(q)^{-1}) = \pi(g) \cdot \pi(s(q))^{-1} = q \cdot q^{-1} = e_Q$, so $g \cdot s(q)^{-1} \in \ker(\pi) = \iota(N)$. Thus $g \cdot s(q)^{-1} = \iota(n)$ for some $n \in N$, giving $g = \iota(n) \cdot s(q)$.

**Uniqueness:** Suppose $\iota(n_1) \cdot s(q_1) = \iota(n_2) \cdot s(q_2)$. Apply $\pi$: $e \cdot q_1 = e \cdot q_2$, so $q_1 = q_2$. Then $\iota(n_1) = \iota(n_2)$, and since $\iota$ is injective, $n_1 = n_2$. $\square$

### 3.5.4 The extension problem

The short exact sequence perspective leads to a fundamental question in group theory.

**Definition 3.5.4: Group extension**

Given groups $N$ and $Q$, an **extension of $Q$ by $N$** is a group $G$ fitting into a short exact sequence $1 \to N \to G \to Q \to 1$.

The **extension problem** asks: given $N$ and $Q$, classify all groups $G$ that are extensions of $Q$ by $N$. This is a deep problem that leads to:

1. Semidirect products (when the sequence splits);
2. Group cohomology $H^2(Q, N)$ (classifying non-split extensions);
3. The structure theory of finite groups (building groups from simple groups).

**Example 3.5.4** (Extensions of $\mathbb{Z}/2$ by $\mathbb{Z}/2$)

What groups $G$ fit into $1 \to \mathbb{Z}/2 \to G \to \mathbb{Z}/2 \to 1$? Such $G$ has order 4. We know there are exactly two groups of order 4: $\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$. Both work:

- $\mathbb{Z}/4$: the sequence $0 \to \mathbb{Z}/2 \xrightarrow{\times 2} \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$ (non-split).
- $\mathbb{Z}/2 \times \mathbb{Z}/2$: the sequence $0 \to \mathbb{Z}/2 \to \mathbb{Z}/2 \times \mathbb{Z}/2 \to \mathbb{Z}/2 \to 0$ (split).

So there are exactly two extensions, corresponding to the two groups of order 4.

**Question 5**

1. Verify that $1 \to \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{\text{mod } n} \mathbb{Z}/n \to 1$ is exact.
2. Show that the sequence $1 \to \langle r \rangle \to D_n \to \mathbb{Z}/2 \to 1$ is exact, where $\langle r \rangle \cong \mathbb{Z}/n$ is the rotation subgroup. Does it split?
3. Let $G$ be a group with normal subgroup $N$. Prove that the sequence $1 \to N \to G \to G/N \to 1$ is exact (with the obvious maps).
4. Suppose $1 \to A \xrightarrow{f} B \xrightarrow{g} C \to 1$ and $1 \to C \xrightarrow{h} D \xrightarrow{k} E \to 1$ are exact. Prove that $1 \to A \xrightarrow{f} B \xrightarrow{h \circ g} D \xrightarrow{k} E \to 1$ is exact.

**Solution**

**(1)** Let $\iota : \mathbb{Z} \to \mathbb{Z}$ be $\iota(k) = nk$ and $\pi : \mathbb{Z} \to \mathbb{Z}/n$ be reduction mod $n$.
Exactness at first $\mathbb{Z}$: $\ker(\iota) = \{k : nk = 0\} = \{0\}$, and $\mathrm{Im}(1 \to \mathbb{Z}) = \{0\}$. ✓
Exactness at second $\mathbb{Z}$: $\mathrm{Im}(\iota) = n\mathbb{Z}$ and $\ker(\pi) = n\mathbb{Z}$. ✓
Exactness at $\mathbb{Z}/n$: $\mathrm{Im}(\pi) = \mathbb{Z}/n$ and $\ker(\mathbb{Z}/n \to 1) = \mathbb{Z}/n$. ✓
**(2)** The rotation subgroup $\langle r \rangle = \{e, r, r^2, \ldots, r^{n-1}\}$ is normal in $D_n$ (we showed this earlier). The quotient $D_n/\langle r \rangle$ has order $2n/n = 2$, so $D_n/\langle r \rangle \cong \mathbb{Z}/2$. The sequence $1 \to \langle r \rangle \xrightarrow{\iota} D_n \xrightarrow{\pi} \mathbb{Z}/2 \to 1$ is exact.
Does it split? Yes! Define $s : \mathbb{Z}/2 \to D_n$ by $s(0) = e$ and $s(1) = s$ (a reflection). Then $\pi(s(1)) = \pi(s) = 1$ (since $s \notin \langle r \rangle$), so $\pi \circ s = \mathrm{id}$. Thus $D_n \cong \mathbb{Z}/n \rtimes \mathbb{Z}/2$.
**(3)** Let $\iota : N \hookrightarrow G$ be inclusion and $\pi : G \to G/N$ be the quotient map.
Exactness at $N$: $\iota$ is injective, so $\ker(\iota) = \{e\} = \mathrm{Im}(1 \to N)$. ✓
Exactness at $G$: $\mathrm{Im}(\iota) = N$ and $\ker(\pi) = N$. ✓
Exactness at $G/N$: $\pi$ is surjective, so $\mathrm{Im}(\pi) = G/N = \ker(G/N \to 1)$. ✓
**(4)** We check exactness at each position.
At $A$: Same as the original sequence—$f$ is injective.
At $B$: We need $\mathrm{Im}(f) = \ker(h \circ g)$. We have $\mathrm{Im}(f) = \ker(g)$ (exactness of first sequence). If $b \in \ker(g)$, then $g(b) = e_C$, so $(h \circ g)(b) = h(e_C) = e_D$, giving $b \in \ker(h \circ g)$. Conversely, if $(h \circ g)(b) = e_D$, then $h(g(b)) = e_D$, so $g(b) \in \ker(h) = \{e_C\}$ (since $h$ is injective). Thus $g(b) = e_C$, so $b \in \ker(g) = \mathrm{Im}(f)$.
At $D$: We need $\mathrm{Im}(h \circ g) = \ker(k)$. From exactness of second sequence, $\mathrm{Im}(h) = \ker(k)$. Since $g$ is surjective, $\mathrm{Im}(h \circ g) = h(g(B)) = h(C) = \mathrm{Im}(h) = \ker(k)$. ✓
At $E$: Same as the original sequence—$k$ is surjective.

**Note:-**

Exact sequences are the beginning of homological algebra. The ideas extend to longer exact sequences, to chain complexes (sequences where consecutive compositions are zero, not necessarily exact), and to derived functors that measure the failure of exactness. In representation theory, exact sequences of representations encode how representations decompose. In topology, the long exact sequence of a fibration is a powerful tool. The language you've learned here will reappear throughout mathematics.

## 3.6 Free groups and presentations

Every group we've seen so far has been defined concretely: permutations, matrices, integers mod $n$. But there's a fundamentally different approach: describe a group by generators and relations. This leads to group presentations, which are both computationally useful and theoretically profound. The key player is the free group—the "most general" group on a given set of generators.

### 3.6.1 Words and reduced words

**Definition 3.6.1: Alphabet and words**

Let $S$ be a set (the **alphabet** or **generating set**). Define $S^{-1} = \{s^{-1} : s \in S\}$, a formal set of "inverse symbols" disjoint from $S$. A **word** in $S$ is a finite sequence

$$w = a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$$

where each $a_i \in S$ and $\epsilon_i \in \{1, -1\}$. The **empty word**, denoted $e$ or $\varepsilon$, is the word with no letters.

**Example 3.6.1**

If $S = \{a, b\}$, then $ab^{-1}a^{-1}ba$, $aaab$, $b^{-1}b^{-1}a$, and $e$ are all words. We write $a^3$ for $aaa$ and $a^{-2}$ for $a^{-1}a^{-1}$.

**Definition 3.6.2: Concatenation**

The **concatenation** of words $w_1 = a_1 \cdots a_m$ and $w_2 = b_1 \cdots b_n$ is $w_1 w_2 = a_1 \cdots a_m b_1 \cdots b_n$. This operation is associative with identity $e$.

Now, we want words to form a group. The issue: $aa^{-1}$ should equal $e$, but as sequences they're different. We need to identify words that "should be equal."

**Definition 3.6.3: Elementary reduction**

An **elementary reduction** of a word $w$ is the operation of removing a subword of the form $ss^{-1}$ or $s^{-1}s$ (for $s \in S$). A word is **reduced** if it contains no such subwords—i.e., no letter is immediately followed by its inverse.

**Example 3.6.2**

The word $ab^{-1}ba^{-1}a$ reduces as follows:

$$ab^{-1}ba^{-1}a \to ab^{-1}b \cdot e \cdot a^{-1}a \to ab^{-1}b \to a \cdot e = a.$$

**Theorem 3.6.1** Unique reduced form

Every word can be reduced to a unique reduced word by a sequence of elementary reductions. The order of reductions doesn't matter.

***Proof:*** Sketch of unique reduction

**Termination:** Each reduction decreases the length by 2, so the process terminates.

**Uniqueness (sketch):** This is the key point. Suppose $w$ reduces to both $u$ and $v$. We use the diamond lemma: if we can perform two different reductions $w \to w_1$ and $w \to w_2$, then there exists $w'$ reachable from both $w_1$ and $w_2$.

The only interesting case is when the two reductions overlap, e.g., $w = \cdots ss^{-1}s \cdots$. Reducing $ss^{-1}$ gives $\cdots s \cdots$; reducing $s^{-1}s$ also gives $\cdots s \cdots$. They agree! So all reduction paths lead to the same result. □

### 3.6.2 The free group

**Definition 3.6.4: Free group**

The **free group on** $S$, denoted $F(S)$ or $F_n$ (where $n = |S|$), is the set of reduced words in $S$ with the group operation:

$$w_1 \cdot w_2 = \mathrm{reduce}(w_1 w_2).$$

The identity is the empty word $e$, and the inverse of $a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}$ is $a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$.

**Proposition 3.6.1** Free group is a group

The free group $F(S)$ satisfies the group axioms.

***Proof:*** Verification of group axioms for $F(S)$

**Closure:** The product of two reduced words, after reduction, is a reduced word.

**Associativity:** This is subtle! We need $(w_1 \cdot w_2) \cdot w_3 = w_1 \cdot (w_2 \cdot w_3)$. Both equal the reduction of $w_1 w_2 w_3$—but the reductions might happen in different orders. By uniqueness of reduced form, they give the same result.

**Identity:** $e \cdot w = \mathrm{reduce}(ew) = \mathrm{reduce}(w) = w$ (since $w$ is already reduced). Similarly $w \cdot e = w$.

**Inverses:** Let $w = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}$ and $w^{-1} = a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$. Then $ww^{-1} = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n} a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$. Reducing: $a_n^{\epsilon_n} a_n^{-\epsilon_n}$ cancels, then $a_{n-1}^{\epsilon_{n-1}} a_{n-1}^{-\epsilon_{n-1}}$ cancels, etc., until we get $e$. □

**Example 3.6.3** (Free group on one generator)

$F(\{a\}) = F_1$ consists of reduced words in $a, a^{-1}$: these are $\ldots, a^{-2}, a^{-1}, e, a, a^2, \ldots$. This is just $\{a^n : n \in \mathbb{Z}\}$ with $a^m \cdot a^n = a^{m+n}$. So $F_1 \cong \mathbb{Z}$.

**Example 3.6.4** (Free group on two generators)

$F_2 = F(\{a, b\})$ is much more interesting. Elements include $e, a, b, a^{-1}, b^{-1}, ab, ba, a^2, ab^{-1}, aba, bab^{-1}a^{-1}, \ldots$—infinitely many distinct reduced words. Notably, $ab \neq ba$ (both are reduced but different), so $F_2$ is non-abelian. In fact, $F_2$ contains copies of every countable free group!

### 3.6.3 The universal property

The free group is "free" because it has no relations other than the group axioms. This is captured by a universal property.

**Theorem 3.6.2** Universal property of free groups

Let $F(S)$ be the free group on $S$, and let $\iota : S \to F(S)$ be the inclusion $s \mapsto s$ (the one-letter word). For any group $G$ and any function $f : S \to G$, there exists a unique homomorphism $\tilde{f} : F(S) \to G$ such that $\tilde{f} \circ \iota = f$.

$$\begin{tikzcd} S \arrow[r, "\iota"] \arrow[dr, "f"'] & F(S) \arrow[d, dashed, "\exists! \tilde{f}"] \\ & G \end{tikzcd}$$

***Proof:*** Proof of universal property

**Existence:** Define $\tilde{f}(a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}) = f(a_1)^{\epsilon_1} \cdots f(a_n)^{\epsilon_n}$. This is well-defined on reduced words and is a homomorphism: $\tilde{f}(w_1 w_2) = \tilde{f}(w_1)\tilde{f}(w_2)$ because the reductions in $F(S)$ correspond to the group relations $f(s)f(s)^{-1} = e_G$ in $G$.

**Uniqueness:** Any homomorphism $\varphi : F(S) \to G$ with $\varphi(s) = f(s)$ for all $s \in S$ must satisfy $\varphi(a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}) = \varphi(a_1)^{\epsilon_1} \cdots \varphi(a_n)^{\epsilon_n} = f(a_1)^{\epsilon_1} \cdots f(a_n)^{\epsilon_n} = \tilde{f}(w)$. □

**Note:-**

The universal property says: to define a homomorphism out of $F(S)$, you just need to say where the generators go—there are no constraints! Any choice of images for $S$ in any group $G$ extends uniquely to a homomorphism. This is why $F(S)$ is "free"—you have complete freedom in defining maps from it. Categorically, the free group functor is left adjoint to the forgetful functor from groups to sets.

### 3.6.4 Group presentations

The universal property immediately gives us a way to describe groups by generators and relations.

**Definition 3.6.5: Group presentation**

A **presentation** of a group $G$ is an expression

$$G = \langle S \mid R \rangle$$

where $S$ is a set of **generators** and $R \subseteq F(S)$ is a set of **relators**. This means $G \cong F(S)/N$, where $N = \langle\langle R \rangle\rangle$ is the **normal closure** of $R$—the smallest normal subgroup of $F(S)$ containing $R$. Equivalently, if $R = \{r_1, r_2, \ldots\}$, we write $G = \langle S \mid r_1 = e, r_2 = e, \ldots \rangle$ or $G = \langle S \mid r_1, r_2, \ldots \rangle$.

**Note:-**

The normal closure $\langle\langle R \rangle\rangle$ consists of all products of conjugates of elements of $R$ and their inverses: $\langle\langle R \rangle\rangle = \{g_1 r_1^{\pm 1} g_1^{-1} \cdots g_k r_k^{\pm 1} g_k^{-1} : g_i \in F(S), r_i \in R\}$. We need normal closure (not just the subgroup generated by $R$) because we're quotienting, and quotients require normal subgroups.

**Example 3.6.5** (Presentations of familiar groups)

1. **Cyclic group:** $\mathbb{Z}/n = \langle a \mid a^n \rangle$. One generator, one relation.
2. **Infinite cyclic:** $\mathbb{Z} = \langle a \mid \rangle$ (no relations). The free group on one generator.
3. **Klein four-group:** $\mathbb{Z}/2 \times \mathbb{Z}/2 = \langle a, b \mid a^2, b^2, aba^{-1}b^{-1} \rangle$. The last relation says $ab = ba$ (commutativity).
4. **Dihedral group:** $D_n = \langle r, s \mid r^n, s^2, srs^{-1}r \rangle$. The last relation says $srs^{-1} = r^{-1}$, i.e., $sr = r^{-1}s$.
5. **Symmetric group:** $S_n = \langle s_1, \ldots, s_{n-1} \mid s_i^2, (s_i s_{i+1})^3, (s_i s_j)^2 \text{ for } |i - j| \geqslant 2 \rangle$, where $s_i = (i\ i+1)$. These are the Coxeter relations.
6. **Free abelian group:** $\mathbb{Z}^n = \langle a_1, \ldots, a_n \mid [a_i, a_j] \text{ for all } i, j \rangle$, where $[a, b] = aba^{-1}b^{-1}$ is the commutator.

**Proposition 3.6.2** Every group has a presentation

For any group $G$, there exist $S$ and $R$ such that $G \cong \langle S \mid R \rangle$.

***Proof:*** Existence of presentation

Take $S = G$ (yes, all of $G$!) and define $f : S \to G$ by $f(g) = g$. By the universal property, this extends to a surjective homomorphism $\tilde{f} : F(G) \to G$. Let $R = \ker(\tilde{f})$. Then $G \cong F(G)/R = \langle G \mid R \rangle$.

This is a "stupid" presentation (huge!), but it exists. Finding nice presentations is an art. □

### 3.6.5 The word problem

Presentations raise a fundamental computational question.

**Definition 3.6.6: Word problem**

Given a presentation $G = \langle S \mid R \rangle$ and two words $w_1, w_2$ in $S$, the **word problem** asks: do $w_1$ and $w_2$ represent the same element of $G$? Equivalently: is $w_1 w_2^{-1}$ in the normal closure of $R$?

**Theorem 3.6.3** Undecidability of the word problem (Novikov-Boone)

There exists a finitely presented group $G = \langle S \mid R \rangle$ (with $S, R$ finite) for which the word problem is **undecidable**—no algorithm can determine, for all pairs of words, whether they're equal in $G$.

**Note:-**

This is a somewhat saddening (or perhaps, better said sobering) result. Group presentations look like a nice computational tool, but they're actually extremely powerful—so powerful, in facct, that they can encode arbitrary computation. The word problem for $G$ can simulate a Turing machine (I needn't expound on the insanity and awesomeness of this)! This doesn't mean presentations are useless; for specific groups (like $S_n$, $D_n$, free groups), the word problem is decidable. But there's no universal algorithm.

**Question 6**

1. Prove that $F_2$ contains a subgroup isomorphic to $F_3$.
2. Show that $\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$.
3. Find a presentation for the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.
4. Prove that $\langle a, b \mid a^2, b^3, (ab)^2 \rangle \cong S_3$.

**Solution**

**(1)** In $F_2 = F(\{a, b\})$, consider the subgroup $H = \langle a^2, ab, b^2 \rangle$. We claim $H \cong F_3$.
To see this, define $f : \{x, y, z\} \to F_2$ by $f(x) = a^2, f(y) = ab, f(z) = b^2$. By the universal property, this extends to a homomorphism $\tilde{f} : F_3 \to F_2$ with image $H$. To show $\tilde{f}$ is injective, we must show no nontrivial reduced word in $x, y, z$ maps to $e$.
Suppose $w(x, y, z) \mapsto e$ in $F_2$. Substituting: $w(a^2, ab, b^2) = e$ as a reduced word in $a, b$. The key observation: $a^2, ab, b^2$ have the property that no product of them (with exponents $\pm 1$) reduces to $e$ unless the product was empty. (This can be shown by analyzing how cancellation works—the words have distinct "shapes.") This is the Schreier basis approach. Thus $\tilde{f}$ is injective, and $H \cong F_3$.
**(2)** The presentation $G = \langle a, b \mid aba^{-1}b^{-1} \rangle$ has one relation: $aba^{-1}b^{-1} = e$, i.e., $ab = ba$. So $G$ is the free group on $a, b$ modulo the relation that $a$ and $b$ commute. This is exactly the free abelian group on two generators, which is $\mathbb{Z}^2$.
Explicitly: define $\varphi : G \to \mathbb{Z}^2$ by $\varphi(a) = (1, 0)$ and $\varphi(b) = (0, 1)$. This is well-defined (the relation $ab = ba$ holds in $\mathbb{Z}^2$). It's surjective since $(1, 0), (0, 1)$ generate $\mathbb{Z}^2$. For injectivity: any element of $G$ can be written as $a^m b^n$ (using commutativity to collect), and $\varphi(a^m b^n) = (m, n)$; if this equals $(0, 0)$, then $m = n = 0$.
**(3)** The quaternion group $Q_8$ has elements $\{1, -1, i, -i, j, -j, k, -k\}$ with relations $i^2 = j^2 = k^2 = ijk = -1$.
A presentation: $Q_8 = \langle i, j \mid i^4, i^2 j^{-2}, iji^{-1}j \rangle$.
Let's verify: $i^4 = 1$ says $i$ has order dividing 4 (actually 4). $i^2 = j^2$ says both square to the same element (which will be $-1$). $iji^{-1}j = e$ means $ij = j^{-1}i^{-1} = j^{-1}i^3 = j^{-1}(i^{-1}) = (ji)^{-1}$, which combined with $i^2 = j^2$ gives $ij = k$ where $k = i^{-1}j^{-1}$, and one can derive all the quaternion relations.

Alternative (cleaner): $Q_8 = \langle x, y \mid x^4, x^2y^{-2}, yxy^{-1}x \rangle$.
**(4)** Let $G = \langle a, b \mid a^2, b^3, (ab)^2 \rangle$. We have $|a| \mid 2$, $|b| \mid 3$, and $(ab)^2 = e$ means $abab = e$, so $ab = b^{-1}a^{-1} = b^{-1}a = b^2a$ (using $a^2 = e$).
So $ab = b^2a$, i.e., $aba^{-1} = b^2 = b^{-1}$. This is exactly the dihedral relation! Compare with $D_3 = \langle r, s \mid r^3, s^2, srs^{-1} = r^{-1} \rangle$. Setting $r = b, s = a$, we get the same relations. Thus $G \cong D_3 \cong S_3$.

Lecture 4.

# Vector Spaces and Linear Algebra

This lecture marks our transition from the purely algebraic study of groups to the more geometric realm of linear algebra. Vector spaces are arguably the most tractable algebraic structures—unlike groups, where classification is hopelessly complex, finite-dimensional vector spaces over a fixed field are completely classified by a single integer: the dimension. We begin by establishing the algebraic prerequisites (ring and field homomorphisms, polynomial rings) before developing the theory of vector spaces systematically.

## 4.1 Ring and Field Homomorphisms

Just as group homomorphisms preserve the group structure, ring and field homomorphisms preserve both operations simultaneously. These maps are fundamental for understanding how algebraic structures relate to one another.

**Definition 4.1.1: Ring homomorphism**

A **ring homomorphism** is a map $\varphi : R \to S$ between rings that respects both operations:

1. $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in R$;
2. $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$;
3. $\varphi(1_R) = 1_S$.

A **field homomorphism** is a ring homomorphism between fields.

**Note:-**

Condition (1) implies that $\varphi$ is a group homomorphism $(R, +) \to (S, +)$, so we automatically get $\varphi(0) = 0$ and $\varphi(-a) = -\varphi(a)$. However, condition (3) does not follow from condition (2), even for fields! Consider $\varphi = 0$: this satisfies $\varphi(ab) = 0 = 0 \cdot 0 = \varphi(a)\varphi(b)$, but $\varphi(1) = 0 \neq 1$. We must include (3) as a separate axiom.

**Proposition 4.1.1** Field homomorphisms are injective

If $\varphi : k \to K$ is a field homomorphism, then $\varphi$ is injective.

***Proof:*** Let $a \in k$ with $a \neq 0$. Since $k$ is a field, there exists $b = a^{-1}$ such that $ab = 1_k$. Applying $\varphi$:

$$\varphi(a)\varphi(b) = \varphi(ab) = \varphi(1_k) = 1_K \neq 0_K.$$

This implies $\varphi(a) \neq 0_K$ (if $\varphi(a) = 0$, then $\varphi(a)\varphi(b) = 0 \neq 1_K$).

Thus $\ker(\varphi) = \{a \in k : \varphi(a) = 0_K\} = \{0_k\}$, so $\varphi$ is injective. $\square$

**Note:-**

This means every field homomorphism is an embedding. There are no "collapsing" maps between fields. This stands in sharp contrast to group and ring homomorphisms, which can have large kernels. The reason is that fields have no nontrivial ideals (a consequence of having multiplicative inverses), so the kernel of any field homomorphism must be either $\{0\}$ or the entire field—but the latter is ruled out by $\varphi(1) = 1 \neq 0$.

## 4.2 Polynomial Rings and Field Extensions

Polynomials are central objects connecting algebra to analysis, number theory, and geometry. Given a field $k$, we construct the ring of polynomials $k[x]$ and explore how to enlarge $k$ to contain roots of polynomials.

### 4.2.1 The polynomial ring $k[x]$

**Definition 4.2.1: Polynomial ring**

Given a field $k$, the **polynomial ring in one variable** is

$$k[x] \coloneqq \left\{ a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \;\middle|\; a_i \in k,\, n \in \mathbb{N} \right\}.$$

Here $x$ is a **formal variable**—it is not an element of $k$ or any particular set. A polynomial is simply a finite sequence of coefficients, and we add and multiply term by term using the familiar rules.

**Note:-**

The adjective "formal" is important. When we write $x^2 + 1 \in \mathbb{R}^x$, we are not thinking of $x$ as a real number—we are thinking of $x^2 + 1$ as a symbolic expression. However, we can evaluate a polynomial at any element of $k$ (or any field extension of $k$): given $f(x) = a_0 + a_1 x + \cdots + a_n x^n$ and $r \in k$, we define $f(r) = a_0 + a_1 r + \cdots + a_n r^n \in k$.

The polynomial ring $k[x]$ is indeed a ring: it is closed under addition and multiplication, has identity 1, and satisfies all the ring axioms. However, it is not a field.

**Proposition 4.2.1** $k[x]$ is not a field

The polynomial ring $k[x]$ has no multiplicative inverses for non-constant polynomials. For instance, $x$ has no inverse in $k[x]$.

***Proof:*** Suppose $f(x) \in k[x]$ satisfies $x \cdot f(x) = 1$. Then $\deg(x \cdot f(x)) = 1 + \deg(f(x)) \geqslant 1$, but $\deg(1) = 0$. Contradiction. □

### 4.2.2 The field of rational functions

To make $k[x]$ into a field, we must allow "fractions" of polynomials, just as we pass from $\mathbb{Z}$ to $\mathbb{Q}$.

**Definition 4.2.2: Field of rational functions**

The **field of rational functions** over $k$ is

$$k(x) \coloneqq \left\{ \frac{p}{q} \;\middle|\; p, q \in k[x],\; q \neq 0 \right\} \Big/ \sim$$

where $\frac{p}{q} \sim \frac{p'}{q'}$ iff $pq' = p'q$. Addition and multiplication are defined as for ordinary fractions.

This construction generalizes naturally to polynomials in any number of variables, giving $k(x_1, \ldots, x_n)$, the field of rational functions in $n$ variables.

### 4.2.3 Formal power series

Another way to extend $k[x]$ is to allow infinite sums—formal power series.

**Definition 4.2.3: Formal power series**

The **ring of formal power series** over $k$ is

$$k[\![x]\!] := \left\{ \sum_{i=0}^{\infty} a_i x^i \;\middle|\; a_i \in k \right\}.$$

Addition and multiplication are defined term by term: given $f = \sum a_i x^i$ and $g = \sum b_i x^i$, we have $f + g = \sum (a_i + b_i) x^i$ and $fg = \sum c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$.

**Note:-**

The product formula makes sense because each coefficient $c_k$ is a finite sum—only finitely many pairs $(i, j)$ satisfy $i + j = k$ with $i, j \geqslant 0$. This is why we call these "formal" power series: we never worry about convergence. The ring $k[\![x]\!]$ contains $k[x]$ as a subring (polynomials are power series with only finitely many nonzero coefficients).

Unlike $k[x]$, the ring $k[\![x]\!]$ has many more invertible elements.

**Lemma 4.2.1** Invertibility in $k[\![x]\!]$

A power series $f = \sum_{i=0}^{\infty} a_i x^i \in k[\![x]\!]$ has a multiplicative inverse if and only if $a_0 \neq 0$.

***Proof:*** ($\Rightarrow$) If $fg = 1$ for some $g = \sum b_i x^i$, then comparing constant terms gives $a_0 b_0 = 1$, so $a_0 \neq 0$.
($\Leftarrow$) Assume $a_0 \neq 0$. We construct $g = \sum b_i x^i$ satisfying $fg = 1$ inductively.
The constant term equation is $a_0 b_0 = 1$, so $b_0 = a_0^{-1}$ (which exists since $k$ is a field).
The coefficient of $x^1$ equation is $a_0 b_1 + a_1 b_0 = 0$, so $b_1 = -a_0^{-1}(a_1 b_0) = -a_0^{-1} a_1 a_0^{-1}$.
In general, the coefficient of $x^n$ gives $\sum_{i=0}^{n} a_i b_{n-i} = 0$ for $n \geqslant 1$, i.e.,

$$a_0 b_n + \sum_{i=1}^{n} a_i b_{n-i} = 0 \implies b_n = -a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i}.$$

Each $b_n$ depends only on $b_0, \ldots, b_{n-1}$, so we can solve inductively. $\square$

**Example 4.2.1**

The geometric series: in $k[\![x]\!]$, we have $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \cdots = \sum_{i=0}^{\infty} x^i$. This is verified directly: $(1 - x)(1 + x + x^2 + \cdots) = 1$ since all higher terms cancel.

### 4.2.4 Laurent series and field of fractions

Every nonzero element of $k[\![x]\!]$ can be written as $x^m(a_m + a_{m+1}x + \cdots)$ where $a_m \neq 0$. To get a field, we need to invert $x^m$—i.e., allow negative powers.

**Definition 4.2.4: Field of Laurent series**

The **field of Laurent series** over $k$ is

$$k(\!(x)\!) := \left\{ \sum_{i=m}^{\infty} a_i x^i \;\middle|\; m \in \mathbb{Z},\ a_i \in k \right\}.$$

Elements are "power series starting at some (possibly negative) power of $x$."

**Proposition 4.2.2** $k((x))$ is a field

Every nonzero Laurent series has a multiplicative inverse.

***Proof:*** Let $f = \sum_{i=m}^{\infty} a_i x^i$ with $a_m \neq 0$. Then $f = x^m g$ where $g = a_m + a_{m+1}x + \cdots \in k[\![x]\!]$ has nonzero constant term. By the lemma, $g^{-1}$ exists in $k[\![x]\!]$. Thus $f^{-1} = x^{-m} g^{-1} \in k((x))$. □

### 4.2.5 Roots of polynomials and field extensions

Given a polynomial $f \in k[x]$ with $\deg(f) \geqslant 1$, we can ask: does $f$ have a root in $k$? If not, can we extend $k$ to a larger field $K$ containing a root?

**Definition 4.2.5: Root and field extension**

Let $f \in k[x]$ and $K$ be a field containing $k$. An element $r \in K$ is a **root** of $f$ if $f(r) = 0$. We call $K$ a **field extension** of $k$, written $K/k$ or $k \subset K$.

**Example 4.2.2**

The polynomial $x^2 - 2 \in \mathbb{Q}^x$ has no roots in $\mathbb{Q}$, but we can form the extension

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\},$$

which is a field containing $\mathbb{Q}$ and the root $\sqrt{2}$. To verify this is a field, we need to check multiplicative inverses: for $a + b\sqrt{2} \neq 0$,

$$\frac{1}{a+b\sqrt{2}} = \frac{a-b\sqrt{2}}{a^2-2b^2} = \frac{a}{a^2-2b^2} - \frac{b}{a^2-2b^2}\sqrt{2} \in \mathbb{Q}(\sqrt{2}).$$

Note that $a^2 - 2b^2 \neq 0$ when $(a,b) \neq (0,0)$, since $\sqrt{2} \notin \mathbb{Q}$.

**Example 4.2.3**

Similarly, $x^2 + 1 \in \mathbb{R}^x$ has no real roots. The extension

$$\mathbb{R}(i) = \mathbb{R}(\sqrt{-1}) = \{a + bi : a, b \in \mathbb{R}\} = \mathbb{C}$$

is the field of complex numbers, which contains roots of $x^2 + 1$ (namely $\pm i$).

**Definition 4.2.6: Algebraically closed field**

A field $k$ is **algebraically closed** if every non-constant polynomial $f \in k[x]$ has a root in $k$. Equivalently, every polynomial factors completely into linear factors over $k$.

**Note:-**

The Fundamental Theorem of Algebra states that $\mathbb{C}$ is algebraically closed. Over an algebraically closed field, every non-constant polynomial has a root, and there are no further algebraic extensions. This is why $\mathbb{C}$ is often called the "algebraic closure" of $\mathbb{R}$. In contrast, $\mathbb{Q}$ is very far from algebraically closed—there are infinitely many distinct algebraic extensions of $\mathbb{Q}$.

## 4.3 Characteristic of a Field

Every field $k$ has a distinguished ring homomorphism from the integers, whose kernel determines an important invariant.

**Proposition 4.3.1** Universal map from $\mathbb{Z}$

For any field $k$, there is a unique ring homomorphism $\varphi : \mathbb{Z} \to k$ defined by $\varphi(1) = 1_k$. Explicitly, $\varphi(n) = n \cdot 1_k$ (add $1_k$ to itself $n$ times for $n > 0$; negate for $n < 0$).

***Proof:*** Uniqueness: Any ring homomorphism must satisfy $\varphi(1) = 1_k$, and since $\mathbb{Z} = \langle 1 \rangle$ as an additive group, $\varphi$ is determined by its value on 1.

Existence: Define $\varphi(n) = n \cdot 1_k$. Then $\varphi(m+n) = (m+n) \cdot 1_k = m \cdot 1_k + n \cdot 1_k = \varphi(m) + \varphi(n)$ and $\varphi(mn) = (mn) \cdot 1_k = (m \cdot 1_k)(n \cdot 1_k) = \varphi(m)\varphi(n)$ (verify by induction). Also $\varphi(1) = 1_k$. $\square$

The kernel of $\varphi$ is a subgroup of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ for some $n \geqslant 0$.

**Proposition 4.3.2** Kernel is zero or prime

The kernel $\ker(\varphi)$ equals $n\mathbb{Z}$ where either $n = 0$ or $n = p$ is prime.

***Proof:*** Suppose $n > 0$ and $n$ is not prime, so $n = ab$ with $1 < a, b < n$. Then

$$\varphi(n) = \varphi(ab) = \varphi(a)\varphi(b) = 0_k.$$

In a field, $xy = 0$ implies $x = 0$ or $y = 0$. So $\varphi(a) = 0$ or $\varphi(b) = 0$, meaning $a \in \ker(\varphi)$ or $b \in \ker(\varphi)$. But $\ker(\varphi) = n\mathbb{Z}$ and $a, b < n$, so $a$ or $b$ is in $n\mathbb{Z}$ only if $a = 0$ or $b = 0$—contradicting $a, b > 1$.

Thus if $n > 0$, then $n$ must be prime. $\square$

**Definition 4.3.1: Characteristic**

The <u>**characteristic**</u> of a field $k$, denoted $\mathrm{char}(k)$, is:

- $0$ if $\ker(\varphi) = \{0\}$, i.e., $\varphi : \mathbb{Z} \hookrightarrow k$ is injective;
- $p$ if $\ker(\varphi) = p\mathbb{Z}$ for a prime $p$.

Equivalently, $\mathrm{char}(k) = p$ means $\underbrace{1_k + 1_k + \cdots + 1_k}_{p \text{ times}} = 0_k$, and $p$ is the smallest such positive integer.

**Example 4.3.1**

- $\mathrm{char}(\mathbb{Q}) = \mathrm{char}(\mathbb{R}) = \mathrm{char}(\mathbb{C}) = 0$. The map $\varphi : \mathbb{Z} \to \mathbb{Q}$ is injective.
- $\mathrm{char}(\mathbb{Z}/p) = p$ for prime $p$. Indeed, $p \cdot 1 = 0$ in $\mathbb{Z}/p$.
- $\mathrm{char}(k(x)) = \mathrm{char}(k[\![x]\!]) = \mathrm{char}(k(\!(x)\!)) = \mathrm{char}(k)$.

**Note:-**

The field $\mathbb{Z}/p$ is often denoted $\mathbb{F}_p$ and called the <u>**prime field**</u> of characteristic $p$. Every field of characteristic $p$ contains a copy of $\mathbb{F}_p$; every field of characteristic 0 contains a copy of $\mathbb{Q}$. Thus $\mathbb{F}_p$ and $\mathbb{Q}$ are the "minimal" fields in their respective characteristics.

**Theorem 4.3.1** Existence and uniqueness of finite fields

For every prime $p$ and $n \geqslant 1$, there exists a unique (up to isomorphism) field with exactly $p^n$ elements, denoted $\mathbb{F}_{p^n}$ or $\mathrm{GF}(p^n)$. These are <u>all</u> the finite fields.

**Note:-**

We will not prove this theorem here. The finite field $\mathbb{F}_{p^n}$ can be constructed as a splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$. There are also infinite fields of characteristic $p$—for example, $\mathbb{F}_p(x)$, the field of rational functions over $\mathbb{F}_p$.

## 4.4 Vector Spaces

We now arrive at the central structure of linear algebra. A vector space is a set with two operations—addition and scalar multiplication—satisfying axioms that generalize familiar properties of $\mathbb{R}^n$.

**Definition 4.4.1: Vector space**

Fix a field $k$. A **vector space over $k$** (or **$k$-vector space**) is a set $V$ equipped with two operations:

1. **Vector addition:** $+ : V \times V \to V$;
2. **Scalar multiplication:** $\cdot : k \times V \to V$;

satisfying the following axioms:

1. $(V, +)$ is an abelian group with identity $0$ (the **zero vector**);
2. $1 \cdot v = v$ for all $v \in V$ (identity for scalar multiplication);
3. $(ab) \cdot v = a \cdot (b \cdot v)$ for all $a, b \in k$ and $v \in V$ (associativity);
4. $(a + b) \cdot v = a \cdot v + b \cdot v$ for all $a, b \in k$ and $v \in V$ (distributivity over scalar addition);
5. $a \cdot (v + w) = a \cdot v + a \cdot w$ for all $a \in k$ and $v, w \in V$ (distributivity over vector addition).

Elements of $V$ are called **vectors**; elements of $k$ are called **scalars**.

**Note:-**

The distributive laws (4) and (5) connect the two operations. Note that axiom (1) implies $0_V + v = v$ and $v + (-v) = 0_V$, while axiom (2)-(5) give us things like $0_k \cdot v = 0_V$ (proof: $0_k \cdot v = (0_k + 0_k) \cdot v = 0_k \cdot v + 0_k \cdot v$, so $0_k \cdot v = 0_V$ by cancellation). We also get $(-1) \cdot v = -v$.

### 4.4.1 Fundamental examples

**Example 4.4.1**

**The standard vector space $k^n$:** For any $n \geqslant 1$, the set

$$k^n = \{(a_1, a_2, \dots, a_n) : a_i \in k\}$$

is a vector space with componentwise operations:

$$\begin{aligned}(a_1, \dots, a_n) + (b_1, \dots, b_n) &= (a_1 + b_1, \dots, a_n + b_n),\\ c \cdot (a_1, \dots, a_n) &= (ca_1, \dots, ca_n).\end{aligned}$$

This is the prototypical $n$-dimensional vector space.

**Example 4.4.2**

**Infinite sequences $k^\infty$:** The set of all sequences in $k$,

$$k^\infty = \{(a_i)_{i \in \mathbb{N}} : a_i \in k\} = \{(a_1, a_2, a_3, \dots) : a_i \in k\},$$

is a vector space with componentwise operations. It contains the subspace of **eventually zero sequences**—those with $a_i = 0$ for all sufficiently large $i$.

**Example 4.4.3**

**Function spaces:** For any set $S$, the set

$$k^S = \{f : S \to k\}$$

of all functions from $S$ to $k$ is a vector space with pointwise operations: $(f+g)(s) = f(s) + g(s)$ and $(c \cdot f)(s) = c \cdot f(s)$.
When $S = \mathbb{N}$, we recover $k^\infty$. When $S = \{1, \ldots, n\}$, we recover $k^n$.

**Example 4.4.4**

**Polynomial and power series spaces:**

- $k[x]$ is a $k$-vector space (addition is polynomial addition; scalar multiplication is multiplying each coefficient by the scalar).
- $k[\![x]\!]$ is a $k$-vector space.
- For each $d \geqslant 0$, the set $k[x]_{\leqslant d} = \{f \in k[x] : \deg(f) \leqslant d\}$ of polynomials of degree at most $d$ is a $k$-vector space.

**Example 4.4.5**

**Spaces of maps $\mathbb{R} \to \mathbb{R}$:** Consider $V = \{f : \mathbb{R} \to \mathbb{R}\}$, all functions from $\mathbb{R}$ to $\mathbb{R}$. This contains many interesting subspaces:

- Continuous functions $C(\mathbb{R})$;
- Differentiable functions $C^1(\mathbb{R})$;
- Smooth (infinitely differentiable) functions $C^\infty(\mathbb{R})$;
- Polynomials $\mathbb{R}^x$.

Each is closed under addition and scalar multiplication, forming a vector space.

### 4.4.2 Subspaces

**Definition 4.4.2: Subspace**

A **subspace** of a vector space $V$ is a nonempty subset $W \subseteq V$ that is closed under addition and scalar multiplication:

1. $w_1, w_2 \in W \implies w_1 + w_2 \in W$;
2. $w \in W$, $a \in k \implies a \cdot w \in W$.

Equivalently, $W$ is a subspace if it is a vector space under the inherited operations.

**Proposition 4.4.1** Subspace criterion

A nonempty subset $W \subseteq V$ is a subspace if and only if $aw_1 + bw_2 \in W$ for all $a, b \in k$ and $w_1, w_2 \in W$.

***Proof:*** ($\Rightarrow$) If $W$ is a subspace, then $aw_1 \in W$ and $bw_2 \in W$ by scalar multiplication closure, and $aw_1 + bw_2 \in W$ by addition closure.

($\Leftarrow$) Taking $a = b = 1$ gives closure under addition. Taking $b = 0$ gives closure under scalar multiplication. $\square$

**Note:-**

The condition "nonempty" is required. Alternatively, we can replace it with "$0 \in W$," which follows from taking $a = 0$ in the scalar multiplication closure. So a subspace is precisely a nonempty subset closed under linear combinations.

**Example 4.4.6**

- $\{0\}$ and $V$ itself are always subspaces (the **trivial subspaces**).
- In $\mathbb{R}^3$, subspaces are: $\{0\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$.
- In $k[x]$, the polynomials of degree $\leqslant n$ form a subspace. The polynomials of degree exactly $n$ do not form a subspace (not closed under addition: $x^n + (-x^n) = 0$).

### 4.4.3 Direct Sums of Subspaces

When can we decompose a vector space into "independent" pieces? The notion of direct sum captures exactly this.

**Definition 4.4.3: Sum of subspaces**

Given subspaces $U, W \subseteq V$, their **sum** is

$$U + W = \{u + w : u \in U,\ w \in W\}.$$

This is the smallest subspace of $V$ containing both $U$ and $W$.

**Proposition 4.4.2** Sum is a subspace

For subspaces $U, W \subseteq V$, the sum $U + W$ is a subspace of $V$.

***Proof:*** Let $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ be elements of $U + W$, and let $\lambda \in k$. Then

$$v_1 + v_2 = (u_1 + u_2) + (w_1 + w_2) \in U + W$$

since $u_1 + u_2 \in U$ and $w_1 + w_2 \in W$. Similarly, $\lambda v_1 = \lambda u_1 + \lambda w_1 \in U + W$. The zero vector $0 = 0 + 0 \in U + W$. □

**Definition 4.4.4: Direct sum of subspaces**

The sum $U + W$ is called a **direct sum**, written $U \oplus W$, if $U \cap W = \{0\}$. Equivalently, every element of $U + W$ can be written uniquely as $u + w$ with $u \in U$ and $w \in W$.

**Proposition 4.4.3** Characterizations of direct sum

For subspaces $U, W \subseteq V$, the following are equivalent:

1. $U + W$ is a direct sum (i.e., $U \cap W = \{0\}$).
2. Every $v \in U + W$ has a unique representation $v = u + w$ with $u \in U$, $w \in W$.
3. $u + w = 0$ with $u \in U$, $w \in W$ implies $u = w = 0$.

***Proof:*** (1) $\Rightarrow$ (2): Suppose $v = u_1 + w_1 = u_2 + w_2$. Then $u_1 - u_2 = w_2 - w_1$. The left side is in $U$, the right side is in $W$, so both are in $U \cap W = \{0\}$. Thus $u_1 = u_2$ and $w_1 = w_2$.

(2) $\Rightarrow$ (3): If $u + w = 0$, then $u + w = 0 + 0$. By uniqueness, $u = 0$ and $w = 0$.

(3) $\Rightarrow$ (1): If $v \in U \cap W$, then $v + (-v) = 0$ with $v \in U$ and $-v \in W$. By (3), $v = 0$. □

**Theorem 4.4.1** Dimension formula for sums

For finite-dimensional subspaces $U, W \subseteq V$:

$$\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

In particular, if $U + W = U \oplus W$ is a direct sum, then $\dim(U \oplus W) = \dim(U) + \dim(W)$.

***Proof:*** Let $\{v_1, \dots, v_k\}$ be a basis of $U \cap W$. Extend to a basis $\{v_1, \dots, v_k, u_1, \dots, u_m\}$ of $U$ and to a basis $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ of $W$.

**Claim:** $\{v_1, \dots, v_k, u_1, \dots, u_m, w_1, \dots, w_n\}$ is a basis of $U + W$.

Spanning: Any $x \in U + W$ is $x = u + w$ with $u \in U$, $w \in W$. Express $u$ and $w$ in terms of the bases above; the result is a linear combination of all the listed vectors.

Independence: Suppose $\sum a_i v_i + \sum b_j u_j + \sum c_\ell w_\ell = 0$. Then

$$\sum c_\ell w_\ell = -\sum a_i v_i - \sum b_j u_j \in U.$$

But $\sum c_\ell w_\ell \in W$, so $\sum c_\ell w_\ell \in U \cap W = \mathrm{Span}\{v_1, \dots, v_k\}$. Since $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ is independent, all $c_\ell = 0$. Similarly, all $b_j = 0$, and then all $a_i = 0$.

Thus $\dim(U+W) = k + m + n = (k+m) + (k+n) - k = \dim(U) + \dim(W) - \dim(U \cap W)$. ☐

**Definition 4.4.5: Internal direct sum decomposition**

We say $V$ is the **(internal) direct sum** of subspaces $U$ and $W$, written $V = U \oplus W$, if:

1. $V = U + W$ (every vector is a sum of elements from $U$ and $W$), and
2. $U \cap W = \{0\}$ (the decomposition is unique).

More generally, $V = U_1 \oplus U_2 \oplus \cdots \oplus U_n$ if every $v \in V$ can be written uniquely as $v = u_1 + u_2 + \cdots + u_n$ with $u_i \in U_i$.

**Example 4.4.7**

- In $\mathbb{R}^3$, let $U = \{(x, 0, 0) : x \in \mathbb{R}\}$ (the $x$-axis) and $W = \{(0, y, z) : y, z \in \mathbb{R}\}$ (the $yz$-plane). Then $\mathbb{R}^3 = U \oplus W$.
- Let $V = k^n$ and $U = \mathrm{Span}\{e_1, \dots, e_m\}$, $W = \mathrm{Span}\{e_{m+1}, \dots, e_n\}$. Then $k^n = U \oplus W$.
- If $V$ is the space of $n \times n$ matrices, let $S$ = symmetric matrices and $A$ = antisymmetric matrices. Then $V = S \oplus A$ (any matrix $M = \frac{1}{2}(M + M^T) + \frac{1}{2}(M - M^T)$).

**Note:-**

When we diagonalize a linear operator, we decompose the vector space as a direct sum of eigenspaces. When we study quotient spaces, we often use the fact that $V = U \oplus W$ implies $V/U \cong W$; if that doesn't make sense right now, don't worry, we'll get to it. The direct sum structure makes computations tractable by breaking a problem into independent pieces.

### 4.4.4 External Direct Sums and Products

We can also construct new vector spaces from old ones by taking products or direct sums "externally."

**Definition 4.4.6: Direct product (external direct sum)**

Given vector spaces $V_1, \ldots, V_n$ over the same field $k$, their **direct product** (or **external direct sum**) is

$$V_1 \times V_2 \times \cdots \times V_n = \{(v_1, v_2, \ldots, v_n) : v_i \in V_i\}$$

with componentwise operations:

$$(v_1, \ldots, v_n) + (w_1, \ldots, w_n) = (v_1 + w_1, \ldots, v_n + w_n),$$
$$\lambda(v_1, \ldots, v_n) = (\lambda v_1, \ldots, \lambda v_n).$$

This is also denoted $V_1 \oplus V_2 \oplus \cdots \oplus V_n$ or $\bigoplus_{i=1}^n V_i$.

**Proposition 4.4.4** Dimension of direct product

If each $V_i$ is finite-dimensional, then

$$\dim(V_1 \times \cdots \times V_n) = \dim(V_1) + \cdots + \dim(V_n).$$

***Proof:*** If $\mathcal{B}_i = \{v_{i,1}, \ldots, v_{i,d_i}\}$ is a basis of $V_i$, then

$$\{(v_{1,1}, 0, \ldots, 0), \ldots, (v_{1,d_1}, 0, \ldots, 0), (0, v_{2,1}, 0, \ldots, 0), \ldots, (0, \ldots, 0, v_{n,d_n})\}$$

is a basis of $V_1 \times \cdots \times V_n$. This has $d_1 + \cdots + d_n$ elements. □

**Note:-**

The external direct sum $V_1 \times V_2$ and the internal direct sum $U \oplus W \subseteq V$ are related as follows: if $V = U \oplus W$, then the map

$$U \times W \to V, \quad (u, w) \mapsto u + w$$

is an isomorphism. Conversely, $V_1 \times V_2$ contains copies of $V_1$ and $V_2$ as subspaces (embedded as $V_1 \times \{0\}$ and $\{0\} \times V_2$), and $V_1 \times V_2$ is their internal direct sum.

**Example 4.4.8**

The space $\mathbb{R}^{m+n}$ is isomorphic to $\mathbb{R}^m \times \mathbb{R}^n$ via $(x_1, \ldots, x_m, y_1, \ldots, y_n) \mapsto ((x_1, \ldots, x_m), (y_1, \ldots, y_n))$. This is why we freely identify $\mathbb{R}^m \oplus \mathbb{R}^n$ with $\mathbb{R}^{m+n}$.

## 4.5 Span, Linear Independence, and Basis

These three concepts—span, linear independence, and basis—are the building blocks of linear algebra. They allow us to describe vector spaces in terms of generating sets and coordinates.

### 4.5.1 Linear combinations and span

**Definition 4.5.1: Linear combination**

A **linear combination** of vectors $v_1, \ldots, v_n \in V$ is any vector of the form

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

where $a_1, \ldots, a_n \in k$ are scalars.

**Definition 4.5.2: Span**

Given vectors $v_1, \dots, v_n \in V$, the **span** of $\{v_1, \dots, v_n\}$ is the set of all linear combinations:

$$\mathrm{Span}(v_1, \dots, v_n) = \{a_1 v_1 + \cdots + a_n v_n : a_i \in k\}.$$

This is the **smallest subspace** of $V$ containing $v_1, \dots, v_n$.
We say $v_1, \dots, v_n$ **span** $V$ (or are a **spanning set** for $V$) if $\mathrm{Span}(v_1, \dots, v_n) = V$.

**Proposition 4.5.1** Span is a subspace

For any $v_1, \dots, v_n \in V$, the span $\mathrm{Span}(v_1, \dots, v_n)$ is a subspace of $V$.

***Proof:*** Let $W = \mathrm{Span}(v_1, \dots, v_n)$. Clearly $0 = 0 \cdot v_1 + \cdots + 0 \cdot v_n \in W$, so $W \neq \emptyset$. For closure: if $w = \sum a_i v_i$ and $w' = \sum a_i' v_i$ are in $W$, and $c, c' \in k$, then

$$cw + c'w' = \sum (ca_i + c'a_i')v_i \in W.$$

$\square$

The span can also be defined for infinite (or even arbitrary) sets of vectors.

**Definition 4.5.3: Span of an arbitrary set**

For $S \subseteq V$, the **span** of $S$ is the set of all finite linear combinations of elements of $S$:

$$\mathrm{Span}(S) = \left\{ \sum_{i=1}^{n} a_i v_i : n \geqslant 0,\ a_i \in k,\ v_i \in S \right\}.$$

When $n = 0$, the empty sum is 0, so $0 \in \mathrm{Span}(S)$ always.

### 4.5.2 Linear independence

**Definition 4.5.4: Linear independence**

Vectors $v_1, \dots, v_n \in V$ are **linearly independent** if the only way to write 0 as a linear combination is the trivial way:

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0 \implies a_1 = a_2 = \cdots = a_n = 0.$$

Otherwise, they are **linearly dependent**—there exist scalars, not all zero, such that $\sum a_i v_i = 0$.

**Note:-**

Linear dependence means some vector can be expressed as a linear combination of the others. If $a_1 v_1 + \cdots + a_n v_n = 0$ with (say) $a_1 \neq 0$, then

$$v_1 = -\frac{a_2}{a_1} v_2 - \cdots - \frac{a_n}{a_1} v_n.$$

So $v_1$ is "redundant"—it's already in the span of the others.

**Proposition 4.5.2** Equivalent characterizations of linear independence

The following are equivalent for $v_1, \dots, v_n \in V$:

1. $v_1, \dots, v_n$ are linearly independent.

2. No $v_i$ is a linear combination of the others.

3. The linear map $\phi : k^n \to V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is injective.

***Proof:*** (1) $\Leftrightarrow$ (3): $\ker(\phi) = \{(a_1, \dots, a_n) : \sum a_i v_i = 0\}$. Linear independence says $\ker(\phi) = \{0\}$, which is equivalent to $\phi$ being injective.

(1) $\Rightarrow$ (2): Suppose $v_j = \sum_{i \neq j} b_i v_i$. Then $\sum_{i \neq j} b_i v_i + (-1)v_j = 0$ is a nontrivial linear combination (coefficient of $v_j$ is $-1 \neq 0$), contradicting independence.

(2) $\Rightarrow$ (1): Suppose $\sum a_i v_i = 0$ with some $a_j \neq 0$. Then $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$, contradicting (2). □

Similarly, we can characterize spanning:

**Proposition 4.5.3** Characterization of spanning

The following are equivalent:

1. $v_1, \dots, v_n$ span $V$.

2. Every $v \in V$ can be written as $\sum a_i v_i$ for some $a_i \in k$.

3. The linear map $\phi : k^n \to V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is surjective.

### 4.5.3 Basis

**Definition 4.5.5: Basis**

A **basis** for $V$ is a set $\{v_1, \dots, v_n\}$ of vectors that are both linearly independent and span $V$. Equivalently, the linear map $\phi : k^n \to V$ is bijective.

**Proposition 4.5.4** Unique representation

If $\{v_1, \dots, v_n\}$ is a basis for $V$, then every vector $v \in V$ can be written uniquely as

$$v = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

for some $a_i \in k$. The scalars $(a_1, \dots, a_n)$ are called the **coordinates** of $v$ with respect to this basis.

***Proof:*** Existence: Since $\{v_i\}$ spans $V$, every $v$ is some linear combination.

Uniqueness: If $v = \sum a_i v_i = \sum a_i' v_i$, then $\sum (a_i - a_i')v_i = 0$. By linear independence, $a_i - a_i' = 0$ for all $i$, so $a_i = a_i'$. □

**Example 4.5.1**

**Standard basis of $k^n$:** The vectors

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad e_n = (0, \dots, 0, 1)$$

form the **standard basis** of $k^n$. Any $(a_1, \dots, a_n) = a_1 e_1 + \cdots + a_n e_n$.

**Example 4.5.2**

**Non-standard basis of $k^2$:** In $k^2$, the vectors $(1, 0)$ and $(0, 1)$ form the standard basis. But so do $(1, 1)$ and $(1, -1)$ (for most $k$—we need $\mathrm{char}(k) \neq 2$ for these to be independent).

**Example 4.5.3**

**Basis of $k[x]_{\leqslant n}$:** The polynomials $\{1, x, x^2, \ldots, x^n\}$ form a basis for $k[x]_{\leqslant n}$, the space of polynomials of degree $\leqslant n$.

**Question 1**

1. Does $k[x]$ have a finite basis? What is a basis for $k[x]$?
2. Does $k[\![x]\!]$ have a basis?

**Solution**

**(1)** The polynomial ring $k[x]$ does not have a finite basis, because any finite set $\{p_1, \ldots, p_n\}$ of polynomials has bounded degree, so cannot span polynomials of higher degree.
However, $k[x]$ has the infinite basis $\{1, x, x^2, x^3, \ldots\}$. Every polynomial is a finite linear combination of these monomials, and they are linearly independent (a nontrivial linear combination would give a nonzero polynomial equaling 0).
**(2)** By the axiom of choice, $k[\![x]\!]$ has a basis (every vector space does). But this basis cannot be explicitly described—it is uncountably infinite and highly non-constructive. Unlike $k[x]$, the monomials $\{1, x, x^2, \ldots\}$ do not form a basis of $k[\![x]\!]$: while they are linearly independent, they do not span (a power series like $\sum_{i=0}^{\infty} x^i$ is not a finite linear combination of monomials).

## 4.6 Finite-Dimensional Vector Spaces and Dimension

The notion of dimension will appear constantly throughout linear algebra and beyond. Unlike groups, where classification is intractable, finite-dimensional vector spaces are completely determined by a single integer—their dimension. We establish this remarkable fact through a sequence of lemmas about the interplay between spanning sets and linearly independent sets.

**Definition 4.6.1: Finite-dimensional**

A vector space $V$ is **finite-dimensional** if there exists a finite spanning set for $V$, i.e., vectors $v_1, \ldots, v_n$ such that $\mathrm{Span}(v_1, \ldots, v_n) = V$. Otherwise, $V$ is **infinite-dimensional**.

### 4.6.1 Extracting bases from spanning sets

Our first key result shows that every spanning set contains a basis.

**Lemma 4.6.1** Spanning sets contain bases

If $\{v_1, \ldots, v_n\}$ spans $V$, then some subset of $\{v_1, \ldots, v_n\}$ is a basis for $V$.

***Proof:*** We construct a basis by iteratively removing redundant vectors.

If $\{v_1, \ldots, v_n\}$ is linearly independent, we are done—it is already a basis.

Otherwise, there exists a nontrivial relation $\sum_{i=1}^{n} a_i v_i = 0$ with some $a_j \neq 0$. Then

$$v_j = -\frac{1}{a_j} \sum_{i \neq j} a_i v_i \in \mathrm{Span}(\{v_i : i \neq j\}).$$

So $\{v_i : i \neq j\}$ still spans $V$ (any linear combination involving $v_j$ can be rewritten using the other vectors).

Remove $v_j$ and repeat. Since we start with finitely many vectors, this process terminates. When it does, the remaining vectors are linearly independent (else we could remove another) and still span $V$, hence form a basis. $\square$

**Corollary 4.6.1** Finite-dimensional spaces have bases

Every finite-dimensional vector space has a basis.

### 4.6.2 Extending linearly independent sets to bases

The dual result shows that any linearly independent set can be enlarged to a basis.

**Lemma 4.6.2** Linearly independent sets extend to bases

If $V$ is finite-dimensional and $\{v_1, \ldots, v_m\}$ is linearly independent, then there exists a basis of $V$ containing $\{v_1, \ldots, v_m\}$.

***Proof:*** Let $\{w_1, \ldots, w_r\}$ be a finite spanning set for $V$ (exists by finite-dimensionality). We enlarge $\{v_1, \ldots, v_m\}$ to a basis using vectors from $\{w_j\}$.

Define $W_0 = \mathrm{Span}(v_1, \ldots, v_m)$, with basis $\{v_1, \ldots, v_m\}$ (our independent set).

**Induction step:** Suppose we have constructed a basis $\mathcal{B}_{j-1}$ for $W_{j-1} = \mathrm{Span}(v_1, \ldots, v_m, w_{i_1}, \ldots, w_{i_{j-1}})$.

Consider $w_j$:

- If $w_j \in W_{j-1}$, set $W_j = W_{j-1}$ and $\mathcal{B}_j = \mathcal{B}_{j-1}$.
- If $w_j \notin W_{j-1}$, then $\mathcal{B}_{j-1} \cup \{w_j\}$ is linearly independent (otherwise $w_j$ would be a linear combination of $\mathcal{B}_{j-1}$, contradicting $w_j \notin W_{j-1}$). Set $W_j = \mathrm{Span}(\mathcal{B}_{j-1} \cup \{w_j\})$ and $\mathcal{B}_j = \mathcal{B}_{j-1} \cup \{w_j\}$.

After processing all $w_j$, we have $W_r = \mathrm{Span}(v_1, \ldots, v_m, w_1, \ldots, w_r) \supseteq \mathrm{Span}(w_1, \ldots, w_r) = V$. So $\mathcal{B}_r$ is a linearly independent set spanning $V$, i.e., a basis containing $\{v_1, \ldots, v_m\}$. $\square$

### 4.6.3 Invariance of basis size

The following theorem shows that the size of a basis is intrinsic to the vector space, not dependent on the choice of basis.

**Theorem 4.6.1** All bases have the same size

If $V$ has a basis with $n$ elements and another basis with $m$ elements, then $m = n$.

***Proof:*** Let $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_m\}$ be bases for $V$. We show $m \leqslant n$ and $n \leqslant m$, hence $m = n$.

**Claim:** $\{v_1, \ldots, v_{m-1}, w_j\}$ is a basis for some choice of reordering, for each $j$ we process.

We proceed by a "replacement" argument. Start with the basis $\{v_1, \ldots, v_n\}$.

Since $w_1 \in V = \mathrm{Span}(v_1, \ldots, v_n)$, we can write $w_1 = \sum_{i=1}^n a_i v_i$. Not all $a_i = 0$ (else $w_1 = 0$, contradicting independence of $\{w_j\}$). By reordering, assume $a_n \neq 0$.

Then

$$v_n = a_n^{-1}(w_1 - \sum_{i=1}^{n-1} a_i v_i) \in \mathrm{Span}(v_1, \ldots, v_{n-1}, w_1).$$

So

$$\mathrm{Span}(v_1, \ldots, v_{n-1}, w_1) \supseteq \mathrm{Span}(v_1, \ldots, v_n) = V,$$

and thus $\{v_1, \ldots, v_{n-1}, w_1\}$ spans $V$. Since it has $n$ elements and spans, it contains a basis—but removing any element would give $n-1$ vectors. We verify it's actually linearly independent: if

$$\sum_{i=1}^{n-1} b_i v_i + c w_1 = 0$$

with $c \neq 0$, then $w_1 \in \mathrm{Span}(v_1, \dots, v_{n-1})$, so $v_n \in \mathrm{Span}(v_1, \dots, v_{n-1})$, contradicting independence of $\{v_i\}$. Thus $c = 0$, and then all $b_i = 0$. So $\{v_1, \dots, v_{n-1}, w_1\}$ is a basis.

Repeat: $w_2 = \sum_{i=1}^{n-1} a_i v_i + c w_1$. Some $a_i \neq 0$ (else $w_2 \in \mathrm{Span}(w_1)$, contradicting independence). Replace that $v_i$ with $w_2$.

Continue this process, replacing one $v$ with one $w$ each time. After $m$ steps, we have a basis

$$\{v_{i_1}, \dots, v_{i_{n-m}}, w_1, \dots, w_m\}$$

(assuming $m \leqslant n$).

If $m > n$, the process would terminate before incorporating all $w_j$, but then some $w_j$ would be in the span of earlier $w$'s, contradicting independence. Hence $m \leqslant n$.

By symmetry, $n \leqslant m$. Thus $m = n$. □

**Definition 4.6.2: Dimension**

The **dimension** of a finite-dimensional vector space $V$, denoted $\dim(V)$ or $\dim_k(V)$, is the number of elements in any basis. The theorem above shows this is well-defined.
By convention, $\dim(\{0\}) = 0$ (the empty set is a basis for the zero space).

**Corollary 4.6.2** Isomorphism with $k^n$

Every finite-dimensional vector space $V$ with $\dim(V) = n$ is isomorphic to $k^n$.

***Proof:*** Let $\{v_1, \dots, v_n\}$ be a basis for $V$. The map

$$\phi : k^n \to V, \quad (a_1, \dots, a_n) \mapsto \sum_{i=1}^{n} a_i v_i$$

is linear (easily verified), injective (by linear independence), and surjective (by spanning). Hence $\phi$ is an isomorphism. □

**Note:-**

The isomorphism $\phi : k^n \xrightarrow{\sim} V$ depends on the choice of basis. Different bases give different isomorphisms. This is the key insight of linear algebra: choosing a basis amounts to choosing coordinates, and we represent elements of $V$ by **column vectors** $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in k^n$.

**Example 4.6.1**

**Dimensions:**

- $\dim(k^n) = n$. The standard basis has $n$ elements.
- $\dim(k[x]_{\leqslant d}) = d + 1$. The monomials $\{1, x, \dots, x^d\}$ form a basis.
- $k[x]$ is infinite-dimensional: any finite set of polynomials has bounded degree and cannot span.
- $k[\![x]\!]$ is infinite-dimensional (and in a "larger" sense—its bases are uncountable).

Lecture 5.

# Linear Maps

Having established the structure of vector spaces, we now study the morphisms between them. Just as group homomorphisms preserve the group operation, linear maps preserve both vector space operations.

**Definition 5.0.1: Linear map**

Let $V$ and $W$ be vector spaces over $k$. A **linear map** (or **$k$-linear map**, or **homomorphism of vector spaces**) is a function $\varphi : V \to W$ satisfying:

1. $\varphi(u+v) = \varphi(u) + \varphi(v)$ for all $u, v \in V$ (additivity);
2. $\varphi(\lambda v) = \lambda\varphi(v)$ for all $\lambda \in k$ and $v \in V$ (homogeneity).

Equivalently, $\varphi$ preserves all linear combinations:

$$\varphi\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i \varphi(v_i).$$

**Note:-**

A linear map is completely determined by its values on a basis. If $\{v_1, \dots, v_n\}$ is a basis for $V$, and we specify $\varphi(v_i) = w_i \in W$ for each $i$, then for any $v = \sum a_i v_i$:

$$\varphi(v) = \varphi\left(\sum a_i v_i\right) = \sum a_i \varphi(v_i) = \sum a_i w_i.$$

This is the **universal property** of bases: specifying a linear map from $V$ is equivalent to choosing where to send the basis vectors.

**Definition 5.0.2: Space of linear maps**

The set of all linear maps from $V$ to $W$ is denoted

$$\mathrm{Hom}_k(V, W) \quad \text{or simply} \quad \mathrm{Hom}(V, W).$$

When $V = W$, we write $\mathrm{End}(V) = \mathrm{Hom}(V, V)$ and call elements **endomorphisms**.

**Proposition 5.0.1** $\mathrm{Hom}(V, W)$ is a vector space

The set $\mathrm{Hom}(V, W)$ is itself a vector space over $k$, with operations:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$ for all $v \in V$;
- **Scalar multiplication:** $(\lambda\varphi)(v) = \lambda \cdot \varphi(v)$ for all $v \in V$.

***Proof:*** We must verify that $\varphi + \psi$ and $\lambda\varphi$ are linear maps when $\varphi, \psi$ are.

For $\varphi + \psi$: Let $u, v \in V$ and $\mu \in k$.

$$\begin{aligned}
(\varphi+\psi)(u+v) &= \varphi(u+v) + \psi(u+v) = \varphi(u) + \varphi(v) + \psi(u) + \psi(v) \\
&= (\varphi(u) + \psi(u)) + (\varphi(v) + \psi(v)) = (\varphi+\psi)(u) + (\varphi+\psi)(v). \\
(\varphi+\psi)(\mu v) &= \varphi(\mu v) + \psi(\mu v) = \mu\varphi(v) + \mu\psi(v) = \mu(\varphi(v) + \psi(v)) = \mu(\varphi+\psi)(v).
\end{aligned}$$

For $\lambda\varphi$: Similarly,

$$(\lambda\varphi)(u+v) = \lambda(\varphi(u+v)) = \lambda(\varphi(u)+\varphi(v)) = \lambda\varphi(u)+\lambda\varphi(v) = (\lambda\varphi)(u)+(\lambda\varphi)(v).$$
$$(\lambda\varphi)(\mu v) = \lambda(\varphi(\mu v)) = \lambda(\mu\varphi(v)) = \mu(\lambda\varphi(v)) = \mu(\lambda\varphi)(v).$$

The vector space axioms for $\mathrm{Hom}(V, W)$ follow from those of $W$ (operations are defined pointwise). The zero vector is the zero map $v \mapsto 0_W$, and additive inverses are $(-\varphi)(v) = -\varphi(v)$. □

**Note:-**

The verification that $\varphi + \psi$ and $\lambda\varphi$ are linear is "rather boring, but worth checking if you're not sure." The operations on $\mathrm{Hom}(V, W)$ are inherited pointwise from $W$, so the axioms follow routinely.

**Theorem 5.0.1** Dimension of $\mathrm{Hom}(V, W)$

If $\dim(V) = n$ and $\dim(W) = m$, then

$$\dim(\mathrm{Hom}(V, W)) = mn.$$

The proof of this theorem will emerge naturally when we establish the correspondence between linear maps and matrices in the next section.

**Note:-**

We will soon see: if $\dim(V) = n$ and $\dim(W) = m$, then $\dim(\mathrm{Hom}(V, W)) = mn$. In terms of bases for $V$ and $W$, linear maps become $m \times n$ matrices!

### 5.0.1 Dependence on the scalar field

The notions of basis, dimension, and linearity depend crucially on the choice of scalar field $k$. The same set can be a vector space over different fields, with different dimensions.

**Example 5.0.1**

**$\mathbb{C}$ as a vector space over different fields:** Consider $\mathbb{C}$ as a vector space:

- Over $\mathbb{C}$: $\dim_{\mathbb{C}}(\mathbb{C}) = 1$, with basis $\{1\}$. Every complex number $z$ is $z \cdot 1$.
- Over $\mathbb{R}$: $\dim_{\mathbb{R}}(\mathbb{C}) = 2$, with basis $\{1, i\}$. Every complex number $a+bi$ is the $\mathbb{R}$-linear combination $a \cdot 1 + b \cdot i$.

**Note:-**

More generally, if $k' \subset k$ is a subfield (e.g., $\mathbb{R} \subset \mathbb{C}$ or $\mathbb{Q} \subset \mathbb{R}$), then any $k$-vector space $V$ is automatically a $k'$-vector space by "restriction of scalars"—we simply forget that we can multiply by elements of $k \setminus k'$. The $k'$-dimension is always at least as large as the $k$-dimension.

**Proposition 5.0.2** Linearity over subfields

If $V$ and $W$ are $k$-vector spaces and $k' \subset k$ is a subfield, then:

1. $V$ is a $k'$-vector space (by restriction of scalars).
2. Any $k$-linear map $\varphi : V \to W$ is also $k'$-linear.
3. The converse fails: not every $k'$-linear map is $k$-linear.

Thus $\mathrm{Hom}_k(V, W) \subsetneq \mathrm{Hom}_{k'}(V, W)$ in general.

**Example 5.0.2**

**Complex conjugation is $\mathbb{R}$-linear but not $\mathbb{C}$-linear:** Consider the map

$$\bar{\ }: \mathbb{C} \to \mathbb{C}, \quad z = a + bi \mapsto \bar{z} = a - bi.$$

**$\mathbb{R}$-linearity:** For $z_1, z_2 \in \mathbb{C}$ and $r \in \mathbb{R}$:

- $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ ✓
- $\overline{r \cdot z} = r \cdot \bar{z}$ (since $\bar{r} = r$ for $r \in \mathbb{R}$) ✓

**Not $\mathbb{C}$-linear:** For $\lambda = i$ and $z = 1$:

$$\overline{i \cdot 1} = \bar{i} = -i, \quad \text{but} \quad i \cdot \bar{1} = i \cdot 1 = i.$$

Since $-i \neq i$, complex conjugation is not $\mathbb{C}$-linear.

This example illustrates a profound point: the "natural" maps between spaces can depend on what scalars we consider. Complex conjugation is an $\mathbb{R}$-linear automorphism of $\mathbb{C}$ that is not $\mathbb{C}$-linear—it does not commute with multiplication by $i$.

## 5.1 Matrix Representation of Linear Maps

The correspondence between linear maps and matrices is one of the most powerful ideas in linear algebra. It reduces abstract questions about linear maps to concrete computations with arrays of numbers.

### 5.1.1 The matrix of a linear map

Fix bases $\mathcal{B} = (v_1, \ldots, v_n)$ for $V$ and $\mathcal{C} = (w_1, \ldots, w_m)$ for $W$. Given a linear map $\varphi : V \to W$, we can express each image $\varphi(v_j)$ as a linear combination of the basis vectors of $W$:

$$\varphi(v_j) = \sum_{i=1}^{m} a_{ij} w_i = a_{1j} w_1 + a_{2j} w_2 + \cdots + a_{mj} w_m.$$

**Definition 5.1.1: Matrix of a linear map**

The **matrix of $\varphi$** with respect to bases $\mathcal{B}$ and $\mathcal{C}$ is the $m \times n$ matrix

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathcal{M}_{m \times n}(k).$$

The $j$-th **column** of $A$ is the coordinate vector of $\varphi(v_j)$ in the basis $\mathcal{C}$.

**Note:-**

The key insight: **columns of $A$ are the images of basis vectors, expressed in coordinates.** This is worth memorizing: the $j$-th column of the matrix tells you where the $j$-th basis vector goes.

### 5.1.2 Matrix-vector multiplication

The power of the matrix representation is that applying $\varphi$ corresponds to matrix multiplication.

Represent a vector $x \in V$ by its coordinate column vector: if $x = \sum_{j=1}^{n} x_j v_j$, we write

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in k^n.$$

Similarly, $y = \varphi(x) \in W$ has coordinates $Y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$ where $y = \sum_{i=1}^{m} y_i w_i$.

**Proposition 5.1.1** Linear maps as matrix multiplication
With the notation above, $Y = AX$, i.e.,

$$y_i = \sum_{j=1}^{n} a_{ij} x_j.$$

***Proof:*** Compute:

$$\begin{aligned} \varphi(x) = \varphi\left(\sum_{j=1}^{n} x_j v_j\right) &= \sum_{j=1}^{n} x_j \varphi(v_j) = \sum_{j=1}^{n} x_j \left(\sum_{i=1}^{m} a_{ij} w_i\right) \\ &= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j\right) w_i. \end{aligned}$$

Reading off the coefficient of $w_i$, we get $y_i = \sum_{j=1}^{n} a_{ij} x_j$, which is exactly the $i$-th component of $AX$. □

**Note:-**

As a memory aid: the isomorphism $k^n \xrightarrow{\sim} V$ given by a basis can be written symbolically as

$$(v_1\ v_2\ \cdots\ v_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{j=1}^{n} x_j v_j.$$

Similarly, $\varphi$ acts as

$$\varphi((v_1\ \cdots\ v_n)X) = (w_1\ \cdots\ w_m)AX.$$

This notation makes the matrix formula $Y = AX$ visually clear.

### 5.1.3 The isomorphism $\mathrm{Hom}(V, W) \cong \mathcal{M}_{m \times n}(k)$

**Theorem 5.1.1** Linear maps correspond to matrices
Fix bases $\mathcal{B}$ for $V$ and $\mathcal{C}$ for $W$. The map

$$\Phi : \mathrm{Hom}(V, W) \to \mathcal{M}_{m \times n}(k), \quad \varphi \mapsto [\varphi]_{\mathcal{B}}^{\mathcal{C}}$$

sending a linear map to its matrix is an **isomorphism of vector spaces**.

***Proof:*** We verify $\Phi$ is linear, injective, and surjective.

**Linearity:** Let $\varphi, \psi \in \mathrm{Hom}(V, W)$ and $\lambda \in k$. The $j$-th column of $[\varphi + \psi]$ is the coordinate vector of $(\varphi + \psi)(v_j) = \varphi(v_j) + \psi(v_j)$, which is the sum of the $j$-th columns of $[\varphi]$ and $[\psi]$. Hence $[\varphi + \psi] = [\varphi] + [\psi]$. Similarly $[\lambda\varphi] = \lambda[\varphi]$.

**Injectivity:** If $[\varphi] = 0$ (the zero matrix), then every column is zero, meaning $\varphi(v_j) = 0$ for all basis vectors $v_j$. Since $\varphi$ is linear and determined by its values on a basis, $\varphi = 0$.

**Surjectivity:** Given any matrix $A \in \mathcal{M}_{m\times n}(k)$, define $\varphi : V \to W$ by specifying $\varphi(v_j) = \sum_i a_{ij} w_i$ (the j-th column of $A$ gives the coordinates of $\varphi(v_j)$). Extend linearly. Then $[\varphi] = A$. $\square$

**Corollary 5.1.1** Dimension formula

$$\dim(\mathrm{Hom}(V, W)) = \dim(\mathcal{M}_{m\times n}(k)) = mn.$$

***Proof:*** The space $\mathcal{M}_{m\times n}(k)$ has the standard basis $\{E_{ij}\}$, where $E_{ij}$ is the matrix with 1 in position $(i, j)$ and 0 elsewhere. There are $mn$ such matrices. $\square$

**Note:-**

This result says: **linear maps are matrices, once we choose coordinates**. The abstract theory of linear maps reduces to the concrete theory of matrix computations. However, the matrix depends on the choice of bases—changing bases changes the matrix (we will study this "change of basis" in the near future).

The interplay between the abstract (linear maps) and the concrete (matrices) is what makes linear algebra, well linear algebra (duh). Our goal is to understand how much of a linear map's structure is intrinsic (independent of basis choice) versus coordinate-dependent.

## 5.2 Change of basis

What if we want to choose a different basis for $V$ and/ or $W$? That is to say, what if we want to take vectors expressable in one basis to the same vectors expressed in another basis? Well, say that we are given two bases: $(v_i)$ and $(v_i')$ in $V$. If given an expression of $(v_i)$ we want to plug in and get an expression in terms of $v_i'$, we write this as

$$v_j' = \sum_{n}^{i=1} p_{ij} v_i.$$

So we get an $n \times m$ matrix $P$ with entries $(p_{ij})$, whose $j^{\text{th}}$ column is the expression of $v_j'$ in the basis $(v_i)$ as a column vector. So symbolically, we can think of this as $(v_i') = (v_i)P$. So $(v_i')X' = (v_i)PX'$, i.e. the element of $V$ represented by a column vector $X'$ in new basis $(v_i')$ is represented by $X = PX'$ in the old basis $(v_i)$ . We can think of this more conceptually as $P$ being a linear map—i.e. a matrix transformation—by $P = \mathcal{M}(\mathrm{id}_V, (v_i'), (v_i))$. Breaking this down, we have the identity mapping taking the vector to itself (from one basis to another, but the same vector(s)); thus we are taking things from $(v_i')$, here, to the targets of $(v_i)$. We can additionally do the same for $W$ in reverse! Let $Q = \mathcal{M}(\mathrm{id}_W, (w_i)(w_i'))$, i.e. $(w_i) = (w_i')Q$. Hence:

$$\varphi((v_i'))X' = \varphi((v_i)PX') = (w_i)APX' = (w_i')QAPX'.$$

That is to say, $\mathcal{M}(\varphi, (v'), (w')) = QAP$.

**Note:-**

Matrix of identity from $(v_i)$ to $(v_i')$ would be $P^{-1}$ which is the matrix such that $pp^{-1}$ is the identity matrix.

If $V = W$ and we change basis using a single new basis $(v_i')$ for both domain and codomain, then a remarkable simplification occurs. Let $\varphi \in \mathrm{End}(V) = \mathrm{Hom}(V, V)$ be an **endomorphism** (a linear map from a space to itself). With $P = \mathcal{M}(\mathrm{id}_V, (v_i'), (v_i))$ as above, and using the same basis change for both domain and codomain, we have $Q = P$ (since we're using $(v_i)$ and $(v_i')$ for $W$ as well). Thus the formula $\mathcal{M}(\varphi, (v'), (w')) = QAP$ becomes:

**Theorem 5.2.1** Change of basis for endomorphisms

Let $\varphi \in \mathrm{End}(V)$, let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{B}' = (v'_1, \dots, v'_n)$ be two bases of $V$, and let $P = \mathcal{M}(\mathrm{id}_V, \mathcal{B}', \mathcal{B})$ be the change-of-basis matrix. If $A = \mathcal{M}(\varphi, \mathcal{B})$ and $A' = \mathcal{M}(\varphi, \mathcal{B}')$, then

$$A' = P^{-1}AP.$$

***Proof:*** We have established that $\mathcal{M}(\varphi, \mathcal{B}', \mathcal{B}') = QAP$ where $Q = \mathcal{M}(\mathrm{id}_V, (v_i), (v'_i))$. But $Q = P^{-1}$ since applying $P$ then $Q$ (or vice versa) gives the identity transformation on coordinate vectors. Explicitly: if $X' = P^{-1}X$ converts old coordinates to new, and $X = PX'$ converts new to old, then composing identity maps gives $\mathcal{M}(\mathrm{id}_V, (v_i), (v'_i)) = P^{-1}$.

Thus $A' = P^{-1}AP$. □

This formula tells us that the matrices representing the same linear operator in different bases are related by **conjugation** by an invertible matrix.

### 5.2.1 Similar matrices

**Definition 5.2.1: Similar matrices**

Two $n \times n$ matrices $A$ and $B$ over a field $k$ are **similar** if there exists an invertible matrix $P \in \mathrm{GL}_n(k)$ such that

$$B = P^{-1}AP.$$

We write $A \sim B$.

**Proposition 5.2.1** Similarity is an equivalence relation

The similarity relation on $\mathcal{M}_{n\times n}(k)$ is an equivalence relation:

1. **Reflexive:** $A \sim A$ (take $P = I$).
2. **Symmetric:** If $A \sim B$, then $B \sim A$ (if $B = P^{-1}AP$, then $A = PBP^{-1} = (P^{-1})^{-1}B(P^{-1})$).
3. **Transitive:** If $A \sim B$ and $B \sim C$, then $A \sim C$ (if $B = P^{-1}AP$ and $C = Q^{-1}BQ$, then $C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ)$).

As we can see, similarity classes of matrices correspond exactly to linear operators on $k^n$: two matrices are similar if and only if they represent the same linear operator in different bases.

**Note:-**

From the categorical perspective, similar matrices represent the same morphism in $\mathbf{Vect}_k$—they are simply different coordinate descriptions of one intrinsic object. This is why similarity-invariant properties (trace, determinant, eigenvalues, rank) are so important: they are properties of the operator itself, not of any particular matrix representation.

### 5.2.2 Invariants under similarity

What properties of a matrix are preserved under similarity? These are called **similarity invariants**—they depend only on the underlying linear operator, not on the choice of basis.

**Definition 5.2.2: Trace of a matrix**

The **trace** of an $n \times n$ matrix $A = (a_{ij})$ is the sum of its diagonal entries:

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

**Lemma 5.2.1** Cyclic property of trace

For any matrices $A$ and $B$ such that both $AB$ and $BA$ are defined and square, we have $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

***Proof:*** Let $A$ be $m \times n$ and $B$ be $n \times m$. Then

$$\mathrm{Tr}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ji}.$$

Similarly,

$$\mathrm{Tr}(BA) = \sum_{j=1}^{n} (BA)_{jj} = \sum_{j=1}^{n} \sum_{i=1}^{m} b_{ji} a_{ij}.$$

These double sums are identical (just reorder the summation), so $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. $\square$

**Proposition 5.2.2** Trace is a similarity invariant

If $A \sim B$, then $\mathrm{Tr}(A) = \mathrm{Tr}(B)$. In particular, the trace of a linear operator $\varphi \in \mathrm{End}(V)$ is well-defined: $\mathrm{Tr}(\varphi) := \mathrm{Tr}(\mathcal{M}(\varphi, \mathcal{B}))$ for any basis $\mathcal{B}$.

***Proof:*** If $B = P^{-1}AP$, then using the cyclic property:

$$\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}((P^{-1}A)P) = \mathrm{Tr}(P(P^{-1}A)) = \mathrm{Tr}((PP^{-1})A) = \mathrm{Tr}(A).$$

$\square$

**Proposition 5.2.3** Determinant is a similarity invariant

If $A \sim B$, then $\det(A) = \det(B)$. Hence the determinant of a linear operator is well-defined.

***Proof:*** If $B = P^{-1}AP$, then by the multiplicativity of determinants:

$$\det(B) = \det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

$\square$

**Proposition 5.2.4** Eigenvalues are similarity invariants

If $A \sim B$, then $A$ and $B$ have the same eigenvalues (with the same algebraic multiplicities). In particular, the eigenvalues of a linear operator are intrinsic to the operator, not dependent on the choice of basis.

***Proof:*** A scalar $\lambda$ is an eigenvalue of $A$ if and only if $\det(A - \lambda I) = 0$. If $B = P^{-1}AP$, then

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}IP = P^{-1}(A - \lambda I)P.$$

Thus $B - \lambda I \sim A - \lambda I$, so $\det(B - \lambda I) = \det(A - \lambda I)$ by the previous result. Hence $A$ and $B$ have identical characteristic polynomials, and therefore the same eigenvalues with the same multiplicities. $\square$

**Note:-**

Rank is also a similarity invariant: $\operatorname{rank}(P^{-1}AP) = \operatorname{rank}(A)$ since multiplying by invertible matrices doesn't change rank. More sophisticated invariants include the **minimal polynomial**, the **Jordan normal form**, and the **rational canonical form**—these completely classify operators up to similarity over algebraically closed fields. We will encounter these when we study eigenvalues in depth.

**Example 5.2.1**

Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ over $\mathbb{R}$. Both have $\operatorname{Tr} = 4$, $\det = 4$, and eigenvalue $\lambda = 2$ with multiplicity 2. Yet $A \not\sim B$! The reason: $B = 2I$, so $P^{-1}BP = P^{-1}(2I)P = 2I = B$ for any $P$. Since $A \neq B$, they cannot be similar.

This shows that trace, determinant, and eigenvalues are necessary but not sufficient conditions for similarity. The difference lies in the **geometric multiplicity** of the eigenvalue 2: for $B$, the eigenspace $E_2 = \ker(B - 2I) = \mathbb{R}^2$ has dimension 2; for $A$, $E_2 = \ker(A - 2I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \operatorname{Span}\{(1, 0)\}$ has dimension 1. Geometric multiplicity is also a similarity invariant!

**Question 1**

1. Let $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map with matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ with respect to the standard basis. Find the matrix of $\varphi$ with respect to the basis $\mathcal{B}' = \{(1, 1), (1, -1)\}$.
2. Prove that an $n \times n$ matrix $A$ is similar to a diagonal matrix if and only if there exists a basis of $k^n$ consisting of eigenvectors of $A$.
3. Show that $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ but in general $\det(A + B) \neq \det(A) + \det(B)$. Give a counterexample.

**Solution**

**(1)** First, find the change-of-basis matrix $P$ from $\mathcal{B}'$ to the standard basis $\mathcal{E}$. The columns of $P$ are the vectors of $\mathcal{B}'$ expressed in $\mathcal{E}$:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then $P^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$.

The matrix of $\varphi$ with respect to $\mathcal{B}'$ is:

$$A' = P^{-1}AP = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Computing $AP$ first: $AP = \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix}$.

Then $P^{-1}(AP) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$.

So $A' = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$. We can verify: $\operatorname{Tr}(A') = 5 = \operatorname{Tr}(A)$ and $\det(A') = 0 - 2 = -2 = \det(A)$. ✓

**(2)** ($\Rightarrow$) Suppose $A$ is similar to a diagonal matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, so $D = P^{-1}AP$ for some invertible $P$. Let $v_i$ be the $i$-th column of $P$. Then $AP = PD$, which gives $Av_i = \lambda_i v_i$. So each column of $P$ is an eigenvector of $A$. Since $P$ is invertible, its columns form a basis of $k^n$.

($\Leftarrow$) Suppose $\{v_1, \ldots, v_n\}$ is a basis of eigenvectors with $Av_i = \lambda_i v_i$. Let $P$ be the matrix with columns $v_i$. Then $AP = PD$ where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, so $D = P^{-1}AP$.

**(3)** We proved $\mathsf{Tr}(AB) = \mathsf{Tr}(BA)$ above. For the determinant counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $\det(A) = 0$, $\det(B) = 0$, but $A + B = I$, so $\det(A + B) = 1 \neq 0 = \det(A) + \det(B)$.

Lecture 6.

# Kernals, images, and more

## 6.1 Kernel and Image

Every linear map carries two intrinsic subspaces that reveal its fundamental structure: the kernel (what gets annihilated) and the image (what gets hit). These are the linear-algebraic analogues of concepts we've seen before: the kernel of a group homomorphism and the image of any function. The intermingling betwixt (what a fun word) kernel and image is governed by the **Rank-Nullity Theorem**.

**Note:-**

From the categorical viewpoint, kernel and image are two ways to "measure" a morphism. The kernel measures the failure of injectivity—how much information the map destroys. The image measures the failure of surjectivity—how much of the codomain the map misses. The Rank-Nullity Theorem says these two failures are complementary: the more a linear map kills, the less it can produce, and vice versa. A linear map $\varphi : V \to W$ is completely determined by: (1) which subspace it kills (the kernel), and (2) which subspace it hits (the image). Once you know these, there's essentially only one way to fill in the rest—the map restricted to any complement of the kernel is an isomorphism onto the image. This is different from nonlinear maps, where knowing the zero set tells you almost nothing about the map's behavior elsewhere.

### 6.1.1 The Kernel

**Definition 6.1.1: Kernel of a linear map**

Let $\varphi : V \to W$ be a linear map. The **kernel** of $\varphi$ is

$$\ker(\varphi) = \{v \in V : \varphi(v) = 0_W\}.$$

This is the set of all vectors that $\varphi$ maps to zero.

**Note:-**

The terminology "kernel" comes from algebra (group homomorphisms), while "null space" is more common in applied linear algebra and matrix theory to refer to the generalized kernal (we will get to this later). In categorical language, the kernel is the equalizer of $\varphi$ and the zero map.

**Example 6.1.1** (Examples of kernels)

1. The zero map $0 : V \to W$ has $\ker(0) = V$.
2. The identity map $\mathrm{id}_V : V \to V$ has $\ker(\mathrm{id}_V) = \{0\}$.
3. Let $\varphi : \mathbb{R}^3 \to \mathbb{R}$ be defined by $\varphi(x, y, z) = x + 2y + 3z$. Then $\ker(\varphi) = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + 3z = 0\}$, a plane through the origin.
4. Let $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ be differentiation, $Dp = p'$. Then $\ker(D)$ is the set of constant polynomials.
5. Let $T : \mathcal{M}_{n\times n}(k) \to k$ be the trace map, $T(A) = \mathrm{Tr}(A)$. Then $\ker(T)$ is the space of traceless matrices.

**Proposition 6.1.1** The kernel is a subspace

Let $\varphi : V \to W$ be a linear map. Then $\ker(\varphi)$ is a subspace of $V$.

***Proof:*** We verify the subspace criteria:

1. **Contains zero:** Since $\varphi(0_V) = 0_W$ (every linear map preserves zero), we have $0_V \in \ker(\varphi)$.
2. **Closed under addition:** If $u, v \in \ker(\varphi)$, then $\varphi(u+v) = \varphi(u)+\varphi(v) = 0+0 = 0$, so $u+v \in \ker(\varphi)$.
3. **Closed under scalar multiplication:** If $v \in \ker(\varphi)$ and $\lambda \in k$, then $\varphi(\lambda v) = \lambda\varphi(v) = \lambda \cdot 0 = 0$, so $\lambda v \in \ker(\varphi)$.

□

**Note:-**

Compare this with group theory: the kernel of a group homomorphism $f : G \to H$ is a normal subgroup of $G$. For linear maps, every subspace is "normal" in the appropriate sense—vector spaces are abelian, so there's no distinction between left and right cosets.

The kernel provides a clean characterization of injectivity:

**Theorem 6.1.1** Injectivity via the kernel

Let $\varphi : V \to W$ be a linear map. Then

$$\varphi \text{ is injective} \iff \ker(\varphi) = \{0\}.$$

***Proof:*** $(\Rightarrow)$ Suppose $\varphi$ is injective. Let $v \in \ker(\varphi)$. Then $\varphi(v) = 0 = \varphi(0)$. Since $\varphi$ is injective, $v = 0$. Thus $\ker(\varphi) \subseteq \{0\}$, and since $0 \in \ker(\varphi)$ always, we have $\ker(\varphi) = \{0\}$.

$(\Leftarrow)$ Suppose $\ker(\varphi) = \{0\}$. Let $u, v \in V$ with $\varphi(u) = \varphi(v)$. Then

$$\varphi(u - v) = \varphi(u) - \varphi(v) = 0,$$

so $u - v \in \ker(\varphi) = \{0\}$. Thus $u - v = 0$, i.e., $u = v$. Hence $\varphi$ is injective. □

**Corollary 6.1.1**

An injective linear map from $V$ to $W$ embeds $V$ as a subspace of $W$. More precisely, if $\varphi : V \to W$ is injective, then $V \cong \mathrm{Im}(\varphi) \subseteq W$.

**Note:-**

Here the kernal has a role more that merely verifying injectivity—it measures the failure of injectivity in a precise sense. Two vectors $u, v \in V$ have the same image under $\varphi$ if and only if $u - v \in \ker(\varphi)$. Thus the fibers of $\varphi$ (preimages of points) are exactly the cosets $v + \ker(\varphi)$. The map $\varphi$ is injective precisely when these cosets are singletons, i.e., when $\ker(\varphi) = \{0\}$. This is why we can "mod out by the kernel" to obtain an injective map—the quotient $V/\ker(\varphi)$ collapses each fiber to a point.

### 6.1.2 The Image (Range)

**Definition 6.1.2: Image of a linear map**

Let $\varphi : V \to W$ be a linear map. The **image** (or **range**) of $\varphi$ is

$$\mathrm{Im}(\varphi) = \{\varphi(v) : v \in V\} = \{w \in W : w = \varphi(v) \text{ for some } v \in V\}.$$

This is the set of all vectors that $\varphi$ "hits."

**Example 6.1.2** (Examples of images)

1. The zero map $0 : V \to W$ has $\mathrm{Im}(0) = \{0_W\}$.
2. The identity map $\mathrm{id}_V : V \to V$ has $\mathrm{Im}(\mathrm{id}_V) = V$.
3. Let $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ be projection $\pi(x, y, z) = (x, y)$. Then $\mathrm{Im}(\pi) = \mathbb{R}^2$.
4. Let $D : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_{n-1}(\mathbb{R})$ be differentiation. Then $\mathrm{Im}(D) = \mathcal{P}_{n-1}(\mathbb{R})$ (every polynomial of degree $\leqslant n-1$ is the derivative of some polynomial of degree $\leqslant n$).
5. Let $A \in \mathcal{M}_{m\times n}(k)$ define $L_A : k^n \to k^m$. Then $\mathrm{Im}(L_A) = \mathrm{ColSpace}(A)$, the column space of $A$.

**Proposition 6.1.2** The image is a subspace

Let $\varphi : V \to W$ be a linear map. Then $\mathrm{Im}(\varphi)$ is a subspace of $W$.

***Proof:*** We verify the subspace criteria:

1. **Contains zero:** $\varphi(0_V) = 0_W$, so $0_W \in \mathrm{Im}(\varphi)$.
2. **Closed under addition:** If $w_1, w_2 \in \mathrm{Im}(\varphi)$, then $w_1 = \varphi(v_1)$ and $w_2 = \varphi(v_2)$ for some $v_1, v_2 \in V$. Thus $w_1 + w_2 = \varphi(v_1) + \varphi(v_2) = \varphi(v_1 + v_2) \in \mathrm{Im}(\varphi)$.
3. **Closed under scalar multiplication:** If $w \in \mathrm{Im}(\varphi)$ and $\lambda \in k$, then $w = \varphi(v)$ for some $v \in V$, so $\lambda w = \lambda\varphi(v) = \varphi(\lambda v) \in \mathrm{Im}(\varphi)$.

$\square$

**Definition 6.1.3: Surjectivity**

A linear map $\varphi : V \to W$ is **surjective** (or **onto**) if $\mathrm{Im}(\varphi) = W$, i.e., every element of $W$ is hit by some element of $V$.

**Proposition 6.1.3** Characterization of surjectivity

Let $\varphi : V \to W$ be a linear map. The following are equivalent:

1. $\varphi$ is surjective.
2. $\mathrm{Im}(\varphi) = W$.
3. If $\mathcal{B}$ spans $V$, then $\varphi(\mathcal{B}) = \{\varphi(v) : v \in \mathcal{B}\}$ spans $W$.

### 6.1.3 The Rank-Nullity Theorem

We now come to the central result connecting kernel and image. This theorem quantifies the fundamental trade-off in linear algebra: the more a map “kills,” the less it can “produce.”

**Definition 6.1.4: Rank and nullity**

Let $\varphi : V \to W$ be a linear map between finite-dimensional vector spaces. The **rank** of $\varphi$ is

$$\operatorname{rank}(\varphi) = \dim(\operatorname{Im}(\varphi)),$$

and the **nullity** of $\varphi$ is

$$\operatorname{null}(\varphi) = \dim(\ker(\varphi)).$$

For a matrix $A$, we define $\operatorname{rank}(A)$ and $\operatorname{null}(A)$ to be the rank and nullity of the corresponding linear map.

**Theorem 6.1.2** The Rank-Nullity Theorem (Fundamental Theorem of Linear Maps)

Let $\varphi : V \to W$ be a linear map with $V$ finite-dimensional. Then $\operatorname{Im}(\varphi)$ is finite-dimensional and

$$\dim(V) = \dim(\ker(\varphi)) + \dim(\operatorname{Im}(\varphi)).$$

Equivalently: $\dim(V) = \operatorname{null}(\varphi) + \operatorname{rank}(\varphi)$.

***Proof:*** Let $\{u_1, \dots, u_m\}$ be a basis for $\ker(\varphi)$. By the basis extension theorem, we can extend this to a basis

$$\{u_1, \dots, u_m, v_1, \dots, v_r\}$$

of $V$, where $\dim(V) = m + r$.

**Claim:** $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\operatorname{Im}(\varphi)$.

Spanning: Let $w \in \operatorname{Im}(\varphi)$, say $w = \varphi(x)$ for some $x \in V$. Write

$$x = a_1u_1 + \cdots + a_mu_m + b_1v_1 + \cdots + b_rv_r.$$

Applying $\varphi$:

$$w = \varphi(x) = a_1\varphi(u_1) + \cdots + a_m\varphi(u_m) + b_1\varphi(v_1) + \cdots + b_r\varphi(v_r).$$

Since each $u_i \in \ker(\varphi)$, we have $\varphi(u_i) = 0$, so

$$w = b_1\varphi(v_1) + \cdots + b_r\varphi(v_r).$$

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ spans $\operatorname{Im}(\varphi)$.

Linear independence: Suppose

$$c_1\varphi(v_1) + \cdots + c_r\varphi(v_r) = 0.$$

Then $\varphi(c_1v_1 + \cdots + c_rv_r) = 0$, so $c_1v_1 + \cdots + c_rv_r \in \ker(\varphi)$. Since $\{u_1, \dots, u_m\}$ is a basis for $\ker(\varphi)$, we can write

$$c_1v_1 + \cdots + c_rv_r = d_1u_1 + \cdots + d_mu_m$$

for some scalars $d_j$. Rearranging:

$$d_1u_1 + \cdots + d_mu_m - c_1v_1 - \cdots - c_rv_r = 0.$$

Since $\{u_1, \dots, u_m, v_1, \dots, v_r\}$ is linearly independent (it's a basis of $V$), all coefficients are zero. In particular, $c_1 = \cdots = c_r = 0$.

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\operatorname{Im}(\varphi)$, so $\dim(\operatorname{Im}(\varphi)) = r$. We conclude:

$$\dim(V) = m + r = \dim(\ker(\varphi)) + \dim(\operatorname{Im}(\varphi)).$$

□

**Note:-**

The Rank-Nullity Theorem is the linear algebra version of the **First Isomorphism Theorem** from group theory. Recall that for a group homomorphism $f : G \to H$, we have $G/\ker(f) \cong \mathrm{Im}(f)$. The linear version states $V/\ker(\varphi) \cong \mathrm{Im}(\varphi)$, and "counting dimensions" gives $\dim(V) - \dim(\ker(\varphi)) = \dim(\mathrm{Im}(\varphi))$.

### 6.1.4 Consequences of the Rank-Nullity Theorem

The Rank-Nullity Theorem has powerful implications for understanding linear maps.

**Corollary 6.1.2** Dimension constraints on injectivity

Let $\varphi : V \to W$ be a linear map with $\dim(V) = n$ and $\dim(W) = m$.

1. If $n > m$, then $\varphi$ cannot be injective.
2. If $n < m$, then $\varphi$ cannot be surjective.

***Proof:*** (1) If $\varphi$ were injective, then $\dim(\ker(\varphi)) = 0$, so by Rank-Nullity:

$$\dim(\mathrm{Im}(\varphi)) = \dim(V) = n > m = \dim(W).$$

But $\mathrm{Im}(\varphi) \subseteq W$, so $\dim(\mathrm{Im}(\varphi)) \leqslant \dim(W) = m$. Contradiction.
(2) If $\varphi$ were surjective, then $\dim(\mathrm{Im}(\varphi)) = \dim(W) = m$, so by Rank-Nullity:

$$\dim(\ker(\varphi)) = \dim(V) - \dim(\mathrm{Im}(\varphi)) = n - m < 0.$$

But dimensions are non-negative. Contradiction. $\square$

**Corollary 6.1.3** Equal dimensions: bijectivity criteria

Let $\varphi : V \to W$ be a linear map with $\dim(V) = \dim(W) = n$. The following are equivalent:

1. $\varphi$ is injective.
2. $\varphi$ is surjective.
3. $\varphi$ is bijective (an isomorphism).
4. $\ker(\varphi) = \{0\}$.
5. $\mathrm{Im}(\varphi) = W$.
6. $\mathrm{rank}(\varphi) = n$.

***Proof:*** By Rank-Nullity, $\mathrm{null}(\varphi) + \mathrm{rank}(\varphi) = n$. Since $\mathrm{rank}(\varphi) \leqslant \dim(W) = n$:

- $\varphi$ injective $\iff$ $\mathrm{null}(\varphi) = 0$ $\iff$ $\mathrm{rank}(\varphi) = n$.
- $\varphi$ surjective $\iff$ $\mathrm{rank}(\varphi) = n$ $\iff$ $\mathrm{null}(\varphi) = 0$.

Thus injectivity, surjectivity, and bijectivity are equivalent when dimensions match. $\square$

**Note:-**

For linear maps between equal-dimensional spaces, "one-to-one" and "onto" are the same condition. This fails for general functions and even for linear maps between infinite-dimensional spaces. Consider the **right shift** $R : k^\infty \to k^\infty$ defined by $R(x_1, x_2, x_3, \ldots) = (0, x_1, x_2, \ldots)$. This is injective (kernel is trivial) but not surjective (the sequence $(1, 0, 0, \ldots)$ is not in the image). Conversely, the **left shift** $L(x_1, x_2, x_3, \ldots) = (x_2, x_3, \ldots)$ is surjective but not injective. In infinite dimensions, there's "room" for one without the other. The finite-dimensional equivalence is a manifestation of the pigeonhole principle applied to basis vectors.

**Corollary 6.1.4** Existence of right/left inverses

Let $\varphi : V \to W$ be a linear map between finite-dimensional spaces.

1. $\varphi$ has a left inverse if and only if $\varphi$ is injective.
2. $\varphi$ has a right inverse if and only if $\varphi$ is surjective.

### 6.1.5 Matrix Interpretation

For a matrix $A \in \mathcal{M}_{m\times n}(k)$, the associated linear map is $L_A : k^n \to k^m$ given by $L_A(x) = Ax$. The Rank-Nullity Theorem becomes:

**Proposition 6.1.4** Rank-Nullity for matrices

For $A \in \mathcal{M}_{m\times n}(k)$:

1. $\ker(L_A) = \{x \in k^n : Ax = 0\}$ is the null space of $A$.
2. $\mathrm{Im}(L_A) = \{Ax : x \in k^n\}$ is the column space of $A$ (the span of the columns).
3. $\mathrm{null}(A) + \mathrm{rank}(A) = n$ (the number of columns).

**Note:-**

The rank of a matrix equals the number of linearly independent columns, which equals the number of linearly independent rows. The column space has dimension equal to the rank, and the null space has dimension equal to $n - \mathrm{rank}(A)$.
Looking ahead: when we study eigenvalues, the kernel will reappear in a crucial role. For a linear operator $T : V \to V$ and scalar $\lambda$, the **eigenspace** $E_\lambda = \ker(T - \lambda I)$ is precisely the kernel of the shifted operator. The dimension of this eigenspace (the geometric multiplicity of $\lambda$) measures "how degenerate" the eigenvalue is. The interplay between algebraic and geometric multiplicities—and the failure of these to match—is what makes Jordan normal form necessary.

**Example 6.1.3** (Computing kernel and image)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix}$ over $\mathbb{R}$.

**Finding** $\ker(A)$**:** Solve $Ax = 0$. Row reduce:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Back-substituting: $x_2 = -2x_3$, $x_1 = -2x_2 - 3x_3 = 4x_3 - 3x_3 = x_3$. So

$$\ker(A) = \mathrm{Span}\{(1, -2, 1)^T\}, \quad \mathrm{null}(A) = 1.$$

**Finding** $\mathrm{Im}(A)$**:** The non-zero rows in RREF correspond to pivot columns. Columns 1 and 2 are pivot columns, so

$$\mathrm{Im}(A) = \mathrm{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right\}, \quad \mathrm{rank}(A) = 2.$$

**Verification:** $\mathrm{null}(A) + \mathrm{rank}(A) = 1 + 2 = 3 = n$. ✓

**Question 1**

1. Let $D : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be the differentiation operator. Find $\ker(D)$, $\mathrm{Im}(D)$, $\mathrm{null}(D)$, and $\mathrm{rank}(D)$. Verify the Rank-Nullity Theorem.
2. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be defined by $T(x_1, x_2, x_3, x_4) = (x_1 - x_2, x_2 - x_3, x_3 - x_4)$. Determine whether $T$ is injective, surjective, or neither.
3. Prove that if $\varphi : V \to V$ is a linear operator on a finite-dimensional space with $\varphi^2 = \varphi$, then $V = \ker(\varphi) \oplus \mathrm{Im}(\varphi)$.
4. Let $V$ and $W$ be finite-dimensional vector spaces with $\dim(V) = \dim(W)$. Prove that a linear map $\varphi : V \to W$ is injective if and only if for every linearly independent list in $V$, its image under $\varphi$ is linearly independent in $W$.

**Solution**

**(1)** The space $\mathcal{P}_3(\mathbb{R})$ has basis $\{1, x, x^2, x^3\}$, so $\dim(\mathcal{P}_3(\mathbb{R})) = 4$.
**Kernel:** $Dp = 0$ if and only if $p$ is a constant polynomial. So $\ker(D) = \mathrm{Span}\{1\}$, which are the constant functions. Thus $\mathrm{null}(D) = 1$.
**Image:** Differentiating: $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. The non-zero outputs span $\mathrm{Span}\{1, x, x^2\} = \mathcal{P}_2(\mathbb{R})$. Thus $\mathrm{Im}(D) = \mathcal{P}_2(\mathbb{R})$ and $\mathrm{rank}(D) = 3$.
**Verification:** $\mathrm{null}(D) + \mathrm{rank}(D) = 1 + 3 = 4 = \dim(\mathcal{P}_3(\mathbb{R}))$. ✓
**(2) Injectivity:** Find $\ker(T)$. Solve $T(x_1, x_2, x_3, x_4) = (0, 0, 0)$:

$$x_1 - x_2 = 0, \quad x_2 - x_3 = 0, \quad x_3 - x_4 = 0.$$

This gives $x_1 = x_2 = x_3 = x_4$. So $\ker(T) = \{(t, t, t, t) : t \in \mathbb{R}\} = \mathrm{Span}\{(1, 1, 1, 1)\}$.
Since $\ker(T) \neq \{0\}$, $T$ is **not injective**.
**Surjectivity:** We have $\mathrm{null}(T) = 1$, so $\mathrm{rank}(T) = 4 - 1 = 3 = \dim(\mathbb{R}^3)$. Thus $\mathrm{Im}(T) = \mathbb{R}^3$, and $T$ is **surjective**.
**(3)** We show $V = \ker(\varphi) + \mathrm{Im}(\varphi)$ and $\ker(\varphi) \cap \mathrm{Im}(\varphi) = \{0\}$.
Sum equals $V$: Let $v \in V$. Write $v = (v - \varphi(v)) + \varphi(v)$. Note that $\varphi(v) \in \mathrm{Im}(\varphi)$. Also:

$$\varphi(v - \varphi(v)) = \varphi(v) - \varphi^2(v) = \varphi(v) - \varphi(v) = 0,$$

so $v - \varphi(v) \in \ker(\varphi)$. Thus $v \in \ker(\varphi) + \mathrm{Im}(\varphi)$.
Intersection is trivial: Let $w \in \ker(\varphi) \cap \mathrm{Im}(\varphi)$. Then $\varphi(w) = 0$ and $w = \varphi(u)$ for some $u \in V$. So:

$$w = \varphi(u) = \varphi^2(u) = \varphi(\varphi(u)) = \varphi(w) = 0.$$

Thus $V = \ker(\varphi) \oplus \mathrm{Im}(\varphi)$.
**(4)** $(\Rightarrow)$ Suppose $\varphi$ is injective and $\{v_1, \ldots, v_m\}$ is linearly independent. Suppose $\sum c_i \varphi(v_i) = 0$. Then $\varphi(\sum c_i v_i) = 0$, so $\sum c_i v_i \in \ker(\varphi) = \{0\}$. Thus $\sum c_i v_i = 0$, and by linear independence of the $v_i$, all $c_i = 0$. Hence $\{\varphi(v_1), \ldots, \varphi(v_m)\}$ is linearly independent.
$(\Leftarrow)$ Suppose $\varphi$ maps every linearly independent set to a linearly independent set. If $v \neq 0$, then $\{v\}$ is linearly independent, so $\{\varphi(v)\}$ is linearly independent, meaning $\varphi(v) \neq 0$. Thus $\ker(\varphi) = \{0\}$, so $\varphi$ is injective.

## 6.2 Duality

The dual space of a vector space $V$ consists of all linear maps from $V$ to the base field $k$—these are called linear functionals. While this might seem like a minor variation on Hom-spaces, duality reveals useful structure: every basis has a “dual basis,” every linear map has a “dual map,” and the matrix of the dual map is the transpose of the original. In more advanced contexts (differential geometry, functional analysis, representation theory), duality becomes indispensable.

**Note:-**

The dual space construction is the first example of a **contravariant functor** you'll encounter. When we pass from $V$ to $V'$, linear maps $T : V \to W$ don't induce maps $V' \to W'$—instead, they induce maps $W' \to V'$, going the "wrong way." This reversal of arrows is the hallmark of contravariance. In category theory, the dual space functor $(-)' : \mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$ is representable: $V' = \mathrm{Hom}(V, k)$ is "the functor represented by $k$."

### 6.2.1 Linear Functionals and the Dual Space

**Definition 6.2.1: Linear functional**

A **linear functional** on a vector space $V$ over a field $k$ is a linear map $\ell : V \to k$. In other words, a linear functional is an element of $\mathrm{Hom}(V, k)$.

**Example 6.2.1** (Examples of linear functionals)

1. **Coordinate projections:** On $k^n$, define $\pi_j(x_1, \ldots, x_n) = x_j$. Each $\pi_j$ is a linear functional.
2. **Weighted sums:** Fix $(c_1, \ldots, c_n) \in k^n$. Define $\ell(x_1, \ldots, x_n) = c_1x_1 + \cdots + c_nx_n$. This is a linear functional on $k^n$, and every linear functional on $k^n$ has this form.
3. **Evaluation:** On $\mathcal{P}(\mathbb{R})$, define $\mathrm{ev}_a(p) = p(a)$ for fixed $a \in \mathbb{R}$. This is a linear functional.
4. **Integration:** On $\mathcal{P}(\mathbb{R})$, define $\ell(p) = \int_0^1 p(x)\, dx$. Linearity of the integral makes this a linear functional.
5. **Trace:** On $\mathcal{M}_{n\times n}(k)$, the trace $\mathrm{Tr}(A) = \sum_{i=1}^n a_{ii}$ is a linear functional.

**Definition 6.2.2: Dual space**

The **dual space** of $V$, denoted $V^*$ or $V'$, is the vector space of all linear functionals on $V$:

$$V^* = \mathrm{Hom}(V, k).$$

This is a vector space under pointwise addition and scalar multiplication: $(\ell_1 + \ell_2)(v) = \ell_1(v) + \ell_2(v)$ and $(\lambda\ell)(v) = \lambda \cdot \ell(v)$.

**Note:-**

We use $V^*$ notation (common in differential geometry and physics) interchangeably with $V'$ (common in algebra). Both mean the same thing. Some texts reserve $V^*$ for the continuous dual in functional analysis, but for finite-dimensional spaces this distinction is irrelevant.

**Proposition 6.2.1** Dimension of the dual space

If $V$ is finite-dimensional with $\dim(V) = n$, then $\dim(V^*) = n$. Thus $V^* \cong V$ as vector spaces (non-canonically).

***Proof:*** By our earlier result on Hom-spaces:

$$\dim(V^*) = \dim(\mathrm{Hom}(V, k)) = \dim(V) \cdot \dim(k) = n \cdot 1 = n.$$

□

**Note:-**

The isomorphism $V \cong V^*$ is non-canonical—it depends on a choice of basis. There's no "natural" way to identify vectors with linear functionals without extra structure. However, the double dual $V^{**} = (V^*)^*$

does have a canonical isomorphism with $V$, given by evaluation: $v \mapsto (\ell \mapsto \ell(v))$.

### 6.2.2 The Dual Basis

Given a basis of $V$, there's a natural associated basis of $V^*$.

**Definition 6.2.3: Dual basis**

Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of $V$. The **dual basis** is the list $\mathcal{B}^* = \{\varphi_1, \ldots, \varphi_n\}$ of linear functionals in $V^*$ defined by

$$\varphi_j(v_i) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

In other words, $\varphi_j$ "picks out" the $j$-th coordinate: if $v = c_1v_1 + \cdots + c_nv_n$, then $\varphi_j(v) = c_j$.

**Proposition 6.2.2** Dual basis is a basis

The dual basis $\{\varphi_1, \ldots, \varphi_n\}$ is indeed a basis of $V^*$.

***Proof:*** **Linear independence:** Suppose $a_1\varphi_1 + \cdots + a_n\varphi_n = 0$ (the zero functional). Evaluating at $v_k$:

$$0 = (a_1\varphi_1 + \cdots + a_n\varphi_n)(v_k) = a_1\varphi_1(v_k) + \cdots + a_n\varphi_n(v_k) = a_k.$$

So all $a_k = 0$.
**Spanning:** Since we have $n$ linearly independent vectors in $V^*$ and $\dim(V^*) = n$, they form a basis. $\square$

**Proposition 6.2.3** Dual basis extracts coordinates

Let $\{v_1, \ldots, v_n\}$ be a basis of $V$ with dual basis $\{\varphi_1, \ldots, \varphi_n\}$. For any $v \in V$:

$$v = \varphi_1(v)v_1 + \varphi_2(v)v_2 + \cdots + \varphi_n(v)v_n.$$

That is, $\varphi_j(v)$ is the $j$-th coordinate of $v$ with respect to the basis $\{v_i\}$.

***Proof:*** Write $v = c_1v_1 + \cdots + c_nv_n$. Applying $\varphi_j$:

$$\varphi_j(v) = c_1\varphi_j(v_1) + \cdots + c_n\varphi_j(v_n) = c_j.$$

Substituting back: $v = \varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n$. $\square$

**Example 6.2.2** (Dual basis of the standard basis)

Let $\{e_1, \ldots, e_n\}$ be the standard basis of $k^n$. The dual basis consists of the coordinate projections:

$$\varphi_j(x_1, \ldots, x_n) = x_j.$$

These are the "row vectors" $\varphi_j = (0, \ldots, 0, 1, 0, \ldots, 0)$ (with 1 in position $j$).

### 6.2.3 The Dual Map

Just as a linear map $T : V \to W$ induces a map on Hom-spaces, it induces a map on dual spaces—but in the opposite direction.

**Definition 6.2.4: Dual map (transpose)**

Let $T : V \to W$ be a linear map. The **dual map** (or **transpose**) is the linear map $T^* : W^* \to V^*$ defined by

$$T^*(\psi) = \psi \circ T$$

for each $\psi \in W^*$. In other words, $T^*(\psi)$ is the linear functional on $V$ given by $v \mapsto \psi(T(v))$.

**Note:-**

The dual map goes "backwards": $T : V \to W$ yields $T^* : W^* \to V^*$. This is because we're precomposing with $T$. If you think of linear functionals as "measurements," then $T^*(\psi)$ is the measurement you get by first applying $T$, then measuring with $\psi$. The reversal of direction is the essence of contravariance.

**Proposition 6.2.4** The dual map is linear

For $T : V \to W$, the dual map $T^* : W^* \to V^*$ is indeed a linear map.

***Proof:*** For $\psi_1, \psi_2 \in W^*$ and $\lambda \in k$:

$$\begin{aligned} T^*(\psi_1 + \psi_2) &= (\psi_1 + \psi_2) \circ T = \psi_1 \circ T + \psi_2 \circ T = T^*(\psi_1) + T^*(\psi_2), \\ T^*(\lambda\psi) &= (\lambda\psi) \circ T = \lambda(\psi \circ T) = \lambda T^*(\psi). \end{aligned}$$

□

**Proposition 6.2.5** Algebraic properties of the dual map

Let $S : V \to W$ and $T : W \to U$ be linear maps.

1. $(S + T)^* = S^* + T^*$ (when both are defined).
2. $(\lambda T)^* = \lambda T^*$ for $\lambda \in k$.
3. $(T \circ S)^* = S^* \circ T^*$ (order reverses!).
4. $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

***Proof:*** (3) is the key property. For $\psi \in U^*$:

$$(T \circ S)^*(\psi) = \psi \circ (T \circ S) = (\psi \circ T) \circ S = S^*(\psi \circ T) = S^*(T^*(\psi)) = (S^* \circ T^*)(\psi).$$

The other properties are similar. □

**Note:-**

Property (3) says the dual is a contravariant functor: it reverses the order of composition. Compare with the transpose of matrices: $(AB)^T = B^T A^T$. This is not a coincidence—the matrix of the dual map is the transpose!

### 6.2.4 Matrix of the Dual Map

The connection between dual maps and matrix transposes clarifies why the transpose operation is natural.

**Theorem 6.2.1** Matrix of the dual map is the transpose

Let $T : V \to W$ be a linear map between finite-dimensional spaces. Fix bases $\mathcal{B} = \{v_1, \ldots, v_n\}$ of $V$ and $\mathcal{C} = \{w_1, \ldots, w_m\}$ of $W$, with dual bases $\mathcal{B}^* = \{\varphi_1, \ldots, \varphi_n\}$ and $\mathcal{C}^* = \{\psi_1, \ldots, \psi_m\}$. If

$$A = \mathcal{M}(T; \mathcal{B}, \mathcal{C}) \in \mathcal{M}_{m \times n}(k)$$

is the matrix of $T$, then the matrix of $T^*$ with respect to the dual bases is the transpose:

$$\mathcal{M}(T^*; \mathcal{C}^*, \mathcal{B}^*) = A^T \in \mathcal{M}_{n\times m}(k).$$

***Proof:*** We need to show that $T^*(\psi_j) = \sum_{i=1}^{n} A_{ji}\varphi_i$ for each $j$.
By definition of the matrix $A$, we have $T(v_k) = \sum_{i=1}^{m} A_{ik}w_i$. Now compute:

$$T^*(\psi_j)(v_k) = \psi_j(T(v_k)) = \psi_j\left(\sum_{i=1}^{m} A_{ik}w_i\right) = \sum_{i=1}^{m} A_{ik}\psi_j(w_i) = A_{jk}.$$

On the other hand, if $T^*(\psi_j) = \sum_{i=1}^{n} c_i\varphi_i$, then

$$T^*(\psi_j)(v_k) = \sum_{i=1}^{n} c_i\varphi_i(v_k) = c_k.$$

Comparing: $c_k = A_{jk}$. Thus $T^*(\psi_j) = \sum_{k=1}^{n} A_{jk}\varphi_k$, which means the $j$-th column of $\mathcal{M}(T^*)$ is the $j$-th row of $A$. Hence $\mathcal{M}(T^*) = A^T$. □

**Note:-**

This theorem explains why the transpose is natural: it's not just a matrix operation, it's the coordinate representation of the dual map. The reversal of dimensions ($m \times n$ becomes $n \times m$) reflects the contravariance of the dual construction.

**Corollary 6.2.1** Rank of transpose equals rank

For any matrix $A$, $\text{rank}(A^T) = \text{rank}(A)$.

***Proof:*** The rank of $A$ is $\dim(\text{Im}(T))$ where $T$ is the associated linear map. By a result we'll see shortly, $\dim(\text{Im}(T^*)) = \dim(\text{Im}(T))$. Since the matrix of $T^*$ is $A^T$, we get $\text{rank}(A^T) = \text{rank}(A)$. □

### 6.2.5 Kernel and Image of the Dual Map

The dual map interchanges the roles of kernel and image in a precise way.

**Proposition 6.2.6** Relationships between $T$ and $T^*$

Let $T : V \to W$ be a linear map between finite-dimensional spaces.

1. $\ker(T^*) = (\text{Im}(T))^\circ$, the annihilator of the image.
2. $\text{Im}(T^*) = (\ker(T))^\circ$, the annihilator of the kernel.
3. $\dim(\ker(T^*)) = \dim(W) - \text{rank}(T)$.
4. $\dim(\text{Im}(T^*)) = \text{rank}(T)$.

Here, for a subspace $U \subseteq V$, the **annihilator** is $U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}$.

**Note:-**

The annihilator $U^\circ$ consists of all linear functionals that vanish on $U$. If $\dim(V) = n$ and $\dim(U) = k$, then $\dim(U^\circ) = n - k$. The annihilator provides a duality between subspaces of $V$ and subspaces of $V^*$: bigger subspaces have smaller annihilators.

**Question 2**

1. Let $V = \mathbb{R}^3$ with standard basis $\{e_1, e_2, e_3\}$. Write down the dual basis explicitly as functions $\mathbb{R}^3 \to \mathbb{R}$.
2. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, y + z)$. Find the matrix of $T$ with respect to the standard bases, then find the matrix of $T^*$ with respect to the dual bases.
3. Prove that if $T : V \to W$ is surjective, then $T^* : W^* \to V^*$ is injective.
4. Prove that if $V$ is finite-dimensional, the map $\Phi : V \to V^{**}$ defined by $\Phi(v)(\ell) = \ell(v)$ is an isomorphism.

**Solution**

**(1)** The dual basis consists of the coordinate projections:

$$\varphi_1(x, y, z) = x, \quad \varphi_2(x, y, z) = y, \quad \varphi_3(x, y, z) = z.$$

These are "row vectors" $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.
**(2)** The matrix of $T$ is:

$$A = \mathcal{M}(T) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

The matrix of $T^*$ is the transpose:

$$A^T = \mathcal{M}(T^*) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Explicitly, if $\psi_1, \psi_2$ are the dual basis of $\mathbb{R}^2$ and $\varphi_1, \varphi_2, \varphi_3$ are the dual basis of $\mathbb{R}^3$:

$$T^*(\psi_1) = \varphi_1 + \varphi_2, \quad T^*(\psi_2) = \varphi_2 + \varphi_3.$$

**(3)** Suppose $T$ is surjective and $T^*(\psi) = 0$, i.e., $\psi \circ T = 0$. For any $w \in W$, since $T$ is surjective, $w = T(v)$ for some $v \in V$. Then:

$$\psi(w) = \psi(T(v)) = (\psi \circ T)(v) = 0.$$

Since this holds for all $w \in W$, we have $\psi = 0$. Thus $\ker(T^*) = \{0\}$, so $T^*$ is injective.
**(4) Well-defined and linear:** For $v \in V$, $\Phi(v) : V^* \to k$ is defined by $\Phi(v)(\ell) = \ell(v)$. This is linear in $\ell$ (check!), so $\Phi(v) \in V^{**}$. Moreover, $\Phi$ is linear: $\Phi(v_1 + v_2)(\ell) = \ell(v_1 + v_2) = \ell(v_1) + \ell(v_2) = \Phi(v_1)(\ell) + \Phi(v_2)(\ell)$.
**Injective:** Suppose $\Phi(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis $\{v, v_2, \ldots, v_n\}$. The dual basis element $\varphi_1$ satisfies $\varphi_1(v) = 1 \neq 0$, contradiction. So $v = 0$.
**Surjective:** Since $\dim(V) = \dim(V^*) = \dim(V^{**})$, injectivity implies surjectivity.

### 6.2.6 The Double Dual and Natural Isomorphisms

We've seen that $V \cong V^*$ for finite-dimensional $V$, but this isomorphism required a choice of basis. The double dual $V^{**} = (V^*)^*$ enjoys a more canonical relationship with $V$.

**Definition 6.2.5: Evaluation map**

The **evaluation map** is $\mathrm{ev} : V \to V^{**}$ defined by

$$\mathrm{ev}(v)(\ell) = \ell(v)$$

for $v \in V$ and $\ell \in V^*$. That is, $\mathrm{ev}(v)$ is the linear functional on $V^*$ that "evaluates at $v$."

**Proposition 6.2.7** Properties of the evaluation map

The evaluation map $\mathrm{ev} : V \to V^{**}$ satisfies:

1. ev is linear.
2. ev is injective.
3. If $V$ is finite-dimensional, ev is an isomorphism.

***Proof:*** **(1)** For $v, w \in V$, $\lambda \in k$, and $\ell \in V^*$:

$$\mathrm{ev}(v + \lambda w)(\ell) = \ell(v + \lambda w) = \ell(v) + \lambda \ell(w) = \mathrm{ev}(v)(\ell) + \lambda\, \mathrm{ev}(w)(\ell).$$

**(2)** Suppose $\mathrm{ev}(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis and let $\ell$ be the dual basis element with $\ell(v) = 1$. Contradiction. So $v = 0$.

**(3)** For finite-dimensional $V$: $\dim(V^{**}) = \dim(V^*) = \dim(V)$. Since ev is injective, it's an isomorphism. □

**Note:-**

The crucial point: ev is natural—it doesn't depend on any choices. Given bases $\{e_1, \ldots, e_n\}$ of $V$, dual basis $\{e_1^*, \ldots, e_n^*\}$ of $V^*$, and double dual basis $\{e_1^{**}, \ldots, e_n^{**}\}$ of $V^{**}$, one can verify that $\mathrm{ev}(e_i) = e_i^{**}$. The evaluation map sends each basis element to its "double dual counterpart."

**Note:-**

For infinite-dimensional $V$, the situation is strikingly different. If $V$ has a basis $\{e_i\}_{i \in I}$ indexed by an infinite set $I$, then every element of $V$ is a finite linear combination $\sum_{i \in I} x_i e_i$ (with only finitely many $x_i \neq 0$). However, elements of $V^*$ can be arbitrary: given any function $a : I \to k$, the map $\ell_a(\sum x_i e_i) = \sum x_i a_i$ is a well-defined linear functional. Thus $V^* \cong k^I = \prod_{i \in I} k$, the full product, which is much larger than the direct sum $V \cong \bigoplus_{i \in I} k$. The evaluation map $\mathrm{ev} : V \to V^{**}$ is still injective, but far from surjective.

Lecture 7.

# More maps and spaces

### 7.0.1 Annihilators

The annihilator provides a precise duality between subspaces of $V$ and subspaces of $V^*$.

**Definition 7.0.1: Annihilator**

For a subspace $U \subseteq V$, the **annihilator** of $U$ is

$$\mathrm{Ann}(U) = U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}.$$

**Theorem 7.0.1** Dimension of the annihilator

If $V$ is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(\mathrm{Ann}(U)) = \dim(V) - \dim(U).$$

***Proof:*** The restriction map $\rho : V^* \to U^*$ defined by $\rho(\ell) = \ell|_U$ is a linear map with $\ker(\rho) = \mathrm{Ann}(U)$. We claim $\rho$ is surjective: given $\psi \in U^*$, extend a basis of $U$ to a basis of $V$, define $\ell$ to agree with $\psi$ on $U$ and be zero on the complementary basis vectors. Then $\rho(\ell) = \psi$.

By rank-nullity:

$$\dim(V^*) = \dim(\mathrm{Ann}(U)) + \dim(U^*).$$

Since $\dim(V^*) = \dim(V)$ and $\dim(U^*) = \dim(U)$, we get $\dim(\mathrm{Ann}(U)) = \dim(V) - \dim(U)$. $\square$

**Theorem 7.0.2** Annihilator isomorphism

For a subspace $U \subseteq V$, there is a natural isomorphism

$$\mathrm{Ann}(U) \cong (V/U)^*.$$

***Proof:*** Define $\Phi : \mathrm{Ann}(U) \to (V/U)^*$ by $\Phi(\ell)(v + U) = \ell(v)$.

**Well-defined:** If $v + U = v' + U$, then $v - v' \in U$, so $\ell(v) - \ell(v') = \ell(v - v') = 0$ since $\ell \in \mathrm{Ann}(U)$.

**Injective:** If $\Phi(\ell) = 0$, then $\ell(v) = 0$ for all $v \in V$, so $\ell = 0$.

**Surjective:** Given $\psi \in (V/U)^*$, define $\ell = \psi \circ q$ where $q : V \to V/U$ is the quotient map. Then $\ell \in \mathrm{Ann}(U)$ and $\Phi(\ell) = \psi$. $\square$

## 7.1 Quotient Spaces

Quotient spaces provide a systematic way to "collapse" a subspace to zero, creating a smaller vector space that captures the structure "perpendicular" to the subspace. This construction is fundamental in linear algebra and pervades abstract algebra.

### 7.1.1 Construction of the Quotient Space

**Definition 7.1.1: Cosets and quotient space**

Let $V$ be a vector space over $k$ and $U \subseteq V$ a subspace. For $v \in V$, the **coset** of $v$ modulo $U$ is

$$v + U = \{v + u : u \in U\}.$$

The **quotient space** is the set of all cosets:

$$V/U = \{v + U : v \in V\}.$$

**Note:-**

Two cosets $v + U$ and $w + U$ are equal if and only if $v - w \in U$. Equivalently, the cosets partition $V$ into equivalence classes under the relation $v \sim w \Leftrightarrow v - w \in U$.

**Proposition 7.1.1** Vector space structure on the quotient

The quotient $V/U$ becomes a vector space over $k$ under the operations:

1. **Addition:** $(v + U) + (w + U) = (v + w) + U$.
2. **Scalar multiplication:** $\lambda(v + U) = (\lambda v) + U$.

The zero element is $0 + U = U$, and $-(v + U) = (-v) + U$.

***Proof:*** The key is well-definedness: if $v+U = v'+U$ and $w+U = w'+U$, then $v-v' \in U$ and $w-w' \in U$, so $(v + w) - (v' + w') = (v - v') + (w - w') \in U$, meaning $(v + w) + U = (v' + w') + U$.

Similarly for scalar multiplication: if $v + U = v' + U$, then $v - v' \in U$, so $\lambda(v - v') = \lambda v - \lambda v' \in U$, giving $\lambda v + U = \lambda v' + U$.

The vector space axioms follow from those in $V$. $\square$

### 7.1.2 The Quotient Map and Exact Sequences

**Definition 7.1.2: Quotient map**

The **quotient map** (or **canonical projection**) is $q : V \to V/U$ defined by $q(v) = v + U$.

**Proposition 7.1.2** Properties of the quotient map

The quotient map $q : V \to V/U$ satisfies:

1. $q$ is linear.
2. $q$ is surjective.
3. $\ker(q) = U$.

***Proof:*** (1) $q(v + \lambda w) = (v + \lambda w) + U = (v + U) + \lambda(w + U) = q(v) + \lambda q(w)$.

(2) Every coset $v + U$ is $q(v)$.

(3) $v \in \ker(q) \Leftrightarrow v + U = 0 + U \Leftrightarrow v \in U$. $\square$

**Note:-**

This gives a **short exact sequence**:

$$0 \longrightarrow U \xrightarrow{\iota} V \xrightarrow{q} V/U \longrightarrow 0$$

where $\iota : U \hookrightarrow V$ is the inclusion. "Exact" means: $\mathrm{Im}(\iota) = \ker(q)$, $\iota$ is injective, and $q$ is surjective. Short exact sequences are a central organizing principle in algebra.

**Theorem 7.1.1** Dimension of the quotient space

If $V$ is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(V/U) = \dim(V) - \dim(U).$$

***Proof:*** Apply rank-nullity to $q : V \to V/U$:

$$\dim(V) = \dim(\ker(q)) + \dim(\mathrm{Im}(q)) = \dim(U) + \dim(V/U).$$

$\square$

**Example 7.1.1** (Concrete example)

Let $V = \mathbb{R}^3$ and $U = \mathrm{Span}\{(1, 0, 0)\}$, the $x$-axis. Then $V/U \cong \mathbb{R}^2$:
The coset $(a, b, c) + U$ consists of all points $\{(a + t, b, c) : t \in \mathbb{R}\}$—a horizontal line through $(a, b, c)$. Two points have the same coset iff they have the same $y$ and $z$ coordinates. The map $(a, b, c) + U \mapsto (b, c)$ is an isomorphism $V/U \xrightarrow{\sim} \mathbb{R}^2$.

### 7.1.3 Universal Property of the Quotient

The quotient space is characterized by a universal property that explains its importance.

**Theorem 7.1.2** Universal property of the quotient

Let $\varphi : V \to W$ be a linear map with $U \subseteq \ker(\varphi)$. Then there exists a unique linear map $\bar{\varphi} : V/U \to W$ such that $\varphi = \bar{\varphi} \circ q$:

$$\begin{tikzcd} V \arrow[r, "\varphi"] \arrow[d, "q"'] & W \\ V/U \arrow[ur, dashed, "\exists!\bar{\varphi}"'] & \end{tikzcd}$$

Moreover, $\bar{\varphi}$ is defined by $\bar{\varphi}(v + U) = \varphi(v)$.

***Proof:*** **Well-defined:** If $v + U = v' + U$, then $v - v' \in U \subseteq \ker(\varphi)$, so $\varphi(v) = \varphi(v')$.
**Linearity:** $\bar{\varphi}((v+U)+\lambda(w+U)) = \bar{\varphi}((v+\lambda w)+U) = \varphi(v+\lambda w) = \varphi(v)+\lambda\varphi(w) = \bar{\varphi}(v+U)+\lambda\bar{\varphi}(w+U)$.
**Uniqueness:** Any $\psi : V/U \to W$ with $\varphi = \psi \circ q$ must satisfy $\psi(v + U) = \psi(q(v)) = \varphi(v) = \bar{\varphi}(v + U)$. $\square$

**Corollary 7.1.1** First Isomorphism Theorem

For any linear map $\varphi : V \to W$:

$$V/\ker(\varphi) \cong \mathrm{Im}(\varphi).$$

The isomorphism is $\bar{\varphi} : v + \ker(\varphi) \mapsto \varphi(v)$.

***Proof:*** By the universal property, $\bar{\varphi} : V/\ker(\varphi) \to W$ is well-defined. Its image is exactly $\mathrm{Im}(\varphi)$. Injectivity: if $\bar{\varphi}(v + \ker(\varphi)) = 0$, then $\varphi(v) = 0$, so $v \in \ker(\varphi)$, meaning $v + \ker(\varphi) = 0 + \ker(\varphi)$. $\square$

**Note:-**

This theorem is remarkably useful. It says that every surjective linear map $\varphi : V \twoheadrightarrow W$ is "really" a quotient map: $V \to V/\ker(\varphi) \cong W$. Dually, every injective map $\iota : U \hookrightarrow V$ makes $U$ isomorphic to a subspace of $V$.

### 7.1.4 Correspondence Between Subspaces

The quotient construction establishes a bijection between certain subspaces.

**Theorem 7.1.3** Lattice correspondence

Let $U \subseteq V$ be a subspace. There is a bijection

$$\{\text{subspaces } W \subseteq V \text{ with } U \subseteq W\} \longleftrightarrow \{\text{subspaces of } V/U\}$$

given by:

- Forward: $W \mapsto W/U = \{w + U : w \in W\}$.
- Backward: $\bar{W} \mapsto q^{-1}(\bar{W}) = \{v \in V : v + U \in \bar{W}\}$.

Moreover, this bijection preserves inclusions: $W_1 \subseteq W_2 \Leftrightarrow W_1/U \subseteq W_2/U$.

***Proof:*** We verify the maps are inverses.

**Forward then backward:** Starting with $W \supseteq U$, we get $W/U$, then $q^{-1}(W/U) = \{v \in V : v + U \in W/U\}$. But $v + U \in W/U$ iff $v + U = w + U$ for some $w \in W$, iff $v - w \in U \subseteq W$, iff $v \in W$. So $q^{-1}(W/U) = W$.

**Backward then forward:** Starting with $\bar{W} \subseteq V/U$, let $W = q^{-1}(\bar{W})$. Note $U = q^{-1}(0) \subseteq q^{-1}(\bar{W}) = W$. Then $W/U = q(W) = q(q^{-1}(\bar{W})) = \bar{W}$ (surjectivity of $q$ ensures $q(q^{-1}(S)) = S$ for any $S \subseteq V/U$). $\square$

**Note:-**

Maximal proper subspaces of $V$ containing $U$ correspond to maximal proper subspaces of $V/U$—i.e., hyperplanes (codimension-1 subspaces). This perspective becomes useful in algebraic geometry and representation theory.

## 7.2 Linear Operators and Endomorphisms

When the domain and codomain of a linear map coincide, we gain additional algebraic structure: the ability to compose maps with themselves.

### 7.2.1 The Ring of Endomorphisms

**Definition 7.2.1: Linear operator / Endomorphism**

A **linear operator** or **endomorphism** of a vector space $V$ is a linear map $\varphi : V \to V$. We write

$$\mathrm{End}(V) = \mathrm{Hom}(V, V)$$

for the set of all endomorphisms of $V$.

**Proposition 7.2.1** Ring structure on $\mathrm{End}(V)$

$\mathrm{End}(V)$ is a **ring** (with identity) under:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$.

- **Multiplication:** $(\varphi \circ \psi)(v) = \varphi(\psi(v))$ (composition).
- **Identity:** The identity map $\mathrm{id}_V$.

Moreover, $\mathrm{End}(V)$ is a $k$-algebra: multiplication distributes over scalar multiplication.

***Proof:*** Addition makes $\mathrm{End}(V)$ an abelian group (it's a subgroup of $\mathrm{Hom}(V, V)$ as a vector space). Composition is associative with identity $\mathrm{id}_V$. Distributivity: $\varphi \circ (\psi + \rho) = \varphi \circ \psi + \varphi \circ \rho$ and $(\varphi + \psi) \circ \rho = \varphi \circ \rho + \psi \circ \rho$. These follow from linearity. □

**Note:-**

The ring $\mathrm{End}(V)$ is noncommutative unless $\dim(V) \leqslant 1$. For $V = k^n$, $\mathrm{End}(V) \cong M_n(k)$, the ring of $n \times n$ matrices. The noncommutativity of matrix multiplication reflects the noncommutativity of $\mathrm{End}(V)$.

### 7.2.2 Polynomials of Operators

Since endomorphisms can be composed, we can form "powers" and evaluate polynomials.

**Definition 7.2.2: Powers and polynomials of operators**

For $\varphi \in \mathrm{End}(V)$ and $n \in \mathbb{Z}_{\geqslant 0}$:

$$\varphi^0 = \mathrm{id}_V, \quad \varphi^n = \underbrace{\varphi \circ \varphi \circ \cdots \circ \varphi}_{n \text{ times}}.$$

For a polynomial $p(x) = a_0 + a_1 x + \cdots + a_n x^n \in k[x]$, define

$$p(\varphi) = a_0 \mathrm{id}_V + a_1 \varphi + \cdots + a_n \varphi^n \in \mathrm{End}(V).$$

**Proposition 7.2.2** Evaluation homomorphism

The map $\mathrm{ev}_\varphi : k[x] \to \mathrm{End}(V)$ defined by $p(x) \mapsto p(\varphi)$ is a ring homomorphism. That is:

1. $(p + q)(\varphi) = p(\varphi) + q(\varphi)$.
2. $(pq)(\varphi) = p(\varphi) \circ q(\varphi)$.
3. $1(\varphi) = \mathrm{id}_V$.

***Proof:*** (1) and (3) are immediate. For (2), let $p(x) = \sum a_i x^i$ and $q(x) = \sum b_j x^j$. Then $(pq)(x) = \sum_k c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$. So

$$(pq)(\varphi) = \sum_k c_k \varphi^k = \sum_k \left( \sum_{i+j=k} a_i b_j \right) \varphi^k = \left( \sum_i a_i \varphi^i \right) \circ \left( \sum_j b_j \varphi^j \right) = p(\varphi) \circ q(\varphi).$$

□

**Note:-**

This makes $\mathrm{End}(V)$ a $k[x]$-module: $p(x) \cdot v = p(\varphi)(v)$. The study of this module structure (particularly the kernel of $\mathrm{ev}_\varphi$, called the minimal polynomial) is central to the theory of canonical forms.

## 7.3 Invariant Subspaces

Understanding an operator often begins with finding subspaces on which its behavior is simpler.

**Definition 7.3.1: Invariant subspace**

A subspace $W \subseteq V$ is **invariant** under $\varphi \in \mathrm{End}(V)$ (or **$\varphi$-invariant**) if

$$\varphi(W) \subseteq W,$$

i.e., $\varphi(w) \in W$ for all $w \in W$.

**Example 7.3.1** (Basic examples of invariant subspaces)

For any $\varphi \in \mathrm{End}(V)$:

- $\{0\}$ and $V$ are always $\varphi$-invariant.
- $\ker(\varphi)$ is $\varphi$-invariant: if $\varphi(v) = 0$, then $\varphi(\varphi(v)) = \varphi(0) = 0$.
- $\mathrm{Im}(\varphi)$ is $\varphi$-invariant: $\varphi(\mathrm{Im}(\varphi)) = \mathrm{Im}(\varphi^2) \subseteq \mathrm{Im}(\varphi)$.
- More generally, $\ker(\varphi^n)$ and $\mathrm{Im}(\varphi^n)$ are $\varphi$-invariant for all $n \geqslant 0$.

**Proposition 7.3.1** Restriction to an invariant subspace

If $W \subseteq V$ is $\varphi$-invariant, the **restriction** $\varphi|_W : W \to W$ is a well-defined linear operator on $W$.

### 7.3.1 Block Diagonal Structure

Invariant subspaces lead to simpler matrix representations.

**Theorem 7.3.1** Direct sum of invariant subspaces

Suppose $V = V_1 \oplus V_2$ where both $V_1$ and $V_2$ are $\varphi$-invariant. Choose bases $\mathcal{B}_1 = \{e_1, \dots, e_m\}$ of $V_1$ and $\mathcal{B}_2 = \{e_{m+1}, \dots, e_n\}$ of $V_2$, forming a basis $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ of $V$. Then the matrix of $\varphi$ with respect to $\mathcal{B}$ is **block diagonal**:

$$\mathcal{M}(\varphi; \mathcal{B}) = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

where $A_1 = \mathcal{M}(\varphi|_{V_1}; \mathcal{B}_1)$ and $A_2 = \mathcal{M}(\varphi|_{V_2}; \mathcal{B}_2)$.

***Proof:*** For $i \leqslant m$: $\varphi(e_i) \in V_1$ (by $\varphi$-invariance), so $\varphi(e_i)$ is a linear combination of $e_1, \dots, e_m$ only. For $i > m$: $\varphi(e_i) \in V_2$, so $\varphi(e_i)$ is a linear combination of $e_{m+1}, \dots, e_n$ only. This gives the block structure. $\square$

**Note:-**

Conversely, if a matrix is block diagonal with blocks $A_1$ and $A_2$, then the spans of the corresponding basis vectors are invariant subspaces. The goal of much of spectral theory is to decompose $V$ into a direct sum of invariant subspaces on which $\varphi$ acts simply.

**Corollary 7.3.1** Block triangular from one invariant subspace

If $V_1$ is $\varphi$-invariant (but we don't assume $V_1$ has an invariant complement), then extending a basis of $V_1$ to a basis of $V$ gives a **block upper-triangular** matrix:

$$\mathcal{M}(\varphi) = \begin{pmatrix} A_1 & * \\ 0 & A_2 \end{pmatrix}.$$

The upper-right block $*$ may be nonzero; it measures the "interaction" between $V_1$ and $V/V_1$.

## 7.4 Eigenvalues and Eigenvectors

The simplest invariant subspaces are one-dimensional. These give rise to the fundamental notion of eigenvalues.

### 7.4.1 Definitions and Basic Properties

**Definition 7.4.1: Eigenvector and eigenvalue**

Let $\varphi \in \mathrm{End}(V)$. A nonzero vector $v \in V$ is an **eigenvector** of $\varphi$ if

$$\varphi(v) = \lambda v$$

for some scalar $\lambda \in k$. The scalar $\lambda$ is called the **eigenvalue** corresponding to $v$.

**Note:-**

The condition $\varphi(v) = \lambda v$ says that $\varphi$ acts on the line spanned by $v$ as scaling by $\lambda$. Equivalently, $\mathrm{Span}\{v\}$ is a one-dimensional $\varphi$-invariant subspace. Conversely, any one-dimensional invariant subspace $U = \mathrm{Span}\{v\}$ consists of eigenvectors (with the same eigenvalue) plus zero.

**Proposition 7.4.1** Characterization of eigenvalues

For $\varphi \in \mathrm{End}(V)$ and $\lambda \in k$, the following are equivalent:

1. $\lambda$ is an eigenvalue of $\varphi$.
2. $\varphi - \lambda \mathrm{id}_V$ is not injective.
3. $\ker(\varphi - \lambda \mathrm{id}_V) \neq \{0\}$.
4. (If $V$ is finite-dimensional) $\varphi - \lambda \mathrm{id}_V$ is not surjective.
5. (If $V$ is finite-dimensional) $\det(\varphi - \lambda \mathrm{id}_V) = 0$.

***Proof:*** (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3): $\lambda$ is an eigenvalue iff there exists nonzero $v$ with $\varphi(v) = \lambda v$, iff $(\varphi - \lambda \mathrm{id})(v) = 0$ for some nonzero $v$, iff $\ker(\varphi - \lambda \mathrm{id}) \neq \{0\}$.

(3) $\Leftrightarrow$ (4) $\Leftrightarrow$ (5): In finite dimensions, a linear operator is injective iff surjective iff invertible iff has nonzero determinant. So $\ker(\varphi - \lambda \mathrm{id}) \neq \{0\}$ iff $\det(\varphi - \lambda \mathrm{id}) = 0$. ☐

**Definition 7.4.2: Eigenspace**

For an eigenvalue $\lambda$ of $\varphi$, the **eigenspace** is

$$E(\lambda, \varphi) = \ker(\varphi - \lambda \mathrm{id}_V) = \{v \in V : \varphi(v) = \lambda v\}.$$

This is a subspace of $V$ (it includes 0, though 0 is not an eigenvector).

**Theorem 7.4.1** Eigenvectors with distinct eigenvalues are linearly independent

Let $v_1, \ldots, v_m$ be eigenvectors of $\varphi$ with distinct eigenvalues $\lambda_1, \ldots, \lambda_m$. Then $v_1, \ldots, v_m$ are linearly independent.

***Proof:*** Induction on $m$. Base case $m = 1$: a single eigenvector is nonzero, hence linearly independent.

Suppose the result holds for $m-1$ eigenvectors. Assume $\sum_{i=1}^m c_i v_i = 0$. Apply $\varphi$:

$$\sum_{i=1}^{m} c_i \lambda_i v_i = 0.$$

Multiply the original equation by $\lambda_m$ and subtract:

$$\sum_{i=1}^{m-1} c_i(\lambda_i - \lambda_m) v_i = 0.$$

By the induction hypothesis (since $v_1, \ldots, v_{m-1}$ are eigenvectors with distinct eigenvalues $\lambda_1, \ldots, \lambda_{m-1}$), we have $c_i(\lambda_i - \lambda_m) = 0$ for $i < m$. Since $\lambda_i \neq \lambda_m$, we get $c_i = 0$. Substituting back: $c_m v_m = 0$, so $c_m = 0$ (since $v_m \neq 0$). ☐

**Corollary 7.4.1** Bound on number of eigenvalues

A linear operator on an $n$-dimensional space has at most $n$ distinct eigenvalues.

***Proof:*** If $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues with eigenvectors $v_1, \ldots, v_m$, then $v_1, \ldots, v_m$ are linearly independent. In an $n$-dimensional space, any linearly independent set has at most $n$ elements. ☐

**Corollary 7.4.2** Sum of eigenspaces is direct

If $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of $\varphi$, then

$$E(\lambda_1, \varphi) + \cdots + E(\lambda_m, \varphi) = E(\lambda_1, \varphi) \oplus \cdots \oplus E(\lambda_m, \varphi).$$

is a direct sum. Furthermore, $\sum_i \dim E(\lambda_i, \varphi) \leqslant \dim V$.

### 7.4.2 Diagonalizable Operators

The best-case scenario: $V$ has a basis of eigenvectors.

**Definition 7.4.3: Diagonalizable operator**

An operator $\varphi \in \mathrm{End}(V)$ is **diagonalizable** if there exists a basis of $V$ consisting entirely of eigenvectors of $\varphi$. Equivalently, $\varphi$ has a diagonal matrix with respect to some basis.

**Note:-**

If $\{v_1, \ldots, v_n\}$ is a basis of eigenvectors with $\varphi(v_i) = \lambda_i v_i$, then the matrix of $\varphi$ is

$$\mathcal{M}(\varphi) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

The eigenvalues appear on the diagonal (possibly with repetition).

**Theorem 7.4.2** Characterizations of diagonalizability

For $\varphi \in \mathrm{End}(V)$ with $\dim(V) = n$ and distinct eigenvalues $\lambda_1, \ldots, \lambda_m$, the following are equivalent:

1. $\varphi$ is diagonalizable.
2. $V$ has a basis of eigenvectors of $\varphi$.
3. $V = E(\lambda_1, \varphi) \oplus \cdots \oplus E(\lambda_m, \varphi)$.
4. $\dim E(\lambda_1, \varphi) + \cdots + \dim E(\lambda_m, \varphi) = n$.

***Proof:*** (1) $\Leftrightarrow$ (2): By definition.

(2) $\Rightarrow$ (3): If $V$ has a basis of eigenvectors, partition it by eigenvalue. Each part is a basis of the corresponding eigenspace, and together they span $V$.

(3) $\Rightarrow$ (4): Take dimensions: $\dim V = \dim(E_1 \oplus \cdots \oplus E_m) = \sum \dim E_i$.

(4) $\Rightarrow$ (2): The sum $E_1 + \cdots + E_m$ is direct (earlier corollary), and has dimension $\sum \dim E_i = n = \dim V$. So $E_1 \oplus \cdots \oplus E_m = V$. Combining bases of each $E_i$ gives a basis of $V$ consisting of eigenvectors. $\square$

**Corollary 7.4.3** Enough distinct eigenvalues implies diagonalizability

If $\varphi \in \mathrm{End}(V)$ has $\dim(V)$ distinct eigenvalues, then $\varphi$ is diagonalizable.

***Proof:*** Each eigenspace has dimension at least 1 (it contains an eigenvector). With $n = \dim V$ distinct eigenvalues, $\sum \dim E(\lambda_i) \geqslant n$. But this sum is also $\leqslant n$, so equality holds, and criterion (4) above is satisfied. $\square$

### 7.4.3 Examples: Diagonalizable and Non-Diagonalizable

**Example 7.4.1** (Diagonal matrices)

The identity matrix $I_n$ has eigenvalue 1 with eigenspace all of $k^n$. More generally, a diagonal matrix $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ has the standard basis vectors as eigenvectors with eigenvalues $\lambda_1, \ldots, \lambda_n$. Such a matrix is (trivially) diagonalizable—it's already diagonal!

**Example 7.4.2** (A non-diagonalizable matrix)

Consider $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues satisfy $\det(A - \lambda I) = (1-\lambda)^2 = 0$, so $\lambda = 1$ is the only eigenvalue.

The eigenspace: $E(1, \varphi) = \ker(A - I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \mathrm{Span}\{e_1\}$.

This has dimension 1, but $\dim(\mathbb{R}^2) = 2$. So $\varphi$ is **not diagonalizable**.
Geometrically, $\varphi$ is a "shear": it fixes the $x$-axis pointwise and slides other points parallel to the $x$-axis. The only eigenvector direction is along the $x$-axis.

**Example 7.4.3** (No eigenvectors over $\mathbb{R}$)

Consider rotation by 90° in $\mathbb{R}^2$:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic polynomial is $\det(R - \lambda I) = \lambda^2 + 1$, which has no real roots. So $R$ has no eigenvalues over $\mathbb{R}$, hence no eigenvectors. Over $\mathbb{C}$, the eigenvalues are $\pm i$ with eigenvectors $(1, \mp i)^T$, and $R$ is diagonalizable as a complex matrix. The eigenfairy doesn't visit $\mathbb{R}$ (sad).
This illustrates that diagonalizability depends on the scalar field!

**Note:-**

The rotation example shows that not every operator is diagonalizable. Even over $\mathbb{C}$, where every polynomial has roots, an operator may fail to be diagonalizable (like the shear above). Understanding non-diagonalizable operators requires the theory of Jordan normal form, which we'll develop later.

**Question 1: Exercises on quotient spaces, operators, and eigenvalues**

1. Let $V = \mathbb{R}^4$ and $U = \{(x, y, z, w) : x + y = 0, z = w\}$. Find $\dim(U)$, $\dim(V/U)$, and describe $V/U$ explicitly.
2. Prove that if $\varphi, \psi \in \mathrm{End}(V)$ commute ($\varphi \circ \psi = \psi \circ \varphi$), then every eigenspace of $\varphi$ is $\psi$-invariant.
3. Let $\varphi \in \mathrm{End}(V)$ be diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_m$. Prove that $p(\varphi) = 0$ where $p(x) = \prod_{i=1}^{m}(x - \lambda_i)$.
4. Show that the quotient map $q : V \to V/U$ induces a bijection between {linear maps $V/U \to W$} and {linear maps $V \to W$ vanishing on $U$}.
5. Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Find the eigenvalues and eigenspaces of $A$. Is $A$ diagonalizable? If so, find a matrix $P$ such that $P^{-1}AP$ is diagonal.

**Solution**

**(1)** The constraints $x + y = 0$ and $z = w$ give 2 independent linear equations on $\mathbb{R}^4$, so $\dim(U) = 4 - 2 = 2$. A basis: $\{(-1, 1, 0, 0), (0, 0, 1, 1)\}$.
By the dimension formula: $\dim(V/U) = \dim(V) - \dim(U) = 4 - 2 = 2$.
The map $\pi : \mathbb{R}^4 \to \mathbb{R}^2$ defined by $\pi(x, y, z, w) = (x + y, z - w)$ has $\ker(\pi) = U$. By the first isomorphism theorem, $V/U \cong \mathbb{R}^2$. Explicitly, the coset $(x, y, z, w) + U$ is determined by $(x + y, z - w)$.
**(2)** Let $v$ be an eigenvector of $\varphi$ with eigenvalue $\lambda$: $\varphi(v) = \lambda v$. We need $\psi(v) \in E(\lambda, \varphi)$, i.e., $\varphi(\psi(v)) = \lambda\psi(v)$.
Compute: $\varphi(\psi(v)) = \psi(\varphi(v)) = \psi(\lambda v) = \lambda\psi(v)$. ✓
(We used commutativity in the first equality.)
**(3)** Let $p(x) = \prod_{i=1}^{m}(x - \lambda_i)$. Since $\varphi$ is diagonalizable, $V = \bigoplus_{i=1}^{m} E(\lambda_i, \varphi)$. For any eigenvector $v \in E(\lambda_j, \varphi)$:

$$(\varphi - \lambda_j \mathrm{id})(v) = 0.$$

So $p(\varphi)(v) = (\varphi - \lambda_1) \cdots (\varphi - \lambda_m)(v) = 0$ because one factor gives zero.
Since this holds for all eigenvectors and they span $V$, we have $p(\varphi) = 0$.
**(4)** Define $\Phi : \mathrm{Hom}(V/U, W) \to \{T \in \mathrm{Hom}(V, W) : U \subseteq \ker(T)\}$ by $\Phi(\bar{T}) = \bar{T} \circ q$.
Injective: If $\bar{T} \circ q = 0$, then $\bar{T}(v + U) = \bar{T}(q(v)) = 0$ for all $v$, so $\bar{T} = 0$.
Surjective: Given $T : V \to W$ with $U \subseteq \ker(T)$, define $\bar{T} : V/U \to W$ by $\bar{T}(v + U) = T(v)$. Well-defined because $v - v' \in U \Rightarrow T(v) = T(v')$. Then $\Phi(\bar{T}) = \bar{T} \circ q = T$.
**(5)** Characteristic polynomial: $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$.
Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.
Eigenspaces:

- $\lambda = 5$: $(A - 5I)v = 0$ gives $\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = 2y$. Eigenspace: $\mathrm{Span}\{(2, 1)^T\}$.
- $\lambda = 2$: $(A - 2I)v = 0$ gives $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = -y$. Eigenspace: $\mathrm{Span}\{(1, -1)^T\}$.

Since we have 2 linearly independent eigenvectors in $\mathbb{R}^2$, $A$ is diagonalizable.
Let $P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ (columns are eigenvectors). Then $P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$.

## 7.5 Let's get general up in here!

### 7.5.1 Generalized Kernals (Null Spaces)

We want to look at general eigenvectors: things that are not *actually* eigenvectors but "kinda want to be" in a generalized sense. What in tarnation does that mean!? Simplest Eigenspace is just 0, so let's focus on that to get an understanding:

**Definition 7.5.1: Generalized kernal**

The generalized kernal of $\varphi$ is $\text{gker}(\varphi) = \{v \in V \mid \exists m > 0 \text{ s.t. } \varphi^m(v) = 0\}$. These are all the vecrtors that are evnetually sent to 0 by repeatedly applying $\varphi$.

**Note:-**

We note that $\text{gker}(\varphi)$ is not standard notation. Auroux likes it though, so that is what we are going with! *If you meet a stranger on the street and bring up gker be sure to tell them what you mean, lest they be confused.*

The difference between the kernal and generalized kernal can be summed up by this: you are in the kernal if you go to 0 in one step; if you apply itterative steps—$\varphi^m$—and end up at 0 you are in the generalized kernal. If I look at the kernal of $\varphi$ I notice that it is in the kernal of $\varphi^2$; then too this holds for more! They may or may not get larger but they are subsets:

$$\text{Ker}(\varphi) \subset \text{Ker}(\varphi^2) \subset \text{Ker}\,\varphi \subset \dots$$

**Proposition 7.5.1**

If $\text{Ker}(\varphi^m) = \text{Ker}(\varphi^{m+1})$ then there is no strictly larger kernal.

***Proof:*** $\varphi^{m+2}(v) = \varphi^{m+1}(\varphi(V)) = 0 \iff \varphi(v) = \text{Ker}\,\varphi^m \iff \varphi^m(\varphi(v)) = \varphi^{m+1}(v) = 0$. So $\text{Ker}(\varphi^{m+1}) = \text{Ker}(\varphi^{m+2})$. Since the sequence stops increasing after at most $n = \dim V$ steps, $\text{gker}(\varphi) = \text{Ker}\,\varphi^n$. $\square$

**Example 7.5.1**

$\varphi : k^2 \to k^2$ with $e_1 \mapsto 0$, $e_2 \mapsto e_1$ and $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, then $\text{Ker}(\varphi) = k \cdot e_1$, but $\text{Ker}(\varphi^2) = \text{gker}(\varphi) = k^2$.

**Note:-**

The generalized kernal is often referred to as the null space (this is the terminology that Axler uses, for example). I may use this interchangeably.

Since it is not the case that $V = \text{gker}(\varphi) \oplus \text{Img}(\varphi)$ for every $\varphi \in \text{Hom}(V, V)$, we can use the following proposition as a substitute:

**Proposition 7.5.2** Generalized kernel-image decomposition

Suppose $\varphi \in \text{End}(V)$ and let $m$ be the smallest positive integer such that $\text{Ker}(\varphi^m) = \text{Ker}(\varphi^{m+1})$. Then

$$V = \text{Ker}(\varphi^m) \oplus \text{Im}(\varphi^m).$$

In particular, $V = \text{gker}(\varphi) \oplus \text{Im}(\varphi^{\dim V})$.

***Proof:*** We verify the two conditions for a direct sum decomposition.

**Intersection is trivial:** Suppose $v \in \text{Ker}(\varphi^m) \cap \text{Im}(\varphi^m)$. Write $v = \varphi^m(u)$ for some $u \in V$. Then $\varphi^m(v) = \varphi^{2m}(u) = 0$, so $u \in \text{Ker}(\varphi^{2m})$.

We claim $\mathrm{Ker}(\varphi^{2m}) = \mathrm{Ker}(\varphi^m)$. By the stabilization property, once the kernel sequence stabilizes at step $m$, it stays constant: $\mathrm{Ker}(\varphi^m) = \mathrm{Ker}(\varphi^{m+1}) = \cdots = \mathrm{Ker}(\varphi^{2m})$. Thus $u \in \mathrm{Ker}(\varphi^m)$, which means $v = \varphi^m(u) = 0$.

**Sum is all of $V$:** By rank-nullity applied to $\varphi^m$:

$$\dim V = \dim \mathrm{Ker}(\varphi^m) + \dim \mathrm{Im}(\varphi^m).$$

Since we just showed $\mathrm{Ker}(\varphi^m) \cap \mathrm{Im}(\varphi^m) = \{0\}$, the sum $\mathrm{Ker}(\varphi^m) + \mathrm{Im}(\varphi^m)$ is direct, and its dimension equals $\dim \mathrm{Ker}(\varphi^m) + \dim \mathrm{Im}(\varphi^m) = \dim V$. Hence it equals $V$. $\square$

**Note:-**

This decomposition is the key structural result for understanding non-diagonalizable operators. The generalized kernel $\mathrm{gker}(\varphi)$ captures the "nilpotent part" of $\varphi$, while $\mathrm{Im}(\varphi^m)$ is where $\varphi$ acts invertibly. On $\mathrm{gker}(\varphi)$, repeated application of $\varphi$ eventually kills everything; on $\mathrm{Im}(\varphi^m)$, the restriction of $\varphi$ is an isomorphism.

**Corollary 7.5.1** Restriction to image is an isomorphism

With notation as above, the restriction $\varphi|_{\mathrm{Im}(\varphi^m)} : \mathrm{Im}(\varphi^m) \to \mathrm{Im}(\varphi^m)$ is an isomorphism.

***Proof:*** First, $\mathrm{Im}(\varphi^m)$ is $\varphi$-invariant since $\varphi(\mathrm{Im}(\varphi^m)) = \mathrm{Im}(\varphi^{m+1}) \subseteq \mathrm{Im}(\varphi^m)$. For injectivity: if $v \in \mathrm{Im}(\varphi^m)$ and $\varphi(v) = 0$, then $v \in \mathrm{Ker}(\varphi) \subseteq \mathrm{Ker}(\varphi^m)$. But $\mathrm{Ker}(\varphi^m) \cap \mathrm{Im}(\varphi^m) = \{0\}$, so $v = 0$. Since $\varphi|_{\mathrm{Im}(\varphi^m)}$ is injective on a finite-dimensional space, it's also surjective onto its image, which equals $\mathrm{Im}(\varphi^m)$ by $\varphi$-invariance. $\square$

### 7.5.2 Generalized Eigenspaces

Now we extend the idea of generalized kernels to arbitrary eigenvalues, not just $\lambda = 0$.

**Definition 7.5.2: Generalized eigenspace**

Let $\varphi \in \mathrm{End}(V)$ and let $\lambda$ be an eigenvalue of $\varphi$. The **generalized eigenspace** of $\varphi$ corresponding to $\lambda$ is

$$V_\lambda = \mathrm{Ker}(\varphi - \lambda \mathrm{id})^{\dim V} = \{v \in V : (\varphi - \lambda \mathrm{id})^n(v) = 0 \text{ for some } n \geqslant 1\}.$$

Equivalently, $V_\lambda = \mathrm{gker}(\varphi - \lambda \mathrm{id})$.

**Note:-**

The ordinary eigenspace $E(\lambda, \varphi) = \mathrm{Ker}(\varphi - \lambda \mathrm{id})$ is always contained in $V_\lambda$, but $V_\lambda$ may be strictly larger. The elements of $V_\lambda \setminus E(\lambda, \varphi)$ are called **generalized eigenvectors**—they are not true eigenvectors, but they are "eventually killed" by $\varphi - \lambda \mathrm{id}$.

**Example 7.5.2** (Generalized eigenspace of the shear)

Recall the shear $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The only eigenvalue is $\lambda = 1$, and the eigenspace is $E(1, \varphi) = \mathrm{Ker}(A - I) = \mathrm{Ker} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \mathrm{Span}\{e_1\}$, which is one-dimensional.

However, $(A - I)^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$.

So the generalized eigenspace is $V_1 = \mathrm{Ker}(A - I)^2 = \mathbb{R}^2$. The vector $e_2$ is a generalized eigenvector: $(A - I)e_2 = e_1 \neq 0$, but $(A - I)^2 e_2 = 0$.

**Proposition 7.5.3** Basic properties of generalized eigenspaces

Let $\varphi \in \mathrm{End}(V)$ with eigenvalue $\lambda$. Then:

1. $E(\lambda, \varphi) \subseteq V_\lambda$.
2. $V_\lambda$ is $\varphi$-invariant.
3. On $V_\lambda$, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent.
4. If $\mu \neq \lambda$ are distinct eigenvalues, then $V_\lambda \cap V_\mu = \{0\}$.

***Proof:*** (1) is immediate: if $(\varphi - \lambda\mathrm{id})(v) = 0$, then certainly $(\varphi - \lambda\mathrm{id})^n(v) = 0$ for all $n \geqslant 1$.

(2) Let $v \in V_\lambda$, so $(\varphi - \lambda\mathrm{id})^n(v) = 0$ for some $n$. Since $\varphi$ commutes with $\varphi - \lambda\mathrm{id}$, we have

$$(\varphi - \lambda\mathrm{id})^n(\varphi(v)) = \varphi((\varphi - \lambda\mathrm{id})^n(v)) = \varphi(0) = 0.$$

Thus $\varphi(v) \in V_\lambda$.

(3) By definition, $V_\lambda = \mathrm{Ker}(\varphi - \lambda\mathrm{id})^m$ for some $m \leqslant \dim V$. So $(\varphi - \lambda\mathrm{id})^m$ restricted to $V_\lambda$ is the zero map.

(4) Suppose $v \in V_\lambda \cap V_\mu$ with $\lambda \neq \mu$. Then there exist $j, k \geqslant 1$ such that $(\varphi - \lambda\mathrm{id})^j(v) = 0$ and $(\varphi - \mu\mathrm{id})^k(v) = 0$.

The polynomials $p(x) = (x - \lambda)^j$ and $q(x) = (x - \mu)^k$ are coprime (they share no common roots). By Bézout's identity, there exist polynomials $a(x), b(x) \in k[x]$ such that $a(x)p(x) + b(x)q(x) = 1$.

Evaluating at $\varphi$: $a(\varphi)p(\varphi) + b(\varphi)q(\varphi) = \mathrm{id}$. Applying to $v$:

$$v = a(\varphi)p(\varphi)(v) + b(\varphi)q(\varphi)(v) = a(\varphi)(0) + b(\varphi)(0) = 0.$$

□

**Note:-**

Part (4) shows that generalized eigenspaces for distinct eigenvalues intersect trivially, just like ordinary eigenspaces. The proof uses the coprimality of $(x - \lambda)^j$ and $(x - \mu)^k$—this is a technique that will reappear in the generalized eigenspace decomposition.

### 7.5.3 The Generalized Eigenspace Decomposition

The following theorem is the main structural result for operators on finite-dimensional vector spaces over algebraically closed fields.

**Theorem 7.5.1** Generalized eigenspace decomposition

Let $V$ be a finite-dimensional vector space over an algebraically closed field $k$, and let $\varphi \in \mathrm{End}(V)$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $\varphi$. Then

$$V = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_m}.$$

Moreover, $\sum_{i=1}^m \dim V_{\lambda_i} = \dim V$.

***Proof:*** We proceed in two steps.

**Step 1: The sum is direct.** We already showed that $V_{\lambda_i} \cap V_{\lambda_j} = \{0\}$ for $i \neq j$. More generally, suppose $v_1 + \cdots + v_m = 0$ where $v_i \in V_{\lambda_i}$. We show each $v_i = 0$.

For each $i$, let $n_i$ be such that $(\varphi - \lambda_i\mathrm{id})^{n_i}(v_i) = 0$. Define $p_i(x) = (x - \lambda_i)^{n_i}$ and

$$q_i(x) = \prod_{j \neq i} (x - \lambda_j)^{n_j}.$$

Since $\lambda_i$ is not a root of $q_i(x)$, the polynomials $p_i(x)$ and $q_i(x)$ are coprime. By Bézout, there exist $a_i(x), b_i(x)$ with $a_i(x)p_i(x) + b_i(x)q_i(x) = 1$.

Now, $q_i(\varphi)(v_j) = 0$ for all $j \neq i$ (since $p_j$ divides $q_i$). Applying $q_i(\varphi)$ to $v_1 + \cdots + v_m = 0$:

$$0 = q_i(\varphi)(v_1 + \cdots + v_m) = q_i(\varphi)(v_i).$$

Thus $v_i \in \mathrm{Ker}(q_i(\varphi)) \cap \mathrm{Ker}(p_i(\varphi)) = V_{\lambda_i} \cap \mathrm{Ker}(q_i(\varphi))$.

But from $a_i(\varphi)p_i(\varphi)(v_i) + b_i(\varphi)q_i(\varphi)(v_i) = v_i$ and both terms being zero, we get $v_i = 0$.

**Step 2: The sum equals $V$.** Define $p(x) = \prod_{i=1}^m (x - \lambda_i)^{n_i}$ where $n_i = \dim V_{\lambda_i}$ (we'll justify this choice shortly). We claim $p(\varphi) = 0$, i.e., $p$ annihilates $\varphi$.

Consider the polynomial $q_i(x) = p(x)/(x-\lambda_i)^{n_i}$. These polynomials are pairwise coprime (each $q_i$ excludes exactly the factor $(x - \lambda_i)^{n_i}$). Their gcd is 1, so there exist $r_i(x)$ with $\sum_i r_i(x)q_i(x) = 1$.

For any $v \in V$, write $v = \sum_i r_i(\varphi)q_i(\varphi)(v)$. We claim $r_i(\varphi)q_i(\varphi)(v) \in V_{\lambda_i}$. Indeed:

$$p_i(\varphi)(r_i(\varphi)q_i(\varphi)(v)) = r_i(\varphi)(p_i(\varphi)q_i(\varphi))(v) = r_i(\varphi)p(\varphi)(v).$$

Once we establish that $p(\varphi) = 0$ (which follows from Cayley-Hamilton, proved later, or can be verified by dimension counting), this shows $v$ is a sum of elements from the $V_{\lambda_i}$.

**Alternative dimension argument:** Since $k$ is algebraically closed, $\varphi$ has at least one eigenvalue. Apply the decomposition $V = \mathrm{gker}(\varphi - \lambda_1 \mathrm{id}) \oplus \mathrm{Im}((\varphi - \lambda_1 \mathrm{id})^{n_1})$ where $n_1$ is the stabilization index. The first summand is $V_{\lambda_1}$. On the second summand, $\varphi - \lambda_1\mathrm{id}$ is invertible, so $\lambda_1$ is not an eigenvalue of $\varphi$ restricted there. Proceed by induction on $\dim V$. □

**Corollary 7.5.2** Dimension of generalized eigenspaces

With notation as above, if $\chi_\varphi(x) = \prod_{i=1}^m (x-\lambda_i)^{n_i}$ is the characteristic polynomial, then $\dim V_{\lambda_i} = n_i$.

**Note:-**

The exponent $n_i$ in the characteristic polynomial is called the **algebraic multiplicity** of $\lambda_i$. The dimension of the ordinary eigenspace $\dim E(\lambda_i, \varphi)$ is called the **geometric multiplicity**. We always have:

$$1 \leqslant \dim E(\lambda_i, \varphi) \leqslant \dim V_{\lambda_i} = n_i.$$

The operator $\varphi$ is diagonalizable if and only if geometric multiplicity equals algebraic multiplicity for every eigenvalue.

**Example 7.5.3** (Decomposition for a $3 \times 3$ matrix)

Consider $\varphi : \mathbb{C}^3 \to \mathbb{C}^3$ with matrix

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

The characteristic polynomial is $\chi_A(x) = (x-2)^2(x-3)$, so $\lambda_1 = 2$ has algebraic multiplicity 2 and $\lambda_2 = 3$ has algebraic multiplicity 1.

For $\lambda_1 = 2$:

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (A-2I)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

So $V_2 = \mathrm{Ker}(A - 2I)^2 = \mathrm{Span}\{e_1, e_2\}$, which has dimension 2.

For $\lambda_2 = 3$: $V_3 = \mathrm{Ker}(A - 3I) = \mathrm{Span}\{e_3\}$, dimension 1.

Indeed, $\mathbb{C}^3 = V_2 \oplus V_3$.

## 7.6 Nilpotent Operators

Nilpotent operators are the building blocks for understanding non-diagonalizable maps. On each generalized eigenspace $V_\lambda$, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent, so understanding nilpotent operators is the key to understanding all operators.

### 7.6.1 Definition and Basic Properties

**Definition 7.6.1: Nilpotent operator**

An operator $\varphi \in \mathrm{End}(V)$ is **nilpotent** if $\varphi^m = 0$ for some positive integer $m$. The smallest such $m$ is called the **index of nilpotency** (or **nilpotence index**) of $\varphi$.

**Example 7.6.1** (Basic examples of nilpotent operators)

1. The zero operator is nilpotent with index 1.
2. The shift operator on $k^n$ with matrix $\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$ is nilpotent with index $n$.
3. Any strictly upper-triangular matrix (zeros on and below the diagonal) is nilpotent.

**Proposition 7.6.1** Characterizations of nilpotent operators

For $\varphi \in \mathrm{End}(V)$ with $\dim V = n$, the following are equivalent:

1. $\varphi$ is nilpotent.
2. $\varphi^n = 0$.
3. The only eigenvalue of $\varphi$ is 0 (if $k$ is algebraically closed).
4. $V = \mathrm{gker}(\varphi)$.

***Proof:*** (1) $\Rightarrow$ (2): If $\varphi^m = 0$, then the chain $\{0\} \subseteq \mathrm{Ker}(\varphi) \subseteq \mathrm{Ker}(\varphi^2) \subseteq \cdots$ stabilizes by step $m$, hence by step $n$ (since dimensions are bounded by $n$). So $\mathrm{Ker}(\varphi^n) = \mathrm{gker}(\varphi)$. But $\varphi^m = 0$ means $\mathrm{Ker}(\varphi^m) = V$, so $\mathrm{gker}(\varphi) = V$, which means $\varphi^n = 0$.

(2) $\Rightarrow$ (3): Suppose $\lambda$ is an eigenvalue with eigenvector $v \neq 0$. Then $\varphi(v) = \lambda v$, so $\varphi^n(v) = \lambda^n v$. If $\varphi^n = 0$, then $\lambda^n v = 0$, so $\lambda^n = 0$, hence $\lambda = 0$.

(3) $\Rightarrow$ (4): Over an algebraically closed field, by the generalized eigenspace decomposition, $V = \bigoplus_\lambda V_\lambda$. If the only eigenvalue is 0, then $V = V_0 = \mathrm{gker}(\varphi)$.

(4) $\Rightarrow$ (1): If $V = \mathrm{gker}(\varphi) = \mathrm{Ker}(\varphi^n)$, then $\varphi^n = 0$. $\square$

**Note:-**

Over a non-algebraically closed field, (3) needs modification: $\varphi$ is nilpotent iff $\varphi$ has no eigenvalues in any field extension, or equivalently, iff the characteristic polynomial is $x^n$.

**Corollary 7.6.1** Nilpotent matrices are similar to strictly upper-triangular matrices

If $\varphi$ is nilpotent, then there exists a basis in which the matrix of $\varphi$ is strictly upper-triangular.

***Proof:*** We prove this by induction on $n = \dim V$. If $n = 1$, then $\varphi = 0$ and we're done.

For $n > 1$: since $\varphi$ is nilpotent and nonzero (otherwise trivial), $\mathrm{Ker}(\varphi) \neq \{0\}$. Pick $0 \neq v_1 \in \mathrm{Ker}(\varphi)$ and extend to a basis $\{v_1, v_2, \ldots, v_n\}$. The matrix of $\varphi$ has first column all zeros.

Alternatively, consider $W = \mathrm{Im}(\varphi)$. Since $\varphi$ is nilpotent and nonzero, $W \subsetneq V$ (otherwise $\varphi$ would be surjective, hence bijective, contradicting nilpotency). The restriction $\varphi|_W : W \to W$ is well-defined ($W$ is

$\varphi$-invariant) and nilpotent. By induction, choose a basis of $W$ making $\varphi|_W$ strictly upper-triangular, extend to a basis of $V$, and the result follows. $\square$

### 7.6.2 The Nilpotent Basis Construction

The key to understanding nilpotent operators is building a basis adapted to the "chains" of vectors connected by $\varphi$.

**Definition 7.6.2: Chains and Jordan chains**

Let $\varphi \in \mathrm{End}(V)$ be nilpotent. A **chain** (or **Jordan chain**) of length $m$ is a sequence of vectors $v_1, v_2, \ldots, v_m$ such that:

$$\varphi(v_m) = v_{m-1}, \quad \varphi(v_{m-1}) = v_{m-2}, \quad \ldots, \quad \varphi(v_2) = v_1, \quad \varphi(v_1) = 0.$$

The vector $v_m$ is called the **head** of the chain, and $v_1$ is the **tail**.

**Note:-**

In a chain, $v_1 \in \mathrm{Ker}(\varphi)$, $v_2 \in \mathrm{Ker}(\varphi^2) \setminus \mathrm{Ker}(\varphi)$, and generally $v_j \in \mathrm{Ker}(\varphi^j) \setminus \mathrm{Ker}(\varphi^{j-1})$. The chain encodes how $\varphi$ "moves" vectors toward the kernel.

**Lemma 7.6.1** Key lemma for chain construction

Let $\varphi$ be nilpotent. Suppose $U \subseteq \mathrm{Ker}(\varphi^{k+1})$ is a subspace satisfying $U \cap \mathrm{Ker}(\varphi^k) = \{0\}$. Then:

1. $\varphi|_U$ is injective.
2. $\varphi(U) \subseteq \mathrm{Ker}(\varphi^k)$.
3. $\varphi(U) \cap \mathrm{Ker}(\varphi^{k-1}) = \{0\}$ (if $k \geqslant 1$).

***Proof:*** (1) If $v \in U$ and $\varphi(v) = 0$, then $v \in U \cap \mathrm{Ker}(\varphi) \subseteq U \cap \mathrm{Ker}(\varphi^k) = \{0\}$.

(2) For $v \in U \subseteq \mathrm{Ker}(\varphi^{k+1})$, we have $\varphi^{k+1}(v) = 0$, so $\varphi^k(\varphi(v)) = 0$, meaning $\varphi(v) \in \mathrm{Ker}(\varphi^k)$.

(3) If $\varphi(v) \in \mathrm{Ker}(\varphi^{k-1})$, then $\varphi^k(v) = \varphi^{k-1}(\varphi(v)) = 0$, so $v \in \mathrm{Ker}(\varphi^k)$. By assumption $U \cap \mathrm{Ker}(\varphi^k) = \{0\}$, so $v = 0$, hence $\varphi(v) = 0$. $\square$

**Theorem 7.6.1** Structure theorem for nilpotent operators

Let $\varphi \in \mathrm{End}(V)$ be nilpotent with index $m$ (so $\varphi^m = 0$ but $\varphi^{m-1} \neq 0$). Then there exists a basis of $V$ of the form

$$\{v_{1,1}, \ldots, v_{1,m_1};\ v_{2,1}, \ldots, v_{2,m_2};\ \ldots;\ v_{\ell,1}, \ldots, v_{\ell,m_\ell}\}$$

where each sequence $(v_{i,1}, \ldots, v_{i,m_i})$ is a chain:

$$\varphi(v_{i,j}) = v_{i,j-1} \text{ for } j > 1, \quad \varphi(v_{i,1}) = 0.$$

The chain lengths satisfy $m = m_1 \geqslant m_2 \geqslant \cdots \geqslant m_\ell \geqslant 1$ and $\sum_{i=1}^{\ell} m_i = \dim V$.

***Proof:*** We construct the basis by working down from $\mathrm{Ker}(\varphi^m) = V$.

**Step 1:** Let $U_m$ be a complement of $\mathrm{Ker}(\varphi^{m-1})$ in $V = \mathrm{Ker}(\varphi^m)$. Pick a basis $\{v_{1,m}, \ldots, v_{k_m,m}\}$ of $U_m$. These will be the heads of chains of length $m$.

By the lemma, $\varphi(U_m) \subseteq \mathrm{Ker}(\varphi^{m-1})$ and $\varphi(U_m) \cap \mathrm{Ker}(\varphi^{m-2}) = \{0\}$. Define $v_{i,m-1} = \varphi(v_{i,m})$ for each $i$. These vectors are linearly independent (since $\varphi|_{U_m}$ is injective).

**Step 2:** Now work in $\mathrm{Ker}(\varphi^{m-1})$. We have $\varphi(U_m) \subseteq \mathrm{Ker}(\varphi^{m-1})$ with $\varphi(U_m) \cap \mathrm{Ker}(\varphi^{m-2}) = \{0\}$. Extend $\varphi(U_m)$ to a larger subspace $U_{m-1}$ that complements $\mathrm{Ker}(\varphi^{m-2})$ in $\mathrm{Ker}(\varphi^{m-1})$. Pick additional basis vectors $\{v_{k_m+1,m-1}, \ldots, v_{k_{m-1},m-1}\}$ to span the complement of $\varphi(U_m)$ in $U_{m-1}$. These start new chains of length $m-1$.

**Continue inductively:** At each stage, we have chains of various lengths. Applying $\varphi$ extends existing chains, and we add new chain heads as needed to complement $\mathrm{Ker}(\varphi^{j-1})$ in $\mathrm{Ker}(\varphi^j)$.

**Termination:** At the final step, we're in $\mathrm{Ker}(\varphi)$, and all chains terminate. The collection of all chain vectors forms a basis of $V$. $\square$

**Note:-**

The chain lengths $(m_1, m_2, \ldots, m_\ell)$ form a partition of $n = \dim V$. This partition is an invariant of the nilpotent operator—two nilpotent operators are similar iff they have the same partition.

### 7.6.3 Jordan Blocks for Nilpotent Operators

The chain basis gives the matrix of a nilpotent operator a very specific form.

**Definition 7.6.3: Nilpotent Jordan block**

A **nilpotent Jordan block** of size $m$ is the $m \times m$ matrix

$$J_m(0) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

This has 1's on the superdiagonal and 0's elsewhere.

**Corollary 7.6.2** Matrix form of nilpotent operators

With respect to a chain basis ordered as $(v_{1,1}, \ldots, v_{1,m_1}, v_{2,1}, \ldots, v_{2,m_2}, \ldots)$, the matrix of a nilpotent operator $\varphi$ is block diagonal:

$$\mathcal{M}(\varphi) = \begin{pmatrix} J_{m_1}(0) & & 0 \\ & J_{m_2}(0) & \\ 0 & & \ddots \end{pmatrix}.$$

***Proof:*** Within each chain $(v_{i,1}, \ldots, v_{i,m_i})$, we have $\varphi(v_{i,j}) = v_{i,j-1}$ for $j > 1$ and $\varphi(v_{i,1}) = 0$. In matrix form with this basis ordering, this gives exactly $J_{m_i}(0)$ for the $i$-th block. The blocks are independent since chains span complementary subspaces. $\square$

**Example 7.6.2** (Nilpotent operator with partition $(3, 2, 1)$)

Consider $\varphi : k^6 \to k^6$ nilpotent with chain lengths $3, 2, 1$. A chain basis might be:

$$\{v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{3,1}\}$$

where $\varphi(v_{1,3}) = v_{1,2}$, $\varphi(v_{1,2}) = v_{1,1}$, $\varphi(v_{1,1}) = 0$, etc.
The matrix is:

$$\mathcal{M}(\varphi) = \begin{pmatrix} 0 & 1 & 0 & & & \\ 0 & 0 & 1 & & 0 & \\ 0 & 0 & 0 & & & \\ & & & 0 & 1 & \\ & 0 & & 0 & 0 & \\ & & & & & 0 \end{pmatrix} = J_3(0) \oplus J_2(0) \oplus J_1(0).$$

**Proposition 7.6.2** Uniqueness of the Jordan form for nilpotent operators

The partition $(m_1, m_2, \ldots, m_\ell)$ associated to a nilpotent operator $\varphi$ is uniquely determined by $\varphi$. Equivalently, two nilpotent operators are similar iff they have the same Jordan block sizes (counting multiplicity).

***Proof:*** The partition can be recovered from dimensions of iterated kernels. Let $d_j = \dim \operatorname{Ker}(\varphi^j)$. Then:

$$d_1 - d_0 = d_1 = \text{number of chains} = \ell$$

$$d_2 - d_1 = \text{number of chains of length} \geqslant 2$$

$$d_j - d_{j-1} = \text{number of chains of length} \geqslant j.$$

From this sequence of differences, we can reconstruct the partition: the number of blocks of size exactly $j$ is $(d_j - d_{j-1}) - (d_{j+1} - d_j)$. Since the $d_j$ are determined by $\varphi$, so is the partition. □

**Note:-**

This dimension-counting argument is the key to proving uniqueness of Jordan form in general. The rank of $(\varphi - \lambda I)^j$ determines how many Jordan blocks for $\lambda$ have size $\geqslant j$.

## 7.7 Jordan Normal Form

We now combine the generalized eigenspace decomposition with the structure theory of nilpotent operators to obtain the Jordan normal form—the "best possible" matrix representation for an operator over an algebraically closed field.

### 7.7.1 Jordan Blocks

**Definition 7.7.1: Jordan block**

A **Jordan block** of size $m$ with eigenvalue $\lambda$ is the $m \times m$ matrix

$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix} = \lambda I_m + J_m(0).$$

This has $\lambda$ on the diagonal, 1's on the superdiagonal, and 0's elsewhere.

**Example 7.7.1** (Small Jordan blocks)

$$J_1(\lambda) = (\lambda), \quad J_2(\lambda) = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}, \quad J_3(\lambda) = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}.$$

A $1 \times 1$ Jordan block is just a scalar—this corresponds to a true eigenvector. Larger blocks correspond to generalized eigenvectors that are not true eigenvectors.

**Proposition 7.7.1** Properties of Jordan blocks

Let $J = J_m(\lambda)$. Then:

1. The only eigenvalue of $J$ is $\lambda$, with algebraic multiplicity $m$.

2. The eigenspace $E(\lambda, J) = \mathrm{Ker}(J - \lambda I) = \mathrm{Span}\{e_1\}$ has dimension 1.

3. The generalized eigenspace is all of $k^m$.

4. $J - \lambda I = J_m(0)$ is nilpotent with index $m$.

***Proof:*** (1) The characteristic polynomial is $\det(J - xI) = (\lambda - x)^m$.

(2) $J - \lambda I = J_m(0)$ has kernel spanned by $e_1$ (the first standard basis vector).

(3) Since $(J - \lambda I)^m = J_m(0)^m = 0$, every vector is in $\mathrm{Ker}(J - \lambda I)^m$.

(4) $J_m(0)^{m-1} \neq 0$ (it maps $e_m \mapsto e_1$) but $J_m(0)^m = 0$. □

**Note:-**

A Jordan block $J_m(\lambda)$ represents the action of an operator on a single chain of generalized eigenvectors. The chain is $(e_1, e_2, \ldots, e_m)$ where $Je_j = \lambda e_j + e_{j-1}$ for $j > 1$ and $Je_1 = \lambda e_1$. Equivalently, $(J - \lambda I)e_j = e_{j-1}$ for $j > 1$ and $(J - \lambda I)e_1 = 0$.

### 7.7.2 The Jordan Normal Form Theorem

**Theorem 7.7.1** Jordan normal form

Let $V$ be a finite-dimensional vector space over an algebraically closed field $k$, and let $\varphi \in \mathrm{End}(V)$. Then there exists a basis of $V$ with respect to which the matrix of $\varphi$ is block diagonal:

$$\mathcal{M}(\varphi) = \begin{pmatrix} J_{m_1}(\lambda_1) & & 0 \\ & J_{m_2}(\lambda_2) & \\ 0 & & \ddots \end{pmatrix}$$

where each block is a Jordan block. This matrix is called the **Jordan normal form** (or **Jordan canonical form**) of $\varphi$.

***Proof:*** By the generalized eigenspace decomposition, $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_r}$ where $\lambda_1, \ldots, \lambda_r$ are the distinct eigenvalues. Each $V_{\lambda_i}$ is $\varphi$-invariant, so we can work on each generalized eigenspace separately.

On $V_{\lambda_i}$, the operator $\psi_i = \varphi|_{V_{\lambda_i}} - \lambda_i \mathrm{id}$ is nilpotent (this was proved earlier). By the structure theorem for nilpotent operators, there exists a chain basis of $V_{\lambda_i}$ such that $\psi_i$ has matrix $J_{m_1}(0) \oplus J_{m_2}(0) \oplus \cdots$.

Since $\varphi|_{V_{\lambda_i}} = \psi_i + \lambda_i \mathrm{id}$, the matrix of $\varphi|_{V_{\lambda_i}}$ in this basis is

$$J_{m_1}(0) + \lambda_i I \oplus J_{m_2}(0) + \lambda_i I \oplus \cdots = J_{m_1}(\lambda_i) \oplus J_{m_2}(\lambda_i) \oplus \cdots$$

Combining the bases for all generalized eigenspaces gives a basis of $V$ in which $\varphi$ has the desired block diagonal form. □

**Note:-**

The Jordan form groups blocks by eigenvalue: all $J_m(\lambda_i)$ blocks for a given $\lambda_i$ appear together, corresponding to the decomposition of $V_{\lambda_i}$ into chains. The total size of blocks with eigenvalue $\lambda_i$ equals $\dim V_{\lambda_i}$ = algebraic multiplicity of $\lambda_i$.

### 7.7.3 Existence and Uniqueness

**Theorem 7.7.2** Uniqueness of Jordan normal form

The Jordan normal form of $\varphi$ is unique up to permutation of the blocks. That is, the multiset of Jordan blocks $\{J_{m_1}(\lambda_1), J_{m_2}(\lambda_2), \ldots\}$ is uniquely determined by $\varphi$.

***Proof:*** For each eigenvalue $\lambda$, the Jordan blocks with eigenvalue $\lambda$ correspond to the partition of $\dim V_\lambda$ given by the nilpotent operator $\varphi|_{V_\lambda} - \lambda\mathrm{id}$. We showed this partition is uniquely determined by dimensions of iterated kernels:

$$d_j(\lambda) = \dim \operatorname{Ker}((\varphi - \lambda\mathrm{id})^j).$$

The number of blocks $J_m(\lambda)$ of size exactly $m$ is

$$(d_m(\lambda) - d_{m-1}(\lambda)) - (d_{m+1}(\lambda) - d_m(\lambda)) = 2d_m(\lambda) - d_{m-1}(\lambda) - d_{m+1}(\lambda).$$

Since these dimensions are intrinsic to $\varphi$, the block structure is uniquely determined. □

**Corollary 7.7.1** Similarity criterion

Two operators $\varphi, \psi \in \operatorname{End}(V)$ are similar if and only if they have the same Jordan normal form (up to block ordering). Equivalently, $\varphi \sim \psi$ iff for each $\lambda \in k$ and each $j \geqslant 1$:

$$\dim \operatorname{Ker}((\varphi - \lambda\mathrm{id})^j) = \dim \operatorname{Ker}((\psi - \lambda\mathrm{id})^j).$$

**Note:-**

This gives a complete set of similarity invariants. Unlike trace and determinant alone (which don't distinguish all similarity classes), the full sequence of kernel dimensions completely determines the similarity class.

### 7.7.4 Computing Jordan Form: The Algorithm

Given a matrix $A$, here is the procedure to find its Jordan form:

**Step 1: Find eigenvalues.** Compute $\chi_A(x) = \det(xI - A)$ and factor it as $\prod_i (x - \lambda_i)^{n_i}$. The $\lambda_i$ are eigenvalues with algebraic multiplicities $n_i$.

**Step 2: For each eigenvalue $\lambda$, determine block sizes.** Compute $d_j = \dim \operatorname{Ker}(A - \lambda I)^j$ for $j = 1, 2, \ldots$ until stabilization. The number of blocks of size $\geqslant j$ is $d_j - d_{j-1}$. Alternatively:

- Number of blocks $= d_1 =$ geometric multiplicity.
- Number of blocks of size $\geqslant 2$ is $d_2 - d_1$.
- Number of blocks of size exactly $j$ is $(d_j - d_{j-1}) - (d_{j+1} - d_j)$.

**Step 3: Assemble the Jordan form.** List all Jordan blocks grouped by eigenvalue.

**Example 7.7.2** (Computing Jordan form)

Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

**Step 1:** $\chi_A(x) = (x-2)^3$, so $\lambda = 2$ with multiplicity 3.

**Step 2:**

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A-2I)^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A-2I)^3 = 0.$$

So $d_1 = 1$, $d_2 = 2$, $d_3 = 3$.
Number of blocks $= d_1 = 1$. Size of the single block $= 3$.
**Step 3:** Jordan form is $J_3(2)$. (The matrix is already in Jordan form!)

**Example 7.7.3** (A more interesting example)

Let $A = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.
**Step 1:** $\chi_A(x) = (x-3)^4$, so $\lambda = 3$ with multiplicity 4.
**Step 2:**

$$A - 3I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$\operatorname{Ker}(A-3I) = \operatorname{Span}\{e_1, e_3\}$, so $d_1 = 2$. $(A-3I)^2 = 0$, so $d_2 = 4$.
Number of blocks $= 2$. Blocks of size $\geqslant 2$: $d_2 - d_1 = 2$. So both blocks have size 2.
**Step 3:** Jordan form is $J_2(3) \oplus J_2(3)$.

**Example 7.7.4** (Multiple eigenvalues)

Let $A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$.
**Step 1:** $\chi_A(x) = (x-1)^2(x-2)^2$. Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$, each with multiplicity 2.
**Step 2 for $\lambda = 1$:**

$$A - I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Restricted to generalized eigenspace (first two coordinates): $d_1 = 1$, $d_2 = 2$. One block of size 2: $J_2(1)$.
**Step 2 for $\lambda = 2$:**

$$A - 2I = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Restricted to generalized eigenspace (last two coordinates): $d_1 = 2$. Two blocks of size 1: $J_1(2) \oplus J_1(2)$.

**Step 3:** Jordan form is $J_2(1) \oplus J_1(2) \oplus J_1(2) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$.

**Note:-**

The last example shows that $\lambda = 2$ contributes two $1 \times 1$ blocks because the eigenspace has dimension 2 (geometric = algebraic multiplicity). This is the diagonalizable case for that eigenvalue. In contrast, $\lambda = 1$ has a $2 \times 2$ block because geometric multiplicity (1) is less than algebraic multiplicity (2).

## 7.8 Characteristic and Minimal Polynomials

We now introduce two polynomials canonically associated to a linear operator. The characteristic polynomial encodes eigenvalues and their algebraic multiplicities; the minimal polynomial captures the "smallest" polynomial relation satisfied by the operator.

### 7.8.1 The Characteristic Polynomial

**Definition 7.8.1: Characteristic polynomial**

Let $\varphi \in \mathrm{End}(V)$ with $\dim V = n$, and let $A = \mathcal{M}(\varphi)$ be the matrix of $\varphi$ with respect to some basis. The **characteristic polynomial** of $\varphi$ is

$$\chi_\varphi(x) = \det(xI - \varphi) = \det(xI - A).$$

This is a monic polynomial of degree $n$.

**Proposition 7.8.1** Basic properties of the characteristic polynomial

1. $\chi_\varphi(x)$ is independent of the choice of basis (hence well-defined for the operator $\varphi$).
2. $\chi_\varphi(x) = x^n - (\mathrm{Tr}\,\varphi)x^{n-1} + \cdots + (-1)^n \det(\varphi)$.
3. The roots of $\chi_\varphi(x)$ are exactly the eigenvalues of $\varphi$.
4. If $k$ is algebraically closed, $\chi_\varphi(x) = \prod_{i=1}^m (x - \lambda_i)^{n_i}$ where $\lambda_i$ are the distinct eigenvalues and $n_i = \dim V_{\lambda_i}$.

***Proof:*** (1) If $B = P^{-1}AP$ is the matrix in another basis, then

$$\det(xI - B) = \det(xI - P^{-1}AP) = \det(P^{-1}(xI - A)P) = \det(xI - A).$$

(2) Expand $\det(xI - A)$ using the Leibniz formula. The leading term comes from the diagonal product $\prod_i (x - a_{ii}) = x^n - (\sum_i a_{ii})x^{n-1} + \cdots$. The constant term is $\det(-A) = (-1)^n \det(A)$.

(3) $\lambda$ is a root iff $\det(\lambda I - A) = 0$ iff $\lambda I - A$ is singular iff $\mathrm{Ker}(\varphi - \lambda \mathrm{id}) \neq \{0\}$ iff $\lambda$ is an eigenvalue.

(4) This follows from the generalized eigenspace decomposition: the algebraic multiplicity of $\lambda_i$ equals $\dim V_{\lambda_i}$. ☐

**Example 7.8.1** (Characteristic polynomial of a $2 \times 2$ matrix)

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\chi_A(x) = \det \begin{pmatrix} x - a & -b \\ -c & x - d \end{pmatrix} = (x-a)(x-d) - bc = x^2 - (a+d)x + (ad - bc) = x^2 - (\mathrm{Tr}\,A)x + \det A.$$

### 7.8.2 The Cayley-Hamilton Theorem

The Cayley-Hamilton theorem states that every operator satisfies its own characteristic polynomial.

**Theorem 7.8.1** Cayley-Hamilton

Let $\varphi \in \mathrm{End}(V)$ with characteristic polynomial $\chi_\varphi(x)$. Then

$$\chi_\varphi(\varphi) = 0.$$

That is, substituting $\varphi$ for $x$ in $\chi_\varphi(x)$ gives the zero operator.

***Proof:*** We give two proofs.

**Proof 1 (via Jordan form, over algebraically closed $k$):** It suffices to prove this for Jordan blocks, since the Jordan form is block diagonal and $\chi_\varphi = \prod \chi_{J_i}$.

For a single Jordan block $J = J_m(\lambda)$, we have $\chi_J(x) = (x-\lambda)^m$. Then

$$\chi_J(J) = (J - \lambda I)^m = J_m(0)^m = 0$$

since $J_m(0)$ is nilpotent with index $m$.

**Proof 2 (general, using adjugate matrix):** Let $A$ be the matrix of $\varphi$. The adjugate (classical adjoint) matrix $\operatorname{adj}(xI - A)$ is a matrix whose entries are polynomials in $x$ of degree at most $n-1$. Write

$$\operatorname{adj}(xI - A) = B_{n-1}x^{n-1} + B_{n-2}x^{n-2} + \cdots + B_1 x + B_0$$

where $B_i$ are constant matrices.

By the adjugate identity: $(xI - A) \cdot \operatorname{adj}(xI - A) = \det(xI - A) \cdot I = \chi_A(x) \cdot I$.

Expanding both sides and comparing coefficients of powers of $x$ gives a system of matrix equations. Multiplying the equation for $x^k$ by $A^k$ and summing yields $\chi_A(A) = 0$. (The details involve telescoping—each $B_i$ appears twice with opposite signs.) □

**Note:-**

The Cayley-Hamilton theorem does <u>not</u> follow from "plugging in $A$ for $x$ in $\det(xI - A) = 0$." The determinant is a scalar-valued function, not a matrix-valued function, so this substitution is meaningless. The theorem requires proof.

**Corollary 7.8.1** Bound on nilpotence index

If $\varphi$ is nilpotent on an $n$-dimensional space, then $\varphi^n = 0$.

***Proof:*** The only eigenvalue is 0, so $\chi_\varphi(x) = x^n$. By Cayley-Hamilton, $\varphi^n = 0$. □

**Corollary 7.8.2** Expression for the inverse

If $\varphi$ is invertible, then $\varphi^{-1}$ can be expressed as a polynomial in $\varphi$.

***Proof:*** Write $\chi_\varphi(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$. Since $\varphi$ is invertible, 0 is not an eigenvalue, so $c_0 = (-1)^n \det(\varphi) \neq 0$.

By Cayley-Hamilton: $\varphi^n + c_{n-1}\varphi^{n-1} + \cdots + c_1\varphi + c_0 I = 0$.

Rearranging: $\varphi(\varphi^{n-1} + c_{n-1}\varphi^{n-2} + \cdots + c_1 I) = -c_0 I$.

Thus $\varphi^{-1} = -\frac{1}{c_0}(\varphi^{n-1} + c_{n-1}\varphi^{n-2} + \cdots + c_1 I)$. □

### 7.8.3 The Minimal Polynomial

**Definition 7.8.2: Minimal polynomial**

The **minimal polynomial** $\mu_\varphi(x)$ of $\varphi \in \operatorname{End}(V)$ is the unique monic polynomial of smallest degree such that $\mu_\varphi(\varphi) = 0$.

**Proposition 7.8.2** Existence and uniqueness of the minimal polynomial

1. There exists a nonzero polynomial $p(x)$ with $p(\varphi) = 0$.
2. Among all such polynomials, there is a unique monic one of minimal degree.
3. $\mu_\varphi(x)$ divides every polynomial $p(x)$ satisfying $p(\varphi) = 0$.

***Proof:*** (1) Cayley-Hamilton gives $\chi_\varphi(\varphi) = 0$.

(2) Let $d$ be the minimal degree of a nonzero polynomial annihilating $\varphi$. If $p(x)$ and $q(x)$ are both monic of degree $d$ with $p(\varphi) = q(\varphi) = 0$, then $(p - q)(\varphi) = 0$ and $\deg(p - q) < d$. By minimality, $p - q = 0$.

(3) Given $p(x)$ with $p(\varphi) = 0$, perform polynomial division: $p(x) = q(x)\mu_\varphi(x) + r(x)$ where $\deg r < \deg \mu_\varphi$. Then $r(\varphi) = p(\varphi) - q(\varphi)\mu_\varphi(\varphi) = 0 - 0 = 0$. By minimality of $\mu_\varphi$, we must have $r = 0$, so $\mu_\varphi \mid p$. ☐

**Theorem 7.8.2** Relationship between characteristic and minimal polynomials

Let $\varphi \in \mathrm{End}(V)$.

1. $\mu_\varphi(x)$ divides $\chi_\varphi(x)$.
2. $\mu_\varphi(x)$ and $\chi_\varphi(x)$ have the same roots (i.e., the same irreducible factors over $k$).
3. Over an algebraically closed field, if $\chi_\varphi(x) = \prod_i (x - \lambda_i)^{n_i}$, then $\mu_\varphi(x) = \prod_i (x - \lambda_i)^{m_i}$ where $1 \leqslant m_i \leqslant n_i$.

***Proof:*** (1) By Cayley-Hamilton, $\chi_\varphi(\varphi) = 0$. By the divisibility property of $\mu_\varphi$, we have $\mu_\varphi \mid \chi_\varphi$.

(2) If $\lambda$ is a root of $\chi_\varphi$, then $\lambda$ is an eigenvalue: there exists $v \neq 0$ with $\varphi(v) = \lambda v$. For any polynomial $p$, $p(\varphi)(v) = p(\lambda)v$. If $p(\varphi) = 0$, then $p(\lambda) = 0$. In particular, $\mu_\varphi(\lambda) = 0$.

Conversely, if $\mu_\varphi(\lambda) = 0$, then $(x - \lambda) \mid \mu_\varphi(x)$, so $(x - \lambda) \mid \chi_\varphi(x)$ by (1), meaning $\lambda$ is a root of $\chi_\varphi$.

(3) From (1) and (2), the factorizations have the same roots with $m_i \leqslant n_i$. The exponent $m_i$ is the size of the largest Jordan block with eigenvalue $\lambda_i$ (proof below). ☐

**Proposition 7.8.3** Minimal polynomial and Jordan block sizes

The exponent of $(x - \lambda)$ in $\mu_\varphi(x)$ equals the size of the largest Jordan block with eigenvalue $\lambda$.

***Proof:*** On the generalized eigenspace $V_\lambda$, the operator $\varphi - \lambda\mathrm{id}$ is nilpotent. Its nilpotence index is the size of the largest Jordan block (the longest chain). Call this $m_\lambda$.

For $(\varphi - \lambda\mathrm{id})^k$ to annihilate $V_\lambda$, we need $k \geqslant m_\lambda$. The smallest such $k$ is $m_\lambda$.

Since $V = \bigoplus_\lambda V_\lambda$ and these are $\varphi$-invariant, $p(\varphi) = 0$ iff $p(\varphi)|_{V_\lambda} = 0$ for all $\lambda$. The latter requires $(x - \lambda)^{m_\lambda} \mid p(x)$. Thus $\mu_\varphi(x) = \prod_\lambda (x - \lambda)^{m_\lambda}$. ☐

### 7.8.4 Diagonalizability Criterion

**Theorem 7.8.3** Diagonalizability via minimal polynomial

An operator $\varphi \in \mathrm{End}(V)$ is diagonalizable if and only if $\mu_\varphi(x)$ splits into distinct linear factors:

$$\mu_\varphi(x) = (x - \lambda_1)(x - \lambda_2)\cdots(x - \lambda_m)$$

where $\lambda_1, \ldots, \lambda_m$ are distinct.

***Proof:*** $(\Rightarrow)$ If $\varphi$ is diagonalizable, all Jordan blocks are $1 \times 1$. The largest block for each eigenvalue has size 1, so each $(x - \lambda_i)$ appears with exponent 1 in $\mu_\varphi$.

($\Leftarrow$) If $\mu_\varphi(x) = \prod_i (x-\lambda_i)$ with distinct $\lambda_i$, then each Jordan block has size 1 (since the exponent of $(x-\lambda_i)$ in $\mu_\varphi$ equals the largest block size for $\lambda_i$). So $\varphi$ has a basis of eigenvectors. $\square$

**Corollary 7.8.3** Diagonalizability of operators with distinct eigenvalues

If $\varphi$ has $n = \dim V$ distinct eigenvalues, then $\varphi$ is diagonalizable.

***Proof:*** The characteristic polynomial has $n$ distinct roots, so $\chi_\varphi(x) = \prod_{i=1}^n (x-\lambda_i)$. Since $\mu_\varphi \mid \chi_\varphi$ and they have the same roots, $\mu_\varphi$ also has distinct linear factors. $\square$

**Example 7.8.2** (Minimal polynomial examples)

1. For $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I$: $\chi_A(x) = (x-2)^2$, but $\mu_A(x) = x - 2$ since $(A - 2I) = 0$.
2. For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$: $\chi_A(x) = (x-2)^2$. Since $A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$, we need $\mu_A(x) = (x-2)^2 = \chi_A(x)$.
3. For $A = \operatorname{diag}(1, 1, 2, 2, 2)$: $\chi_A(x) = (x-1)^2(x-2)^3$, but $\mu_A(x) = (x-1)(x-2)$ since $(A-I)(A-2I) = 0$.

### 7.8.5 Exercises

**Question 2: Exercises on advanced spectral theory**

1. Let $\varphi : \mathbb{C}^4 \to \mathbb{C}^4$ have characteristic polynomial $\chi_\varphi(x) = (x-1)^2(x-2)^2$. List all possible Jordan normal forms for $\varphi$.
2. Prove that if $A$ and $B$ are similar matrices, then $\mu_A = \mu_B$.
3. Find the Jordan normal form and minimal polynomial of $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.
4. Let $\varphi \in \operatorname{End}(V)$ satisfy $\varphi^2 = \varphi$. Prove that $\varphi$ is diagonalizable and find its possible eigenvalues.
5. Suppose $\varphi \in \operatorname{End}(V)$ has minimal polynomial $\mu_\varphi(x) = (x-1)(x-2)(x-3)$. What can you conclude about $\varphi$?

**Solution**

**(1)** The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$, each with algebraic multiplicity 2. For each eigenvalue, the Jordan blocks must have sizes summing to 2. Possibilities for each:

- One block of size 2: $J_2(\lambda)$
- Two blocks of size 1: $J_1(\lambda) \oplus J_1(\lambda)$

Combining independently for each eigenvalue, the possible Jordan forms are:

(i) $J_2(1) \oplus J_2(2)$

(ii) $J_2(1) \oplus J_1(2) \oplus J_1(2)$

(iii) $J_1(1) \oplus J_1(1) \oplus J_2(2)$

(iv) $J_1(1) \oplus J_1(1) \oplus J_1(2) \oplus J_1(2)$ (diagonal matrix)

**(2)** If $B = P^{-1}AP$, then for any polynomial $p$:

$$p(B) = p(P^{-1}AP) = P^{-1}p(A)P.$$

So $p(B) = 0$ iff $p(A) = 0$. The set of annihilating polynomials is the same, hence the minimal polynomial (the monic generator of this ideal) is the same.
**(3)** The matrix is already in Jordan form: $A = J_3(0)$.
Characteristic polynomial: $\chi_A(x) = x^3$.
Minimal polynomial: We need the smallest $k$ with $A^k = 0$. Compute $A^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0$ and $A^3 = 0$. So $\mu_A(x) = x^3 = \chi_A(x)$.
**(4)** The condition $\varphi^2 = \varphi$ means $\varphi^2 - \varphi = 0$, so $\varphi(\varphi - \mathrm{id}) = 0$. Thus $p(x) = x(x-1) = x^2 - x$ annihilates $\varphi$.
The minimal polynomial divides $x(x-1)$. Since $x(x-1)$ has distinct roots, so does $\mu_\varphi$. By the diagonalizability criterion, $\varphi$ is diagonalizable.
The possible eigenvalues are roots of $\mu_\varphi$, which divides $x(x-1)$. So eigenvalues are among $\{0, 1\}$.
(Such operators are called **projections** or **idempotents**. They decompose $V = \mathrm{Ker}(\varphi) \oplus \mathrm{Im}(\varphi)$ and project onto $\mathrm{Im}(\varphi)$ along $\mathrm{Ker}(\varphi)$.)
**(5)** Since $\mu_\varphi(x) = (x-1)(x-2)(x-3)$ has three distinct linear factors:

- $\varphi$ is diagonalizable (by the criterion).
- The eigenvalues are exactly $1, 2, 3$ (roots of $\mu_\varphi$).
- $\dim V \geqslant 3$ (at least one eigenvector per eigenvalue).
- $V = E(1, \varphi) \oplus E(2, \varphi) \oplus E(3, \varphi)$.

The eigenspace dimensions can be any positive integers summing to $\dim V$, but we need at least one of each.

Lecture 8.

# Categories: An Interlude

Many otherwise disparate fields of mathematics share fundamental structures. The language of categories systematizes these common patterns. While category theory won't answer questions specific to a particular area, it suggests useful ways to think about problems by analogy with other settings. We've already encountered categorical ideas implicitly: the universal property of quotient spaces, the behavior of dual maps, and the isomorphism theorems all fit naturally into this framework.

## 8.1 The Language of Categories

### 8.1.1 Objects and Morphisms

The basic data of a category consists of objects and arrows between them, subject to axioms that capture the essential features of function composition.

**Definition 8.1.1: Category**

A **category** $\mathcal{C}$ consists of:

(1) A collection of **objects**, denoted $\mathrm{Ob}(\mathcal{C})$.

(2) For every pair of objects $A, B \in \mathrm{Ob}(\mathcal{C})$, a collection of **morphisms** from $A$ to $B$, denoted $\mathrm{Mor}(A, B)$ or $\mathrm{Hom}_{\mathcal{C}}(A, B)$.

(3) A **composition rule**: for every triple $A, B, C \in \mathrm{Ob}(\mathcal{C})$, a map

$$\mathrm{Mor}(A, B) \times \mathrm{Mor}(B, C) \longrightarrow \mathrm{Mor}(A, C)$$

sending $(f, g)$ to $g \circ f$.

These must satisfy two axioms:

(i) **Associativity:** For any $f \in \mathrm{Mor}(A, B)$, $g \in \mathrm{Mor}(B, C)$, $h \in \mathrm{Mor}(C, D)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

(ii) **Identity:** For every object $A$, there exists $\mathrm{id}_A \in \mathrm{Mor}(A, A)$ such that for any $f \in \mathrm{Mor}(A, B)$,

$$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f.$$

**Note:-**

The identity morphism $\mathrm{id}_A$ is unique: if $\mathrm{id}'_A$ also satisfies the identity axiom, then $\mathrm{id}'_A = \mathrm{id}'_A \circ \mathrm{id}_A = \mathrm{id}_A$.

We write $f : A \to B$ to indicate $f \in \mathrm{Mor}(A, B)$, and call $A$ the **domain** (or source) and $B$ the **codomain** (or target) of $f$.

**Example 8.1.1** (The category of sets)

The category **Set** has:

- Objects: all sets.

- Morphisms: $\mathrm{Mor}(A, B) = \{f : A \to B \mid f \text{ is a function}\}$.
- Composition: ordinary function composition.

Variants include:

- **FinSet**: objects are finite sets, morphisms are functions.
- **Set$_*$**: **pointed sets**. Objects are pairs $(A, a)$ with $a \in A$ a distinguished element. Morphisms $(A, a) \to (B, b)$ are functions $f : A \to B$ with $f(a) = b$.

**Example 8.1.2** (The category of groups)
The category **Grp** has:

- Objects: all groups.
- Morphisms: group homomorphisms.
- Composition: composition of homomorphisms.

The identity axiom holds because $\mathrm{id}_G : G \to G$ is a homomorphism. Variants:

- **Ab**: abelian groups and group homomorphisms.
- **FinGrp**: finite groups and homomorphisms.

Similarly, we have categories **Ring** (rings and ring homomorphisms) and **CRing** (commutative rings).

**Example 8.1.3** (The category of vector spaces)
Fix a field $k$. The category **Vect**$_k$ has:

- Objects: vector spaces over $k$.
- Morphisms: $k$-linear maps.
- Composition: composition of linear maps.

The subcategory **FinVect**$_k$ consists of finite-dimensional vector spaces. We've been studying **Vect**$_k$ all along: $\mathrm{Hom}(V, W)$ is exactly $\mathrm{Mor}(V, W)$ in this category.

**Example 8.1.4** (The category of topological spaces)
The category **Top** has:

- Objects: topological spaces.
- Morphisms: continuous maps.
- Composition: composition of continuous maps.

The pointed variant **Top**$_*$ has objects $(X, x_0)$ (a space with basepoint) and morphisms being continuous maps preserving basepoints. This category is fundamental in algebraic topology.

**Note:-**
In all these examples, objects are "decorated sets"—sets equipped with additional structure (a group operation, a topology, a vector space structure)—and morphisms are functions that respect this structure. This is typical of categories arising in practice, though the abstract definition permits more exotic examples. For instance, any poset $(P, \leqslant)$ defines a category: objects are elements of $P$, and $\mathrm{Mor}(a, b)$ contains a single element if $a \leqslant b$ and is empty otherwise. Composition is forced by transitivity.

**Example 8.1.5** (Posets as categories)

Let $(P, \leqslant)$ be a partially ordered set. Define a category $\mathcal{C}_P$ by:

- $\mathrm{Ob}(\mathcal{C}_P) = P$.
- $\mathrm{Mor}(a, b) = \begin{cases} \{*\} & \text{if } a \leqslant b \\ \varnothing & \text{otherwise} \end{cases}$
- Composition: the unique element of $\mathrm{Mor}(a, b) \times \mathrm{Mor}(b, c)$ maps to the unique element of $\mathrm{Mor}(a, c)$ (by transitivity).

The identity axiom uses reflexivity ($a \leqslant a$), and associativity is trivial since all compositions are unique when defined.

### 8.1.2 Isomorphisms

The notion of "sameness" in a category is captured by isomorphisms.

**Definition 8.1.2: Isomorphism**

Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Ob}(\mathcal{C})$. A morphism $f \in \mathrm{Mor}(A, B)$ is an **isomorphism** if there exists $g \in \mathrm{Mor}(B, A)$ such that

$$g \circ f = \mathrm{id}_A \quad \text{and} \quad f \circ g = \mathrm{id}_B.$$

We write $A \cong B$ if such an isomorphism exists, and say $A$ and $B$ are **isomorphic**.

**Proposition 8.1.1** Uniqueness of inverses

If $f : A \to B$ is an isomorphism, then the morphism $g$ in the definition is unique. We denote it $f^{-1}$ and call it the **inverse** of $f$.

***Proof:*** Suppose $g, g' \in \mathrm{Mor}(B, A)$ both satisfy the conditions. Then

$$g = g \circ \mathrm{id}_B = g \circ (f \circ g') = (g \circ f) \circ g' = \mathrm{id}_A \circ g' = g'.$$ □

**Proposition 8.1.2** Properties of isomorphisms

Let $\mathcal{C}$ be a category.

(i) For any object $A$, $\mathrm{id}_A$ is an isomorphism with $\mathrm{id}_A^{-1} = \mathrm{id}_A$.

(ii) If $f : A \to B$ is an isomorphism, then $f^{-1} : B \to A$ is an isomorphism with $(f^{-1})^{-1} = f$.

(iii) If $f : A \to B$ and $g : B \to C$ are isomorphisms, then $g \circ f : A \to C$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

***Proof:*** (i) $\mathrm{id}_A \circ \mathrm{id}_A = \mathrm{id}_A$.
(ii) By definition, $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$. This says exactly that $f$ is the inverse of $f^{-1}$.
(iii) Compute:

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \mathrm{id}_B \circ f = f^{-1} \circ f = \mathrm{id}_A.$$

Similarly, $(g \circ f) \circ (f^{-1} \circ g^{-1}) = \mathrm{id}_C$. □

**Example 8.1.6** (Isomorphisms in familiar categories)

- In **Set**: isomorphisms are bijections.
- In **Grp**: isomorphisms are group isomorphisms (bijective homomorphisms).
- In $\mathbf{Vect}_k$: isomorphisms are invertible linear maps.
- In **Top**: isomorphisms are homeomorphisms (continuous bijections with continuous inverse).

In $\mathbf{FinVect}_k$, two objects are isomorphic iff they have the same dimension. In **FinSet**, two objects are isomorphic iff they have the same cardinality.

**Note:-**

In **Top**, a bijective continuous map need not be an isomorphism—the inverse may fail to be continuous. For example, $\mathrm{id} : (\mathbb{R}, \text{discrete}) \to (\mathbb{R}, \text{standard})$ is a continuous bijection but not a homeomorphism. This illustrates that "isomorphism" in a category means more than just bijection on underlying sets.

### 8.1.3 Automorphism Groups

The self-isomorphisms of an object form a group, connecting category theory back to group theory.

**Definition 8.1.3: Automorphism group**

For an object $A$ in a category $\mathcal{C}$, the **automorphism group** of $A$ is

$$\mathrm{Aut}(A) = \{f \in \mathrm{Mor}(A, A) : f \text{ is an isomorphism}\}.$$

**Theorem 8.1.1** Automorphisms form a group

$\mathrm{Aut}(A)$, equipped with composition, is a group.

***Proof:***

- **Closure:** If $f, g \in \mathrm{Aut}(A)$, then $g \circ f$ is an isomorphism by the previous proposition.
- **Associativity:** Inherited from the category axiom.
- **Identity:** $\mathrm{id}_A \in \mathrm{Aut}(A)$ is the identity element.
- **Inverses:** If $f \in \mathrm{Aut}(A)$, then $f^{-1} \in \mathrm{Aut}(A)$. □

**Example 8.1.7** (Automorphism groups in practice)

- In **FinSet**: if $|A| = n$, then $\mathrm{Aut}(A) \cong S_n$, the symmetric group.
- In $\mathbf{FinVect}_k$: if $\dim V = n$, then $\mathrm{Aut}(V) \cong \mathrm{GL}_n(k)$, the general linear group.
- In **Grp**: $\mathrm{Aut}(G)$ is the group of group automorphisms. For $G = \mathbb{Z}/n$, we have $\mathrm{Aut}(\mathbb{Z}/n) \cong (\mathbb{Z}/n)^\times$.

**Exercise 8.1.1** Isomorphic objects have isomorphic automorphism groups

Let $f : A \to B$ be an isomorphism in a category $\mathcal{C}$. Show that the map

$$c_f : \mathrm{Aut}(A) \to \mathrm{Aut}(B), \quad g \mapsto f \circ g \circ f^{-1}$$

is a group isomorphism.

Solution

**Well-defined:** If $g \in \mathrm{Aut}(A)$, then $f \circ g \circ f^{-1}$ is a composition of isomorphisms, hence an isomorphism $B \to B$.
**Homomorphism:** For $g, h \in \mathrm{Aut}(A)$:

$$c_f(g \circ h) = f \circ (g \circ h) \circ f^{-1} = (f \circ g \circ f^{-1}) \circ (f \circ h \circ f^{-1}) = c_f(g) \circ c_f(h).$$

**Bijective:** The map $c_{f^{-1}} : \mathrm{Aut}(B) \to \mathrm{Aut}(A)$ is an inverse:

$$c_{f^{-1}}(c_f(g)) = f^{-1} \circ (f \circ g \circ f^{-1}) \circ f = g.$$

Similarly, $c_f(c_{f^{-1}}(h)) = h$.

**Note:-**

This exercise shows that the automorphism group is an "isomorphism invariant": isomorphic objects have isomorphic automorphism groups. The converse is false—non-isomorphic objects can have isomorphic automorphism groups. For instance, in **Set**, any two 3-element sets have automorphism group $S_3$, but they are only isomorphic if we consider them as abstract sets (which they always are).

A category where every morphism is an isomorphism has a special name.

**Definition 8.1.4: Groupoid**

A **groupoid** is a category in which every morphism is an isomorphism.

**Note:-**

A groupoid generalizes the notion of a group: a group is precisely a groupoid with exactly one object. In a groupoid, morphisms can be composed only when the target of one matches the source of the other, but when composition is defined, it has inverses. The fundamental groupoid of a topological space, with objects being points and morphisms being homotopy classes of paths, is a key example in algebraic topology.

**Example 8.1.8** (Groupoids)

- The category whose objects are finite sets and whose morphisms are only bijections (not all functions) is a groupoid.
- Any group $G$ defines a groupoid $\mathbf{B}G$ with one object $*$ and $\mathrm{Mor}(*, *) = G$.
- The category whose objects are groups and whose morphisms are only group isomorphisms is a groupoid.

## 8.2 Universal Properties

Many constructions in mathematics are characterized not by explicit formulas but by universal properties—conditions that specify the construction uniquely up to isomorphism. We've already seen this with quotient spaces: the quotient $V/U$ is characterized by factoring any linear map that kills $U$. Products and coproducts provide two more fundamental examples.

### 8.2.1 Products and Coproducts

**Definition 8.2.1: Product**

Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Ob}(\mathcal{C})$. A **product** of $A$ and $B$ is an object $P$ together with morphisms $\pi_1 : P \to A$ and $\pi_2 : P \to B$ (called **projections**) satisfying the following **universal property**:
For any object $T$ and morphisms $\alpha : T \to A$, $\beta : T \to B$, there exists a <u>unique</u> morphism $\varphi : T \to P$ such that $\pi_1 \circ \varphi = \alpha$ and $\pi_2 \circ \varphi = \beta$.

$$\begin{tikzcd}
T \arrow[d, dashed, "\exists!\,\varphi"'] \arrow[dr, "\alpha"] \arrow[drr, bend left, "\beta"] & & \\
P \arrow[r, "\pi_1"'] & A & B \arrow[l, "\pi_2"]
\end{tikzcd}$$

We write $P = A \times B$ and $\varphi = (\alpha, \beta)$ for the unique induced map.

**Note:-**

The universal property says: to give a map into the product is the same as giving a map into each factor. This is the categorical abstraction of the familiar fact that a function $f : T \to A \times B$ is determined by its components $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$.

The dual notion reverses all arrows.

**Definition 8.2.2: Coproduct**

Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Ob}(\mathcal{C})$. A **coproduct** (or **sum**) of $A$ and $B$ is an object $S$ together with morphisms $\iota_1 : A \to S$ and $\iota_2 : B \to S$ (called **inclusions** or **coprojections**) satisfying:
For any object $T$ and morphisms $\alpha : A \to T$, $\beta : B \to T$, there exists a <u>unique</u> morphism $\varphi : S \to T$ such that $\varphi \circ \iota_1 = \alpha$ and $\varphi \circ \iota_2 = \beta$.

$$\begin{tikzcd}
A \arrow[r, "\iota_1"] \arrow[dr, "\alpha"'] & S \arrow[d, dashed, "\exists!\,\varphi"] & B \arrow[l, "\iota_2"'] \arrow[dl, "\beta"] \\
& T &
\end{tikzcd}$$

We write $S = A \amalg B$ or $A + B$ and $\varphi = [\alpha, \beta]$ for the unique induced map.

**Note:-**

The universal property of the coproduct says: to give a map out of the coproduct is the same as giving a map out of each summand. This duality between products and coproducts—reversing all arrows—is a general phenomenon in category theory.

**Theorem 8.2.1** Uniqueness of products and coproducts

If a product (resp. coproduct) of $A$ and $B$ exists, it is unique up to unique isomorphism. That is, if $(P, \pi_1, \pi_2)$ and $(P', \pi_1', \pi_2')$ are both products of $A$ and $B$, then there is a unique isomorphism $\psi : P \to P'$ with $\pi_1' \circ \psi = \pi_1$ and $\pi_2' \circ \psi = \pi_2$.

***Proof:*** By the universal property of $P'$, applied to the maps $\pi_1 : P \to A$ and $\pi_2 : P \to B$, there exists a unique $\psi : P \to P'$ with $\pi_1' \circ \psi = \pi_1$ and $\pi_2' \circ \psi = \pi_2$.

By the universal property of $P$, applied to $\pi_1' : P' \to A$ and $\pi_2' : P' \to B$, there exists a unique $\psi' : P' \to P$ with $\pi_1 \circ \psi' = \pi_1'$ and $\pi_2 \circ \psi' = \pi_2'$.

Now $\psi' \circ \psi : P \to P$ satisfies $\pi_1 \circ (\psi' \circ \psi) = \pi_1$ and $\pi_2 \circ (\psi' \circ \psi) = \pi_2$. But $\mathrm{id}_P$ also satisfies these equations. By uniqueness in the universal property of $P$, we have $\psi' \circ \psi = \mathrm{id}_P$. Similarly, $\psi \circ \psi' = \mathrm{id}_{P'}$.

The proof for coproducts is dual (reverse all arrows). $\square$

**Note:-**

This proof pattern—using universal properties to construct inverse maps—is standard in category theory. The key is that universal properties give uniqueness, which forces the composites to be identities.

Products and coproducts need not exist in every category. When working in a new category, one of the first questions is whether these constructions exist and what form they take.

### 8.2.2 Examples in $\mathbf{Vect}_k$

**Example 8.2.1** (Products and coproducts in **Set**)

In the category of sets:

- The **product** $A \times B$ is the Cartesian product $\{(a, b) : a \in A, b \in B\}$ with projections $\pi_1(a, b) = a$ and $\pi_2(a, b) = b$.
- The **coproduct** $A \amalg B$ is the disjoint union $A \sqcup B = \{(a, 1) : a \in A\} \cup \{(b, 2) : b \in B\}$ with inclusions $\iota_1(a) = (a, 1)$ and $\iota_2(b) = (b, 2)$.

The universal properties are immediate from the definitions of functions on Cartesian products and disjoint unions.

**Example 8.2.2** (Products and coproducts in **Grp**)

In the category of groups:

- The **product** $G \times H$ is the direct product with componentwise operations and projections.
- The **coproduct** $G * H$ is the **free product**—a more complicated construction involving alternating words from $G$ and $H$.

In the subcategory **Ab** of abelian groups, both product and coproduct are the direct sum $G \oplus H$ (with the same underlying group but different universal properties).

**Example 8.2.3** (Products and coproducts in $\mathbf{Vect}_k$)

In the category of finite-dimensional vector spaces over $k$, both the product and coproduct of $V$ and $W$ are realized by the **direct sum** $V \oplus W$.
**As a product:** Define $\pi_1 : V \oplus W \to V$ by $\pi_1(v, w) = v$ and $\pi_2 : V \oplus W \to W$ by $\pi_2(v, w) = w$. Given linear maps $\alpha : T \to V$ and $\beta : T \to W$, define $\varphi : T \to V \oplus W$ by $\varphi(t) = (\alpha(t), \beta(t))$. This is the unique map with $\pi_1 \circ \varphi = \alpha$ and $\pi_2 \circ \varphi = \beta$.
**As a coproduct:** Define $\iota_1 : V \to V \oplus W$ by $\iota_1(v) = (v, 0)$ and $\iota_2 : W \to V \oplus W$ by $\iota_2(w) = (0, w)$. Given linear maps $\alpha : V \to T$ and $\beta : W \to T$, define $\varphi : V \oplus W \to T$ by $\varphi(v, w) = \alpha(v) + \beta(w)$. This is the unique map with $\varphi \circ \iota_1 = \alpha$ and $\varphi \circ \iota_2 = \beta$.

**Note:-**

The coincidence of products and coproducts in $\mathbf{Vect}_k$ (and **Ab**) reflects the additive structure of these categories. In categories without such structure (like **Set** or **Grp**), products and coproducts are genuinely different constructions.

**Exercise 8.2.1** Products and coproducts exercises

(1) Verify explicitly that the Cartesian product satisfies the universal property of products in **Set**.

(2) Show that in **Ab**, the coproduct of $\mathbb{Z}$ and $\mathbb{Z}$ is $\mathbb{Z}^2$, while in **Grp**, the coproduct is the free group $F_2$ on two generators.

(3) Let $V$ and $W$ be finite-dimensional vector spaces over $k$. Show that $\dim(V \oplus W) = \dim V + \dim W$ using the universal property (not by direct computation).

Solution

**(1)** Let $A, B$ be sets and let $T$ be any set with functions $\alpha : T \to A$ and $\beta : T \to B$. We must show there exists a unique function $\varphi : T \to A \times B$ with $\pi_1 \circ \varphi = \alpha$ and $\pi_2 \circ \varphi = \beta$.
Existence: Define $\varphi(t) = (\alpha(t), \beta(t))$. Then $\pi_1(\varphi(t)) = \alpha(t)$ and $\pi_2(\varphi(t)) = \beta(t)$.
Uniqueness: If $\psi : T \to A \times B$ satisfies $\pi_1 \circ \psi = \alpha$ and $\pi_2 \circ \psi = \beta$, then for any $t \in T$, $\psi(t) = (\pi_1(\psi(t)), \pi_2(\psi(t))) = (\alpha(t), \beta(t)) = \varphi(t)$.
**(2)** In **Ab**: The coproduct of $\mathbb{Z}$ and $\mathbb{Z}$ has inclusions $\iota_1, \iota_2 : \mathbb{Z} \to S$ such that any pair of homomorphisms $\alpha, \beta : \mathbb{Z} \to A$ (for $A$ abelian) factors uniquely through $S$. Taking $S = \mathbb{Z}^2$ with $\iota_1(n) = (n, 0)$ and $\iota_2(n) = (0, n)$, the map $\varphi : \mathbb{Z}^2 \to A$ defined by $\varphi(m, n) = m \cdot \alpha(1) + n \cdot \beta(1)$ is the unique such factorization.
In **Grp**: The coproduct must handle non-commuting images. Given any group $G$ and homomorphisms $\alpha, \beta : \mathbb{Z} \to G$, the elements $\alpha(1)$ and $\beta(1)$ generate a subgroup that need not be abelian. The free group $F_2 = \langle a, b \rangle$ with $\iota_1(1) = a$ and $\iota_2(1) = b$ satisfies the universal property: any choice of $g = \alpha(1)$ and $h = \beta(1)$ in $G$ extends uniquely to a homomorphism $F_2 \to G$.
**(3)** By the universal property of coproducts, $\mathrm{Hom}(V \oplus W, T) \cong \mathrm{Hom}(V, T) \times \mathrm{Hom}(W, T)$ for any $T$. Taking $T = k$:

$$(V \oplus W)^* \cong V^* \times W^*.$$

Thus $\dim(V \oplus W)^* = \dim V^* + \dim W^*$. Since $\dim U^* = \dim U$ for finite-dimensional spaces, we get $\dim(V \oplus W) = \dim V + \dim W$.

**Note:-**

The argument in (3) illustrates a general principle: universal properties translate into isomorphisms of Hom-sets. The functor $\mathrm{Hom}(-, T)$ sends coproducts to products, which is why $\mathrm{Hom}(V \oplus W, T) \cong \mathrm{Hom}(V, T) \times \mathrm{Hom}(W, T)$. This is part of a larger story involving adjoint functors.

Lecture 9.

# Inner Product Spaces

So far we have studied vector spaces equipped only with their linear structure. Adding geometric notions—length, angle, orthogonality—requires additional data: an inner product. This structure transforms abstract linear algebra into something closer to Euclidean geometry, while revealing that many geometric intuitions extend far beyond $\mathbb{R}^n$.

We begin with the more general notion of bilinear forms, then specialize to the symmetric positive-definite case (inner products over $\mathbb{R}$) and its complex analogue (Hermitian inner products over $\mathbb{C}$).

## 9.1 Inner Products

### 9.1.1 Bilinear Forms

Before defining inner products, we study the broader class of bilinear forms. This generality clarifies which properties of inner products come from bilinearity alone and which require symmetry or positivity.

**Definition 9.1.1: Bilinear form**

Let $V$ be a vector space over a field $k$. A **bilinear form** on $V$ is a map $b : V \times V \to k$ that is linear in each variable:

(i) $b(\lambda v, w) = \lambda b(v, w) = b(v, \lambda w)$ for all $\lambda \in k$, $v, w \in V$.

(ii) $b(u + v, w) = b(u, w) + b(v, w)$ for all $u, v, w \in V$.

(iii) $b(u, v + w) = b(u, v) + b(u, w)$ for all $u, v, w \in V$.

**Note:-**

A bilinear form is not a linear map $V \times V \to k$. If we view $V \times V$ as a vector space (the external direct sum $V \oplus V$), a linear map would satisfy $b(\lambda(v, w)) = \lambda b(v, w)$, i.e., $b(\lambda v, \lambda w) = \lambda b(v, w)$. But bilinearity gives $b(\lambda v, \lambda w) = \lambda^2 b(v, w)$. The tensor product $V \otimes V$ is the correct domain for linearizing bilinear maps.

**Definition 9.1.2: Symmetric and skew-symmetric forms**

A bilinear form $b$ is:

- **Symmetric** if $b(v, w) = b(w, v)$ for all $v, w \in V$.
- **Skew-symmetric** (or **alternating**) if $b(v, w) = -b(w, v)$ for all $v, w \in V$.

**Example 9.1.1** (The standard dot product)

On $k^n$, the map $b(v, w) = \sum_{i=1}^{n} v_i w_i$ is a symmetric bilinear form. This is the "dot product" familiar from calculus.

**Example 9.1.2** (A skew-symmetric form)

On $k^2$, define $b((x_1, x_2), (y_1, y_2)) = x_1 y_2 - x_2 y_1$. This equals $\det \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}$ and is skew-symmetric. Geometrically, it computes signed area.

### Matrix Representation

Given a basis $\{e_1, \dots, e_n\}$ of $V$, a bilinear form $b$ is determined by the values $a_{ij} = b(e_i, e_j)$. By bilinearity:

$$b\left(\sum_i x_i e_i, \sum_j y_j e_j\right) = \sum_{i,j} x_i y_j \, b(e_i, e_j) = \sum_{i,j} a_{ij} x_i y_j.$$

Writing $x = (x_1, \dots, x_n)^T$ and $y = (y_1, \dots, y_n)^T$ as column vectors, this is:

$$b(x, y) = x^T A y$$

where $A = (a_{ij})$ is the **matrix of** $b$ with respect to the chosen basis.

**Proposition 9.1.1** Symmetry and the matrix

(i) $b$ is symmetric if and only if $A = A^T$.

(ii) $b$ is skew-symmetric if and only if $A = -A^T$.

***Proof:*** We have $b(v, w) = v^T A w$ and $b(w, v) = w^T A v = (w^T A v)^T = v^T A^T w$. So $b(v, w) = b(w, v)$ for all $v, w$ iff $v^T A w = v^T A^T w$ for all $v, w$, iff $A = A^T$. The skew-symmetric case is similar. □

### Bilinear Forms and the Dual Space

There is a natural connection between bilinear forms and linear maps to the dual space.

**Proposition 9.1.2** The map $\varphi_b : V \to V^*$

Given a bilinear form $b : V \times V \to k$, define $\varphi_b : V \to V^*$ by

$$\varphi_b(v) = b(v, \cdot) \in V^*,$$

i.e., $\varphi_b(v)$ is the linear functional $w \mapsto b(v, w)$. Then $\varphi_b$ is a linear map.

***Proof:*** For $v_1, v_2 \in V$ and $\lambda \in k$:

$$\varphi_b(v_1 + \lambda v_2)(w) = b(v_1 + \lambda v_2, w) = b(v_1, w) + \lambda b(v_2, w) = (\varphi_b(v_1) + \lambda \varphi_b(v_2))(w).$$

So $\varphi_b(v_1 + \lambda v_2) = \varphi_b(v_1) + \lambda \varphi_b(v_2)$. □

Conversely, any linear map $\varphi : V \to V^*$ determines a bilinear form $b(v, w) = \varphi(v)(w)$.

**Theorem 9.1.1** Bilinear forms and $\mathrm{Hom}(V, V^*)$

The map $b \mapsto \varphi_b$ defines an isomorphism of vector spaces

$$B(V) \xrightarrow{\sim} \mathrm{Hom}(V, V^*)$$

where $B(V)$ denotes the space of bilinear forms on $V$.

**Definition 9.1.3: Rank and nondegeneracy**

The **rank** of a bilinear form $b$ is $\operatorname{rank}(\varphi_b) = \dim(\operatorname{Im}(\varphi_b))$. Equivalently, it is the rank of the matrix $A$ in any basis.
The form $b$ is **nondegenerate** if $\varphi_b$ is an isomorphism, i.e., if $\ker(\varphi_b) = \{0\}$. Equivalently, $b$ is nondegenerate iff $A$ is invertible.

**Note:-**

For $V$ finite-dimensional, $\dim V = \dim V^*$, so $\varphi_b$ is an isomorphism iff it is injective iff it is surjective. Nondegeneracy means: if $b(v, w) = 0$ for all $w \in V$, then $v = 0$.

**Proposition 9.1.3** Dimension of the space of bilinear forms

If $\dim V = n$, then $\dim B(V) = n^2$.

***Proof:*** We have $B(V) \cong \operatorname{Hom}(V, V^*)$. Since $\dim V^* = n$, we get $\dim \operatorname{Hom}(V, V^*) = n \cdot n = n^2$. Alternatively, a bilinear form is determined by an $n \times n$ matrix. $\square$

### 9.1.2 Inner Products over the reals

To do geometry, we need more than bilinearity. The key additional properties are symmetry (so that $b(v, w) = b(w, v)$) and positive-definiteness (so that "length" is always nonnegative).

**Definition 9.1.4: Inner product (real case)**

Let $V$ be a vector space over $\mathbb{R}$. An **inner product** on $V$ is a bilinear form $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

(i) **Symmetry:** $\langle v, w \rangle = \langle w, v \rangle$ for all $v, w \in V$.

(ii) **Positive-definiteness:** $\langle v, v \rangle \geqslant 0$ for all $v \in V$, with equality iff $v = 0$.

A vector space equipped with an inner product is called an **inner product space**.

**Note:-**

Positive-definiteness only makes sense over an ordered field where "$\geqslant 0$" is meaningful. In practice, this means $\mathbb{R}$. We cannot directly define inner products over $\mathbb{C}$ or finite fields using this definition—we will see the complex workaround shortly.

**Example 9.1.3** (The standard inner product on $\mathbb{R}^n$)

The dot product

$$\langle v, w \rangle = \sum_{i=1}^{n} v_i w_i = v^T w$$

is an inner product. Symmetry is clear, and $\langle v, v \rangle = \sum v_i^2 \geqslant 0$ with equality iff all $v_i = 0$.

**Example 9.1.4** (A weighted inner product)

On $\mathbb{R}^n$, for positive constants $c_1, \ldots, c_n > 0$, define

$$\langle v, w \rangle = \sum_{i=1}^{n} c_i v_i w_i.$$

This is an inner product (check!). Different weights emphasize different coordinates.

**Proposition 9.1.4** Inner products are nondegenerate

Every inner product is nondegenerate.

***Proof:*** Suppose $\langle v, w\rangle = 0$ for all $w \in V$. Taking $w = v$, we get $\langle v, v\rangle = 0$. By positive-definiteness, $v = 0$. $\square$

**Note:-**

The converse is false: a nondegenerate symmetric bilinear form need not be positive-definite. For example, the Minkowski metric $b(v, w) = v_1w_1 - v_2w_2 - v_3w_3 - v_4w_4$ on $\mathbb{R}^4$ is nondegenerate and symmetric, but $b(v, v)$ can be negative.

### 9.1.3 Hermitian Inner Products over the complexes

Why can't we directly use the same definition over $\mathbb{C}$? If $b$ is a bilinear form on a complex vector space with $b(v, v) \geqslant 0$, consider $b(iv, iv) = i^2 b(v, v) = -b(v, v)$. So if $b(v, v) > 0$, then $b(iv, iv) < 0$. Bilinearity and positive-definiteness are incompatible over $\mathbb{C}$.

The solution is to weaken bilinearity: we require linearity in one variable and conjugate-linearity in the other.

**Definition 9.1.5: Sesquilinear form**

A map $H : V \times V \to \mathbb{C}$ on a complex vector space $V$ is **sesquilinear** if:

(i) $H(u + v, w) = H(u, w) + H(v, w)$ and $H(u, v + w) = H(u, v) + H(u, w)$.

(ii) $H(\lambda u, v) = \lambda H(u, v)$ for $\lambda \in \mathbb{C}$.

(iii) $H(u, \lambda v) = \bar{\lambda} H(u, v)$ for $\lambda \in \mathbb{C}$.

That is, $H$ is linear in the first argument and conjugate-linear in the second.

**Note:-**

Some authors (including Axler) use the opposite convention: conjugate-linear in the first argument, linear in the second. The choice is a matter of convention; we follow Artin here.

**Definition 9.1.6: Hermitian form**

A sesquilinear form $H$ is **Hermitian** (or **conjugate-symmetric**) if

$$H(u, v) = \overline{H(v, u)}$$

for all $u, v \in V$.

**Note:-**

Conjugate-symmetry implies $H(v, v) = \overline{H(v, v)}$, so $H(v, v) \in \mathbb{R}$ for all $v$. This makes positive-definiteness meaningful.

**Definition 9.1.7: Hermitian inner product**

A **Hermitian inner product** on a complex vector space $V$ is a sesquilinear form $H : V \times V \to \mathbb{C}$ that is:

(i) **Conjugate-symmetric:** $H(u, v) = \overline{H(v, u)}$.

(ii) **Positive-definite:** $H(v, v) \geqslant 0$ for all $v$, with equality iff $v = 0$.

**Example 9.1.5** (The standard Hermitian inner product on $\mathbb{C}^n$)
Define

$$H(z, w) = \sum_{i=1}^{n} z_i \overline{w_i} = z^* w$$

where $z^* = \overline{z}^T$ is the conjugate transpose. Then:

- $H(w, z) = \sum w_i \overline{z_i} = \overline{\sum z_i \overline{w_i}} = \overline{H(z, w)}$.
- $H(z, z) = \sum |z_i|^2 \geqslant 0$, with equality iff $z = 0$.

**Proposition 9.1.5** The map $\varphi_H : V \to V^*$

For a Hermitian form $H$, the map $\varphi_H(v) = H(v, \cdot)$ is a **conjugate-linear** map $V \to V^*$:

$$\varphi_H(\lambda v) = \overline{\lambda} \varphi_H(v).$$

If $H$ is positive-definite, then $\varphi_H$ is injective (and hence bijective when $V$ is finite-dimensional).

**Note:-**

Various properties carry over from the real case. A positive-definite Hermitian form is nondegenerate. Given a subspace $W \subseteq V$, its orthogonal complement $W^\perp = \{v \in V : H(v, w) = 0 \ \forall w \in W\}$ is a subspace, and $V = W \oplus W^\perp$ when $H$ is positive-definite.

### 9.1.4 Norms and the Cauchy-Schwarz Inequality

An inner product induces a notion of length via the norm.

**Definition 9.1.8: Norm**

Let $V$ be an inner product space (real or Hermitian). The **norm** of a vector $v \in V$ is

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

This is well-defined since $\langle v, v \rangle \geqslant 0$.

**Definition 9.1.9: Orthogonality**

Vectors $v, w \in V$ are **orthogonal**, written $v \perp w$, if $\langle v, w \rangle = 0$.

**Proposition 9.1.6** Basic properties of the norm

(i) $\|v\| \geqslant 0$, with equality iff $v = 0$.

(ii) $\|\lambda v\| = |\lambda| \|v\|$ for $\lambda \in k$.

(iii) (Pythagorean theorem) If $v \perp w$, then $\|v + w\|^2 = \|v\|^2 + \|w\|^2$.

***Proof:*** (i) and (ii) follow from the definitions. For (iii):

$$\|v + w\|^2 = \langle v + w, v + w \rangle = \|v\|^2 + \langle v, w \rangle + \langle w, v \rangle + \|w\|^2 = \|v\|^2 + \|w\|^2$$

since $\langle v, w \rangle = 0 = \langle w, v \rangle$ (using symmetry/conjugate-symmetry). $\square$

The Pythagorean theorem suggests that orthogonal vectors behave like legs of a right triangle. More generally:

$$\|v - w\|^2 = \|v\|^2 + \|w\|^2 - 2\,\mathrm{Re}\langle v, w\rangle.$$

This motivates defining the angle between nonzero vectors. But first we need to know that $|\langle v, w\rangle| \leqslant \|v\|\|w\|$ so that the ratio lies in $[-1, 1]$.

**Theorem 9.1.2** Cauchy–Schwarz inequality

For all $u, v$ in an inner product space,

$$|\langle u, v\rangle| \leqslant \|u\|\|v\|,$$

with equality iff $u$ and $v$ are linearly dependent.

***Proof:*** The inequality is trivial if $u = 0$. Assume $u \neq 0$; by scaling, we may assume $\|u\| = 1$.

Decompose $v$ along $u$: let $S = \mathrm{span}(u)$. Since $\langle \cdot, \cdot\rangle$ is nondegenerate and $\dim S = 1$, we have $V = S \oplus S^\perp$. Write

$$v = v_1 + v_2, \quad v_1 = \langle v, u\rangle u \in S, \quad v_2 = v - \langle v, u\rangle u \in S^\perp.$$

Then $v_1 \perp v_2$, so by the Pythagorean theorem:

$$\|v\|^2 = \|v_1\|^2 + \|v_2\|^2 \geqslant \|v_1\|^2 = |\langle v, u\rangle|^2\|u\|^2 = |\langle v, u\rangle|^2.$$

Taking square roots gives $\|v\| \geqslant |\langle v, u\rangle| = |\langle u, v\rangle|$, which is the inequality for $\|u\| = 1$.

Equality holds iff $v_2 = 0$, i.e., $v \in S = \mathrm{span}(u)$. □

**Definition 9.1.10: Angle between vectors**

For nonzero vectors $v, w$ in a real inner product space, the **angle** $\theta$ between them is defined by

$$\cos\theta = \frac{\langle v, w\rangle}{\|v\|\|w\|}, \quad \theta \in [0, \pi].$$

The Cauchy–Schwarz inequality ensures this ratio lies in $[-1, 1]$.

**Corollary 9.1.1** Triangle inequality

For all $u, v$ in an inner product space,

$$\|u + v\| \leqslant \|u\| + \|v\|.$$

***Proof:***

$$\begin{aligned}
\|u + v\|^2 &= \|u\|^2 + 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2 \\
&\leqslant \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \\
&\leqslant \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2.
\end{aligned}$$

□

### 9.1.5 Orthogonality and Orthogonal Complements

**Definition 9.1.11: Orthogonal complement**

Let $V$ be an inner product space and $S \subseteq V$ a subset. The **orthogonal complement** of $S$ is

$$S^\perp = \{v \in V : \langle v, w\rangle = 0 \ \forall w \in S\}.$$

**Proposition 9.1.7** Basic properties of orthogonal complements

(i) $S^\perp$ is a subspace of $V$ (even if $S$ is not).

(ii) $S \subseteq T \implies T^\perp \subseteq S^\perp$.

(iii) $S \subseteq (S^\perp)^\perp$.

(iv) $\{0\}^\perp = V$ and $V^\perp = \{0\}$.

***Proof:*** (i) If $v_1, v_2 \in S^\perp$ and $\lambda \in k$, then for any $w \in S$:

$$\langle v_1 + \lambda v_2, w\rangle = \langle v_1, w\rangle + \lambda\langle v_2, w\rangle = 0 + \lambda \cdot 0 = 0.$$

(ii)–(iv) are immediate from the definition. □

**Note:-**

For a general bilinear form (not necessarily symmetric), we must distinguish "left-orthogonal" ($b(v, w) = 0$ for all $w \in S$) from "right-orthogonal" ($b(w, v) = 0$ for all $w \in S$). For symmetric forms and inner products, these coincide.

The key result is that an inner product space decomposes as $V = S \oplus S^\perp$ for any subspace $S$.

**Lemma 9.1.1** Dimension of orthogonal complement

Let $b$ be a nondegenerate bilinear form on a finite-dimensional space $V$, and let $S \subseteq V$ be a subspace. Then

$$\dim S^\perp = \dim V - \dim S.$$

If $b$ is an inner product (positive-definite), then $S \cap S^\perp = \{0\}$.

***Proof:*** Consider the composition

$$V \xrightarrow{\varphi_b} V^* \xrightarrow{r} S^*$$

where $\varphi_b(v) = b(v, \cdot)$ and $r : V^* \to S^*$ is restriction. The kernel of $r \circ \varphi_b$ is exactly $S^\perp$.

By rank-nullity: $\dim S^\perp = \dim V - \operatorname{rank}(r \circ \varphi_b)$.

Since $\varphi_b$ is an isomorphism (nondegeneracy) and $r$ is surjective, $r \circ \varphi_b$ is surjective. Thus $\operatorname{rank}(r \circ \varphi_b) = \dim S^* = \dim S$, giving $\dim S^\perp = \dim V - \dim S$.

For the second claim: if $v \in S \cap S^\perp$, then $\langle v, v\rangle = 0$. By positive-definiteness, $v = 0$. □

**Theorem 9.1.3** Orthogonal decomposition

Let $V$ be a finite-dimensional inner product space and $S \subseteq V$ a subspace. Then

$$V = S \oplus S^\perp.$$

***Proof:*** By the lemma, $\dim S + \dim S^\perp = \dim V$ and $S \cap S^\perp = \{0\}$. The dimension formula for sums gives

$$\dim(S + S^\perp) = \dim S + \dim S^\perp - \dim(S \cap S^\perp) = \dim V.$$

So $S + S^\perp = V$, and the sum is direct since $S \cap S^\perp = \{0\}$. □

**Example 9.1.6** (Standard orthogonal complement in $\mathbb{R}^n$)

Let $V = \mathbb{R}^n$ with the standard inner product $\langle v, w\rangle = \sum v_i w_i$. If $S$ is a subspace, then $S^\perp$ is the "usual" orthogonal complement: vectors perpendicular to every vector in $S$.
For instance, if $S = \operatorname{span}((1, 0, 0))$ in $\mathbb{R}^3$, then $S^\perp = \{(0, y, z) : y, z \in \mathbb{R}\}$.

**Example 9.1.7** (Skew-symmetric forms can have $S^\perp = S$)

Let $V = k^2$ with the skew-symmetric form $b((x_1, x_2), (y_1, y_2)) = x_1y_2 - x_2y_1$. Let $S$ be any 1-dimensional subspace spanned by a nonzero vector $v$.
For any $w \in k^2$, we have $b(v, w) = 0 \iff \det(v \mid w) = 0 \iff w \in \operatorname{span}(v) = S$.
So $S^\perp = S$. In particular, $S \cap S^\perp = S \neq \{0\}$, and $V \neq S \oplus S^\perp$. This cannot happen for inner products.

## 9.2 Orthonormal Bases

### 9.2.1 The Gram-Schmidt Process

In an inner product space, we can ask for bases that respect the geometry: bases whose vectors are mutually orthogonal and have unit length.

**Definition 9.2.1: Orthonormal basis**

Let $V$ be a finite-dimensional inner product space. A basis $\{v_1, \dots, v_n\}$ of $V$ is **orthonormal** if

$$\langle v_i, v_j\rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Equivalently, $\|v_i\| = 1$ for all $i$, and $v_i \perp v_j$ for $i \neq j$.

**Note:-**

In an orthonormal basis, the inner product takes a simple form: if $v = \sum x_ie_i$ and $w = \sum y_je_j$, then

$$\langle v, w\rangle = \sum_{i,j} x_i\overline{y_j}\langle e_i, e_j\rangle = \sum_i x_i\overline{y_i}.$$

For real spaces, this is just $x^Ty$. For complex spaces with Hermitian inner product, it is $x^*y$.

**Theorem 9.2.1** Existence of orthonormal bases

Every finite-dimensional inner product space has an orthonormal basis.

We give two proofs: one by induction, one constructive (Gram–Schmidt).

***Proof 1 (Induction on dimension):*** Let $\dim V = n$. If $n = 0$, the empty set is an orthonormal basis. For $n \geqslant 1$:

Pick any nonzero $v \in V$ and set $v_1 = v/\|v\|$, so $\|v_1\| = 1$. Let $S = \operatorname{span}(v_1)$. By the orthogonal decomposition theorem, $V = S \oplus S^\perp$, and $\dim S^\perp = n - 1$.

The restriction of $\langle\cdot, \cdot\rangle$ to $S^\perp$ is still an inner product (check positive-definiteness). By induction, $S^\perp$ has an orthonormal basis $\{v_2, \dots, v_n\}$.

Then $\{v_1, v_2, \dots, v_n\}$ is an orthonormal basis of $V$: the $v_i$ are orthonormal in $S^\perp$, and $v_1 \perp v_i$ for $i \geqslant 2$ since $v_i \in S^\perp$. □

***Proof 2 (Gram–Schmidt):*** Start with any basis $\{w_1, \dots, w_n\}$ of $V$. We inductively construct orthonormal vectors $v_1, \dots, v_n$ such that $\operatorname{span}(v_1, \dots, v_k) = \operatorname{span}(w_1, \dots, w_k)$ for each $k$.

**Step 1:** Set $v_1 = w_1/\|w_1\|$.

**Step j:** Given orthonormal $v_1, \dots, v_{j-1}$, define

$$u_j = w_j - \sum_{i=1}^{j-1}\langle w_j, v_i\rangle v_i.$$

This subtracts from $w_j$ its projections onto each $v_i$, leaving a vector orthogonal to $\operatorname{span}(v_1, \dots, v_{j-1})$. Since $w_j \notin \operatorname{span}(w_1, \dots, w_{j-1}) = \operatorname{span}(v_1, \dots, v_{j-1})$ (linear independence), we have $u_j \neq 0$. Set

$$v_j = \frac{u_j}{\|u_j\|}.$$

By construction, $v_j \perp v_i$ for $i < j$, and $\|v_j\| = 1$. The span condition holds since $v_j \in \operatorname{span}(w_1, \dots, w_j)$ and $w_j \in \operatorname{span}(v_1, \dots, v_j)$. ☐

**Corollary 9.2.1** Isomorphism with standard space

Every $n$-dimensional real inner product space is isomorphic to $(\mathbb{R}^n, \text{standard inner product})$ as an inner product space. Similarly, every $n$-dimensional Hermitian inner product space is isomorphic to $(\mathbb{C}^n, \text{standard Hermitian product})$.

***Proof:*** Let $\{v_1, \dots, v_n\}$ be an orthonormal basis. The coordinate map $\varphi : V \to k^n$ sending $v = \sum a_i v_i$ to $(a_1, \dots, a_n)$ is a linear isomorphism. Moreover,

$$\langle v, w \rangle = \langle \sum a_i v_i, \sum b_j v_j \rangle = \sum_{i,j} a_i \overline{b_j} \delta_{ij} = \sum_i a_i \overline{b_i}$$

which is the standard inner product of the coordinate vectors. ☐

**Note:-**

This says that all $n$-dimensional inner product spaces (over the same field) are "the same." The choice of orthonormal basis corresponds to the choice of isomorphism with standard space. Different orthonormal bases give different coordinate systems, all equally valid.

**Example 9.2.1** (Gram–Schmidt in $\mathbb{R}^3$)

Apply Gram–Schmidt to the basis $w_1 = (1,1,0)$, $w_2 = (1,0,1)$, $w_3 = (0,1,1)$ of $\mathbb{R}^3$.
**Step 1:** $\|w_1\| = \sqrt{2}$, so $v_1 = \frac{1}{\sqrt{2}}(1,1,0)$.
**Step 2:** $\langle w_2, v_1 \rangle = \frac{1}{\sqrt{2}}(1+0+0) = \frac{1}{\sqrt{2}}$.

$$u_2 = w_2 - \frac{1}{\sqrt{2}} v_1 = (1,0,1) - \frac{1}{2}(1,1,0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

$\|u_2\|^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$, so $v_2 = \sqrt{\frac{2}{3}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \frac{1}{\sqrt{6}}(1,-1,2)$.
**Step 3:** Compute projections and proceed similarly to get $v_3$.

### 9.2.2 Orthogonal Projections

The orthogonal decomposition $V = S \oplus S^\perp$ gives a canonical way to project onto $S$.

**Definition 9.2.2: Orthogonal projection**

Let $V$ be an inner product space and $S \subseteq V$ a subspace. The **orthogonal projection** onto $S$ is the linear map $\pi_S : V \to V$ defined by: for $v = v_S + v_{S^\perp}$ with $v_S \in S$ and $v_{S^\perp} \in S^\perp$,

$$\pi_S(v) = v_S.$$

**Proposition 9.2.1** Properties of orthogonal projections

Let $\pi_S : V \to V$ be the orthogonal projection onto $S$.

(i) $\operatorname{Im}(\pi_S) = S$ and $\operatorname{Ker}(\pi_S) = S^\perp$.

(ii) $\pi_S^2 = \pi_S$ (i.e., $\pi_S$ is idempotent).

(iii) $\pi_S^* = \pi_S$ (i.e., $\pi_S$ is self-adjoint; see §9.3).

(iv) $\pi_{S^\perp} = \mathrm{id} - \pi_S$.

***Proof:*** (i) follows from the definition. For (ii): if $v = v_S + v_{S^\perp}$, then $\pi_S(v) = v_S \in S$, so $\pi_S(\pi_S(v)) = \pi_S(v_S) = v_S$. (iv) is immediate from $v = \pi_S(v) + \pi_{S^\perp}(v)$. □

**Proposition 9.2.2** Formula in orthonormal basis

If $\{e_1, \ldots, e_k\}$ is an orthonormal basis of $S$, then

$$\pi_S(v) = \sum_{i=1}^{k} \langle v, e_i \rangle e_i.$$

***Proof:*** Let $w = \sum_{i=1}^{k} \langle v, e_i \rangle e_i \in S$. We show $v - w \in S^\perp$: for any $j \leqslant k$,

$$\langle v - w, e_j \rangle = \langle v, e_j \rangle - \sum_i \langle v, e_i \rangle \langle e_i, e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0.$$

Since $\{e_j\}$ spans $S$, we have $v - w \perp S$, so $v - w \in S^\perp$. Thus $w = \pi_S(v)$. □

**Proposition 9.2.3** Best approximation

For $v \in V$ and $S \subseteq V$ a subspace, the projection $\pi_S(v)$ is the unique vector in $S$ minimizing $\|v - s\|$ over $s \in S$.

***Proof:*** For any $s \in S$, write $s = \pi_S(v) + (s - \pi_S(v))$ where $s - \pi_S(v) \in S$. Then:

$$\|v - s\|^2 = \|(v - \pi_S(v)) - (s - \pi_S(v))\|^2.$$

Since $v - \pi_S(v) \in S^\perp$ and $s - \pi_S(v) \in S$, these are orthogonal:

$$\|v - s\|^2 = \|v - \pi_S(v)\|^2 + \|s - \pi_S(v)\|^2 \geqslant \|v - \pi_S(v)\|^2,$$

with equality iff $s = \pi_S(v)$. □

## 9.3 The Adjoint

### 9.3.1 Definition and Existence

Given a linear operator on an inner product space, we can define its "adjoint" by transposing the role of the two arguments in the inner product.

**Definition 9.3.1: Adjoint operator**

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space and $T : V \to V$ a linear operator. The **adjoint** of $T$ is the unique linear operator $T^* : V \to V$ satisfying

$$\langle v, T(w) \rangle = \langle T^*(v), w \rangle$$

for all $v, w \in V$.

**Theorem 9.3.1** Existence and uniqueness of the adjoint

For any linear operator $T$ on a finite-dimensional inner product space, the adjoint $T^*$ exists and is unique.

***Proof (Existence):*** Fix $v \in V$. The map $w \mapsto \langle v, T(w) \rangle$ is a linear functional on $V$.

Since the inner product is nondegenerate, the map $\varphi : V \to V^*$ defined by $\varphi(u) = \langle u, \cdot \rangle$ is an isomorphism. Thus there exists a unique $u \in V$ with $\langle u, w \rangle = \langle v, T(w) \rangle$ for all $w$. Define $T^*(v) = u$.

We must check $T^*$ is linear. For $v_1, v_2 \in V$ and $\lambda \in k$:

$$\langle T^*(v_1 + \lambda v_2), w \rangle = \langle v_1 + \lambda v_2, T(w) \rangle = \langle v_1, T(w) \rangle + \overline{\lambda} \langle v_2, T(w) \rangle$$

$$= \langle T^*(v_1), w \rangle + \overline{\lambda} \langle T^*(v_2), w \rangle = \langle T^*(v_1) + \lambda T^*(v_2), w \rangle.$$

(In the real case, $\overline{\lambda} = \lambda$.) Since this holds for all $w$, we get $T^*(v_1 + \lambda v_2) = T^*(v_1) + \lambda T^*(v_2)$.

**Uniqueness:** If $S$ also satisfies $\langle v, T(w) \rangle = \langle S(v), w \rangle$, then $\langle T^*(v) - S(v), w \rangle = 0$ for all $w$, so $T^*(v) = S(v)$ by nondegeneracy. □

**Note:-**

Alternatively, the adjoint can be described as follows. The inner product gives an isomorphism $\varphi : V \xrightarrow{\sim} V^*$, $\varphi(v) = \langle v, \cdot \rangle$. The transpose $T^t : V^* \to V^*$ is defined by $T^t(\ell) = \ell \circ T$. Then

$$T^* = \varphi^{-1} \circ T^t \circ \varphi.$$

This shows the adjoint is the "conjugate" of the transpose by the inner-product isomorphism.

### 9.3.2 Properties of the Adjoint

**Proposition 9.3.1** Algebraic properties of the adjoint

Let $S, T : V \to V$ be linear operators on an inner product space, and $\lambda \in k$. Then:

(i) $(T^*)^* = T$.

(ii) $(S + T)^* = S^* + T^*$.

(iii) $(\lambda T)^* = \overline{\lambda} T^*$.

(iv) $(ST)^* = T^* S^*$.

(v) $(\mathrm{id})^* = \mathrm{id}$.

***Proof:*** Each follows from manipulating the defining equation $\langle v, T(w) \rangle = \langle T^*(v), w \rangle$.

(i) $\langle v, T^*(w) \rangle = \overline{\langle T^*(w), v \rangle} = \overline{\langle w, T(v) \rangle} = \langle T(v), w \rangle$. So $(T^*)^* = T$.

(iv) $\langle v, (ST)(w) \rangle = \langle v, S(T(w)) \rangle = \langle S^*(v), T(w) \rangle = \langle T^*(S^*(v)), w \rangle = \langle (T^* S^*)(v), w \rangle$. □

**Note:-**

Property (iv) shows that taking adjoints reverses order, just like taking transposes or inverses. The map $T \mapsto T^*$ is a conjugate-linear anti-automorphism of $\mathrm{End}(V)$.

**Matrix of the Adjoint**

**Theorem 9.3.2** Matrix of the adjoint

Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of $V$, and let $M = M(T)$ be the matrix of $T$ in this basis. Then the matrix of $T^*$ is:

$$M(T^*) = \begin{cases} M^T & \text{(real inner product)} \\ M^* = \overline{M}^T & \text{(Hermitian inner product)} \end{cases}$$

where $M^*$ denotes the conjugate transpose.

***Proof:*** The $(i,j)$-entry of $M(T)$ is the coefficient of $e_i$ in $T(e_j)$, which equals $\langle T(e_j), e_i\rangle = \overline{\langle e_i, T(e_j)\rangle}$.
The $(i,j)$-entry of $M(T^*)$ is $\langle T^*(e_j), e_i\rangle$.
Using the defining property:

$$\langle T^*(e_j), e_i\rangle = \langle e_j, T(e_i)\rangle = \overline{\langle T(e_i), e_j\rangle}.$$

The $(j,i)$-entry of $M(T)$ is $\langle T(e_i), e_j\rangle$, so

$$M(T^*)_{ij} = \overline{M(T)_{ji}} = (M^*)_{ij}.$$

□

**Note:-**

In an orthonormal basis, self-adjoint operators ($T^* = T$) correspond to symmetric matrices (real case) or Hermitian matrices (complex case, $M = M^*$).

**Proposition 9.3.2** Kernel and image of the adjoint

For $T : V \to V$:

(i) $\mathrm{Ker}(T^*) = \mathrm{Im}(T)^\perp$.

(ii) $\mathrm{Im}(T^*) = \mathrm{Ker}(T)^\perp$.

***Proof:*** (i) $v \in \mathrm{Ker}(T^*) \iff T^*(v) = 0 \iff \langle T^*(v), w\rangle = 0\ \forall w \iff \langle v, T(w)\rangle = 0\ \forall w \iff v \perp \mathrm{Im}(T)$.
(ii) Apply (i) to $T^*$: $\mathrm{Ker}((T^*)^*) = \mathrm{Im}(T^*)^\perp$, i.e., $\mathrm{Ker}(T) = \mathrm{Im}(T^*)^\perp$. Taking orthogonal complements: $\mathrm{Ker}(T)^\perp = (\mathrm{Im}(T^*)^\perp)^\perp = \mathrm{Im}(T^*)$ (using $(S^\perp)^\perp = S$ in finite dimensions). □

**Corollary 9.3.1** Rank equality

$\mathrm{rank}(T) = \mathrm{rank}(T^*)$.

***Proof:*** $\mathrm{rank}(T^*) = \dim \mathrm{Im}(T^*) = \dim \mathrm{Ker}(T)^\perp = \dim V - \dim \mathrm{Ker}(T) = \mathrm{rank}(T)$. □

## 9.4 Self-Adjoint and Normal Operators

### 9.4.1 The Spectral Theorem (Real Symmetric Case)

We now study operators with special relationships to their adjoints. The most important case is when $T = T^*$.

**Definition 9.4.1: Self-adjoint operator**

An operator $T : V \to V$ on an inner product space is **self-adjoint** (or **Hermitian**) if $T^* = T$, i.e.,

$$\langle v, T(w)\rangle = \langle T(v), w\rangle$$

for all $v, w \in V$.

**Note:-**

In an orthonormal basis, $T$ is self-adjoint iff its matrix is symmetric (real case) or Hermitian (complex case, $M = \overline{M}^T$). Self-adjoint operators need not be invertible: for example, the zero operator is self-adjoint.

**Proposition 9.4.1** Invariance of orthogonal complements

Let $T : V \to V$ be self-adjoint and $S \subseteq V$ a $T$-invariant subspace (i.e., $T(S) \subseteq S$). Then $S^\perp$ is also $T$-invariant.

***Proof:*** Let $v \in S^\perp$. For any $w \in S$, we have $T(w) \in S$ (since $S$ is $T$-invariant), so:

$$\langle T(v), w\rangle = \langle v, T(w)\rangle = 0$$

since $v \perp T(w)$. Thus $T(v) \perp S$, i.e., $T(v) \in S^\perp$. □

**Note:-**

This is the key property that makes self-adjoint operators diagonalizable: once we find an eigenvector, its orthogonal complement is invariant, and we can proceed by induction.

To prove the spectral theorem, we need to establish that self-adjoint operators have real eigenvalues and that eigenvectors exist over $\mathbb{R}$.

**Lemma 9.4.1** Positivity lemma

Let $T$ be self-adjoint on a real inner product space. For any $a > 0$, the operator $T^2 + a \cdot \mathrm{id}$ is invertible.

***Proof:*** We show $\mathrm{Ker}(T^2 + a \cdot \mathrm{id}) = \{0\}$. For $v \neq 0$:

$$\langle (T^2 + a \cdot \mathrm{id})(v), v\rangle = \langle T^2(v), v\rangle + a\langle v, v\rangle = \langle T(v), T(v)\rangle + a\|v\|^2 = \|T(v)\|^2 + a\|v\|^2 > 0.$$

So $(T^2 + a \cdot \mathrm{id})(v) \neq 0$ for $v \neq 0$, i.e., $T^2 + a \cdot \mathrm{id}$ is injective, hence invertible. □

**Corollary 9.4.1** Quadratics without real roots

If $p(x) = x^2 + bx + c \in \mathbb{R}^x$ has no real roots (i.e., $b^2 - 4c < 0$), and $T$ is self-adjoint, then $p(T) = T^2 + bT + c \cdot \mathrm{id}$ is invertible.

***Proof:*** Complete the square: $T^2 + bT + c \cdot \mathrm{id} = (T + \frac{b}{2}\mathrm{id})^2 + (c - \frac{b^2}{4})\mathrm{id}$.

Since $b^2 - 4c < 0$, we have $a = c - \frac{b^2}{4} > 0$. The operator $S = T + \frac{b}{2}\mathrm{id}$ is self-adjoint (check: $S^* = T^* + \frac{b}{2}\mathrm{id} = T + \frac{b}{2}\mathrm{id} = S$). By the lemma, $S^2 + a \cdot \mathrm{id}$ is invertible. □

**Theorem 9.4.1** Spectral theorem (real self-adjoint)

Let $T : V \to V$ be a self-adjoint operator on a finite-dimensional real inner product space. Then:

(i) All eigenvalues of $T$ are real.

(ii) $T$ is diagonalizable.

(iii) $V$ has an orthonormal basis consisting of eigenvectors of $T$.

***Proof:*** We first show that $T$ has an eigenvector. Since $V$ is finite-dimensional, choose any nonzero $v \in V$. The vectors $v, T(v), T^2(v), \ldots, T^n(v)$ (where $n = \dim V$) are linearly dependent, so there exists a nonzero polynomial $p \in \mathbb{R}^x$ with $p(T)(v) = 0$.

Factor $p$ over $\mathbb{R}$: since $\mathbb{R}^x$ has unique factorization, we can write

$$p(x) = c \prod_i (x - \lambda_i) \prod_j (x^2 + b_j x + c_j)$$

where the linear factors have real roots and the quadratic factors are irreducible (no real roots, i.e., $b_j^2 - 4c_j < 0$).

By the corollary, each $(T^2 + b_j T + c_j \cdot \mathrm{id})$ is invertible. So

$$0 = p(T)(v) = c \prod_i (T - \lambda_i \cdot \mathrm{id}) \prod_j (T^2 + b_j T + c_j \cdot \mathrm{id})(v).$$

Since the quadratic factors are invertible, we can remove them:

$$\prod_i (T - \lambda_i \cdot \mathrm{id})(v') = 0$$

for some nonzero $v'$.

At least one factor $(T - \lambda_i \cdot \mathrm{id})$ must have nontrivial kernel (otherwise their product would be invertible). Thus $T$ has an eigenvalue $\lambda_i \in \mathbb{R}$.

**(i) Eigenvalues are real:** This is automatic since we found the eigenvalue in $\mathbb{R}$.

**(iii) Orthonormal eigenbasis:** We proceed by induction on $\dim V$.

Base case: $\dim V = 1$. Any nonzero vector is an eigenvector; normalize it.

Inductive step: Let $v_1$ be an eigenvector with $T(v_1) = \lambda_1 v_1$, normalized so $\|v_1\| = 1$. Let $S = \mathrm{span}(v_1)$. Since $T(S) \subseteq S$, the orthogonal complement $S^\perp$ is $T$-invariant.

The restriction $T|_{S^\perp} : S^\perp \to S^\perp$ is self-adjoint (the inner product on $S^\perp$ is the restriction of the inner product on $V$). By induction, $S^\perp$ has an orthonormal eigenbasis $\{v_2, \ldots, v_n\}$.

Then $\{v_1, v_2, \ldots, v_n\}$ is an orthonormal eigenbasis of $V$: the $v_i$ are orthonormal, and each is an eigenvector of $T$.

**(ii) Diagonalizable:** In an eigenbasis, $T$ has a diagonal matrix. ☐

**Note:-**

The spectral theorem says that self-adjoint operators on real inner product spaces are "as nice as possible": they are diagonalizable, with real eigenvalues, and the eigenspaces are mutually orthogonal. The matrix in an orthonormal eigenbasis is $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

**Corollary 9.4.2** Eigenvectors for distinct eigenvalues are orthogonal

If $T$ is self-adjoint and $T(v) = \lambda v$, $T(w) = \mu w$ with $\lambda \neq \mu$, then $v \perp w$.

***Proof:*** $\lambda\langle v, w\rangle = \langle \lambda v, w\rangle = \langle T(v), w\rangle = \langle v, T(w)\rangle = \langle v, \mu w\rangle = \mu\langle v, w\rangle$. So $(\lambda - \mu)\langle v, w\rangle = 0$, and $\lambda \neq \mu$ implies $\langle v, w\rangle = 0$. ☐

### Skew-Adjoint Operators

The "opposite" of self-adjoint is skew-adjoint.

**Definition 9.4.2: Skew-adjoint operator**

An operator $T : V \to V$ on an inner product space is **skew-adjoint** (or **anti-self-adjoint**) if $T^* = -T$, i.e.,

$$\langle v, T(w) \rangle = -\langle T(v), w \rangle$$

for all $v, w \in V$.

**Note:-**

In an orthonormal basis, $T$ is skew-adjoint iff its matrix $M$ satisfies $M^T = -M$ (real case) or $M^* = -M$ (complex case). Such matrices are called **skew-symmetric** or **skew-Hermitian**.

**Proposition 9.4.2** Properties of skew-adjoint operators

Let $T : V \to V$ be skew-adjoint on a real inner product space.

(i) $\langle T(v), v \rangle = 0$ for all $v \in V$.

(ii) If $\lambda$ is an eigenvalue of $T$ (in $\mathbb{C}$), then $\lambda$ is purely imaginary.

(iii) $I + T$ is invertible.

***Proof:*** (i) $\langle T(v), v \rangle = -\langle v, T(v) \rangle = -\overline{\langle T(v), v \rangle}$. Over $\mathbb{R}$, this gives $2\langle T(v), v \rangle = 0$.

(ii) If $T(v) = \lambda v$ with $v \neq 0$, then $\lambda \langle v, v \rangle = \langle T(v), v \rangle = 0$ by (i), so $\lambda = 0$. More generally, working in the complexification, if $\lambda = a + bi$ is an eigenvalue, the skew-adjoint condition forces $a = 0$.

(iii) Suppose $(I + T)(v) = 0$, i.e., $T(v) = -v$. Then $-\|v\|^2 = \langle T(v), v \rangle = 0$ by (i), so $v = 0$. Thus $I + T$ is injective, hence invertible. $\square$

**Proposition 9.4.3** Cayley transform

Let $T$ be skew-adjoint on a real inner product space. Then $S = (I - T)(I + T)^{-1}$ is orthogonal (i.e., $S^* = S^{-1}$).

***Proof:*** Since $T^* = -T$, we have $(I + T)^* = I - T$ and $(I - T)^* = I + T$. Also, $I - T$ and $I + T$ commute (both are polynomials in $T$).

Then:

$$S^* = ((I + T)^{-1})^*(I - T)^* = ((I + T)^*)^{-1}(I - T)^* = (I - T)^{-1}(I + T).$$

And:

$$S^{-1} = (I + T)(I - T)^{-1}.$$

Since $I - T$ and $I + T$ commute, $(I - T)^{-1}(I + T) = (I + T)(I - T)^{-1}$. Thus $S^* = S^{-1}$. $\square$

### 9.4.2 Normal Operators on Complex Spaces

Over $\mathbb{C}$, the spectral theorem extends to a larger class of operators: those that commute with their adjoint.

**Definition 9.4.3: Normal operator**

An operator $T : V \to V$ on an inner product space is **normal** if

$$TT^* = T^*T.$$

**Example 9.4.1** (Classes of normal operators)

- Self-adjoint operators ($T^* = T$) are normal: $TT^* = T^2 = T^*T$.

- Skew-adjoint operators ($T^* = -T$) are normal: $TT^* = -T^2 = T^*T$.
- Unitary operators ($T^* = T^{-1}$) are normal: $TT^* = I = T^*T$.

**Lemma 9.4.2** Characterization of normality
An operator $T$ is normal if and only if $\|T(v)\| = \|T^*(v)\|$ for all $v \in V$.

***Proof:*** $\|T(v)\|^2 = \langle T(v), T(v)\rangle = \langle T^*T(v), v\rangle$ and $\|T^*(v)\|^2 = \langle T^*(v), T^*(v)\rangle = \langle TT^*(v), v\rangle$.
So $\|T(v)\| = \|T^*(v)\|$ for all $v$ iff $\langle (T^*T - TT^*)(v), v\rangle = 0$ for all $v$.
For self-adjoint operators (and $T^*T - TT^*$ is self-adjoint), this implies $T^*T - TT^* = 0$. Indeed, if $S$ is self-adjoint with $\langle S(v), v\rangle = 0$ for all $v$, then for any $v, w$:

$$0 = \langle S(v+w), v+w\rangle = \langle S(v), w\rangle + \langle S(w), v\rangle = 2\,\mathrm{Re}\langle S(v), w\rangle,$$

and similarly with $v + iw$ gives the imaginary part, so $\langle S(v), w\rangle = 0$ for all $v, w$, hence $S = 0$. □

**Proposition 9.4.4** Normal operators and eigenspaces
Let $T$ be normal.

(i) If $T(v) = \lambda v$, then $T^*(v) = \bar{\lambda} v$.

(ii) Eigenvectors for distinct eigenvalues are orthogonal.

(iii) $\mathrm{Ker}(T - \lambda I) = \mathrm{Ker}(T^* - \bar{\lambda} I)$.

***Proof:*** (i) Note that $T - \lambda I$ is normal (check: $(T - \lambda I)(T - \lambda I)^* = (T - \lambda I)(T^* - \bar{\lambda} I)$, and these commute since $T$ and $T^*$ commute).
By the lemma, $\|T(v) - \lambda v\| = \|T^*(v) - \bar{\lambda} v\|$. If $T(v) = \lambda v$, the left side is zero, so $T^*(v) = \bar{\lambda} v$.
(ii) If $T(v) = \lambda v$ and $T(w) = \mu w$ with $\lambda \neq \mu$:

$$\lambda\langle v, w\rangle = \langle T(v), w\rangle = \langle v, T^*(w)\rangle = \langle v, \bar{\mu} w\rangle = \mu\langle v, w\rangle.$$

So $(\lambda - \mu)\langle v, w\rangle = 0$, giving $v \perp w$.
(iii) Follows from (i): $v \in \mathrm{Ker}(T - \lambda I) \iff T(v) = \lambda v \iff T^*(v) = \bar{\lambda} v \iff v \in \mathrm{Ker}(T^* - \bar{\lambda} I)$. □

**Theorem 9.4.2** Complex spectral theorem
Let $T : V \to V$ be a normal operator on a finite-dimensional complex inner product space (with Hermitian inner product). Then $V$ has an orthonormal basis consisting of eigenvectors of $T$.

***Proof:*** Since $\mathbb{C}$ is algebraically closed, $T$ has an eigenvalue $\lambda$. Let $v_1$ be a unit eigenvector with $T(v_1) = \lambda v_1$.
Let $S = \mathrm{span}(v_1)$. We show $S^\perp$ is $T$-invariant. By part (i) of the proposition, $T^*(v_1) = \bar{\lambda} v_1$, so $S$ is also $T^*$-invariant.
For $w \in S^\perp$ and any $s \in S$:

$$\langle T(w), s\rangle = \langle w, T^*(s)\rangle.$$

Since $T^*(s) \in S$ and $w \perp S$, we get $\langle T(w), s\rangle = 0$. Thus $T(w) \perp S$, i.e., $T(w) \in S^\perp$.
The restriction $T|_{S^\perp}$ is normal (the inner product restricts, and $(T|_{S^\perp})^* = T^*|_{S^\perp}$). By induction on dimension, $S^\perp$ has an orthonormal eigenbasis $\{v_2, \ldots, v_n\}$. Then $\{v_1, \ldots, v_n\}$ is an orthonormal eigenbasis of $V$. □

**Note:-**

The complex spectral theorem is stronger than the real one: any normal operator (not just self-adjoint) is diagonalizable. The price is working over $\mathbb{C}$ with Hermitian inner products.
For self-adjoint $T$, the eigenvalues are real. For skew-adjoint $T$, they are purely imaginary. For unitary $T$, they lie on the unit circle $|\lambda| = 1$.

## 9.5 Orthogonal and Unitary Operators

### 9.5.1 Definitions and Characterizations

Orthogonal (real) and unitary (complex) operators are the isometries of inner product spaces: they preserve lengths and angles.

**Definition 9.5.1: Orthogonal operator**

Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space. An operator $T : V \to V$ is **orthogonal** if

$$\langle T(u), T(v)\rangle = \langle u, v\rangle$$

for all $u, v \in V$.

**Definition 9.5.2: Unitary operator**

Let $(V, H)$ be a complex inner product space (with Hermitian inner product $H$). An operator $T : V \to V$ is **unitary** if

$$H(T(u), T(v)) = H(u, v)$$

for all $u, v \in V$.

**Note:-**

We use "orthogonal" over $\mathbb{R}$ and "unitary" over $\mathbb{C}$, though the definitions are parallel. Some authors use "orthogonal" for both.

**Theorem 9.5.1** Characterizations of orthogonal/unitary operators

For an operator $T : V \to V$ on a finite-dimensional inner product space, the following are equivalent:

(i) $T$ is orthogonal (resp. unitary).

(ii) $T^*T = I$ (equivalently, $T^* = T^{-1}$).

(iii) $TT^* = I$.

(iv) $\|T(v)\| = \|v\|$ for all $v \in V$.

(v) $T$ maps some orthonormal basis to an orthonormal basis.

(vi) $T$ maps every orthonormal basis to an orthonormal basis.

***Proof:*** (i) $\Rightarrow$ (ii): $\langle T^*T(u), v\rangle = \langle T(u), T(v)\rangle = \langle u, v\rangle = \langle I(u), v\rangle$ for all $u, v$. By nondegeneracy, $T^*T = I$.
(ii) $\Rightarrow$ (iii): $T^*T = I$ implies $T$ is invertible with $T^{-1} = T^*$. Then $TT^* = T(T^{-1}) = I$.
(ii) $\Rightarrow$ (iv): $\|T(v)\|^2 = \langle T(v), T(v)\rangle = \langle T^*T(v), v\rangle = \langle v, v\rangle = \|v\|^2$.
(iv) $\Rightarrow$ (i): Use the polarization identity. Over $\mathbb{R}$:

$$\langle u, v\rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2).$$

If $\mathsf{T}$ preserves norms: $\langle \mathsf{T}(u), \mathsf{T}(v)\rangle = \frac{1}{4}(\|\mathsf{T}(u)+\mathsf{T}(v)\|^2 - \|\mathsf{T}(u)-\mathsf{T}(v)\|^2) = \frac{1}{4}(\|\mathsf{T}(u+v)\|^2 - \|\mathsf{T}(u-v)\|^2) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) = \langle u, v\rangle$.

(i) $\Rightarrow$ (vi): If $\{e_1, \dots, e_n\}$ is orthonormal, then $\langle \mathsf{T}(e_i), \mathsf{T}(e_j)\rangle = \langle e_i, e_j\rangle = \delta_{ij}$.

(vi) $\Rightarrow$ (v): Trivial.

(v) $\Rightarrow$ (ii): Let $\{e_i\}$ be orthonormal with $\{\mathsf{T}(e_i)\}$ orthonormal. For any $v = \sum a_i e_i$:

$$\mathsf{T}^*\mathsf{T}(v) = \mathsf{T}^*(\sum a_i \mathsf{T}(e_i)).$$

Since $\{\mathsf{T}(e_i)\}$ is orthonormal, $\langle \mathsf{T}(e_i), \mathsf{T}(e_j)\rangle = \delta_{ij}$, so $\mathsf{T}^*(\mathsf{T}(e_i)) = e_i$ (the dual basis property in an orthonormal basis). Thus $\mathsf{T}^*\mathsf{T}(v) = \sum a_i e_i = v$. $\square$

**Corollary 9.5.1** Orthogonal/unitary operators are invertible

If $\mathsf{T}$ is orthogonal or unitary, then $\mathsf{T}^{-1} = \mathsf{T}^*$, which is also orthogonal/unitary.

**Corollary 9.5.2** Orthogonal/unitary operators form a group

The set of orthogonal (resp. unitary) operators on $V$ forms a subgroup of $\mathrm{GL}(V)$.

***Proof:*** Identity: $I^* = I$, so $I^*I = I$. Closure: if $S^*S = I$ and $\mathsf{T}^*\mathsf{T} = I$, then $(S\mathsf{T})^*(S\mathsf{T}) = \mathsf{T}^*S^*S\mathsf{T} = \mathsf{T}^*\mathsf{T} = I$. Inverses: $(\mathsf{T}^{-1})^* = (\mathsf{T}^*)^* = \mathsf{T}$, so $(\mathsf{T}^{-1})^*(\mathsf{T}^{-1}) = \mathsf{T}\mathsf{T}^{-1} = I$. $\square$

**Matrix Characterization**

**Proposition 9.5.1** Orthogonal and unitary matrices

Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of $V$, and let $M$ be the matrix of $\mathsf{T}$ in this basis.

(i) (Real case) $\mathsf{T}$ is orthogonal iff $M^{\mathsf{T}}M = I$, i.e., $M$ is an **orthogonal matrix**.

(ii) (Complex case) $\mathsf{T}$ is unitary iff $M^*M = I$, i.e., $M$ is a **unitary matrix**.

Equivalently: the columns of $M$ form an orthonormal set (with respect to the standard inner product on $k^n$).

***Proof:*** In an orthonormal basis, $M(\mathsf{T}^*) = M(\mathsf{T})^*$. So $\mathsf{T}^*\mathsf{T} = I$ iff $M^*M = I$.

The columns of $M$ are $Me_1, \dots, Me_n$ (standard basis vectors). The $(i, j)$-entry of $M^*M$ is $(M^*M)_{ij} = \sum_k \overline{M_{ki}}M_{kj} = \langle \mathrm{col}_i(M), \mathrm{col}_j(M)\rangle$. So $M^*M = I$ iff columns are orthonormal. $\square$

**Corollary 9.5.3** Rows of orthogonal/unitary matrices

If $M$ is orthogonal (resp. unitary), then the rows of $M$ also form an orthonormal set.

***Proof:*** $M^*M = I$ implies $M$ is invertible with $M^{-1} = M^*$. Thus $MM^* = I$, which says the columns of $M^*$ ($=$ rows of $M$) are orthonormal. $\square$

**Proposition 9.5.2** Determinant of orthogonal/unitary operators

If $\mathsf{T}$ is orthogonal or unitary, then $|\det(\mathsf{T})| = 1$.
For orthogonal $\mathsf{T}$ (real case), $\det(\mathsf{T}) = \pm 1$.

***Proof:*** $\det(\mathsf{T}^*\mathsf{T}) = \det(I) = 1$. But $\det(\mathsf{T}^*\mathsf{T}) = \det(\mathsf{T}^*)\det(\mathsf{T}) = \overline{\det(\mathsf{T})}\det(\mathsf{T}) = |\det(\mathsf{T})|^2$.

Over $\mathbb{R}$, $\det(\mathsf{T}^{\mathsf{T}}) = \det(\mathsf{T})$, so $\det(\mathsf{T})^2 = 1$, giving $\det(\mathsf{T}) = \pm 1$. $\square$

### 9.5.2 Reflections

A particularly simple class of orthogonal operators are reflections across hyperplanes.

**Definition 9.5.3: Reflection**

Let $V$ be a real inner product space. A **reflection** is an orthogonal operator $P : V \to V$ such that $P^2 = I$ and $\dim \operatorname{Ker}(P + I) = 1$.

**Note:-**

The condition $P^2 = I$ says $P$ is an involution. Since $P$ is orthogonal, it is also self-adjoint ($P^* = P^{-1} = P$). The eigenvalues of $P$ are $\pm 1$ (since $P^2 = I$), and $\dim \operatorname{Ker}(P + I) = 1$ means the $-1$-eigenspace is 1-dimensional.

**Proposition 9.5.3** Formula for reflections

Every reflection can be written as

$$P(x) = x - 2\langle x, v\rangle v$$

where $v \in V$ with $\|v\| = 1$. Geometrically, $P$ reflects across the hyperplane $v^\perp$.

***Proof:*** Define $P(x) = x - 2\langle x, v\rangle v$. We verify $P$ is a reflection.

**Linear:** Clear from the formula.

**Orthogonal:**

$$\begin{aligned}\langle P(x), P(y)\rangle &= \langle x - 2\langle x, v\rangle v, y - 2\langle y, v\rangle v\rangle \\ &= \langle x, y\rangle - 2\langle x, v\rangle\langle v, y\rangle - 2\langle y, v\rangle\langle x, v\rangle + 4\langle x, v\rangle\langle y, v\rangle\|v\|^2 \\ &= \langle x, y\rangle - 4\langle x, v\rangle\langle y, v\rangle + 4\langle x, v\rangle\langle y, v\rangle = \langle x, y\rangle.\end{aligned}$$

**Involution:** $P(P(x)) = P(x) - 2\langle P(x), v\rangle v = x - 2\langle x, v\rangle v - 2(\langle x, v\rangle - 2\langle x, v\rangle)v = x$.

**Eigenspaces:** $P(v) = v - 2\|v\|^2 v = -v$, so $v$ spans the $-1$-eigenspace. If $w \perp v$, then $P(w) = w$, so $v^\perp$ is the $+1$-eigenspace.

Conversely, any reflection $P$ has $\operatorname{Ker}(P + I)$ one-dimensional. Let $v$ span this kernel with $\|v\| = 1$. Then $P(v) = -v$, and on $v^\perp$, $P$ acts as $+I$ (since $P$ is diagonalizable with eigenvalues $\pm 1$, and the $-1$-eigenspace is spanned by $v$). For $x = \langle x, v\rangle v + (x - \langle x, v\rangle v)$ with the second term in $v^\perp$:

$$P(x) = -\langle x, v\rangle v + (x - \langle x, v\rangle v) = x - 2\langle x, v\rangle v. \qquad \square$$

**Proposition 9.5.4** Reflections map equal-length vectors

Given any two vectors $x, y \in V$ with $\|x\| = \|y\|$, there exists a reflection $P$ with $P(x) = y$.

***Proof:*** If $x = y$, take any reflection fixing $x$ (e.g., reflect across $x^\perp$ if $x \neq 0$, or the identity hyperplane if $x = 0$... but then $P$ wouldn't be a reflection. Actually if $x = y = 0$, any reflection works.)

Assume $x \neq y$. Let $v = \frac{x-y}{\|x-y\|}$ and $P(z) = z - 2\langle z, v\rangle v$.

Compute:

$$P(x) = x - 2\langle x, v\rangle v = x - 2\frac{\langle x, x - y\rangle}{\|x - y\|^2}(x - y).$$

Since $\|x\| = \|y\|$, we have $\langle x, x\rangle = \langle y, y\rangle$, so

$$\langle x, x - y\rangle = \|x\|^2 - \langle x, y\rangle = \frac{1}{2}(\|x\|^2 + \|y\|^2 - 2\langle x, y\rangle) = \frac{1}{2}\|x - y\|^2.$$

Thus $P(x) = x - (x - y) = y$. $\square$

**Theorem 9.5.2** Orthogonal operators as products of reflections

Every orthogonal operator on $\mathbb{R}^n$ is a product of at most $n$ reflections.

***Proof:*** Induction on $n$. For $n = 0$, the identity is the empty product.

For $n \geqslant 1$: Let $T$ be orthogonal. Pick a unit vector $e_1$. Then $\|T(e_1)\| = \|e_1\| = 1$.

By the previous proposition, there exists a reflection $P_1$ with $P_1(T(e_1)) = e_1$. Let $T_1 = P_1T$. Then $T_1$ is orthogonal and $T_1(e_1) = e_1$.

Since $T_1(e_1) = e_1$, the subspace $e_1^\perp$ is $T_1$-invariant (if $w \perp e_1$, then $\langle T_1(w), e_1\rangle = \langle w, T_1^*(e_1)\rangle = \langle w, T_1^{-1}(e_1)\rangle = \langle w, e_1\rangle = 0$).

The restriction $T_1|_{e_1^\perp}$ is orthogonal on the $(n-1)$-dimensional space $e_1^\perp$. By induction, $T_1|_{e_1^\perp} = Q_{n-1} \cdots Q_2$ for reflections $Q_i$ on $e_1^\perp$.

Extend each $Q_i$ to $V$ by setting $\tilde{Q}_i(e_1) = e_1$ and $\tilde{Q}_i|_{e_1^\perp} = Q_i$. Then $\tilde{Q}_i$ is a reflection on $V$ (the $-1$-eigenspace of $Q_i$ in $e_1^\perp$ becomes the $-1$-eigenspace of $\tilde{Q}_i$ in $V$).

We have $T_1 = \tilde{Q}_{n-1} \cdots \tilde{Q}_2$, so $P_1T = \tilde{Q}_{n-1} \cdots \tilde{Q}_2$, giving

$$T = P_1^{-1}\tilde{Q}_{n-1} \cdots \tilde{Q}_2 = P_1\tilde{Q}_{n-1} \cdots \tilde{Q}_2$$

(since $P_1^2 = I$). This is a product of at most $1 + (n-1) = n$ reflections. $\square$

### 9.5.3 The Classical Matrix Groups

The orthogonal and unitary operators form important matrix groups.

**Definition 9.5.4: Orthogonal and special orthogonal groups**

- $O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I\}$, the **orthogonal group**.
- $SO(n) = \{A \in O(n) : \det(A) = 1\}$, the **special orthogonal group**.

**Definition 9.5.5: Unitary and special unitary groups**

- $U(n) = \{A \in GL_n(\mathbb{C}) : A^*A = I\}$, the **unitary group**.
- $SU(n) = \{A \in U(n) : \det(A) = 1\}$, the **special unitary group**.

**Proposition 9.5.5** Index of $SO(n)$ in $O(n)$

$SO(n)$ is a normal subgroup of $O(n)$ of index 2. We have an exact sequence

$$1 \to SO(n) \to O(n) \xrightarrow{\det} \{\pm 1\} \to 1.$$

***Proof:*** The determinant map $\det : O(n) \to \{\pm 1\}$ is a surjective homomorphism (surjective since $\det(I) = 1$ and $\det(-I) = (-1)^n$, or use a reflection with $\det = -1$). Its kernel is $SO(n)$. $\square$

**Note:-**

Elements of $SO(n)$ are called **rotations** or **proper orthogonal transformations**. They preserve orientation. Elements of $O(n) \setminus SO(n)$ reverse orientation.

**Example 9.5.1** (Low-dimensional cases)

- $O(1) = \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$, $SO(1) = \{1\}$.

- $SO(2)$ consists of rotation matrices $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Thus $SO(2) \cong S^1$ (the circle group) via $\theta \mapsto e^{i\theta}$.
- $U(1) = \{z \in \mathbb{C} : |z| = 1\} \cong S^1$.

**Theorem 9.5.3** Structure of orthogonal operators

Let $T \in O(V)$ where $V$ is a finite-dimensional real inner product space. Then $V$ decomposes as an orthogonal direct sum

$$V = V_1 \oplus V_{-1} \oplus W_1 \oplus \cdots \oplus W_k$$

where $V_1 = \operatorname{Ker}(T - I)$ is the $+1$-eigenspace, $V_{-1} = \operatorname{Ker}(T + I)$ is the $-1$-eigenspace, and each $W_j$ is a 2-dimensional $T$-invariant subspace on which $T$ acts as rotation by angle $\theta_j \neq 0, \pi$.

***Proof:*** The characteristic polynomial of $T$ has real coefficients. Its roots (possibly complex) come in conjugate pairs. If $\lambda = e^{i\theta}$ is an eigenvalue with $|\lambda| = 1$ and $\lambda \notin \mathbb{R}$, then $\bar{\lambda} = e^{-i\theta}$ is also an eigenvalue.

Let $p(x) = (x - e^{i\theta})(x - e^{-i\theta}) = x^2 - 2\cos\theta \cdot x + 1$, an irreducible quadratic over $\mathbb{R}$. By the real Jordan form theory (or direct argument), $\operatorname{Ker}(p(T))$ is an even-dimensional $T$-invariant subspace.

Decompose $V$ into generalized eigenspaces for real eigenvalues ($\pm 1$) and for conjugate pairs of complex eigenvalues. Since $T$ is orthogonal and hence normal, it is diagonalizable over $\mathbb{C}$, so generalized eigenspaces are actual eigenspaces.

For the conjugate pair $e^{\pm i\theta}$, the real subspace $W = \operatorname{Ker}((T - e^{i\theta} I)(T - e^{-i\theta} I)) \cap V$ is $T$-invariant. Choose a unit eigenvector $v \in W_{\mathbb{C}}$ for $e^{i\theta}$. Then $v + \bar{v}$ and $i(v - \bar{v})$ are real vectors spanning a 2-dimensional $T$-invariant subspace on which $T$ acts as rotation by $\theta$.

Repeating this decomposition and using orthogonality of distinct eigenspaces gives the result. □

**Note:-**

In matrix form, choosing an orthonormal basis adapted to this decomposition, $T$ is block-diagonal:

$$T \sim \begin{pmatrix} I_a & & & & \\ & -I_b & & & \\ & & R_{\theta_1} & & \\ & & & \ddots & \\ & & & & R_{\theta_k} \end{pmatrix}$$

where $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.
This is the "real canonical form" for orthogonal matrices.

### Hermitian Forms and Real Forms

When a complex vector space is viewed as a real vector space, a Hermitian form decomposes into real and imaginary parts with specific properties.

**Proposition 9.5.6** Decomposition of Hermitian forms

Let $V$ be a complex vector space with Hermitian form $H$, and let $W$ denote $V$ viewed as a real vector space (of twice the dimension). Define $G, B : W \times W \to \mathbb{R}$ by

$$G(u, v) = \operatorname{Re} H(u, v), \qquad B(u, v) = \operatorname{Im} H(u, v).$$

Let $J : W \to W$ be scalar multiplication by $i$. Then:

(i) $G$ is a symmetric bilinear form on $W$.

(ii) $B$ is a skew-symmetric bilinear form on $W$.

(iii) $G(u, v) = B(u, Jv)$ for all $u, v \in W$.

(iv) $H$ is nondegenerate iff $G$ is nondegenerate iff $B$ is nondegenerate.

***Proof:*** (i) $G$ is $\mathbb{R}$-bilinear since $H$ is sesquilinear and Re is $\mathbb{R}$-linear. For symmetry:

$$G(v, u) = \operatorname{Re} H(v, u) = \operatorname{Re} \overline{H(u, v)} = \operatorname{Re} H(u, v) = G(u, v).$$

(ii) Similarly, $B(v, u) = \operatorname{Im} H(v, u) = \operatorname{Im} \overline{H(u, v)} = -\operatorname{Im} H(u, v) = -B(u, v)$.

(iii) $B(u, Jv) = \operatorname{Im} H(u, iv) = \operatorname{Im}(\bar{i}H(u, v)) = \operatorname{Im}(-iH(u, v)) = \operatorname{Re} H(u, v) = G(u, v)$.

(iv) Suppose $H$ is nondegenerate and $G(u, v) = 0$ for all $v$. Then $B(u, Jv) = G(u, v) = 0$ for all $v$, so $B(u, w) = 0$ for all $w$ (since $J$ is surjective). Thus $H(u, v) = G(u, v) + iB(u, v) = 0$ for all $v$, giving $u = 0$. The other implications are similar. $\square$

**Note:-**

A linear operator $T : W \to W$ is $\mathbb{C}$-linear (i.e., $T \circ J = J \circ T$) iff it comes from a $\mathbb{C}$-linear operator on $V$. The proposition implies: for nondegenerate $H$, any two of the following imply the third:

(i) $T$ preserves $G$.

(ii) $T$ preserves $B$.

(iii) $T$ is $\mathbb{C}$-linear.

This gives the relation $U(n) = O(2n) \cap GL_n(\mathbb{C}) = Sp(2n, \mathbb{R}) \cap GL_n(\mathbb{C}) = O(2n) \cap Sp(2n, \mathbb{R})$.

**Exercise 9.5.1** Inner product spaces exercises

(1) Let $V \subset \mathbb{R}^x$ be the space of polynomials of degree at most 2, with inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$. Apply Gram–Schmidt to the basis $\{1, x, x^2\}$ to find an orthonormal basis.

(2) Let $T : V \to V$ be self-adjoint on a real inner product space, and suppose $v \in V$ with $\|v\| = 1$ maximizes the Rayleigh quotient $R(w) = \langle T(w), w\rangle/\|w\|^2$ among unit vectors. Prove that $v$ is an eigenvector of $T$. (Do not use the spectral theorem.)

(3) Let $T : V \to V$ be an invertible linear operator on a finite-dimensional real inner product space. Show that $T$ maps the unit sphere $S = \{v : \|v\| = 1\}$ to an ellipsoid. (Hint: consider $T^*T$.)

(4) For an invertible operator $T$, define the **dilatation** $K(T) = \frac{\sup_{\|v\|=1}\|Tv\|}{\inf_{\|v\|=1}\|Tv\|}$. Show that $K(T)^2$ equals the ratio of the largest to smallest eigenvalues of $T^*T$.

(5) Let $A$ be the matrix of a (symmetric positive-definite) inner product on $\mathbb{R}^n$ with respect to the standard basis, so the inner product is $\langle x, y\rangle = x^TAy$. For $n = 2$, prove that the largest entry of $A$ occurs on the diagonal.

Solution

**(1)** Start with $w_1 = 1$, $w_2 = x$, $w_3 = x^2$.
$\langle w_1, w_1\rangle = \int_0^1 1\,dx = 1$, so $v_1 = 1$.
$\langle w_2, v_1\rangle = \int_0^1 x\,dx = \frac{1}{2}$. $u_2 = x - \frac{1}{2}\cdot 1 = x - \frac{1}{2}$. $\|u_2\|^2 = \int_0^1 (x-\frac{1}{2})^2 dx = \int_0^1 (x^2 - x + \frac{1}{4})dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}$.
So $v_2 = \sqrt{12}(x - \frac{1}{2}) = 2\sqrt{3}(x - \frac{1}{2})$.
$\langle w_3, v_1\rangle = \int_0^1 x^2 dx = \frac{1}{3}$. $\langle w_3, v_2\rangle = 2\sqrt{3}\int_0^1 x^2(x - \frac{1}{2})dx = 2\sqrt{3}(\frac{1}{4} - \frac{1}{6}) = 2\sqrt{3} \cdot \frac{1}{12} = \frac{\sqrt{3}}{6}$. $u_3 = x^2 - \frac{1}{3} - \frac{\sqrt{3}}{6} \cdot 2\sqrt{3}(x - \frac{1}{2}) = x^2 - \frac{1}{3} - (x - \frac{1}{2}) = x^2 - x + \frac{1}{6}$. $\|u_3\|^2 = \int_0^1 (x^2 - x + \frac{1}{6})^2 dx = \frac{1}{180}$. So $v_3 = \sqrt{180}(x^2 - x + \frac{1}{6}) = 6\sqrt{5}(x^2 - x + \frac{1}{6})$.

The orthonormal basis is $\{1, 2\sqrt{3}(x-\frac{1}{2}), 6\sqrt{5}(x^2-x+\frac{1}{6})\}$.
**(2)** Let $f(w) = \langle T(w), w\rangle/\|w\|^2$. For $u \perp v$ and $t \in \mathbb{R}$ small, consider $w_t = v + tu$. Since $\|v\| = 1$ and $u \perp v$:

$$\|w_t\|^2 = 1 + t^2\|u\|^2.$$

$$\langle T(w_t), w_t\rangle = \langle T(v), v\rangle + t(\langle T(v), u\rangle + \langle T(u), v\rangle) + t^2\langle T(u), u\rangle.$$

Since $T$ is self-adjoint, $\langle T(v), u\rangle = \langle v, T(u)\rangle = \overline{\langle T(u), v\rangle} = \langle T(u), v\rangle$ (real), so:

$$\langle T(w_t), w_t\rangle = \langle T(v), v\rangle + 2t\langle T(v), u\rangle + t^2\langle T(u), u\rangle.$$

Thus:

$$f(w_t) = \frac{\langle T(v), v\rangle + 2t\langle T(v), u\rangle + O(t^2)}{1 + t^2\|u\|^2}.$$

At $t = 0$, $f(v) = \langle T(v), v\rangle$ is a maximum, so $\frac{d}{dt}|_{t=0} f(w_t) = 0$.
Differentiating: $\frac{d}{dt}|_{t=0} f(w_t) = 2\langle T(v), u\rangle$. So $\langle T(v), u\rangle = 0$ for all $u \perp v$.
Thus $T(v) \perp v^\perp$, meaning $T(v) \in (v^\perp)^\perp = \text{span}(v)$. So $T(v) = \lambda v$ for some $\lambda$.
**(3)** We have $T(S) = \{w : \|T^{-1}(w)\| = 1\} = \{w : \langle T^{-1}(w), T^{-1}(w)\rangle = 1\}$.
Let $P = (T^{-1})^* T^{-1} = (T^*)^{-1} T^{-1} = (TT^*)^{-1}$. Equivalently, $P = (T^*T)^{-1}$ if we set $P = (T^{-1})^* T^{-1}$.
Actually: $\|T^{-1}(w)\|^2 = \langle T^{-1}(w), T^{-1}(w)\rangle = \langle (T^{-1})^* T^{-1}(w), w\rangle$.
The operator $Q = (T^{-1})^* T^{-1}$ is self-adjoint and positive-definite. By the spectral theorem, there exists an orthonormal eigenbasis $\{u_1, \ldots, u_n\}$ with $Q(u_i) = \mu_i u_i$, where $\mu_i > 0$.
For $w = \sum x_i u_i$: $\langle Q(w), w\rangle = \sum \mu_i x_i^2$.
So $T(S) = \{w : \sum \mu_i x_i^2 = 1\}$. Setting $a_i = 1/\sqrt{\mu_i}$, this is $\sum x_i^2/a_i^2 = 1$, an ellipsoid with semi-axes $a_i$ along $u_i$.
**(4)** We have $\|T(v)\|^2 = \langle T(v), T(v)\rangle = \langle T^*T(v), v\rangle$. The operator $T^*T$ is self-adjoint and positive-definite (since $T$ is invertible).
By the spectral theorem, let $\{e_i\}$ be an orthonormal eigenbasis with $T^*T(e_i) = \lambda_i e_i$, $\lambda_i > 0$. For $v = \sum a_i e_i$ with $\|v\| = 1$ (i.e., $\sum a_i^2 = 1$):

$$\|T(v)\|^2 = \langle T^*T(v), v\rangle = \sum \lambda_i a_i^2.$$

This is maximized when all weight is on the largest eigenvalue: $\sup = \lambda_{\max}$. It is minimized when all weight is on the smallest: $\inf = \lambda_{\min}$.
Thus $K(T)^2 = \frac{\lambda_{\max}}{\lambda_{\min}}$.
**(5)** For $n = 2$, let $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ with $a, c > 0$ (positive-definiteness requires positive diagonal) and $ac > b^2$.
We show $\max\{a, c\} \geqslant |b|$. Suppose not: $|b| > a$ and $|b| > c$. Then $b^2 > ac$, contradicting $ac > b^2$.
So the largest entry in magnitude is on the diagonal. Since $a, c > 0$ and $ac > b^2 \geqslant 0$, the largest entry is $\max\{a, c\}$, on the diagonal.